HP Insight Remote Support

Monitored Devices Configuration Guide

Software Version: 7.1

Document Release Date: September 2014

Software Release Date: July 2014

Legal Notices

Warranty

The only warranties for HP products and services are set forth in the express warranty statements accompanying such products and services. Nothing herein should be construed as constituting an additional warranty. HP shall not be liable for technical or editorial errors or omissions contained herein.

The information contained herein is subject to change without notice.

Restricted Rights Legend

Confidential computer software. Valid license from HP required for possession, use or copying. Consistent with FAR 12.211 and 12.212, Commercial Computer Software, Computer Software Documentation, and Technical Data for Commercial Items are licensed to the U.S. Government under vendor's standard commercial license.

Copyright Notice

Trademark Notices

Microsoft® and Windows® are U.S. registered trademarks of Microsoft Corporation.

UNIX® is a registered trademark of The Open Group.

Contents

Preface

Document Purpose and Audience

This document provides the necessary information to configure your monitored devices before discovering the devices with HP Insight Remote Support (RS).

This document is for HP Customers and HP Support Personnel who are installing, configuring, and using Insight RS.

Product Overview

Insight RS is a software solution that enables reactive and proactive remote support to improve the availability of supported servers, storage systems, and other devices in your data center. Insight RS relies on several HP components and communication between various software applications within the customer enterprise and between the customer enterprise and HP to deliver support services. Software components may be installed on the Hosting Device or on monitored devices depending on their purpose.

Insight RS can be used on its own or can be integrated with HP Systems Insight Manager (SIM).

For more information about Insight RS, go to: http://www.hp.com/go/insightremotesupport.

Important: To configure Insight RS correctly, it is essential that you read this document thoroughly *before* proceeding with the installation of Insight RS.

Related Documents

For additional Insight RS documentation, go to: http://www.hp.com/go/insightremotesupport/docs.

- *HP Insight Remote Support Release Notes*

  This document provides product details and information about which monitored devices and Hosting Devices are supported for use with the Insight RS solution.
- *HP Insight Remote Support Quick Installation Guide*

  This document provides a checklist for installing and configuring Insight RS.
- *HP Insight Remote Support Installation and Configuration Guide*

  This document provides detailed information about installing and configuring Insight RS.
- *HP Insight Remote Support Monitored Devices Configuration Guide*

  This document provides information to configure the devices that will be monitored by Insight RS.
- *HP Insight Remote Support Security White Paper*

  This document provides an overview of the security features available in Insight RS.
- *HP Insight Remote Support Upgrade Guide*

  This document provides information about upgrading Insight RS to version 7.1.
- *HP Insight Online Getting Started Guide*

  This document provides information about the prerequisites for using HP Insight Online, and explains how to use Insight Online to manage your company's HP devices, contracts and warranties.

Document Revision History

Edition	Software Version	Publication Date	Change Summary
1.2	7.1	September 2014	• Updated D2D to remove references to collection support • Clarified required MSA 1040 communication protocols
1.1	7.1	September 2014	Updated the ProLiant Gen8 server section to include information for ProLiant Gen9 servers.
1.0	7.1	July 2014	Initial release.

Sign Up to Receive Insight Remote Support Communications

The HP Insight Remote Support product team uses HP's Support Communication process to communicate important news such as Engineering Advisories, Customer Advisories, Engineering Notices, and Customer Notices.

To sign up to receive Support Communications using HP Subscribers Choice, go to: https://h30046.www3.hp.com/SubChoice/country/us/en/signin.aspx.

When you subscribe, search for *Insight Remote Support*.

HP Support Information

HP recommends you consult the Insight RS documentation to resolve issues. The documentation is designed to guide you through a successful installation and configuration. However, if you need further support for Insight RS, help is available through HP's local Response Centers. For contact details, go to: http://www.hp.com/go/rstechsupport.

Before contacting support, you can check if your issue has a solution available. Note that a valid contract and HP Passport log on is required to view issue solution documents.

To view Insight RS issue solutions, complete the following steps:

1. Go to http://www.hp.com and browse to **Support** → **Support & troubleshooting**.
2. In the **Find by product** field, type `Insight Remote Support` and click **Go**.
3. In the search results, click **HP Insight Remote Support Next Gen Software**.
4. In the **Most viewed solutions** pane, click **View all** to see the solutions.

We Appreciate Your Feedback!

If you have comments about this document, you can contact the documentation team by email. If an email client is configured on this system, click the link above and an email window opens with the following information in the subject line:

Feedback on Insight Remote Support, 7.1 Monitored Devices Configuration Guide

Just add your feedback to the email and click send.

If no email client is available, copy the information above to a new message in a web mail client, and send your feedback to techdocs_feedback@hp.com.

Chapter 1: Understanding Monitored Devices Prerequisites

To use Insight Remote Support (RS) you need at least two pieces of hardware: a supported Windows ProLiant server to be used as the Hosting Device and one or more devices to be monitored by Insight RS. Use the table below to identify what protocols and software components are necessary for your monitored devices and then go to the information for your specific devices for complete configuration details. Before you get started, verify Insight RS supports your monitored devices by consulting the *HP Insight Remote Support Release Notes*.

Insight RS checks every device's warranty and contract to make sure it has a valid HP warranty, CarePack or contract. If a device has no HP warranty or contract, the monitoring health indicator in the Insight RS Console will be red. If this is red, then no service events will be analyzed or sent to HP.

Identify Required Communication Protocols and Software Components

Monitored devices can communicate with the Hosting Device using one or more of the following communication protocols and software components:

- Event Log Monitoring Collector (ELMC)
- HP P6000 Command View
- Hypertext Transfer Protocol Secure (https)
- Hypertext Transport Protocol (http)
- iLO Remote Insight Board Command Language Protocol (RIBCL)
- P4000 SAN Solution (SAN/iQ)
- Secure Shell (SSH)
- Simple Network Management Protocol Version 1 (SNMPv1)
- Simple Network Management Protocol Version 2 (SNMPv2)
- Simple Network Management Protocol Version 3 (SNMPv3)
- Telnet
- VMWare VirtualCenter Web Service Interface
- Web Services Management Protocol (WSMAN)

- Web-Based Enterprise Management (WBEM)
- Windows Management Instrumentation (WMI)

If you do not configure the required protocols and software for your monitored devices, the devices will *not* communicate with the Hosting Device and your events and configuration collections will *not* be sent to HP for support.

Important: Not only do you need to configure the protocol on the monitored device, you must also add the protocol within the Insight RS Console so Insight Remote Support can access the device.

Important: If you change the username/password that Insight RS uses to connect to your monitored devices, or if the credentials expire as part of your security policies, you *must* modify those corresponding credentials in the Insight RS Console to continue monitoring your monitored devices.

The required protocols and software are determined by your monitored device type. Read the appropriate monitored device section to understand what configuration is necessary for your specific monitored device. Some monitored device types can use multiple communication protocols. Which protocol you choose depends on your security strategy and preferences.

In some cases more than one protocol will be required. Some protocols may be used by the Insight RS Console to discover monitored devices while other protocols may be used to monitor hardware events or retrieve configuration collection information from the monitored devices. The following table highlights the requirements, but read the section for your monitored device type for more detail.

Assign devices to SAN collections on the **Collection Services** → **Collection Schedules** tab of the Insight RS Console in the SAN Configuration Collection Schedule. See the Help for more information about setting up your SAN collections.

Table 1.1 ***Required Monitored Device Communication Protocols and Components for Insight RS Services***

	Discovery	Monitoring	Basic Collections	SAN Collections	Notes
HP Servers					
ProLiant Gen8 and Gen9	RIBCL	RIBCL	RIBCL		HP ProLiant Gen8 and Gen9 servers can be monitored using the embedded Remote Support functionality available through iLO 4 and Intelligent Provisioning or using a diagnostic agent installed on the operating system. Note, if you want to monitor attached devices you need to install the HP Service Pack for ProLiant as detailed in the ProLiant server section that corresponds to the operating system below. RIBCL credentials are required for all HP ProLiant Gen8 and Gen9 servers regardless if the embedded capabilities or the diagnostic agent are used. See "Configuring ProLiant Gen8 and Gen9 Servers" on page 39.
Windows on ProLiant	SNMP or WMI	SNMP or WMI	SNMP or WMI		Must install the HP Service Pack for ProLiant and all providers that ship with it. You can choose to use either SNMP or WMI, however to gather SAN Collections for a ProLiant Windows server, you must use WMI. Insight RS requires Insight Management WBEM Providers version 2.8 or later, but HP recommends installing the most recent version available for your device. See "Configuring ProLiant Windows Servers" on page 48. For configuring using WMI, see "Configuring ProLiant Windows Servers Using WMI" on page 48. For configuring using SNMP, see "Configuring ProLiant Windows Servers Using SNMP" on page 53.
				WMI	
Linux on ProLiant	SNMP	SNMP	SNMP		Must install the HP Service Pack for ProLiant and all providers that ship with it. See "Configuring ProLiant Linux Servers" on page 62.
				Telnet or SSH	

Table 1.1 ***Required Monitored Device Communication Protocols and Components for Insight RS Services, continued***

	Discovery	Monitoring	Basic Collections	SAN Collections	Notes
ESX on ProLiant	SNMP	SNMP	SNMP		See "Configuring ProLiant VMware ESX Servers" on page 69.
				SSH	
ESXi on ProLiant	WBEM	WBEM	WBEM	WBEM	See "Configuring ProLiant VMware ESXi Servers" on page 75.
Citrix Xen Server on ProLiant	SNMP	SNMP	SNMP		See "Configuring ProLiant Citrix Xen Servers" on page 79.
				Telnet or SSH	
Hyper-V on ProLiant	(See Note)				Configuration collections are not supported. Hyper-V is a role on a Windows server, so you should follow the configuration instructions for a ProLiant Windows. See "Configuring ProLiant Windows Servers" on page 48.
HP ProLiant Superdome Onboard Administrator (OA)	WS-Man	WS-Man	WS-Man		The Linux partitions are monitored through the OA using WS-Man. Collections on the partitions also use WS-Man, but Insight RS communicates directly with the partition. See "Configuring ProLiant Superdome Servers" on page 86.
Windows 2003 on Integrity	WMI	WMI	WMI	WMI	Must install the Integrity Support Pack and all providers that ship with it. This includes the installation of SNMP and WMI. See "Configuring Integrity Windows 2003 Servers" on page 92.
	SNMP	SNMP	SNMP		
		ELMC			
Windows 2008 on Integrity	WMI	WMI	WMI	WMI	Must install the HP Integrity Support Pack and all providers that ship with it. See "Configuring Integrity Windows 2008 Servers" on page 100.
Linux on Integrity	WBEM	WBEM			Configuration collections are not supported, except when the server is part of a SAN collection. You can choose to use either Telnet or SSH to gather SAN collections for an Integrity Linux server. See "Configuring Integrity Linux Servers" on page 105.
				Telnet or SSH	

Table 1.1 ***Required Monitored Device Communication Protocols and Components for Insight RS Services, continued***

	Discovery	Monitoring	Basic Collections	SAN Collections	Notes
HP-UX on Integrity	WBEM	WBEM	WBEM		Install the proper WBEM providers on HP-UX. These generally are part of the feature set of HP-UX and will update through patches and software updates on the device as appropriate.
				Telnet or SSH	You can choose to use either Telnet or SSH to gather SAN collections for an Integrity HP-UX server. See "Configuring Integrity HP-UX Servers" on page 109.
HP Integrity Superdome 2 Onboard Administrator (OA)	WS-Man	WS-Man			Discovery uses WS-Man and collections use HTTPS. The OA is monitored using WS-Man, but the Superdome 2 partitions are monitored using WBEM. See "Configuring Integrity Superdome 2 Servers" on page 125.
			HTTPS		
OpenVMS on Integrity	WBEM	WBEM	WBEM	WBEM	Must install the OpenVMS WBEM provider. Configuration collections and discovery only works when the WBEM providers are installed and the credentials are properly configured. See "Configuring OpenVMS Servers" on page 131.
		ELMC			
OpenVMS on AlphaServer	SNMP	SNMP			Configuration collections are not supported, except when the OpenVMS server is part of a SAN collection. See "Configuring OpenVMS Servers" on page 131 and the special note on support for this platform.
		ELMC			
				Telnet	
Tru64 UNIX on AlphaServer	SNMP	SNMP			Configuration collections are not supported, except when the Tru64 server is part of a SAN collection. You can choose to use either Telnet or SSH to gather SAN Collections for a Tru64 Unix server. See "Configuring Tru64 UNIX Servers" on page 141 and the special note on support for this platform.
		ELMC			
				Telnet or SSH	

Table 1.1 ***Required Monitored Device Communication Protocols and Components for Insight RS Services, continued***

	Discovery	Monitoring	Basic Collections	SAN Collections	Notes
E5000 Messaging Systems	SNMP or WMI	SNMP or WMI	SNMP or WMI		These devices should be configured the same as a Windows ProLiant server.
				WMI	You can choose to use SNMP or WMI, however to gather SAN Collections, you must use WMI.
			HTTP		Configuration collections for the chassis connector use HTTP. See "Configuring ProLiant Windows Servers" on page 48.
Multivendor Servers					
Windows on IBM Servers	SNMP	SNMP			Configuration collections are not supported. See "Configuring IBM Servers" on page 148.
Windows on Dell PowerEdge	SNMP	SNMP			Configuration collections are not supported. See "Configuring Dell PowerEdge Servers" on page 158.
HP BladeSystems					
HP Onboard Administrator for c-Class Enclosures	SNMP	SNMP			HP BladeSystem c-Class Enclosures can be monitored using the embedded management capabilities in the HP Onboard Administrator (OA) or by configuring SNMP in the OA. See "Configuring BladeSystem c-Class Enclosures" on page 164.
			HTTPS		For configuring the OA, see "Configuring BladeSystem c-Class Enclosures Through the OA" on page 164. For configuring using SNMP, see "Configuring BladeSystem c-Class Enclosures Using SNMP" on page 168.

Table 1.1 ***Required Monitored Device Communication Protocols and Components for Insight RS Services, continued***

	Discovery	Monitoring	Basic Collections	SAN Collections	Notes
HP ProLiant BladeSystem Servers	(See Notes)				Supported ProLiant BladeSystem servers are monitored independent of the c-Class Enclosure in which they are installed. ProLiant BladeSystem servers can be configured using the instructions for ProLiant servers. For configuration details, see the ProLiant server section that corresponds to the operating system installed on the ProLiant BladeSystem server.
HP Integrity BladeSystem Servers	(See Notes)				Supported Integrity BladeSystem servers are monitored independent of the c-Class Enclosure in which they are installed. Integrity BladeSystem servers can be configured using the instructions for Integrity servers. For configuration details, see the Integrity server section that corresponds to the operating system installed on the Integrity BladeSystem server.
Storage Blades	(See Notes)				Configuration collections are not supported. Storage Blades are monitored through the Storage Agents installed on the partner server blade. The partner server blade is the adjacent server blade connected directly to the storage blade.
Tape Blades	(See Notes)				Configuration collections are notsupported. Tape Blades are monitored through the Storage Agents installed on the partner server blade. The partner server blade is the adjacent server blade connected directly to the tape blade.
HP Disk Arrays					
P6000 Enterprise Virtual Arrays SBM	HP P6000 Command View	HP P6000 Command View	HP P6000 Command View	HP P6000 Command View	For configuring with Server Based Management, see "Configuring P6000 and Enterprise Virtual Arrays Using Server Based Management" on page 173.
		ELMC			

Table 1.1 ***Required Monitored Device Communication Protocols and Components for Insight RS Services, continued***

	Discovery	Monitoring	Basic Collections	SAN Collections	Notes
P6000 Enterprise Virtual Arrays ABM	HP P6000 Command View	HP P6000 Command View	HP P6000 Command View	HP P6000 Command View	For configuring with Array Based Management, see "Configuring P6000 and Enterprise Virtual Arrays Using Array Based Management" on page 186.
HP StoreVirtual 4xxx (formerly P4000 SAN Solutions/ LeftHand Storage)	SNMP	SNMP			See "Configuring StoreVirtual P4000 Storage Systems" on page 189.
		LeftHand OS	LeftHand OS	LeftHand OS	
HP MSA 1040 HP MSA 2040	SNMP	SNMP			Both SNMP and WBEM must be installed and configured. See "Configuring P2000 G3, MSA 1040, and MSA 2040 Modular Smart Arrays" on page 205.
	WBEM	WBEM	WBEM	WBEM	
HP P2000 G3 MSA	SNMP	SNMP			Both SNMP and WBEM must be installed and configured. See "Configuring P2000 G3, MSA 1040, and MSA 2040 Modular Smart Arrays" on page 205.
	WBEM	WBEM	WBEM	WBEM	
HP MSA23xx G2	SNMP	SNMP			Install the WBEM SMI-S Proxy Provider on the host server and not the MSA itself. See "Configuring MSA23xx G2 Modular Smart Arrays" on page 211.
	WBEM		WBEM	WBEM	
HP MSA2xxx G1	SNMP	SNMP		SNMP	Configuration collections are not supported, except when the MSA2xxx G1 is part of a SAN collection. See "Configuring MSA2xxx G1 Modular Smart Arrays" on page 217.
HP MSA15xx	SNMP	SNMP	SNMP	SNMP	See "Configuring MSA15xx Modular Smart Arrays" on page 223.
StoreEasy Commercial NAS	WMI	WMI	WMI	WMI	See "Configuring StoreEasy Storage Systems" on page 227.
HP StoreAll Network Storage Systems (formerly HP IBRIX Storage)	SNMP	SNMP	SNMP		See "Configuring StoreAll Network Storage Systems" on page 232.

Table 1.1 ***Required Monitored Device Communication Protocols and Components for Insight RS Services, continued***

	Discovery	Monitoring	Basic Collections	SAN Collections	Notes
Data Protection					
HP StoreOnce Backup (D2D)	SNMP	SNMP			See "Configuring StoreOnce Backup (D2D) Systems" on page 237.
HP Virtual Library System (VLS)	SNMP	SNMP			Configuration collections are not supported, except when the VLS is part of a SAN collection. See "Configuring Virtual Library Systems" on page 241.
				SSH	
HP StoreEver MSL (formerly Modular Systems Library)	SNMP	SNMP	SNMP		See "Configuring StoreEver Tape Libraries" on page 245.
HP StoreEver ESL (formerly Enterprise Systems Library)	SNMP	SNMP			WBEM is supplied through Command View for Tape Libraries, installed on the server managing the tape libraries. See "Configuring StoreEver Tape Libraries" on page 245.
			WBEM		
				Telnet or SSH	
HP StoreEver EML (formerly Enterprise Modular Library)	SNMP	SNMP			WBEM is supplied through Command View for Tape Libraries, installed on the server managing the tape libraries. See "Configuring StoreEver Tape Libraries" on page 245.
			WBEM		
				Telnet or SSH	

Table 1.1 ***Required Monitored Device Communication Protocols and Components for Insight RS Services, continued***

	Discovery	Monitoring	Basic Collections	SAN Collections	Notes
HP StoreFabric SAN Switches					
B-Series SAN Switch	SNMP	SNMP			See "Configuring StoreFabric B-Series Switches" on page 255.
			Telnet or SSH	Telnet or SSH	
C-Series SAN Switch	SNMP	SNMP	SNMP	SNMP	See "Configuring StoreFabric C-Series Switches" on page 262.
H-Series SAN Switch	SNMP	SNMP	SNMP	SNMP	See "Configuring StoreFabric H-Series Switches" on page 266.
HP Networking Switches					
HP ProVision-based Switches (formerly E-Series/ProCurve)	SNMP	SNMP			See "Configuring ProVision-based Networking Switches" on page 269.
			Telnet or SSH		
HP ComWare-based Switches (formerly A-Series or H3C/3Com)	SNMP	SNMP			See "Configuring ComWare-based Networking Switches" on page 278.
			Telnet or SSH		
Rack and Power					
HP Rack-mountable and Tower UPS	SNMP	SNMP			Configuration collections are not supported. If your UPS uses a Management Module, see "Configuring UPS Management Modules" on page 285. If your UPS uses a Network Module, see "Configuring UPS Network Modules" on page 288.

Table 1.1 ***Required Monitored Device Communication Protocols and Components for Insight RS Services, continued***

	Discovery	Monitoring	Basic Collections	SAN Collections	Notes
HP Modular Cooling System	SNMP	SNMP			Configuration collections are not supported. See "Configuring Modular Cooling Systems" on page 293.
Applications					
VMware vCenter Server on ProLiant	VMware VirtualCenter Web Service Interface		VMware VirtualCenter Web Service Interface		Gathers configuration collections on the ESX and ESXi virtual machines in the VMware vCenter cluster. See "Configuring VMware vCenter Servers" on page 297.

Functionality Not Supported in Insight RS 7.1

The following devices and functionality were supported in 5.x versions of Insight RS, but are not supported in version 7.1:

Table 1.2 ***Insight RS 5.x Functionality Not Available in Insight RS 7.1***

Functionality	Details
Hosting Device	• All Microsoft Windows 2003 operating system versions. Note that this will *not* be supported on any Insight RS 7.x versions.
Product coverage for monitored devices	• HP Superdome SD-A and SD-B (Insight RSA only) • P9000/HP XP Disk Arrays (Insight RSA only) • HP Enterprise Secure Key Manager (Insight RSA only) • HP Secure Key Manager (Insight RSA only) • HP NonStop servers (Insight RSA only) • IBM AIX servers (Insight RSA only) • Sun Solaris servers (Insight RSA only)
Monitored environment size	• Environments with more than 2,500 devices
Mission Critical Service Delivery (Insight RSA only)	Capabilities to deliver mission critical services such as: • Revision and configuration management for HP-UX • Assessments including patch, firmware, configuration analysis for HP-UX • HP-UX System Health Check assessments • Availability analysis for HP-UX • TAM-S and CCMon Services • Unreachable Device Notification

Table 1.2 ***Insight RS 5.x Functionality Not Available in Insight RS 7.1, continued***

Functionality	Details
Products that will *not* be supported by any 7.x version	• HP Dynamic Smart Cooling • HP SAN Virtualization Services Platform • HP Modular Array • HP Enterprise Modular Array • HP Raid Array • HP Enterprise Storage Array • M-series Switches (McData)

Chapter 2: Configuring ProLiant Gen8 and Gen9 Servers

The Insight Remote Support (RS) feature available through the iLO 4 web interface provides intelligent event diagnosis and automatic secure submission of event notifications to HP and your HP Authorized Channel Partner through the Insight RS installation on the Hosting Device.

Important: If you do not want to use the Remote Support functionality available through the iLO 4 web interface, you need to configure your ProLiant server using SNMP Agents or WBEM Providers. Consult the information that corresponds to your operating system and follow the configuration instructions (see "Understanding Monitored Devices Prerequisites" on page 26).

For more information about ProLiant Gen8 servers, go to: http://www.hp.com/go/proliantgen8/docs.

Fulfill Configuration Requirements

To configure your ProLiant Gen8 and Gen9 servers to be monitored by Insight RS, complete the following sections:

Table 2.1 ***ProLiant Gen8 and Gen9 Configuration Steps***

Task	Required/ Optional	Complete?
Make sure Insight RS supports your ProLiant server by checking the *HP Insight Remote Support Release Notes*.	Required	
Make sure the operating system on the ProLiant server is running.	Required	
Install and configure the Agentless Management Service (AMS) on the ProLiant server if you want your host name or IP displayed instead of the device's serial number. For more information, see Known Issue "Missing IP Address During Discovery" on page 47.	Optional	
Verify a supported version of the iLO firmware is installed on your ProLiant server: • For HP ProLiant Gen8 servers: Version 1.10 or later is required. • For HP ProLiant Gen9 servers: Version 2.00 or later is required. You can download the latest firmware from the HP website: www.hp.com/support/ilo4.	Required	
Add the RIBCL protocol to the Insight RS Console.	Required	
Discover the ProLiant server in the Insight RS Console.	Required	
Send a test event to verify connectivity between your ProLiant server and Insight RS.	Required	

Configure Monitored Devices

To configure your monitored devices, complete the following section:

Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create a RIBCL Protocol Credential in the Insight RS Console

To enable Insight RS to gather Active Health Service collections, you must set up the iLO Remote Insight Board Command Language (RIBCL) protocol credentials in the Insight RS Console.

To configure RIBCL in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the **Select and Configure Protocol** drop-down list, select **iLO Remote Insight Board Command Language Protocol (RIBCL)**.
4. Click **New**. The New Credential dialog box appears.
5. Fill in the required fields. Use the Username and Password of the iLO 4 web interface on the ProLiant Gen8 server.
6. Click **Add**.

Add a RIBCL protocol credential for each unique username and password on your ProLiant Gen8 servers.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the ProLiant Gen8 or Gen9 Server

Enable the ProLiant Gen8 or Gen9 server for Insight Remote Support in one of two ways:

- Discover the ProLiant server from the Insight RS Console.

  See "Discovering the ProLiant Server from the Insight RS Console" on the next page.
- Register Remote Support from the ProLiant server iLO 4 web interface.

  See "Enabling Insight Remote Support for a ProLiant Server" on the next page.

Note: You can also register for remote support through the Intelligent Provisioning interface. For more information, see *HP Insight Remote Support and Insight Online Setup Guide for ProLiant Servers and BladeSystem c-Class Enclosures* at: http://www.hp.com/go/insightremotesupport/docs.

Discovering the ProLiant Server from the Insight RS Console

To discover the ProLiant server from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP address(es) of the devices to be discovered. Use the IP address of the iLO.
   d. Click **Add**.
4. Click **Start Discovery**.

Note: After discovery, the operating system for the ProLiant server will not appear in the Insight RS Console. This is expected behavior.

Enabling Insight Remote Support for a ProLiant Server

With the HP iLO Management Engine, no additional protocols need to be installed on the ProLiant server. You only need to register the ProLiant server with your Hosting Device to enable Insight Remote Support. You can register Insight Remote Support for a ProLiant server in the iLO 4 web interface.

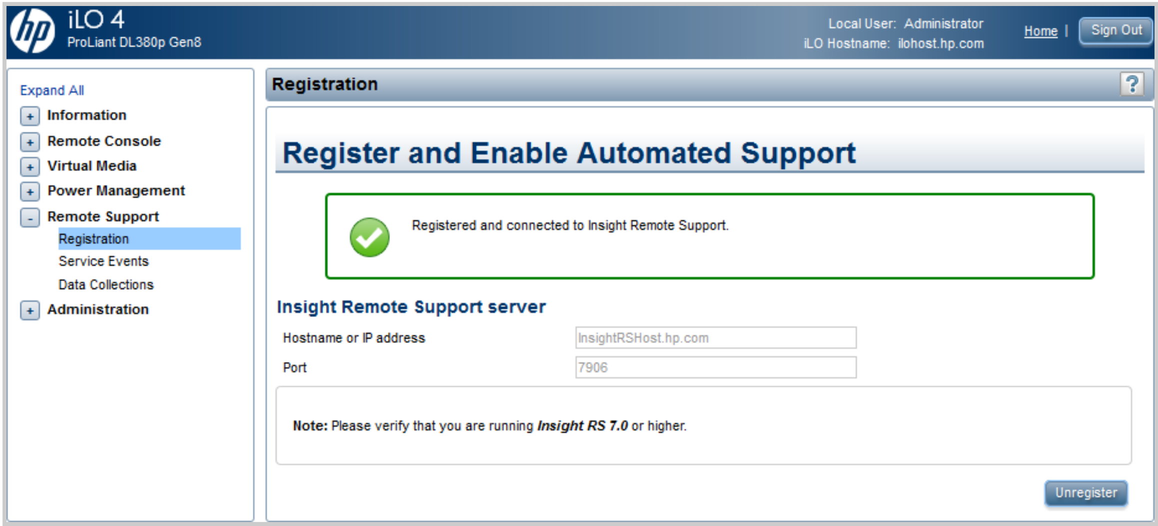
Important: When setting up your iLO 4 web interface, it is important to set the time zone. If the time zone is not set, events sent to Insight Remote Support are displayed with a GMT offset. In the iLO 4 web interface, browse to **Administration** → **Network** → **SNTP Settings** to set your time zone.

Use the **Remote Support** → **Registration** screen to register and enable Remote Support. You must have the Configure iLO Settings privilege to modify the Remote Support settings.

To enable a ProLiant server for Insight Remote Support, complete the following steps:

1. Log on to the iLO 4 web interface (`https://`*`<iLO hostname or IP address>`*).
2. In the navigation menu, click **Remote Support** → **Registration**. The Register and Enable Automated Support screen appears.
3. Complete the Hostname or IP address field.
4. In the Port field, type `7906`.

5. Click **Register**. The iLO 4 web interface registers the device and a confirmation message appears.



Note: After registration, you may see both the ProLiant server and the iLO 4 web interface listed in the Insight RS Console. The iLO 4 web interface will disappear if the ProLiant server and the RIBCL credentials have been configured correctly.

Note: After discovery, the operating system for the ProLiant server will not appear in the Insight RS Console. This is expected behavior.

Test Communication from the ProLiant Gen8 or Gen9 Server to Insight RS

To verify communication between your monitored device and Insight RS, complete the following sections:

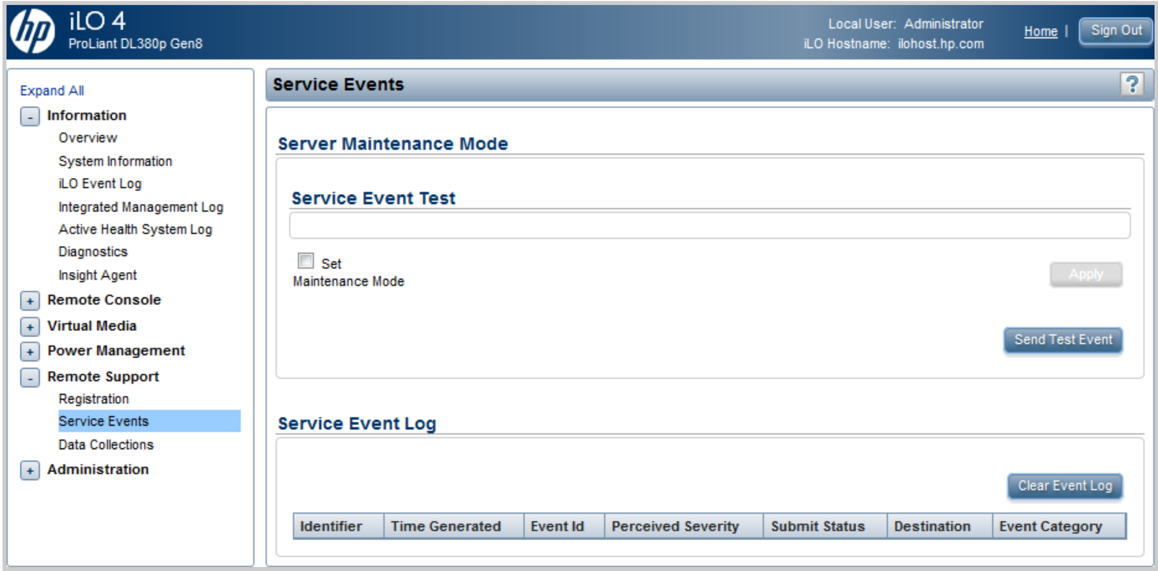
Verify Service Event Monitoring

After registering the ProLiant server with Insight Remote Support, send a test event to confirm the connection.

Note: Sending SNMP test traps from the iLO 4 web interface **Administration** → **Management** page is not supported in Insight Remote Support. To send a test event, do so from the **Remote Support** → **Service Events** page as shown below.

To send a test event, complete the following steps:

1. Log on to the iLO 4 web interface (`https://<iLO hostname or IP address>`).
2. In the navigation menu, click **Remote Support** → **Service Events**. The Service Events screen appears.



3. Click **Send Test Event**. The following message appears:

   `Are you sure you want to send a test event?`

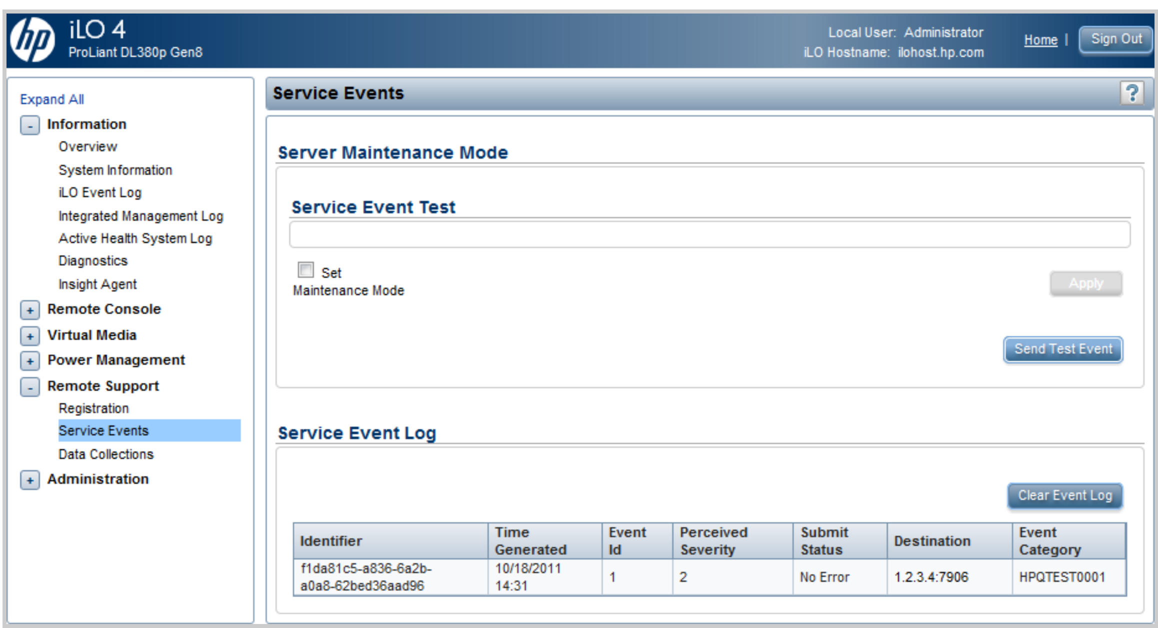
4. Click **OK** to confirm that you want to send the test event. The following messages appear:

   `Test Service Event Transmission has been initiated`

   `Service Event transmission in progress.`

   When the transmission is complete, the test event is listed in the Service Event Log and in the Insight RS Console. If the test is successful, the Submit Status column displays the text *No Error*.

   The Time Generated column in the Service Event Log shows the date and time based on the configured iLO time zone.



5. Check the Insight RS Console to verify that the test event arrived:
   a. Log on to the Insight RS Console.
   b. In the main menu, select **Devices**.
   c. Find the ProLiant server and click the device name.
   d. Click the **Service Events** tab. All service events submitted against the system display in the Service Events screen (even if you clear the Service Event Log).

      Insight RS converts the iLO service event Time Generated value to the time zone of the browser used to access the Insight RS Console.

      Note: Test events are automatically closed by HP since no further actions are required.

Initiate a Data Collection

You can test your collections from the iLO 4 web interface or the Insight RS Console. Collections are automatically generated during Discovery. You do not need to manually run collections unless you want to test your configuration.

Note: Insight Remote Support will run a data collection when the ProLiant server is discovered. If you manually generate a data collection using one of the methods described below, you will have two data collections shown for the device in the Insight RS Console.

Initiating a Data Collection in the iLO 4 Web Interface

To initiate a data collection in the iLO 4 web interface, complete the following steps:

1. Log on to the iLO 4 web interface (`https://<iLO hostname or IP address>`).
2. Navigate to **Remote Support** → **Data Collections**. The Data Collections screen displays.
3. Click **Send Data Collection**.
4. Click **OK** to confirm the action. When the transmission completes, the Last Data Collection Transmission date and time updates.

iLO sends the data collection to Insight RS. In the Insight RS Console you can view the results on the Collection Services → Basic Collection Results screen.

Initiating a Data Collection in Insight Remote Support

To initiate a data collection from Insight Remote Support, complete the following steps:

1. Log on to the Insight RS Console.
2. Navigate to **Collection Services** → **Collection Schedules**.
3. In the List of Collection Schedules pane, select **Server Basic Configuration Collection Schedule**.
4. In the Schedule Information pane, click **Run Now**.

Insight Remote Support runs the schedule. You can view the results on the Basic Collection Results tab.

Send an Active Health System Report

Note: Insight Remote Support runs an Active Health System report when the ProLiant server is discovered. If you send an Active Health System report using one of the methods described below, two Active Health System reports appear for the device in the Insight RS Console.

Sending an Active Health System Report from the iLO 4 Web Interface

To send an Active Health System report from the iLO 4 web interface, complete the following steps:

1. Log on to the iLO web interface (`https://<iLO hostname or IP address>`).
2. Navigate to **Remote Support** → **Data Collections**. The Data Collections screen displays.
3. Click **Send Active Health System Report**.
4. Click **OK** to confirm the action. When the transmission completes, the Last Active Health System Reporting Transmission date and time updates.

iLO sends the Active Health System report to Insight RS. In the Insight RS Console you can view the results on the Collection Services → Basic Collection Results screen.

Sending an Active Health System Report from Insight Remote Support

To send an Active Health System report from the Insight RS Console, complete the following steps:

1. Log on to the Insight RS Console.
2. Navigate to **Collection Services** → **Collection Schedules**.
3. In the List of Collection Schedules pane, select **AHS Collection**.
4. In the Schedule Information pane, click **Run Now**.

Insight Remote Support runs the schedule. You can view the results on the Basic Collection Results tab.

For maintenance and troubleshooting tasks, expand the following section:

Maintenance and Troubleshooting

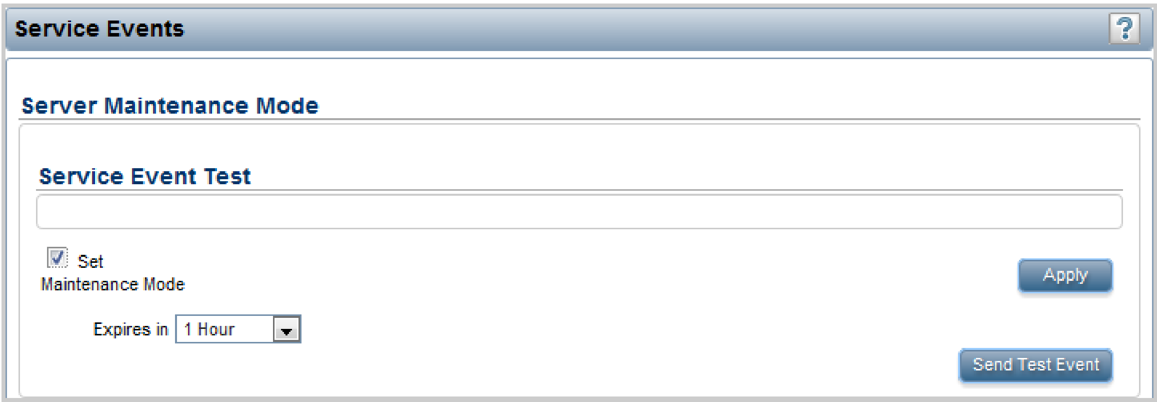
The following maintenance tasks are available for ProLiant Gen8 and Gen9 servers:

Set Maintenance Mode

Make sure to turn Maintenance Mode on when performing maintenance work on the ProLiant erver. When in maintenance mode, any events or messages that are sent to Insight Remote Support are flagged to indicate no action needs to be taken for the event.

To set maintenance mode, complete the following steps:

1. In the navigation menu, click **Remote Support** → **Service Events**. The Service Events screen displays.
2. In the Server Maintenance Mode section, select the **Set Maintenance Mode** check box. The **Expires in** drop-down list appears.



3. In the **Expires in** drop-down list, select the length of time you will work on the ProLiant server.
4. Click **Apply**.

Note: Maintenance Mode ends once the specified time has elapsed. You can end maintenance mode early by selecting the **Clear Maintenance Mode** check box and clicking **Apply**. A confirmation message appears.

Disable Monitoring of a ProLiant Gen8 and Gen9 Server

There may be a reason that you need to disconnect a ProLiant server so that is no longer recognized by Insight Remote Support. For example, if the server's warranty has expired or if you have recently changed your Hosting Device.

If you want to temporarily disable monitoring of your HP ProLiant server, disable the device in the Insight RS Console. If you want to permanently disable monitoring of your HP ProLiant server, delete the device in the Insight RS Console.

To temporarily disable monitoring, complete the following steps:

Note: Unregistering directly from the iLO 4 web interface is the same as temporarily disabling the device in the Insight RS Console.

1. In the Insight RS Console, navigate to **Devices** → **Device Summary**.
2. Select the check box in the far left column for the devices you want to disable.
3. Click **Actions** → **Disable Selected**, and click **OK** in the confirmation dialog box.

To permanently disable monitoring, complete the following steps:

1. In the Insight RS Console, navigate to **Devices → Device Summary**.
2. Select the check box in the far left column for the devices you want to delete.
3. Click **Actions → Delete Selected**, and click **OK** in the confirmation dialog box.

The following known issues apply to ProLiant Gen8 and Gen9 servers:

Missing IP Address During Discovery

After discovery of a ProLiant server through iLO 4, the server might identify itself by its serial number (indicated by an S/N prefix) in the Insight RS Console Device Name column.

To display the ProLiant server by host name or IP address, complete the following steps:

1. Verify that the following prerequisites are met:
   - **iLO firmware**—iLO firmware version 1.10 or later is installed.
   - **AMS**—The Agentless Management Service is enabled and the operating system is running.
   - **RIBCL**—The iLO Remote Insight Board Common Language Protocol (RIBCL) credentials for the server are configured in the Insight RS Console and are assigned to the ProLiant server.
2. Log on to the Insight RS Console.
3. Navigate to **Devices → Device Summary**.
4. Click the serial number in the Device Name column.
5. On the **Device** tab, expand the **Status** section and click **Discover Device**. When the device is rediscovered host name or IP address is displayed.

   Note: The DNS configuration determines whether the host name or IP address is displayed.

Warranty and Contract Discrepancy in Insight Online

A discrepancy in Insight Online can occur between the Support Status in the Linked Warranties section and the Support Status at the top of the screen. The ProLiant Serial Number used has a valid warranty, and the Linked Warranty section confirms the warranty is `Active`. However the Support Status at the top of the screen is listed as `Expired`.

This discrepancy occurs when the site address set during Insight Remote Support configuration is not a valid address. Without a valid address, Insight Online is not able to determine the country code so it attempts to obtain the country code from the HP Passport (HPP) profile to be used for entitlement. However, since the country code is not a requirement in HPP, it might be missing. In this case, when the country code is missing during entitlement checking, a status of `Expired` is returned.

To resolve this issue, ensure that the customer address set in the Insight RS Console on the **Company Information → Sites** tab is valid, with no abbreviations for street or city names, and make sure this site information is applied to the device that is experiencing this discrepancy.

Chapter 3: Configuring ProLiant Windows Servers

Insight Remote Support (RS) must be able to communicate with your ProLiant server before it can be monitored. Insight RS can communicate with ProLiant servers running Windows with either WMI or SNMP. The following information describes how to install and configure the communication protocols and other recommended software components so that it can be monitored by Insight RS.

Use these configuration instructions for Windows 2003, Windows 2008, and Windows 2012.

If both SNMP and WMI are available on the ProLiant Windows monitored device it is recommend that one of the protocols be disabled from monitoring in the Insight RS Console to prevent dual notification of a single failing component. Disable SNMP if there are no smart detached storage devices such as MSAs attached to the monitored device. If smart detached storage devices are installed then disable the WMI protocol in the Insight RS Console. This is done by deleting the protocol for the monitored device in the Insight RS Console.

Important: To gather SAN Collections for a ProLiant Windows server, you must use WMI and Service Pack for ProLiant version 2.8 or later.

To configure your ProLiant Windows server using WMI, see "Configuring ProLiant Windows Servers Using WMI" below.

To configure your ProLiant Windows server using SNMP, see "Configuring ProLiant Windows Servers Using SNMP" on page 53.

Configuring ProLiant Windows Servers Using WMI

Insight Remote Support (RS) must be able to communicate with your ProLiant server before it can be monitored. Insight RS can communicate with ProLiant servers running Windows with WMI. The following information describes how to install and configure WMI and other recommended software components so that it can be monitored by Insight RS.

These configuration instructions can be used for Windows 2003, Windows 2008, and Windows 2012.

Fulfill Configuration Requirements

To configure your ProLiant Windows servers using WMI to be monitored by Insight RS, complete the following sections:

Table 3.1 ***ProLiant Windows Server Using WMI Configuration Steps***

Task	Complete?
Check the *HP Insight Remote Support Release Notes* to make sure your ProLiant Windows server is supported.	
Install WBEM/WMI on the ProLiant Windows server, available from the Service Pack for ProLiant (SPP).	
Add the WMI protocol to the Insight RS Console.	

Table 3.1 ***ProLiant Windows Server Using WMI Configuration Steps, continued***

Task	Complete?
Discover the ProLiant Windows server in the Insight RS Console.	
Send a test event to verify connectivity between your ProLiant Windows server and Insight RS.	

Install and Configure Communication Software on Servers

To configure your monitored devices, complete the following sections:

Install and Configure WMI

Install Service Pack for ProLiant

The Service Pack for ProLiant (SPP) is a software bundle that includes drivers, utilities, and management agents for ProLiant device(s). SPP is included on the HP SmartStart CD that ships with every ProLiant. The most current version is also available at: http://www.hp.com/go/spp/download.

HP ProLiant systems running Microsoft Windows are supported by Insight Management Providers (IM Providers for WBEM/WMI Mapper support). Both providers and agents are available through the same SPP media.

Important: To gather SAN Collections for a ProLiant Windows server, you must use WMI and Service Pack for ProLiant version 2.8 or later.

If SNMP is available on the ProLiant Windows monitored device it is recommend that it be disabled from monitoring in the Insight RS Console to prevent dual notification of a single failing component. This is done by deleting the SNMP protocol for the monitored device in the Insight RS Console.

Install WMI

IM Providers on the SmartStart CD and/or in the SPP version 8.1 and higher are supported for Insight Remote Support.

See the *HP Insight Management WBEM Providers* web site at: http://h18013.www1.hp.com/products/servers/management/wbem/ for more information about WBEM providers, security, and credentials.

If you are using the Windows WBEM Providers (IM Providers), WBEM credentials must be set for each monitored device within the Insight RS Console before it can be monitored. If you change your WBEM credentials at any time, you *must* modify the entry for those credentials in the Insight RS Console.

The WBEM IM Providers are optional in the SPP install. Make sure you select them while installing SPP.

Install System Management Homepage

System Management Homepage (SMH) is also part of the SPP. It provides additional reporting capabilities on the monitored device itself. While not mandatory for Insight Remote Support, SMH can be used to work with the WBEM protocol such as sending test events.

Disabling User Account Control in Windows 2008

In Windows 2008, WMI fails to connect to the namespace root\WMI if User Account Control (UAC) is enabled. Use one of the following procedures to disable UAC, depending on which version of Windows 2008 you are using. To perform this procedure, you must be logged on as a local administrator or provide the credentials of a member of the local Administrators group.

To turn off UAC in Windows 2008, complete the following steps:

1. Click **Start**, and then click **Control Panel**.
2. In Control Panel, click **User Accounts**.
3. In the User Accounts window, click **User Accounts**.
4. In the User Accounts tasks window, click **Turn User Account Control on or off**.
5. If UAC is currently configured in Admin Approval Mode, the User Account Control message appears. Click **Continue**.
6. Clear the **Use User Account Control (UAC) to help protect your computer** check box and click **OK**.
7. Click **Restart Now** to apply the change right away, or click **Restart Later**, then close the User Accounts tasks window.

To turn off UAC in Windows 2008 R2 and Windows 2012, complete the following steps:

1. Click **Start**, and then click **Control Panel**.
2. In Control Panel, click **User Accounts**.
3. In the User Accounts window, click **User Accounts**.
4. In the User Accounts tasks window, click **Change User Account Control settings**.
5. If the User Account Control dialog box appears, make sure that the action it displays is what you want, and then click **Yes**.
6. On the User Account Control Settings page, disable UAC: move the slider to **Never notify** and click **OK**.
7. Restart the server to apply the changes.

Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create a WMI Protocol Credential in Insight RS Console

To configure WMI in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the **Select and Configure Protocol** drop-down list, select **Windows Management Instrumentation (WMI)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Username and Password you have configured on your device.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP address(es) of the devices to be discovered.
   d. Click **Add**.
4. Click **Start Discovery**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon (✔).

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Communication Between Monitored Device and Insight RS

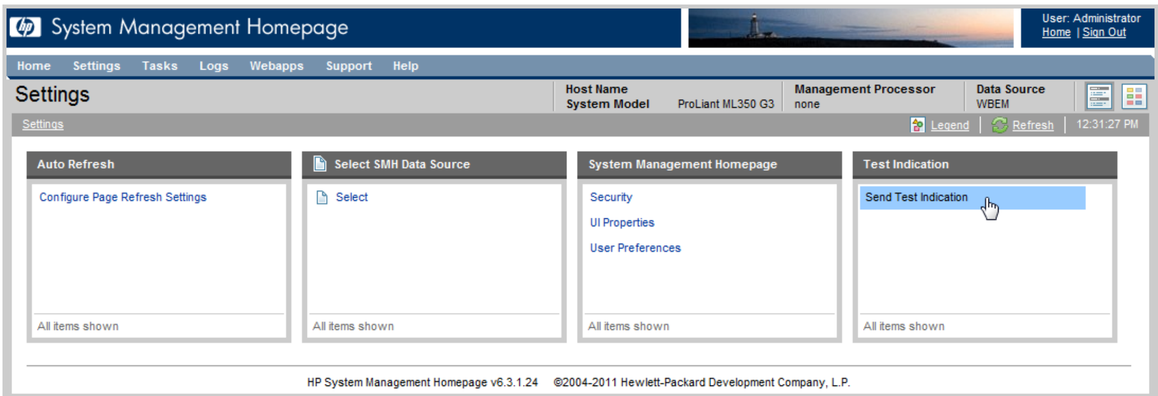
To verify communication between your monitored device and Insight RS, complete the following sections:

Send a WBEM Test Indication to the Hosting Device

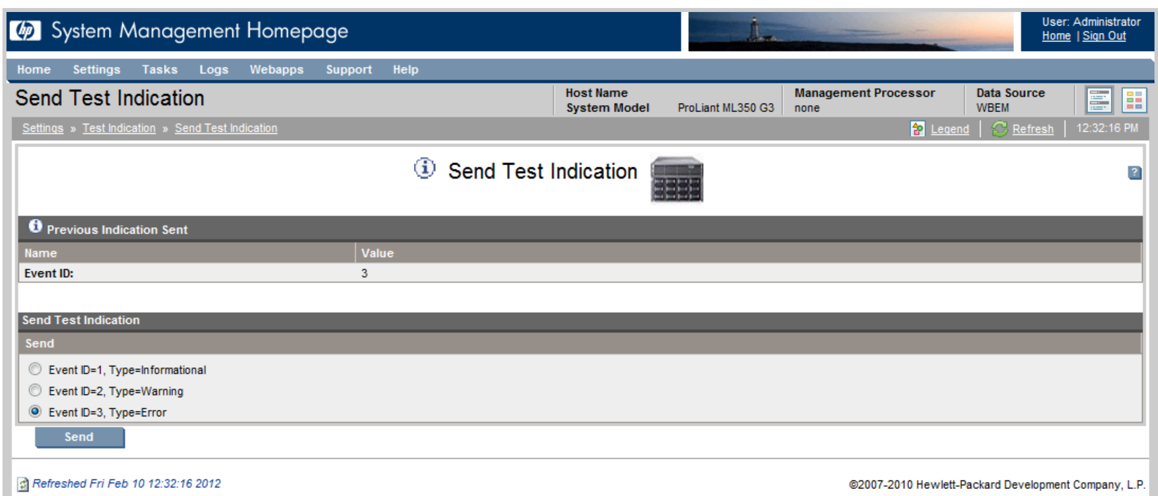
To verify connectivity from the monitored device to the Hosting Device, send a WBEM test indication to the Hosting Device and then verify the test indication was received in the Insight RS Console.

1. In a web browser, access System Management Homepage (SMH) on the monitored device: `https://[ipaddress]:2381`.
2. Log on using the administrator user name and password for the monitored device.

   If you are not prompted for a logon, check the upper right corner of the SMH interface and click the **Sign In** link. If you are not logged in as an administrator for the monitored device you will not have all of the relevant configuration options.
3. In the top menu bar, click **Settings**.



4. If you chose to install WBEM with SPP, it will be set as your Data Source. In the Test Indication pane, click **Send Test Indication**.



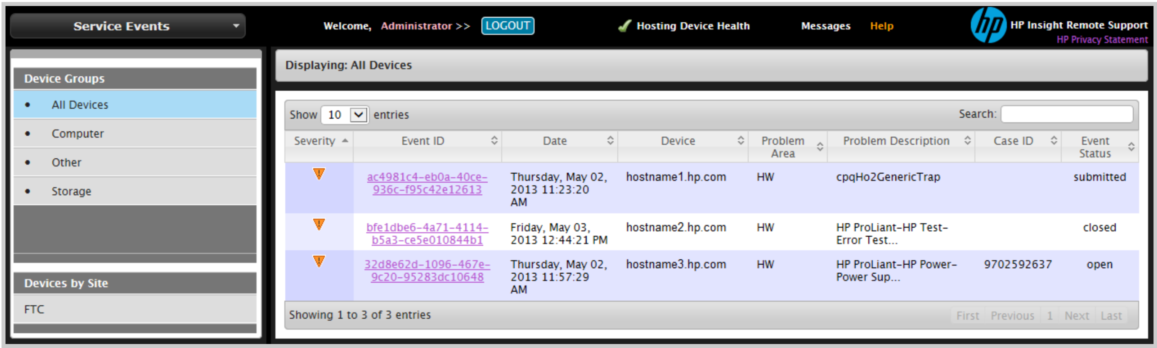
5. In the **Send Test Indication** screen, select an Event ID type (any will work) and click **Send**.

Viewing Test Events in the Insight RS Console

Some monitored device types allow you to send a test event to Insight Remote Support. After you configure your monitored device and send a test event, use the following process to verify the test event arrived.

1. Log on to the Insight RS Console.
2. In the main menu, select **Service Events**. If your monitored device is properly configured, the event

will appear in the Service Events Information pane.



Verify Collections in the Insight RS Console

After discovery, a Server Basic Configuration Collection is automatically run for servers. Verify that this collection ran successfully on the Collection Services → Basic Collection Results tab in the Insight RS Console. Note that there is a delay between when discovery finishes and when the collection runs. For more information about scheduling collections, see the Insight RS Help.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services** and then click the **Basic Collection Results** tab.
3. Expand the Server Basic Configuration Collection section.
4. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears ( ). If it failed, an error icon appears ( ).

Maintenance and Troubleshooting

The following troubleshooting tasks are available for ProLiant Windows servers that use WMI:

Device Health in Insight Remote Support

If you are using SNMP to support your Windows server, and the server has the WMI service turned on, the device health in the Insight RS Console can display an error icon ( ) if invalid WMI credentials have been associated with the server in the Insight RS Console.

If this occurs, then the server's health displays an error icon because Insight RS cannot set up WMI. To resolve this issue, complete one of the following steps:

- Remove the default WMI credentials and only associate them with the devices that are known to have the WMI providers installed.

  or

- Correct the WMI credentials attached to the server in the Insight RS Console.

Configuring ProLiant Windows Servers Using SNMP

Insight Remote Support (RS) must be able to communicate with your ProLiant server before it can be monitored. Insight RS can communicate with ProLiant servers running Windows with SNMP. The following information describes how to install and configure the communication protocols and other recommended software components so that it can be monitored by Insight RS.

Use these configuration instructions for Windows 2003, Windows 2008, and Windows 2012.

Fulfill Configuration Requirements

To configure your ProLiant Windows servers using SNMP to be monitored by Insight RS, complete the following sections:

Table 3.2 ***ProLiant Windows Server Using SNMP Configuration Steps***

Task	Complete?
Check the *HP Insight Remote Support Release Notes* to make sure your ProLiant Windows server is supported.	
Install SNMP on the ProLiant Windows server, available from the Service Pack for ProLiant (SPP).	
Configure SNMP on the ProLiant Windows server.	
Add the SNMP protocol to the Insight RS Console.	
Discover the ProLiant Windows server in the Insight RS Console.	
Send a test event to verify connectivity between your ProLiant Windows server and Insight RS.	

Install and Configure Communication Software on Servers

To configure your monitored devices, complete the following sections:

Install and Configure SNMP Agents

Install Service Pack for ProLiant

The Service Pack for ProLiant (SPP) is a software bundle that includes drivers, utilities, and management agents for ProLiant device(s). SPP is included on the HP SmartStart CD that ships with every ProLiant. The current version is also available at: http://www.hp.com/go/spp/download.

HP ProLiant systems running Microsoft Windows are supported by Insight Management Agents (IM Agents for SNMP support). Both providers and agents are available through the same SPP media.

If WBEM is available on the ProLiant Windows monitored device it is recommend that it be disabled from monitoring in the Insight RS Console to prevent dual notification of a single failing component. This is done by deleting the WMI protocol for the monitored device in the Insight RS Console.

If you are using the Windows SNMP Agents (IM Agents), you must configure SNMP to communicate with the Hosting Device (see "Configuring SNMP").

Install System Management Homepage

System Management Homepage (SMH) is also part of the SPP. It provides additional reporting capabilities on the monitored device itself. While not mandatory for Insight Remote Support, if you have not configured your SNMP services correctly to communicate with the Hosting Device during Insight

Management Agent installation you can use SMH to reconfigure those settings. If you have installed both the default SNMP configuration and the optional WMI configuration while installing the SPP, then SMH will default to the WMI configuration and you may need to reset it to the SNMP configuration if you choose to get your hardware events through SNMP.

Configure SNMP

Your monitored devices must be configured to communicate with the Hosting Device. If you choose to use SNMP the following steps are required to allow the monitored devices to fully communicate with the Hosting Device.

Monitored devices that use SNMP notifications must include the following:

- All monitored devices must have a working intranet connection, such as through an Ethernet adapter, with TCP/IP installed and running. Monitored devices must have two-way communication with the Hosting Device over this connection.
- All monitored devices need the Insight Management Agent software for problem detection and trap generation. The IM agents are distributed by HP and are designed to generate SNMP traps with information that allows for a more complete analysis.
- All monitored devices need to have the IP address of the Hosting Device host defined as a trap destination.

The Hosting Device must be able to communicate with the monitored device, but by default, Windows Server only accepts SNMP packets from the localhost. To configure Windows Server to send traps to the Hosting Device, complete the following steps:

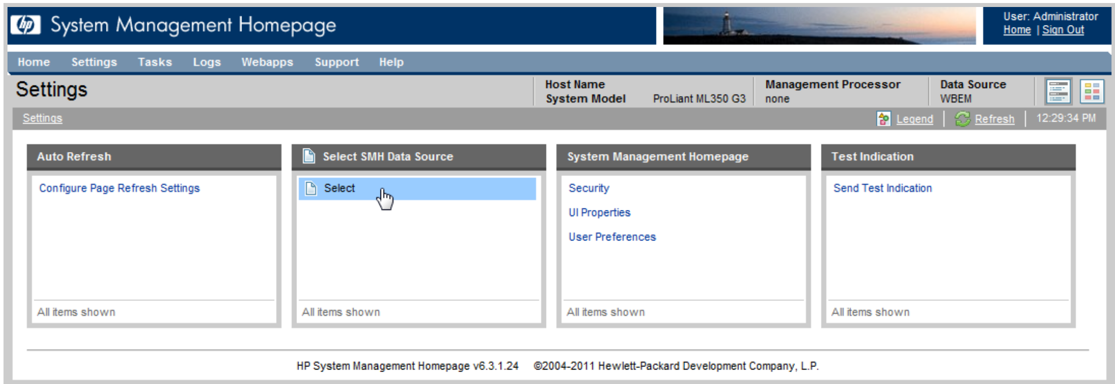
1. In a web browser, access System Management Homepage (SMH) on the monitored device: `https://[ipaddress]:2381`.
2. Log on using the administrator user name and password for the monitored device.

   If you are not prompted for a logon, check the upper right corner of the SMH interface and click the **Sign In** link. If you are not logged in as an administrator you will not have all of the relevant configuration options.
3. In the top menu bar, click **Settings**.

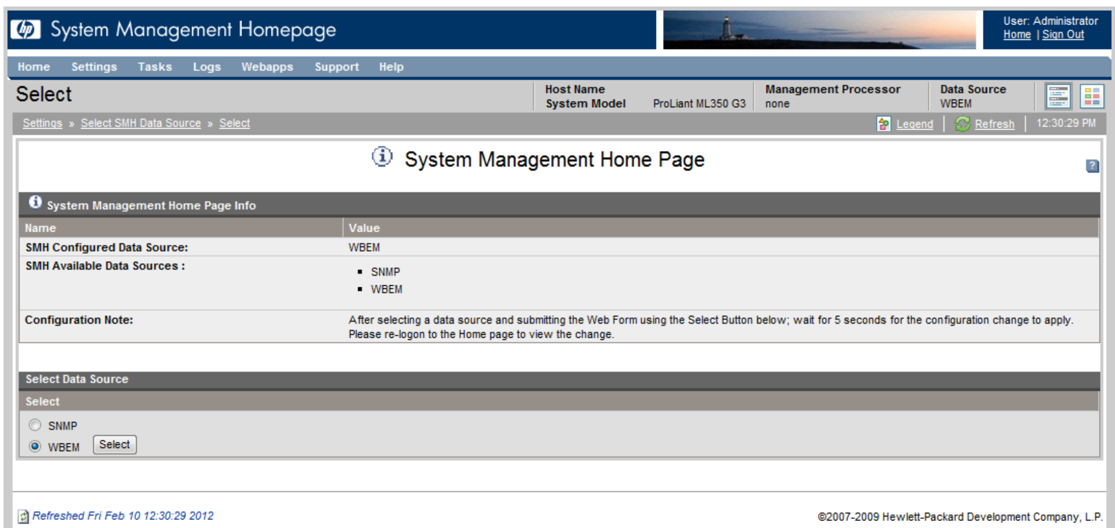
   If **Settings** displays the below image, this means that WBEM is set as the data source. You will need to first set your data source to SNMP. If SNMP is already set as your data source, skip to the next step.

   To change your data source to SNMP, complete the following steps:

a. In the **Select SMH Data Source** pane, click the **Select** link.

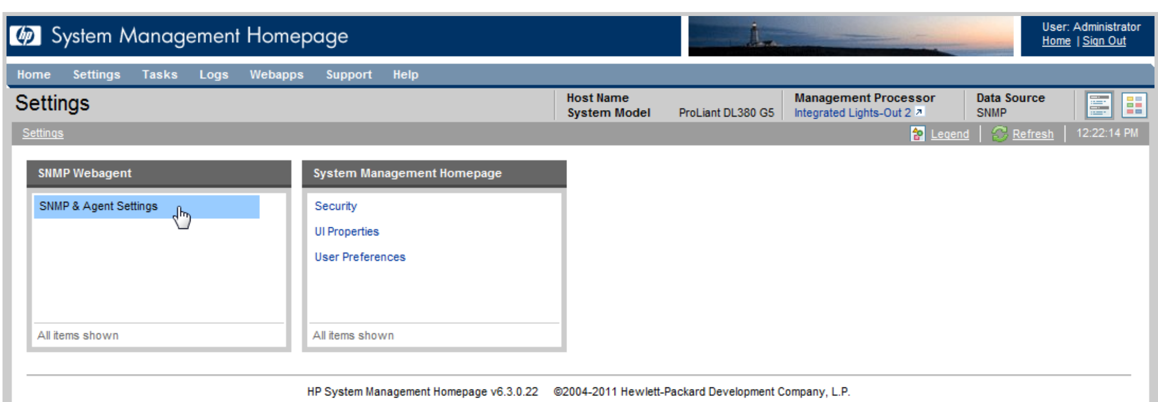


b. In the Select Data Source pane, click the SNMP option and click **Select**.

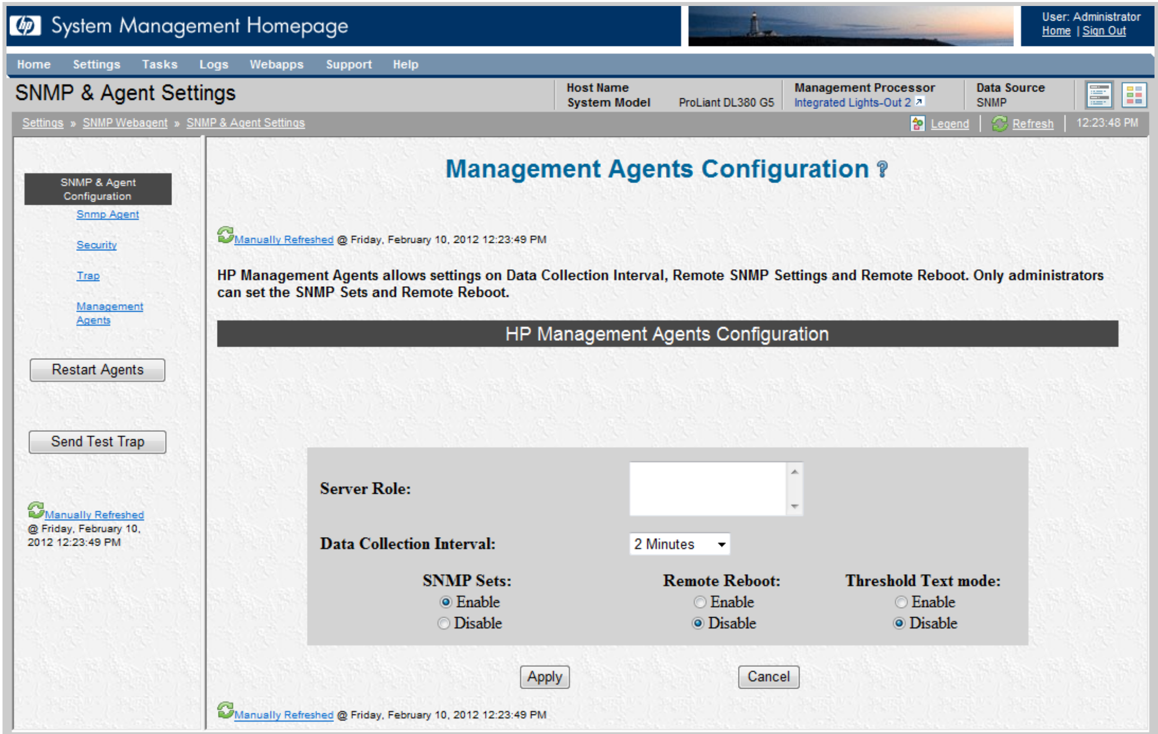


c. In the top menu bar, click **Settings**. If you changed the data source to SNMP while logged into SMH remotely, you will have to log on again and click the **Settings** tab.

4. In the **SNMP Webagent** pane, click the **SNMP & Agent Settings** link.

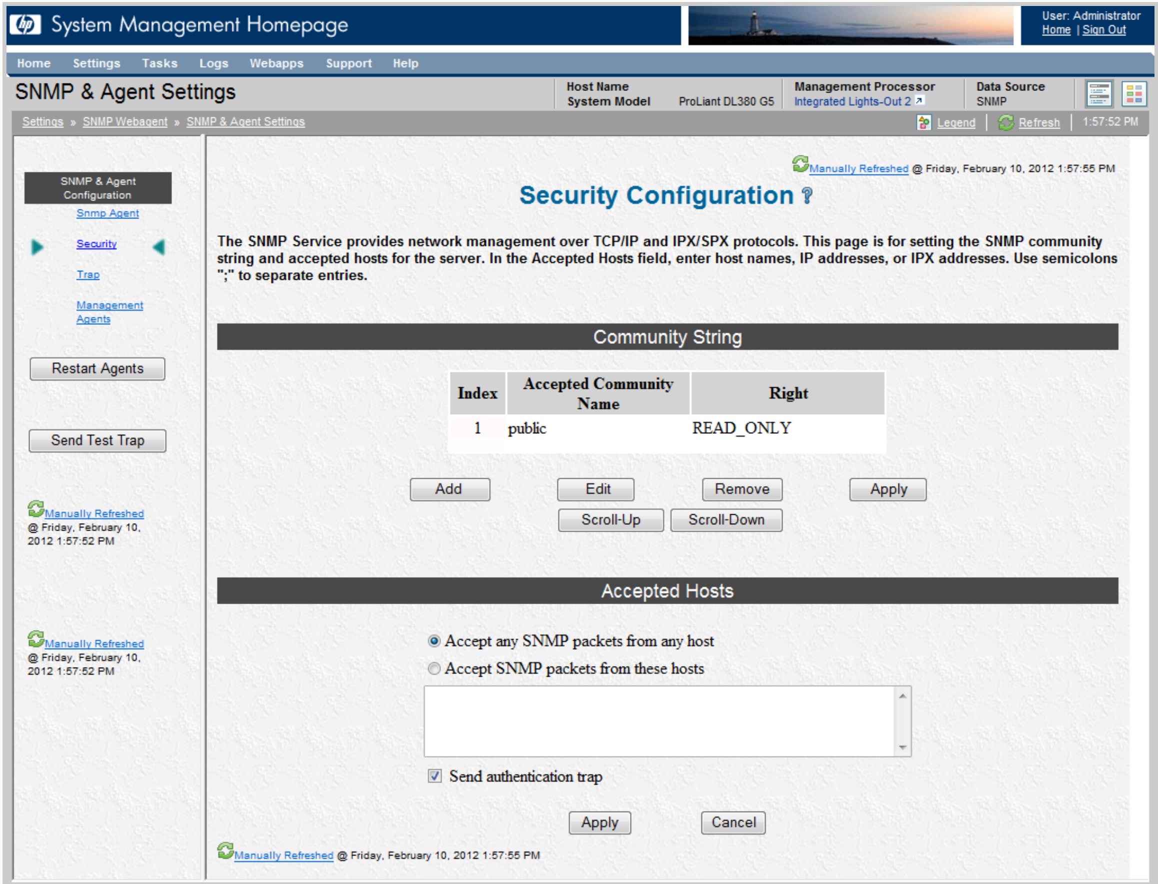


5. In the left menu of the Management Agents Configuration screen, click the **Security** link.



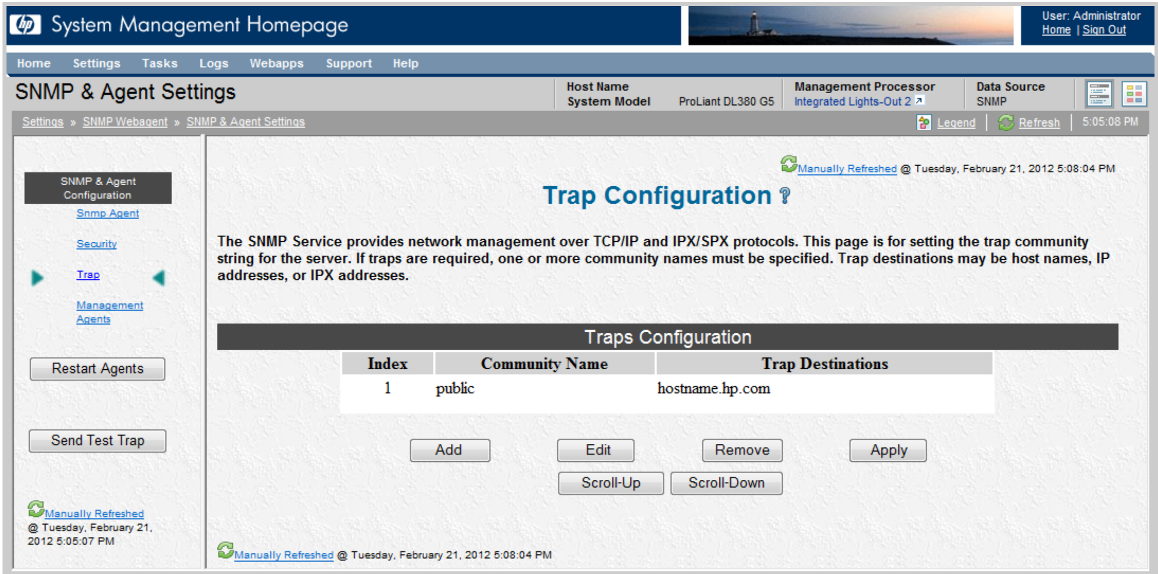
6. In the Security Configuration screen click one of the following options in the Selected Hosts section:
   - **Accept SNMP packets from any host**.

     or
   - **Accept SNMP packets from these hosts** and specify the IP address for your Hosting Device in the text area.



7. In the left menu of the Security Configuration screen, click the **Trap** link.
8. In the Trap Configuration screen, do one of the following:
   - Add a Community Name that includes the Hosting Device as a Trap Destination.

     or
   - Edit the public Community Name to include the Hosting Device as a Trap Destination.



Complete the preceding steps for each ProLiant Windows monitored device that uses SNMP to communicate with the Hosting Device.

Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create an SNMP Protocol Credential in Insight RS Console

If your device's SNMP community string is set to `public` and your community access mode is `read only`, Insight RS automatically associates an SNMP protocol with your device. If you use a different community string or use a nonstandard port, you must create an SNMP protocol credential in the Insight RS Console.

To configure SNMP in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the Select and Configure Protocol drop-down list, select **Simple Network Management Protocol** for the version of SNMP configured on your server.
4. Click **New**. The New Credential dialog box appears.
5. Type the information configured on your server.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP address(es) of the devices to be discovered.
   d. Click **Add**.
4. Click **Start Discovery**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon ().

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following sections:

Send an SNMP Test Trap to the Hosting Device

To verify connectivity from the monitored device to the Hosting Device, send an SNMP test trap to the Hosting Device and then verify the test trap was received in the Insight RS Console.

1. In a web browser, access System Management Homepage (SMH) on the monitored device: `https://[ipaddress]:2381`.
2. Log on using the administrator user name and password for the monitored device.

   If you are not prompted for a logon, check the upper right corner of the SMH interface and click the **Sign In** link. If you are not logged in as an administrator you will not have all of the relevant configuration options.
3. In the top menu bar, click **Settings**.

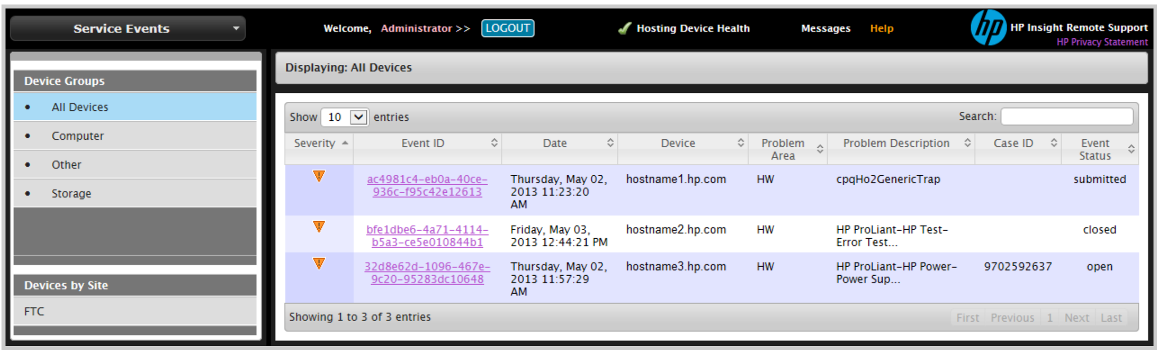
   If you installed WBEM with SPP, it may be set as your data source. If that is the case, refer to "Configure SNMP" on page 55 for information about how to set your data source to SNMP.

4. In the SNMP Webagent pane, click the **SNMP & Agent Settings** link.
5. In the left menu, click **Send Test Trap**.

Viewing Test Events in the Insight RS Console

Some monitored device types allow you to send a test event to Insight Remote Support. After you configure your monitored device and send a test event, use the following process to verify the test event arrived.

1. Log on to the Insight RS Console.
2. In the main menu, select **Service Events**. If your monitored device is properly configured, the event will appear in the Service Events Information pane.



Verify Collections in the Insight RS Console

After discovery, a Server Basic Configuration Collection is automatically run for servers. Verify that this collection ran successfully on the Collection Services → Basic Collection Results tab in the Insight RS Console. Note that there is a delay between when discovery finishes and when the collection runs. For more information about scheduling collections, see the Insight RS Help.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services** and then click the **Basic Collection Results** tab.
3. Expand the Server Basic Configuration Collection section.
4. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears (). If it failed, an error icon appears ().

Chapter 4: Configuring ProLiant Linux Servers

Fulfill Configuration Requirements

To configure your ProLiant Linux servers to be monitored by Insight RS, complete the following sections:

Table 4.1 ***ProLiant Linux Server Configuration Steps***

Task	Complete?
Make sure Insight RS supports your ProLiant Linux server by checking the *HP Insight Remote Support Release Notes*.	
Install SNMP on the ProLiant Linux server, available from the Service Pack for ProLiant (SPP).	
Configure SNMP on the ProLiant Linux server.	
Add the SNMP protocol to the Insight RS Console.	
Discover the ProLiant Linux server in the Insight RS Console.	
Send a test event to verify connectivity between your ProLiant Linux server and Insight RS.	

Install and Configure Communication Software on Servers

To configure your monitored devices, complete the following sections:

Install SNMP Agents

The Service Pack for ProLiant (SPP) is a system-specific software bundle that includes drivers, utilities, and management agents for your ProLiant device(s). The SPP is shipped with every ProLiant on the HP SmartStart CD. The most current version is also available at: http://www.hp.com/go/spp/download.

For more information about SPP, go to: http://www.hp.com/go/spp or http://www.hp.com/go/spp/documentation.

HP ProLiant systems running Linux support Insight Management Agents (IM Agents), which are delivered through the SPP. IM Agents on the SmartStart CD and/or in SPP version 7.1 and higher are supported for Insight Remote Support.

For supported Linux ProLiants, IM Agents version 7.1 or higher, must be installed (or verified if already installed) from the SPP before you can continue.

To check the version of IM Agents installed, run the following command: **`rpm -q hp-snmp-agents`**

Important: SNMP support (including IM Agents and Health Drivers) cannot be supported on the same monitored device as WBEM (including IM Providers and Health Drivers). WBEM, if installed, must be removed according to the SPP documentation available at: http://www.hp.com/go/spp/documentation/.

Note: On your Linux monitored device, the `/etc/snmp/snmpd.conf` file contains the SNMP configuration information. During the IM Agents installation script, watch for information about this file. When the script prompts you about `Configuring SNMP access from remote Management Station(s)`, be sure to include the Hosting Device IP address. If you do not, you will need to reconfigure SNMP.

System Management Homepage (SMH) is also part of the SPP. It provides additional reporting capabilities on the monitored device itself. SMH is not mandatory for Insight Remote Support, but it can be used to reconfigure SNMP if settings made during Insight Management Agent installation need to be changed.

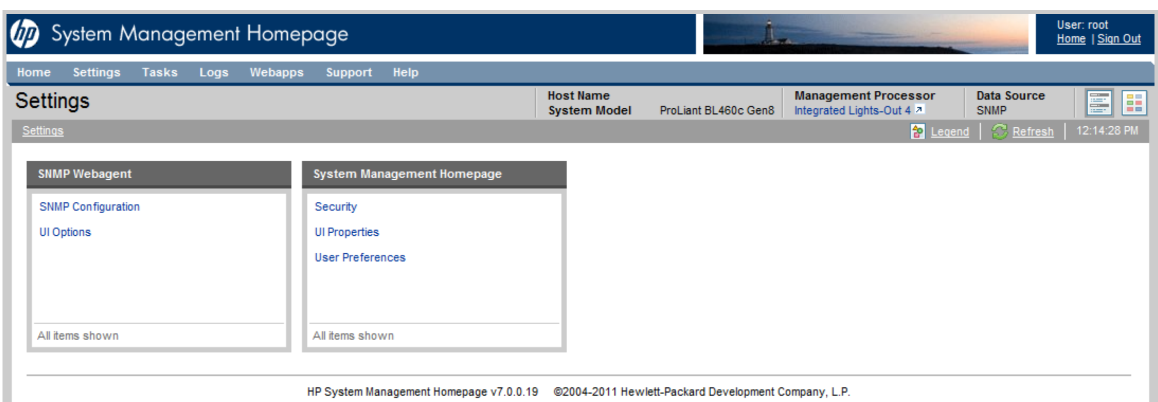
Configure SNMP

Once you have installed the Insight Management Agents on your monitored devices, use SMH to edit the `snmpd.conf` file to add the Hosting Device IP address and the SNMP community string. This enables SNMP communication from the monitored device to the Hosting Device. You will need to do this for each Linux monitored device.

Note: You can also edit the `snmpd.conf` file in a text editor if you are not using SMH.

The Hosting Device must be able to communicate with the monitored device. To configure the monitored device to send traps to the Hosting Device, complete the following steps:

1. In a web browser, access System Management Homepage (SMH) on the monitored device: `https://[`*`ipaddress`*`]:2381`.
2. Log on using the monitored device root username and password.
3. In the top menu bar, click **Settings**.



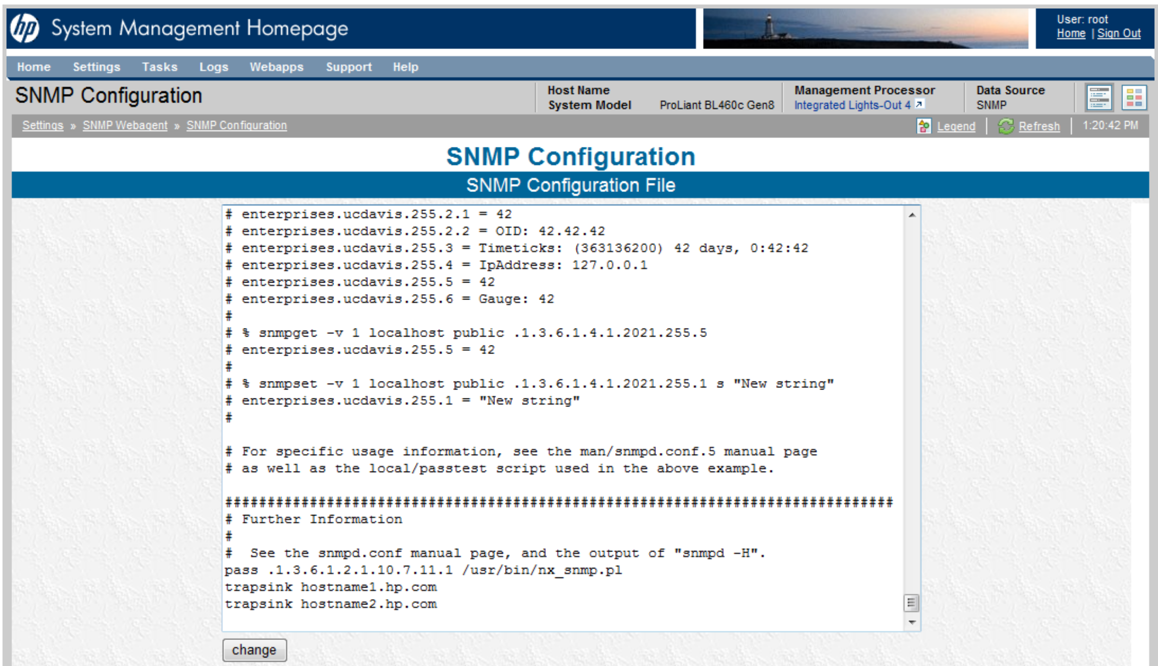
4. Click the **SNMP Configuration** link.
5. In the SNMP Configuration File add a `trapsink` entry for the Hosting Device IP address, for example `trapsink 1.2.3.4 public`, and click **Change**.

   The `trapsink` command is required for events to be sent to Insight Remote Support for analysis. If the `trapsink` command is not configured, Insight Remote Support will not receive traps.

A `rocommunity` directive allows `SNMP GET` and `GETNEXT` access. It is required for discovery and by the analysis rules. The format is: `rocommunity <community_string>`, for example `rocommunity public`. The community string used must be the same read community string configured in the Insight RS Console SNMP protocol assigned to the ProLiant Linux server. The `rocommunity` is used during analysis rules trap processing to retrieve additional information not supplied with traps.

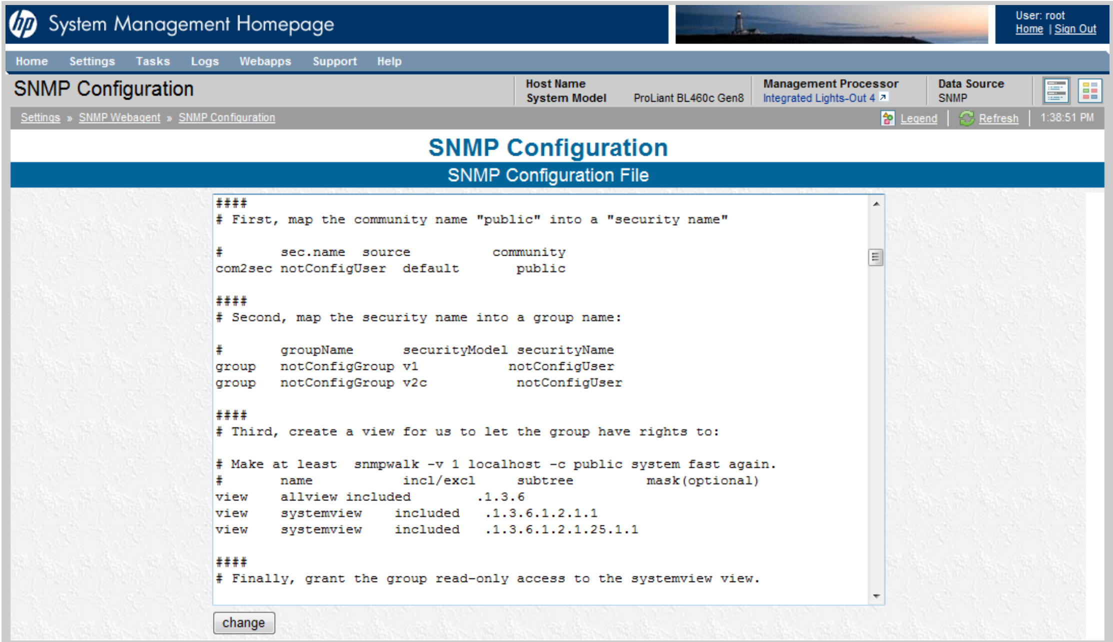
The `rwcommunity` directive allows `SNMP GET`, `GETNEXT`, and `SET` access. It is not required for use by Insight Remote Support, but may be needed by other applications.

Note: If you do not use the `public` community string in the `trapsink` entry, then you must add a new SNMP protocol for this monitored device in the Insight RS Console that uses the same community string.



6. For those Linux installations (for example Red Hat 4 or 5) that support the added security of view statements, add a `view systemview included` entry. The `view systemview included` entry allows Insight Remote Support to read the entire .232 MIB tree and gather all of the analysis data when events occur. Make the following change to the default settings:

a. Scroll down in the **SNMP Configuration File** to the `view systemview included` entries.



b. Comment out the default entries with a # character.

c. Add the following entry: `view systemview included .1 80`

d. Click **Change**.

Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create an SNMP Protocol Credential in Insight RS Console

If your device's SNMP community string is set to `public` and your community access mode is `read only`, Insight RS automatically associates an SNMP protocol with your device. If you use a different community string or use a nonstandard port, you must create an SNMP protocol credential in the Insight RS Console.

To configure SNMP in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.

3. From the Select and Configure Protocol drop-down list, select **Simple Network Management Protocol** for the version of SNMP configured on your server.
4. Click **New**. The New Credential dialog box appears.
5. Type the information configured on your server.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP address(es) of the devices to be discovered.
   d. Click **Add**.
4. Click **Start Discovery**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon ( ).

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following sections:

Verify Connectivity by Sending an SNMP Test Trap to the Hosting Device

To verify connectivity from the monitored device to the Hosting Device, send an SNMP test trap to the Hosting Device and then verify the test trap was received in the Insight RS Console.

Use one of the following methods to send an SNMP test trap to the Hosting Device.

- In SMH:
  a. Click **Settings**.
  b. In the SNMP Webagent pane, click **SNMP Configuration**.
  c. Scroll to the bottom of the SNMP Configuration screen and in the Test Trap section, click **Send Trap**.
- On the command line, type the following command to send a test trap to the Hosting Device:

  `snmptrap -v 1 -c public [`*`Hosting Device IP Address`*`] .1.3.6.1.4.1.232 [`*`Linux monitored device IP Address`*`] 6 11003 1234 .1.3.6.1.2.1.1.5.0 s test .1.3.6.1.4.1.232.11.2.11.1.0 i 0 .1.3.6.1.4.1.232.11.2.8.1.0 s [`*`provide your own identifier and time stamp`*`]`

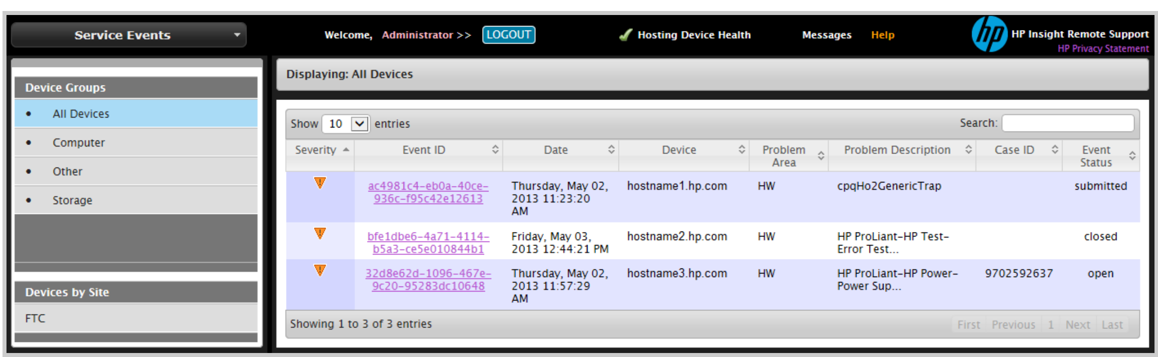
  The resulting text with details of the monitored device and Hosting Device should be returned.

  `Management Agents Test Trap sent - [timestamp]`

Viewing Test Events in the Insight RS Console

Some monitored device types allow you to send a test event to Insight Remote Support. After you configure your monitored device and send a test event, use the following process to verify the test event arrived.

1. Log on to the Insight RS Console.
2. In the main menu, select **Service Events**. If your monitored device is properly configured, the event will appear in the Service Events Information pane.



Verify Collections in the Insight RS Console

After discovery, a Server Basic Configuration Collection is automatically run for servers. Verify that this collection ran successfully on the Collection Services → Basic Collection Results tab in the Insight RS Console. Note that there is a delay between when discovery finishes and when the collection runs. For more information about scheduling collections, see the Insight RS Help.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services** and then click the **Basic Collection Results** tab.
3. Expand the Server Basic Configuration Collection section.
4. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears (). If it failed, an error icon appears ().

Chapter 5: Configuring ProLiant VMware ESX Servers

Fulfill Configuration Requirements

To configure your ProLiant VMware ESX servers to be monitored by Insight RS, complete the following sections:

Table 5.1 ***ProLiant VMware ESX Server Configuration Steps***

Task	Complete?
Make sure Insight RS supports your ProLiant VMware ESX server by checking the *HP Insight Remote Support Release Notes*.	
Configure SNMP on the ProLiant VMware ESX server.	
Add the SNMP protocol credentials to the Insight RS Console.	
Discover the ProLiant VMware ESX server in the Insight RS Console.	
Send a test event to verify connectivity between your ProLiant VMware ESX server and Insight RS.	

Install and Configure Communication Software on Servers

To configure your monitored devices, complete the following sections:

Configure SNMP

Your monitored devices must be configured to communicate with the Hosting Device. VMware ESX uses SNMP for communication to the Hosting Device. SNMP should be installed by default with VMware ESX, but you must configure your monitored device SNMP settings to communicate with the Hosting Device.

Monitored devices that participate in SNMP notifications must include the following:

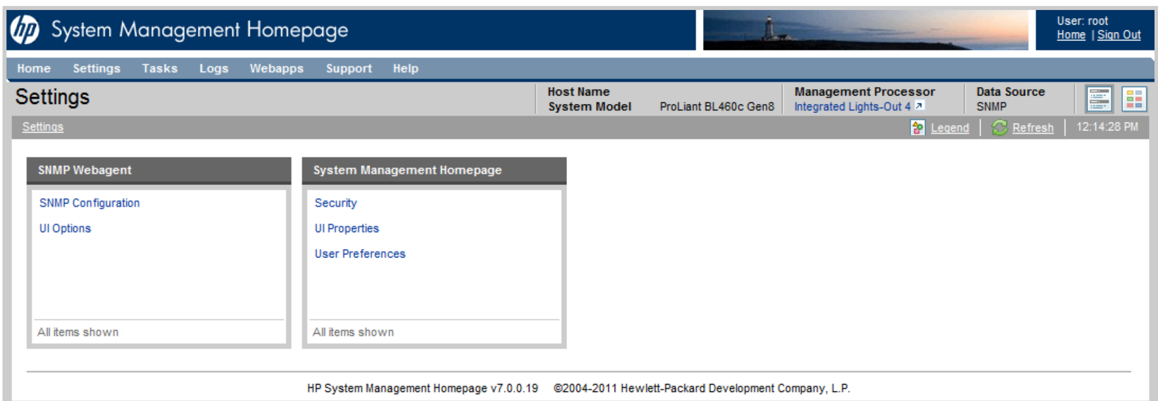
- All monitored devices must have a working intranet connection, such as through an Ethernet adapter, with TCP/IP installed and running. Monitored devices must have two-way communication with the Hosting Device over this connection.
- Monitored devices need the Management Agent software for problem detection and trap generation. The IM agents are distributed by HP and are designed to generate SNMP traps with information that allows for a more complete analysis. With VMware ESX the management agents are part of the VMware ESX bundle. So, if your monitored device is properly configured, the agents will be present on the server.
- Finally, all monitored devices need to have the Hosting Device IP address defined as a trap destination.

SNMP Agents are required for VMware ESX monitored devices. Once you have installed the SNMP Agents on your monitored devices, use SMH to edit the `snmpd.conf` file to add the Hosting Device IP address and the SNMP community string. This enables SNMP communication from the monitored device to the Hosting Device. You will need to do this for each VMware ESX monitored device.

Note: You can also edit the `snmpd.conf` file in a text editor if you are not using SMH.

The Hosting Device must be able to communicate with the monitored device. To configure the monitored device to send traps to the Hosting Device, complete the following steps:

1. In a web browser, access System Management Homepage (SMH) on the monitored device: `https://[`*`ipaddress`*`]:2381`.
2. Log on using the monitored device root user name and password.
3. In the top menu bar, click **Settings**.



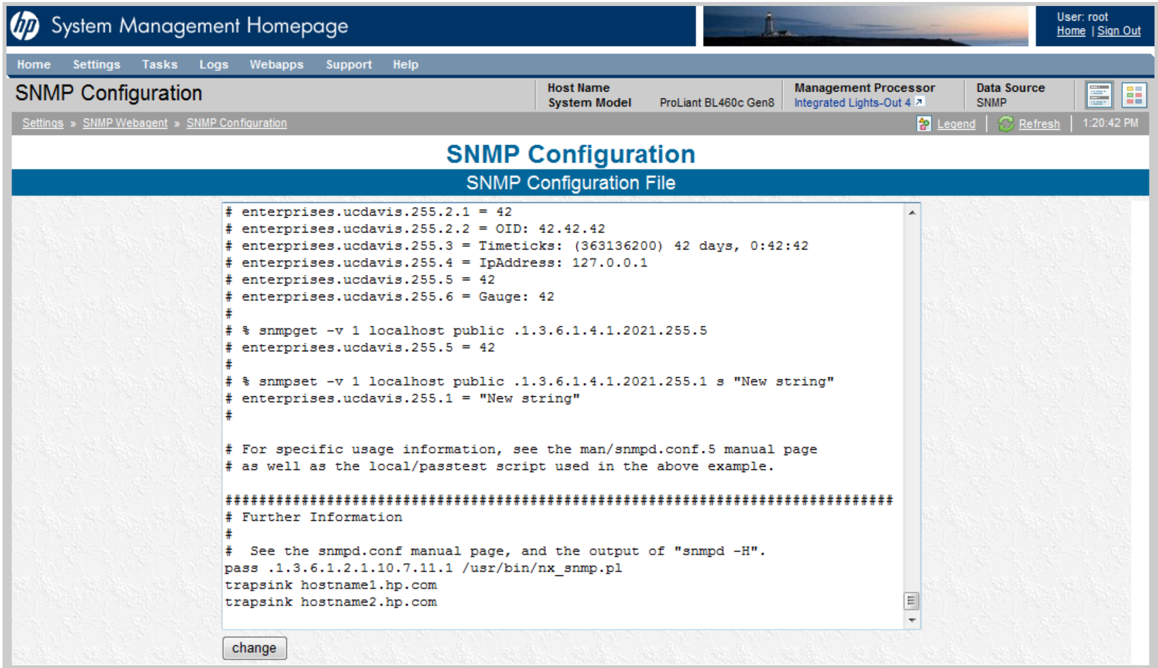
4. Click the **SNMP Configuration** link.
5. In the SNMP Configuration File add a `trapsink` command that includes the IP address for your Hosting Device, for example `trapsink 1.2.3.4 public`, and click **Change**.

   The `trapsink` command is required for events to be sent to Insight Remote Support for analysis. If the `trapsink` command is not configured, Insight Remote Support will not receive traps.

   A `rocommunity` directive allows `SNMP GET` and `GETNEXT` access. It is required for discovery and by the analysis rules. The format is: `rocommunity` <*community_string*>, for example `rocommunity public`. The community string used must be the same read community string configured in the Insight RS Console SNMP protocol assigned to the ProLiant ESX server. The `rocommunity` is used during analysis rules trap processing to retrieve additional information not supplied with traps.

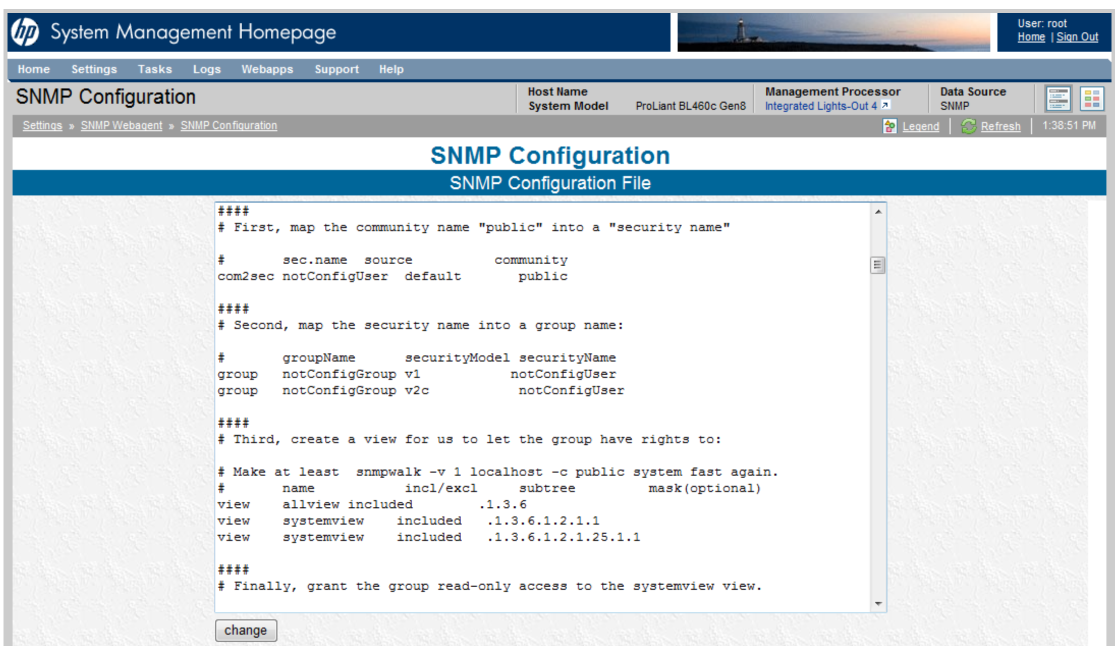
   The `rwcommunity` directive allows `SNMP GET`, `GETNEXT`, and `SET` access. It is not required for use by Insight Remote Support, but may be needed by other applications.

   Note: If you do not use the `public` community string in the `trapsink` entry, then you must add a new SNMP protocol for this monitored device in the Insight RS Console that uses the same community string.



6. For those Linux installations (for example Red Hat 4 or 5) that support the added security of view statements, add a `view systemview included` entry. The `view systemview included` entry allows Insight Remote Support to read the entire .232 MIB tree and gather all of the analysis data when events occur. Make the following change to the default settings:

   a. Scroll down in the **SNMP Configuration File** to the `view systemview included` entries.



   b. Comment out the default entries with a # character.

c. Add the following entry: `view systemview included .1 80`

d. Click **Change**.

Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create an SNMP Protocol Credential in Insight RS Console

If your device's SNMP community string is set to `public` and your community access mode is `read only`, Insight RS automatically associates an SNMP protocol with your device. If you use a different community string or use a nonstandard port, you must create an SNMP protocol credential in the Insight RS Console.

To configure SNMP in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the Select and Configure Protocol drop-down list, select **Simple Network Management Protocol** for the version of SNMP configured on your server.
4. Click **New**. The New Credential dialog box appears.
5. Type the information configured on your server.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP address(es) of the devices to be discovered.
   d. Click **Add**.
4. Click **Start Discovery**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon ().

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following sections:

Verify Connectivity by Sending an SNMP Test Trap to the Hosting Device

To verify connectivity from the monitored device to the Hosting Device, send an SNMP test trap to the Hosting Device and then verify the test trap was received in the Insight RS Console.

Use one of the following methods to send an SNMP test trap to the Hosting Device.

- In SMH:
  a. Click **Settings**.
  b. In the SNMP Webagent pane, click **SNMP Configuration**.
  c. Scroll to the bottom of the SNMP Configuration screen and in the Test Trap section, click **Send Trap**.
- On the command line, type the following command to send a test trap to the Hosting Device:

  ```
  snmptrap -v 1 -c public [Hosting Device IP Address] .1.3.6.1.4.1.232 [Linux monitored device IP Address] 6 11003 1234 .1.3.6.1.2.1.1.5.0 s test .1.3.6.1.4.1.232.11.2.11.1.0 i 0 .1.3.6.1.4.1.232.11.2.8.1.0 s [provide your own identifier and time stamp]
  ```

  The resulting text with details of the monitored device and Hosting Device should be returned.

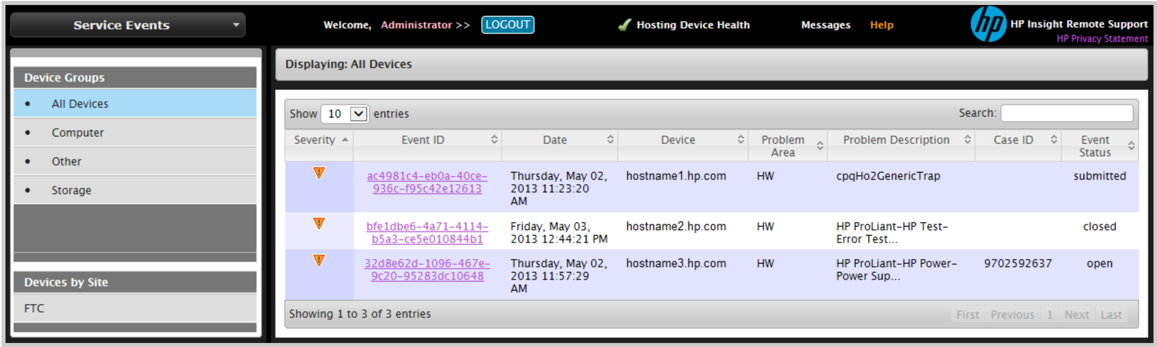
  ```
  Management Agents Test Trap sent - [timestamp]
  ```

Viewing Test Events in the Insight RS Console

Some monitored device types allow you to send a test event to Insight Remote Support. After you configure your monitored device and send a test event, use the following process to verify the test event arrived.

1. Log on to the Insight RS Console.
2. In the main menu, select **Service Events**. If your monitored device is properly configured, the event

will appear in the Service Events Information pane.



Verify Collections in the Insight RS Console

After discovery, a Server Basic Configuration Collection is automatically run for servers. Verify that this collection ran successfully on the Collection Services → Basic Collection Results tab in the Insight RS Console. Note that there is a delay between when discovery finishes and when the collection runs. For more information about scheduling collections, see the Insight RS Help.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services** and then click the **Basic Collection Results** tab.
3. Expand the Server Basic Configuration Collection section.
4. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears (). If it failed, an error icon appears ().

Chapter 6: Configuring ProLiant VMware ESXi Servers

Fulfill Configuration Requirements

To configure your ProLiant VMware ESXi servers to be monitored by Insight RS, complete the following sections:

Table 6.1 ***ProLiant VMware ESXi Server Configuration Steps***

Task	Complete?
Make sure Insight RS supports your ProLiant VMware ESXi server by checking the *HP Insight Remote Support Release Notes*.	
Install and configure WBEM providers on the ESXi server. If using HP's distribution of ESXi, all necessary WBEM providers are included.	
Add the WBEM protocol credentials to the Insight RS Console.	
Discover the ProLiant VMware ESXi server in the Insight RS Console.	
Verify communication between the ESXi server and Insight RS.	

Install and Configure Communication Software on Servers

To configure your monitored devices, complete the following sections:

Install an ESXi Distribution

There are two methods of obtaining ESXi:

- When ESXi is HP's distribution, which contains the WBEM providers (see "Obtain the HP ESXi Distribution" below).
- When ESXi is VMware's distribution, which does not contain the WBEM providers. The WBEM providers need to be installed separately, which are also available from HP (see "Obtain and Configure the VMware ESXi Distribution" on the next page).

Obtain the HP ESXi Distribution

HP's distribution of VMware ESXi includes the necessary WBEM providers to send hardware events to the Hosting Device. Use the root account credentials for the ESXi system when configuring the WBEM protocols for this monitored device on the Hosting Device. The ability to send a test event from the physical host is not possible because the VMware ESXi hypervisor is embedded and therefore the console is not available to the user.

Important: If using the ESXi distribution files distributed by HP, then it is not necessary to download and install the WBEM Providers bundle because the distribution files already have the bundle integrated within.

Download HP ESXi 4 distribution from the following link:

https://h20392.www2.hp.com/portal/swdepot/displayProductInfo.do?productNumber=HPVM06.

Download HP ESXi 5 distribution from the following link:

https://h20392.www2.hp.com/portal/swdepot/displayProductInfo.do?productNumber=HPVM09.

Obtain and Configure the VMware ESXi Distribution

HP has developed WBEM providers for VMware ESXi. The bundle contains the WBEM providers, and it is necessary to download and install the bundle when the distribution files used to install the ESXi operating system were obtained from VMware.

The bundle is available from the following links:

- **VMware ESXi 4.0**

  http://h20000.www2.hp.com/bizsupport/TechSupport/SoftwareDescription.jsp?swItem=MTX-7ba9a9031d7e4a41ad65f038e0
- **VMware ESXi 4.1**

  http://h20000.www2.hp.com/bizsupport/TechSupport/SoftwareDescription.jsp?swItem=MTX-4f1abc63053540bebfd1389f0f
- **VMware ESXi 5.0**

  http://h20000.www2.hp.com/bizsupport/TechSupport/SoftwareDescription.jsp?swItem=MTX-83016c4f7e82421f82ad5c2fcc

Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Add Protocol Credentials and Start Discovery

Important: If you are managing your ESXi hosts using VMware vCenter Server, you can skip the following steps and discover all of your ESXi hosts with a single discovery of the VMware vCenter Server. For more information, see "Configuring VMware vCenter Servers" on page 297.

To discover your monitored devices, complete the following sections:

Create a WBEM Protocol Credential in Insight RS Console

To configure WBEM in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the **Select and Configure Protocol** drop-down list, select **Web-Based Enterprise Management (WBEM)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Username and Password you have configured on your device.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
    a. Click **New**.
    b. Select the **Single Address**, **Address Range**, or **Address List** option.
    c. Type the IP address(es) of the devices to be discovered.
    d. Click **Add**.
4. Click **Start Discovery**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon ( ).

    If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following section:

Verify Collections in the Insight RS Console

After discovery, a Server Basic Configuration Collection is automatically run for servers. Verify that this collection ran successfully on the Collection Services → Basic Collection Results tab in the Insight RS Console. Note that there is a delay between when discovery finishes and when the collection runs. For more information about scheduling collections, see the Insight RS Help.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services** and then click the **Basic Collection Results** tab.
3. Expand the Server Basic Configuration Collection section.
4. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears (). If it failed, an error icon appears ().

Chapter 7: Configuring ProLiant Citrix Xen Servers

Fulfill Configuration Requirements

To configure your ProLiant Citrix Xen servers to be monitored by Insight RS, complete the following sections:

Table 7.1 ***ProLiant Citrix Xen Server Configuration Steps***

Task	Complete?
Make sure Insight RS supports your ProLiant Citrix Xen server by checking the *HP Insight Remote Support Release Notes*.	
Install HP SNMP Agents for Citrix XenServer on the ProLiant Citrix Xen server.	
Configure SNMP on the ProLiant Citrix Xen server.	
Add the SNMP protocol to the Insight RS Console.	
Discover the ProLiant Citrix server in the Insight RS Console.	
Send a test event to verify connectivity between your ProLiant Citrix server and Insight RS.	

Install and Configure Communication Software on Servers

To configure your monitored devices, complete the following sections:

Install SNMP Agents for Citrix XenServer

Your Citrix monitored devices must be configured to communicate with the Hosting Device. Monitored devices that use SNMP notifications must include the following:

- All monitored devices must have a working intranet connection, such as through an Ethernet adapter, with TCP/IP installed and running. Monitored devices must have two-way communication with the Hosting Device over this connection.
- Monitored devices need the SNMP Agent software for problem detection and trap generation. HP distributes the SNMP Agents and they are designed to generate SNMP traps with information that allows for a more complete analysis.
- Finally, all monitored devices need to have the Hosting Device IP address defined as a trap destination.

The HP SNMP Agents for Citrix XenServer is a system-specific software bundle that includes drivers, utilities, and management agents for your Citrix ProLiant. HP SNMP Agents for Citrix XenServer are available from the following link: http://h20000.www2.hp.com/bizsupport/TechSupport/SoftwareDescription.jsp?swItem=MTX-9f349b2b4fc94915ac6cde8b8a.

SNMP Agents support HP ProLiant systems running Citrix, which are included in the HP SNMP Agents for Citrix XenServer software package.

Note: On your Citrix monitored device, the `/etc/snmp/snmpd.conf` file contains the current SNMP configuration. During your configuration of the installation script of SNMP Agents, watch for information about this file. When the script prompts you about `Configuring SNMP access from remote Management Station(s)`, be sure to include the Hosting Device IP address. If you do not, you will need to reconfigure this on the monitored device.

System Management Homepage (SMH) is also included in the HP SNMP Agents for Citrix XenServer software package. It provides additional reporting capabilities on the monitored device itself. While not mandatory for Insight Remote Support, if you configure SNMP services incorrectly to communicate with the Hosting Device during SNMP Agent installation you will need SMH to reconfigure those settings.

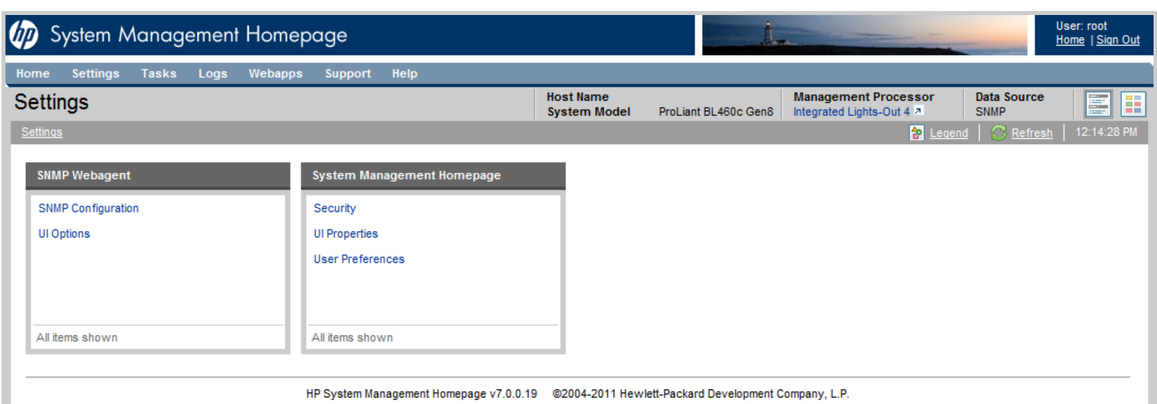
Configure SNMP

SNMP Agents are required for Citrix monitored devices. Once you have installed the SNMP Agents on your monitored devices, use SMH to edit the `snmpd.conf` file to add the Hosting Device IP address and the SNMP community string. This enables SNMP communication from the monitored device to the Hosting Device. You will need to do this for each Citrix monitored device.

Note: You can also edit the `snmpd.conf` file in a text editor if you are not using SMH.

The Hosting Device must be able to communicate with the monitored device. To configure the monitored device to send traps to the Hosting Device, complete the following steps:

1. In a web browser, access System Management Homepage (SMH) on the monitored device: `https://[`*`ipaddress`*`]:2381`.
2. Log on using the monitored device root user name and password.
3. In the top menu bar, click **Settings**.



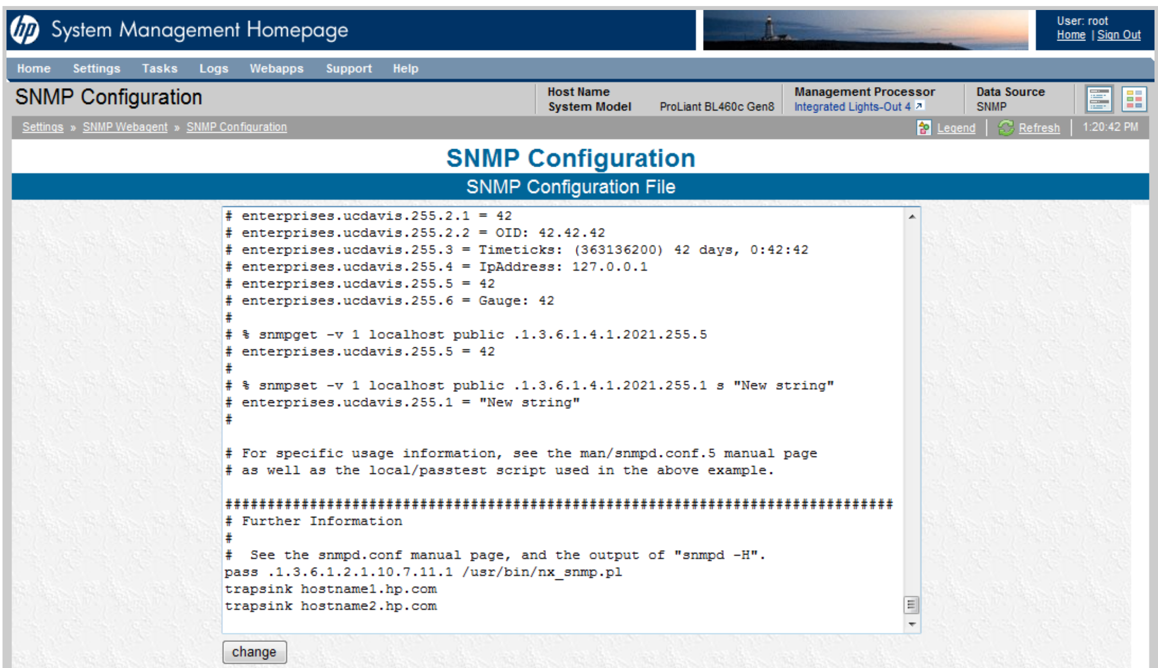
4. Click the **SNMP Configuration** link.
5. In the SNMP Configuration File add a `trapsink` command that includes the Hosting Device IP address, for example `trapsink 1.2.3.4 public`, and click **Change**.

The `trapsink` command is required for events to be sent to Insight Remote Support for analysis. If the `trapsink` command is not configured, Insight Remote Support will not receive traps.

A `rocommunity` directive allows `SNMP GET` and `GETNEXT` access. It is required for discovery and by the analysis rules. The format is: `rocommunity <`*`community_string`*`>`, for example `rocommunity public`. The community string used must be the same read community string configured in the Insight RS Console SNMP protocol assigned to the ProLiant Citrix server. The `rocommunity` is used during analysis rules trap processing to retrieve additional information not supplied with traps.

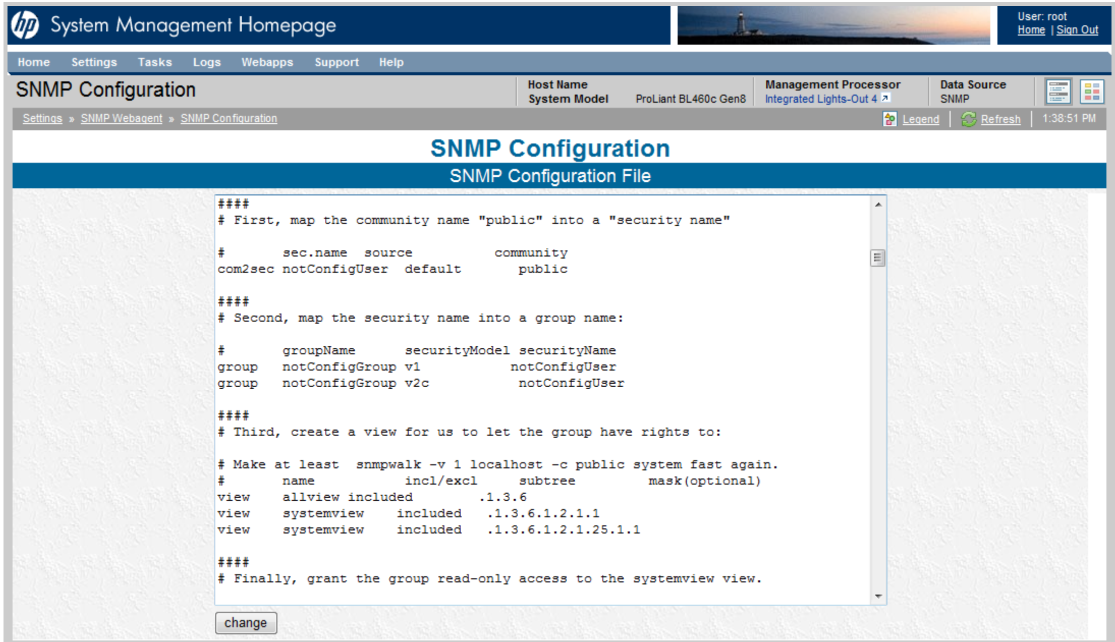
The `rwcommunity` directive allows `SNMP GET`, `GETNEXT`, and `SET` access. It is not required for use by Insight Remote Support, but may be needed by other applications.

Note: If you do not use the `public` community string in the `trapsink` entry, then you must add a new SNMP protocol for this monitored device in the Insight RS Console that uses the same community string.



6. For those Linux installations (for example Red Hat 4 or 5) that support the added security of view statements, add a `view systemview included` entry. The `view systemview included` entry allows Insight Remote Support to read the entire .232 MIB tree and gather all of the analysis data when events occur. Make the following change to the default settings:

a. Scroll down in the **SNMP Configuration File** to the `view systemview included` entries.



b. Comment out the default entries with a # character.

c. Add the following entry: `view systemview included .1 80`

d. Click **Change**.

Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create an SNMP Protocol Credential in Insight RS Console

If your device's SNMP community string is set to `public` and your community access mode is `read only`, Insight RS automatically associates an SNMP protocol with your device. If you use a different community string or use a nonstandard port, you must create an SNMP protocol credential in the Insight RS Console.

To configure SNMP in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the Select and Configure Protocol drop-down list, select **Simple Network Management Protocol** for the version of SNMP configured on your server.
4. Click **New**. The New Credential dialog box appears.

5. Type the information configured on your server.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP address(es) of the devices to be discovered.
   d. Click **Add**.
4. Click **Start Discovery**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon ( ).

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following sections:

Verify Connectivity by Sending an SNMP Test Trap to the Hosting Device

To verify connectivity from the monitored device to the Hosting Device, send an SNMP test trap to the Hosting Device and then verify the test trap was received in the Insight RS Console.

Use one of the following methods to send an SNMP test trap to the Hosting Device.

- In SMH:
  a. Click **Settings**.
  b. In the SNMP Webagent pane, click **SNMP Configuration**.
  c. Scroll to the bottom of the SNMP Configuration screen and in the Test Trap section, click **Send Trap**.
- On the command line, type the following command to send a test trap to the Hosting Device:

  snmptrap -v 1 -c public [*Hosting Device IP Address*] .1.3.6.1.4.1.232 [*Linux monitored device IP Address*] 6 11003 1234 .1.3.6.1.2.1.1.5.0 s test .1.3.6.1.4.1.232.11.2.11.1.0 i 0 .1.3.6.1.4.1.232.11.2.8.1.0 s [*provide your own identifier and time stamp*]

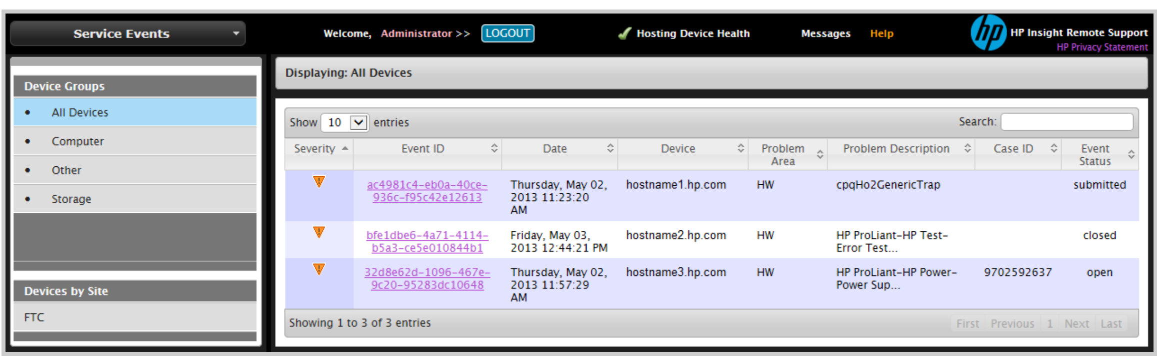
  The resulting text with details of the monitored device and Hosting Device should be returned.

  ```
  Management Agents Test Trap sent - [timestamp]
  ```

Viewing Test Events in the Insight RS Console

Some monitored device types allow you to send a test event to Insight Remote Support. After you configure your monitored device and send a test event, use the following process to verify the test event arrived.

1. Log on to the Insight RS Console.
2. In the main menu, select **Service Events**. If your monitored device is properly configured, the event will appear in the Service Events Information pane.



Verify Collections in the Insight RS Console

After discovery, a Server Basic Configuration Collection is automatically run for servers. Verify that this collection ran successfully on the Collection Services → Basic Collection Results tab in the Insight RS Console. Note that there is a delay between when discovery finishes and when the collection runs. For more information about scheduling collections, see the Insight RS Help.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services** and then click the **Basic Collection Results** tab.
3. Expand the Server Basic Configuration Collection section.

4. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears (✅). If it failed, an error icon appears (❎).

Chapter 8: Configuring ProLiant Superdome Servers

The ProLiant Superdome system has implemented single-source event reporting such that all event indications are reported through the ProLiant Superdome Onboard Administrator (OA). The OA monitors the core system hardware and generates WS-Management alert indications when it determines that an important event occurs. The Linux WS-Man providers monitor partition IO devices and report their events through the OA as well.

Although the OA is the sole source for event reporting for the entire system complex, Insight RS still needs to discover the Linux partitions to support configuration collections.

In addition to the default WS-Man protocol, the ProLiant Superdome OA may optionally be configured to send alerts using email and/or SNMP. It is recommended that SNMP not be enabled unless so required by the customer as it may result in redundant alert indications being forwarded to HP Support.

Fulfill Configuration Requirements

To configure your ProLiant Superdome servers to be monitored by Insight RS, complete the following sections:

Table 8.1 ***ProLiant Superdome Server Configuration Steps***

Task	Completed?
Verify ProLiant Superdome network settings and protocol status.	
Add WS-Man protocol credentials for both the OA and the Linux partitions in the Insight RS Console. WS-Man uses port 443 for the OA and port 5986 for the Linux partitions.	
Discover the ProLiant Superdome server in the Insight RS Console.	
Discover the Linux partitions in the Insight RS Console.	
Verify the ProLiant Superdome OA and the Linux partitions were discovered correctly.	
Send an OA and a Linux test event to verify connectivity to the Hosting Device.	

Configure Communication Software on Servers

To configure your monitored devices, complete the following sections:

ProLiant Superdome OA Configuration Verification

Connect to the ProLiant Superdome OA using Telnet, SSH or a web browser (HTTPS), as detailed below. Both the username and password login credentials are case sensitive.

Verify the Network Settings are correct, including the DNS server addresses for both the active and standby OAs, and verify the Protocol Status indicates that WS-Man is enabled.

HP recommends that Enclosure IP Mode is enabled as it supports the passing of the active OA’s IP address to the Standby OA should an OA failover occur. If Enclosure IP Mode is disabled then IRS should be configured to discover both the active and the standby OA.

Using Telnet or SSH

1. Telnet or SSH to the Onboard Administrator CLI.
2. Verify DNS is configured: type the **show network** command to show the Network Settings and Protocol Status of the OA.

Using a Web Browser

1. In a web browser, log on to the Onboard Administrator web interface. Either HTTP or HTTPS may be used to open the connection to the default port.
2. In the left menu, select **Enclosure Information** → **Enclosure Settings** → **Enclosure TCP/IP Settings**.

Add Protocol Credentials and Start Discovery

You must add the WS-Man protocol login credential for the OA and all Linux partitions to the Insight RS Console so Insight RS may communicate with these endpoints.

To discover your monitored devices, complete the following sections:

Create a WS-Man Protocol Credential for the OA in the Insight RS Console

To configure WS-Man in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the **Select and Configure Protocol** drop-down list, select **Web Services Management Protocol (WSMAN)**.
4. Click **New**. The New Credential dialog box appears.
5. Use the default port: `443`.
6. Type the Username and Password credentials that Insight RS will use to communicate with the OA.
7. Click **Add**.

Create a WS-Man Protocol Credential for the Linux Partitions in the Insight RS Console

To configure WS-Man in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.

3. From the **Select and Configure Protocol** drop-down list, select **Web Services Management Protocol (WSMAN)**.
4. Click **New**. The New Credential dialog box appears.
5. Change the default port to: `5986`.
6. Type the Usernames and Passwords that Insight RS will use to communicate with each of the Linux partitions.
7. Click **Add**.

Note: Create additional WS-Man protocol credentials if your partitions use different credentials.

Discover the ProLiant Superdome Server and Linux Partitions in the Insight RS Console

To discover the ProLiant Superdome server and Linux partitions from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP addresses for the OA and the Linux partitions:
   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP addresses of the OAs and Linux partitions to be discovered.
   d. Click **Add**.
4. Click **Start Discovery**.

Note: If discovery is configured to discover standby OAs, it will populate the Devices screen with minimal information on these devices because although their network interfaces can be active and be reachable, they are not monitoring port 443 used for communication and therefore cannot provide Insight RS with information about themselves. Likewise because they are standby devices Insight RS will not be able to establish event monitoring subscriptions directly with them. If an OA failover should occur, the OA will pass the subscription to the new Monarch and Insight RS will continue to monitor the OA for event indications using that active subscription.

Verify ProLiant Superdome Discovery

After discovery completes, select **Devices** from the main menu. Verify that all the ProLiant Superdome OAs and Linux partitions previously configured for discovery were discovered and that the Status for those devices shows the success icon ().

Verify the ProLiant Superdome OA

To verify the device information for ProLiant Superdome OA devices, complete the following steps:

1. In the main menu, select **Devices**. Locate the ProLiant Superdome OA, and click its Device Name.
2. On the **Device** tab, expand the Status section and verify the Registration Status shows **Registered**, and the Enabled has the **Yes** option selected.
3. Expand the Hardware section and verify the Device Name, Category, Product, Vendor, IP Address, Acquired Serial Number and Acquired Product Number are populated correctly for the device. The category should be reported as ***MANAGEMENT_DEVICE***, the Product should be reported as ***Superdome 2 16s x86*** and the Vendor should be ***hp***. The remaining values may be verified using the Telnet or WEB Browser steps outlined above.
4. Expand the Warranty & Contract section and verify the values listed are correct.

Note: The OS Name and OS Version fields reported for the OA will be blank if viewed from the Operating System section.

5. Expand the Delivery section and verify the information is correct.
6. Click **Save Changes** to complete the discovery process for this device.

Verify the ProLiant Superdome Linux Partitions

To verify the device information for ProLiant Superdome Linux partitions devices, complete the following steps:

1. In the main menu, select **Devices**. Locate a ProLiant Superdome Linux partition, and click its Device Name.
2. On the **Device** tab, expand the Status section and verify the Registration Status shows Registered, and the Enabled has the Yes option selected.
3. Expand the Operating System section. The OS Name and OS Version fields reported for the Linux partition will report an OS Name of Linux and an OS Version number that should match the value reported by the **`uname -r`** command when entered in a SSH or telnet window connection to that partition. The values reported by the OS Hardware section should be the same values as reported in the OA Hardware section detailed above. Expand the Warranty & Contract section and verify the values listed are correct.
4. Expand the Delivery section and verify the information is correct.
5. Click **Save Changes** to complete the discovery process for this device.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following sections:

Generating Test Events

The ProLiant Superdome supports the capability for an administrator to connect to the OA and generate a WS-Man test indication. Use the following procedure to generate a WS-Man test indication:

1. Telnet or SSH to the OA CLI interface and log on using a user account with administrator privileges.
2. Type the **test wsman** command.
3. In a web browser, log on to the Insight RS Console.
4. In the main menu, select **Service Events**. Click on the Date column header twice to sort the most recent event to the top, and verify that Insight RS received the test event.

Test events can be sent from the Linux partitions using System Management Homepage (SMH). When you send a test event, it is sent transmitted by the OS partition provider and also by the OA. Insight RS only uses the event that comes through the OA path. The OS partition provider does not pass Informational events to the OA, so an Informational test event should not be used for this verification step.

To send a test event from each partition, complete the following steps:

1. Open a web browser and connect to the System Management Homepage (SMH) running on the Linux partition on port `2381`. Log on using a user account with administrator privileges.
2. In SMH, browse to **Settings** → **Test Indication**, and click **Send Test Indication**.
3. Select *Warning* or *Error* for the test indication Type.
4. Click **Send**.
5. In a separate web browser, log on to the Insight RS Console.
6. In the main menu, select **Service Events**. Click on the Date column header twice to sort the most recent event to the top, and verify that Insight RS received the test event. Note that partition events are reported through the OA, so you must click on the OA to view events. To identify which partition originated the event, open the Service Event by clicking on the Event ID link, scroll down to the Failing Host Details section and verify the value reported in the Failing Host Preferred FQDN field is the partition name and/or the value reported in the Failing Hos Preferred IP field is the IP address used for that partition.

Verify Collections in the Insight RS Console

After discovery, a Server Basic Configuration Collection is automatically run for servers. Verify that this collection ran successfully on the Collection Services → Basic Collection Results tab in the Insight RS Console. Note that there is a delay between when discovery finishes and when the collection runs. For more information about scheduling collections, see the Insight RS Help.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services** and then click the **Basic Collection Results** tab.
3. Expand the Server Basic Configuration Collection section.
4. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears ( ). If it failed, an error icon appears ( ).

Configure Your ProLiant Superdome as a Solution

To add your ProLiant Superdome to Insight RS as a solution, complete the following steps:

1. In the main menu, select **Solution Manager**.
2. In the List of Solutions pane, click **Add New Solution**. The fields you need to complete appear in the Create New Solution pane.
3. In the Create New Solution pane, complete the following fields:

Field	Action
Solution Name	Type a unique name for the solution.
Solution Type	Select a solution from the drop-down list of supported HP solutions. This selection will populate the Solution Product Number and Solution Product Model fields. If you select Other, you must populate those fields.
Solution Serial Number	Type the serial number of the solution, which can be found on the hardware label or in the order information when you purchased the HP solution.
Solution Product Number	Type the product number of the solution, which can be found on the hardware label or in the order information when you purchased the HP solution.
Solution Product Model	Type the product model of the solution, which can be found on the hardware label or in the order information when you purchased the HP solution.
Custom Delivery ID	Type the optional alphanumeric value that will only be required if it has been supplied to you by your HP representative or in setup instructions for the purpose of customized handling or routing of service events sent to HP.

4. Click **Save**. The system creates and displays the new solution in the List of Solutions pane. The assigned devices pane appears where you can assign devices to the new solution.
5. Assign devices or partitions in the assigned devices pane. The default view shows devices not assigned to the solution. To show additional devices, click the **Not Assigned to this Solution** or **All Devices** options. To show devices in a specific device group, select a device group from the **Filter by device group** drop-down list. To search for a specific device, type the device name into the **Search** box. The table displays the devices based on your filter and search criteria.
   - To add a device, select the check box next to the device in the devices table.
   - To remove a device, clear the check box next to the device in the devices table.
6. Click **Save Devices**.

The system creates and displays the solution in the List of Solutions pane.

Chapter 9: Configuring Integrity Windows 2003 Servers

Event Log Monitoring Collector (ELMC) is responsible for remote support monitoring on Windows 2003 Integrity monitored devices. SNMP is necessary for Discovery, and WBEM is necessary for configuration collections.

Fulfill Configuration Requirements

To configure your Integrity Windows 2003 servers to be monitored by Insight RS, complete the following sections:

Table 9.1 ***Integrity Windows 2003 Server Configuration Steps***

Task	Complete?
Make sure Insight RS supports your Integrity Windows 2003 server by checking the *HP Insight Remote Support Release Notes*.	
Install ELMC to the Integrity Windows 2003 server.	
Verify SNMP is installed on the Integrity Windows 2003 server.	
Verify WBEM is installed on the Integrity Windows 2003 server.	
Add the ELMC protocol to the Insight RS Console.	
Add the SNMP protocol to the Insight RS Console.	
Add the WBEM protocol to the Insight RS Console.	
Discover the Windows 2003 Integrity server in the Insight RS Console.	
Send a test event to verify connectivity between your Windows 2003 Integrity server and Insight RS.	

Install and Configure Communication Software on Servers

To configure your monitored devices, complete the following sections:

Install the ELMC Software Package on the Monitored Device

The monitored device must meet the following basic requirements before you install ELMC:

- Processor architecture: Itanium on Integrity Servers
- Operating system: On Integrity Servers, all supported versions of Windows 2003
- TCP-IP configured on the monitored server

- Required Permissions and Access: To install, upgrade, or uninstall ELMC, you must be logged on as the Administrator or as a user with administrator privileges.

ELMC is required on the Integrity Windows 2003 server so that events can be forwarded to Insight RS.

If ELMC is already installed on the Integrity Windows 2003 server, make sure it is version 6.2 or later. If it's older than version 6.2, it needs to be upgraded. To check the version of ELMC, run the following command: `wccproxy version`.

Note: When upgrading to ELMC version 6.4, the incorrect version number is displayed in the upgrade window. After performing the upgrade, the correct version number of 6.4 will show in the Programs and Features window and when you run the `wccproxy version` command.

To install the ELMC software package, complete the following steps:

1. In the Insight RS Console, navigate to the **Administrator Settings** → **Software Updates** tab and select the *Event Log Monitoring Collector (ELMC)* software package.
2. On the **Available Version** tab, click **Download**.
3. When the download completes, click **Install**. The ELMC packages are placed in the `%HP_RS_DATA%\ELMC` folder. This folder defaults to `C:\ProgramData\HP\RS\DATA\ELMC`.

   Note: The `ProgramData` folder is a hidden folder. To view this folder, set the folder options to show hidden folders.
4. Copy the appropriate Itanium (IA64) or x86/x64 ELMC software package into a temporary directory on the Integrity Windows 2003 server.
5. Double-click the installer file to begin the install process. The kit will install and finish with no user prompting. No user configuration input is required to install the ELMC software package.

Verify SNMP Agent Prerequisites on the Monitored Device

For your Windows 2003 Integrity monitored device to be discovered by the Insight RS Console, you must have HP Integrity SNMP Agents for Windows Server 2003 installed and fully configured on your monitored devices as directed in the HP Smart Setup product documentation for the appropriate package.

Note: WBEM Providers can coexist with SNMP Agents on the monitored device system as long as they have the same version number. Mismatched versions of HP Insight Management WBEM Providers and HP Insight Management SNMP Agents are not supported.

Windows Integrity servers shipped by HP manufacturing have the Smart Setup installation files pre-loaded on the operating system disk. Otherwise the packages can be installed using the Smart Setup CD or reinstallation media. You can download the Smart Setup CD or Integrity Support Pack at: http://h20000.www2.hp.com/bizsupport/TechSupport/SoftwareIndex.jsp?lang=en&cc=us&prodNameId=3346453&prodTypeId=15351&prodSeriesId=3346452&swLang=13&taskId=135&swEnvOID=1060.

For SNMP Agents to communicate to the Hosting Device, you must provide the SNMP credentials within the Insight RS Console. These actions must occur within the Insight RS Console on the Hosting Device when you are completing the Insight Remote Support configuration.

Verify WBEM Provider Prerequisites on the Monitored Device

For your Windows 2003 Integrity monitored device to support configuration collections, you must have HP Integrity WBEM Providers for Windows Server 2003 installed and fully configured on your monitored devices as directed in the HP Smart Setup product documentation for the appropriate package.

Note: WBEM Providers can coexist with SNMP Agents on the monitored device system as long as they have the same version number. Mismatched versions of HP Insight Management WBEM Providers and HP Insight Management SNMP Agents are not supported.

Windows Integrity servers shipped by HP manufacturing have the Smart Setup installation files pre-loaded on the operating system disk. Otherwise the packages can be installed using the Smart Setup CD or reinstallation media. You can download the Smart Setup CD or Integrity Support Pack at: http://h20000.www2.hp.com/bizsupport/TechSupport/SoftwareIndex.jsp?lang=en&cc=us&prodNameId=3346453&prodTypeId=15351&prodSeriesId=3346452&swLang=13&taskId=135&swEnvOID=1060.

For WBEM Providers to communicate to the Hosting Device, you must provide the WBEM credentials as directed in the *HP Insight Remote Support Installation and Configuration Guide*. These actions must occur within the Insight RS Console when you are completing the Insight Remote Support configuration.

Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create an ELMC Protocol in the Insight RS Console

To configure ELMC in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the **Select and Configure Protocol** drop-down list, select **Event Log Monitoring Collector (ELMC)**.
4. Click **New**. The New Credential dialog box appears.

Insight RS creates the protocol credential and it appears in the credentials table.

Create an SNMPv1 Protocol Credential in Insight RS Console

If your device's SNMP community string is set to `public` and your community access mode is `read only`, Insight RS automatically associates an SNMPv1 protocol with your device. If you use a different community string or use a nonstandard port, you must create an SNMPv1 protocol credential in the Insight RS Console.

To configure SNMPv1 in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the Select and Configure Protocol drop-down list, select **Simple Network Management Protocol Version 1 (SNMPv1)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Community String you have configured on your device.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Create a WMI Protocol Credential in Insight RS Console

To configure WMI in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the **Select and Configure Protocol** drop-down list, select **Windows Management Instrumentation (WMI)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Username and Password you have configured on your device.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:

   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP address(es) of the devices to be discovered.
   d. Click **Add**.
4. Click **Start Discovery**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon (✅).

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following sections:

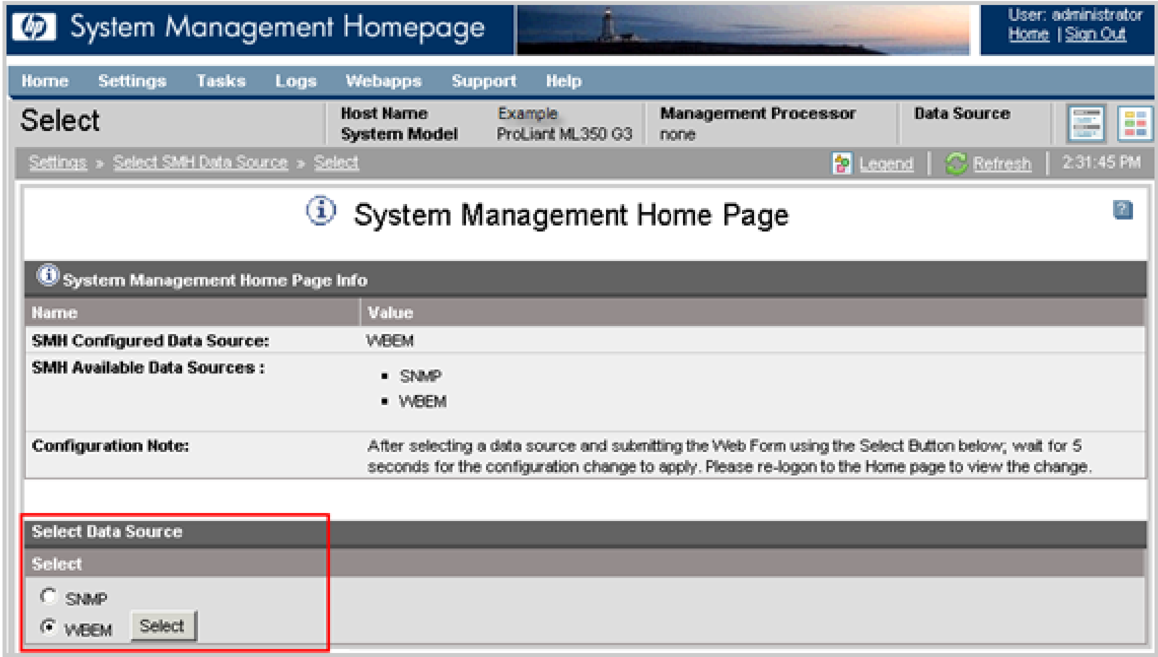
Verify Connectivity by Sending an SNMP Test Trap

Use SNMP Agents to send SNMP Test Traps to verify connectivity.

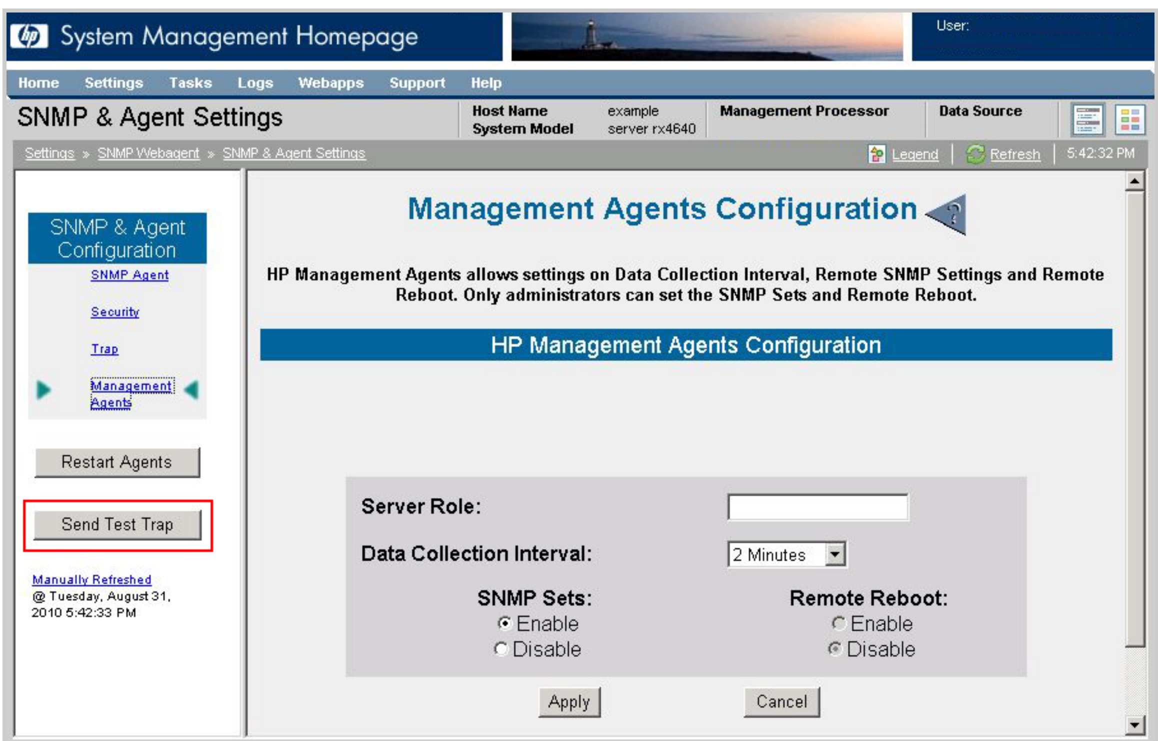
1. In a web browser, access System Management Homepage (SMH) on the monitored device: `https://[`*`ipaddress`*`]:2381`.
2. Log on using the administrator user name and password for the monitored device.

   Note: If you are not prompted to log on, check the upper right corner of the SMH interface and click the Sign In link. If you are not logged in as an administrator for the monitored device you will not have all of the relevant configuration options.

3. If SNMP is not set as your Data Source, click **Settings** in the top menu bar.
4. In the **Select SMH Data Source** box, click the **Select** link.
5. In the Select Data Source section choose the **SNMP** option, and then click **Select**.



6. Once your Data Source is set to SNMP, click the **Settings** option, and then select the **SNMP & Agent Configuration** setting.
7. In the Management Agents Configuration screen, click **Send Test Trap**.



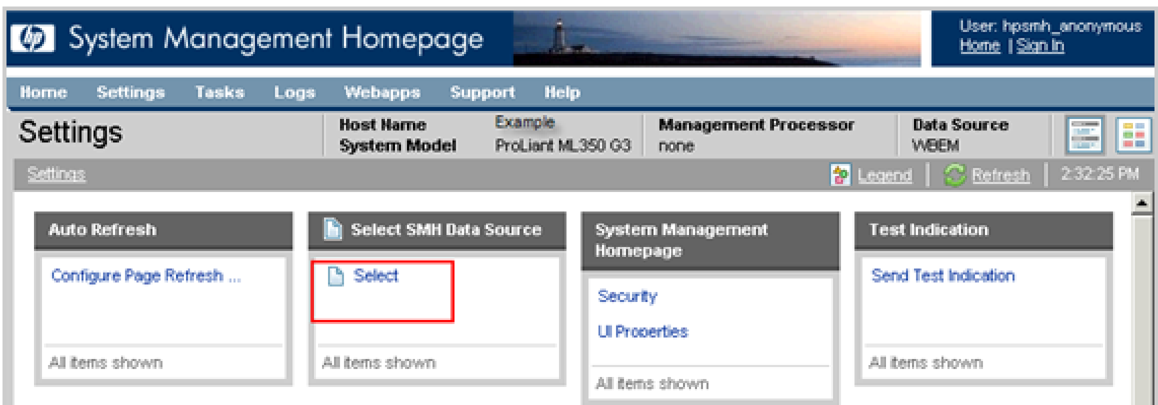
Sending a WBEM Test Indication to Verify Connectivity

Use WBEM Providers to send WBEM Indications to verify connectivity.

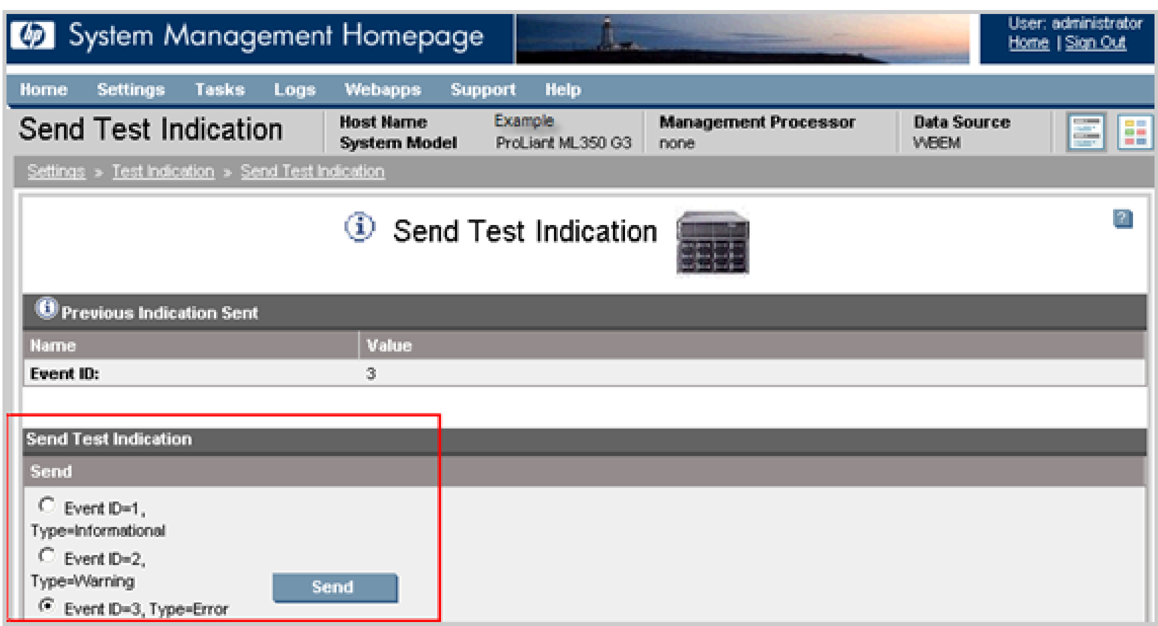
1. In a web browser, access System Management Homepage (SMH) on the monitored device: `https://[ipaddress]:2381`.
2. Log on using the administrator user name and password for the monitored device.

> **Note:** If you are not prompted to log on, check the upper right corner of the SMH interface and click the **Sign In** link. You will not have access to all relevant configuration options if you are not logged in as an administrator for the monitored device.

3. In the top menu bar, click **Settings**.
4. If you chose to install WBEM with the Integrity Support Pack, it will be set as your Data Source. Click the **Send Test Indication** option.



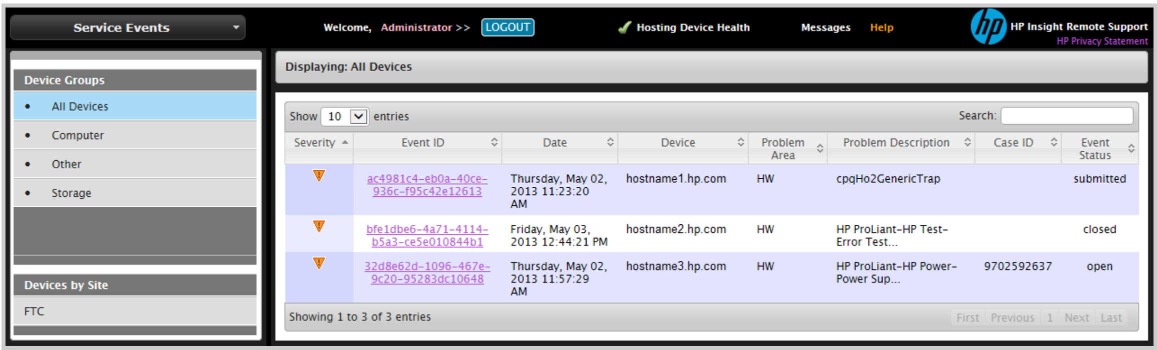
5. In the Send Test Indication screen, select an Event ID type and click **Send**.



Viewing Test Events in the Insight RS Console

Some monitored device types allow you to send a test event to Insight Remote Support. After you configure your monitored device and send a test event, use the following process to verify the test event arrived.

1. Log on to the Insight RS Console.
2. In the main menu, select **Service Events**. If your monitored device is properly configured, the event will appear in the Service Events Information pane.



Verify Collections in the Insight RS Console

After discovery, a Server Basic Configuration Collection is automatically run for servers. Verify that this collection ran successfully on the Collection Services → Basic Collection Results tab in the Insight RS Console. Note that there is a delay between when discovery finishes and when the collection runs. For more information about scheduling collections, see the Insight RS Help.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services** and then click the **Basic Collection Results** tab.
3. Expand the Server Basic Configuration Collection section.
4. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears (). If it failed, an error icon appears ().

Chapter 10: Configuring Integrity Windows 2008 Servers

Fulfill Configuration Requirements

To configure your Integrity Windows 2008 servers to be monitored by Insight RS, complete the following sections:

Table 10.1 ***Integrity Windows 2008 Server Configuration Steps***

Task	Complete?
Make sure Insight RS supports your Integrity Windows 2008 server by checking the *HP Insight Remote Support Release Notes*.	
Install WBEM on the Integrity Windows 2008 server.	
Add the WBEM protocol to the Insight RS Console.	
Discover the Windows 2008 Integrity server in the Insight RS Console.	
Send a test event to verify connectivity between your Windows 2008 Integrity server and Insight RS.	

Install and Configure Communication Software on Servers

To configure your monitored devices, complete the following sections:

Install WBEM Providers on the Monitored Device

For your Windows 2008 Integrity monitored device to be discovered by the Insight RS Console and to support configuration collections, you must have HP Integrity WBEM Providers for Windows Server 2008 installed and fully configured on your monitored devices as directed in the HP Smart Setup product documentation for the appropriate package. Windows 2008 Integrity monitored devices also use WBEM for event submission/remote monitoring.

Windows Integrity servers shipped by HP manufacturing have the Smart Setup installation files pre-loaded on the operating system disk. Otherwise the packages can be installed using the Smart Setup CD or reinstallation media. You can download the Smart Setup CD or Integrity Support Pack at:

- Windows Server 2008: http://h20000.www2.hp.com/bizsupport/TechSupport/SoftwareIndex.jsp?lang=en&cc=us&prodNameId=3346453&prodTypeId=15351&prodSeriesId=3346452&swLang=13&taskId=135&swEnvOID=4023
- Windows Server 2008 R2: http://h20000.www2.hp.com/bizsupport/TechSupport/SoftwareIndex.jsp?lang=en&cc=us&prodNameId=3346453&prodTypeId=15351&prodSeriesId=3346452&swLang=13&taskId=135&swEnvOID=4068

For WBEM Providers to communicate to the Hosting Device, you must provide the WBEM credentials within the Insight RS Console when you are completing the Insight Remote Support configuration.

Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create a WBEM Protocol Credential in Insight RS Console

To configure WBEM in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the **Select and Configure Protocol** drop-down list, select **Web-Based Enterprise Management (WBEM)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Username and Password you have configured on your device.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP address(es) of the devices to be discovered.
   d. Click **Add**.
4. Click **Start Discovery**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon ( ).

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following sections:

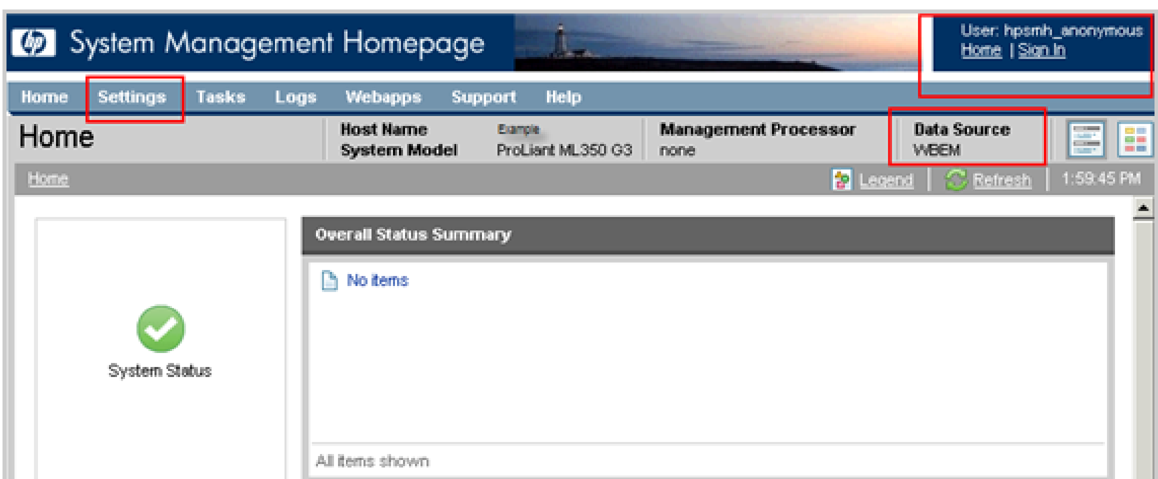
Verify Connectivity by Sending a WBEM Test Indication

WBEM Providers may be used to send WBEM Indications to verify connectivity.

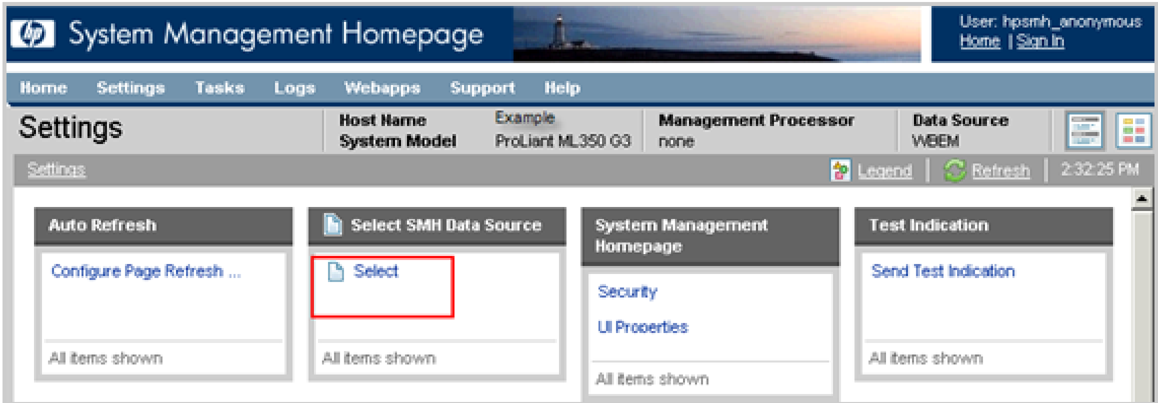
1. In a web browser, access System Management Homepage (SMH) on the monitored device: `https://ipaddress:2381`.
2. Log on using the administrator user name and password for the monitored device.

> **Note:** If you are not prompted for a logon, check the upper right corner of the SMH interface and click the **Sign In** link. If you are not logged in as an administrator for the monitored device you will not have all of the relevant configuration options.

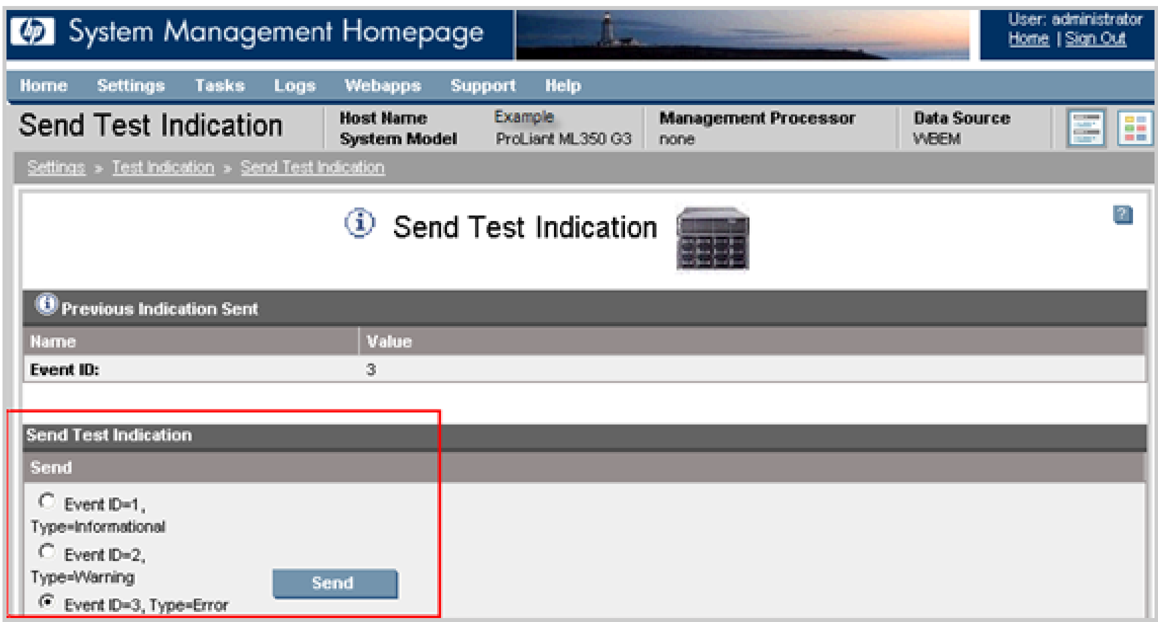
3. In the top menu bar, click **Settings**.



4. If you chose to install WBEM with the Support Pack, it will be set as your Data Source, click the **Send Test Indication** option.



5. In the Send Test Indication screen, select an Event ID type and click **Send**.

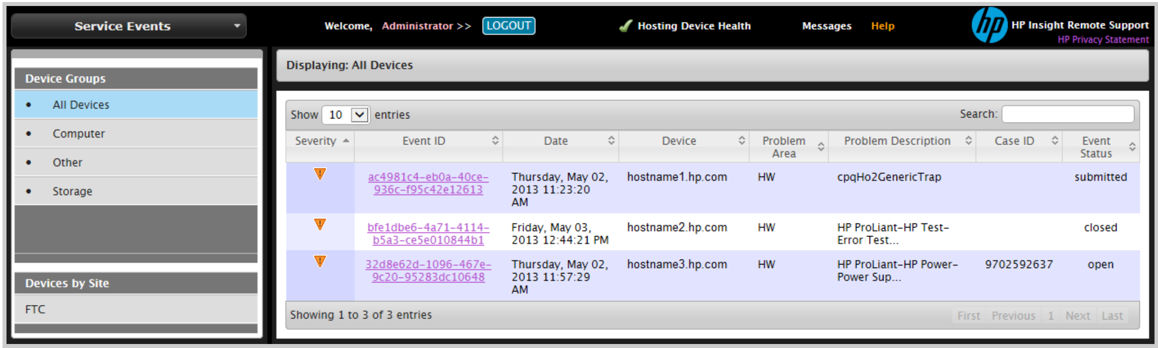


Viewing Test Events in the Insight RS Console

Some monitored device types allow you to send a test event to Insight Remote Support. After you configure your monitored device and send a test event, use the following process to verify the test event arrived.

1. Log on to the Insight RS Console.
2. In the main menu, select **Service Events**. If your monitored device is properly configured, the event

will appear in the Service Events Information pane.



Verify Collections in the Insight RS Console

After discovery, a Server Basic Configuration Collection is automatically run for servers. Verify that this collection ran successfully on the Collection Services → Basic Collection Results tab in the Insight RS Console. Note that there is a delay between when discovery finishes and when the collection runs. For more information about scheduling collections, see the Insight RS Help.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services** and then click the **Basic Collection Results** tab.
3. Expand the Server Basic Configuration Collection section.
4. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears ( ). If it failed, an error icon appears ( ).

Chapter 11: Configuring Integrity Linux Servers

Important: Configuration collections are not supported.

Note: nPars are not supported on Linux Integrity servers.

Fulfill Configuration Requirements

To configure your Integrity Linux servers to be monitored by Insight RS, complete the following sections:

Table 11.1 ***Integrity Linux Server Configuration Steps***

Task	Complete?
Make sure Insight RS supports your Integrity Linux server by checking the *HP Insight Remote Support Release Notes*.	
Verify that WBEM is installed and configured on the Integrity Linux server.	
Add the WBEM protocol to the Insight RS Console.	
Discover the Integrity Linux server in the Insight RS Console.	
Send a test event to verify connectivity between your Integrity Linux server and Insight RS.	

Install and Configure Communication Software on Servers

To configure your monitored devices, complete the following sections:

Verify HP WBEM Providers Installed

Important: When you are configuring Insight Remote Support, you *must* configure the WBEM credentials for your Linux Integrity monitored devices in the Insight RS Console. Take note of these credentials as you will need to provide them later.

The HP WBEM Providers are required on your Integrity Linux monitored device(s). The Integrity Linux HP WBEM Providers and the relevant documentation are part of the Support Pack, which is part of the larger HP Integrity Essentials Foundation Pack for Linux. HP recommends installing the HP Integrity Essentials Foundation Pack rather than installing the HP Insight Management Agent or Insight Management WBEM Provider alone as it contains additional recommended component updates. For more details, visit http://h20341.www2.hp.com/integrity/w1/en/os/linux-on-integrity-certification-matrix-novell-suse.html.

You can download the latest version of the *HP Integrity Essentials Foundation Pack for Linux* software at: http://www.hp.com/go/integritylinuxessentials.

Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create a WBEM Protocol Credential in Insight RS Console

To configure WBEM in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the **Select and Configure Protocol** drop-down list, select **Web-Based Enterprise Management (WBEM)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Username and Password you have configured on your device.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP address(es) of the devices to be discovered.
   d. Click **Add**.
4. Click **Start Discovery**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon ( ).

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following sections:

Send a Test Event to the Hosting Device

To send a test event from the monitored device to the Hosting Device, from a command line on the monitored device type the following command:

```
touch /tmp/SMX.test
```

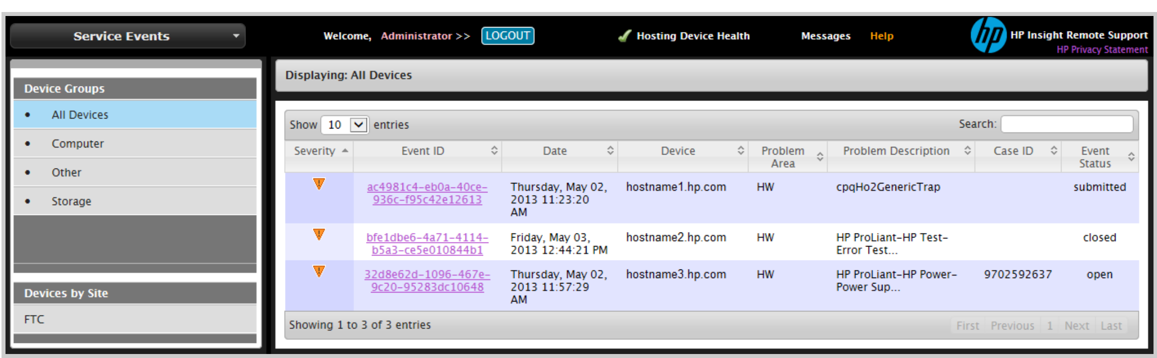
Note: This process is not supported for Red Hat Linux 4 or 5.

This creates a zero-length file called `SMX.test` in the `tmp` directory. The SMX providers will create an indication to send to the Hosting Device and remove the temporary file that you created in the previous step.

Viewing Test Events in the Insight RS Console

Some monitored device types allow you to send a test event to Insight Remote Support. After you configure your monitored device and send a test event, use the following process to verify the test event arrived.

1. Log on to the Insight RS Console.
2. In the main menu, select **Service Events**. If your monitored device is properly configured, the event will appear in the Service Events Information pane.



Verify Collections in the Insight RS Console

Configuration collections are not supported for this device type except when it is part of a SAN collection. If you have added this device to a SAN collection, you can manually run a SAN collection to verify the configuration.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services**.
3. Click the **Collection Schedules** tab.
4. In the List of Collection Schedules pane, select the **SAN Configuration Collection Schedule**. Information about the collection appears in the Collection Information pane. The On These Devices pane lists the devices the collection will run on.
5. In the Schedule Information pane, click **Run Now**.
6. When the collection completes, click the **SAN Storage Collection Results** tab.
7. Expand the SAN Configuration Collection section.
8. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears (). If it failed, an error icon appears ().

Chapter 12: Configuring Integrity HP-UX Servers

Before attempting to configure Insight Remote Support for your HP-UX monitored device, read the following information.

Note: Insight Remote Support version 7.1 only supports HP-UX on Integrity servers.

Fulfill Configuration Requirements

To configure your Integrity HP-UX servers to be monitored by Insight RS, complete the following sections:

Table 12.1 ***Integrity HP-UX Server Configuration Steps***

Task	Completed?
Make sure Insight RS supports your Integrity HP-UX server by checking the *HP Insight Remote Support Release Notes*.	
Install System Fault Management prerequisites on the Integrity HP-UX server: • OS patches • OpenSSL Secure Network Communications Protocol • Online Diagnostics • WBEM Services Core Product • System Management Web	
Install and configure System Fault Management on the Integrity HP-UX server.	
Configure WBEM users on the HP-UX server.	
Set firewall to allow communication on specific ports.	
Add the WBEM protocol to the Insight RS Console.	
Discover the HP-UX server in the Insight RS Console.	
Send a test event to verify connectivity to the Hosting Device.	

Install and Configure Communication Software on Servers

To configure your monitored devices, complete the following sections:

Install System Fault Management

System Fault Management (SFM) is an Insight Remote Support communications requirement on all supported versions of HP-UX. SFM has many patch prerequisites, and those are detailed in the section "Meeting HP-UX Operating System, Software, and Patch Requirements". If your system(s) already meet the minimum requirements identified in that section, then you do not need to remove or reinstall those components.

Note: You may be familiar with the variety of SFM providers available in the different SFM offerings. Insight Remote Support only requires the providers in the standard SFM package(s) as specified in section "Meeting HP-UX Operating System, Software, and Patch Requirements".

Web-Based Enterprise Management (WBEM) is an industry initiative to standardize management information across different platforms. System Fault Management (SFM) is the HP-UX fault management solution that implements WBEM standards. SFM integrates with other manageability applications like Insight Remote Support and HP SMH. SFM requires that HP-UX have *WBEM Services for HP-UX* and other software installed.

Events in WBEM are called indications. Before an indication can be reported to a client system (for example, the Hosting Device), the client system must first subscribe to the event. An event subscription tells the monitored device's *Common Information Model Object Manager* (CIMOM) the Hosting Device is interested in receiving indications from that monitored device. When the CIMOM receives an indication from an indication provider, it sends the indication to those clients that previously subscribed to receive them.

Insight Remote Support subscribes to those indications. Once the subscription is completed, indications are delivered to Insight Remote Support as they occur.

More information about SFM and providers can be found in the *OE/AR Depot Information for HP-UX WBEM Providers* at: http://h10018.www1.hp.com/wwsolutions/misc/hpsim-helpfiles/OEARInformation.pdf.

Consult the following tables to configure your HP-UX monitored devices *before* installing Insight Remote Support on your Hosting Device. Then, on the Hosting Device, use the Insight RS Console to discover your HP-UX monitored device and configure the System Information.

System Fault Management (SFM) is a Remote Support communications requirement on all supported versions of HP-UX. SFM has many patch prerequisites, and those are detailed in the following tables. If your system(s) already meet the minimum requirements, you do not need to remove or reinstall those components. If your systems do not meet the minimum requirements, please upgrade accordingly.

Additionally, you may be familiar with the variety of SFM providers available in the different SFM offerings. Insight Remote Support only requires the providers in the standard SFM package(s) specified in the tables below to support your monitored devices. An overview of these providers exists in the *OE/AR Depot Information for HP-UX WBEM Providers* at: http://h10018.www1.hp.com/wwsolutions/misc/hpsim-helpfiles/OEARInformation.pdf.

Important: On the Hosting Device, discover your HP-UX monitored device and configure the System Information after you have met the HP-UX prerequisites defined in this chapter.

Fulfilling Software and Patch Requirements for HP-UX 11i v2

Important: While not required for HP-UX 11i v2 (11.23), HP strongly recommends installing the latest `QPKBASE` patch bundle. `QPKBASE` fulfills all system patch requirements and provides a stable supported environment. The following table identifies the *minimum* required patches (required if `QPKBASE` is not installed).

Note: *Notes About the Following Table*

- WBEM Services, Online Diagnostics, and SysMgmtWeb are available on the Operating Environment (OE) media and can be selected for install during the SFM installation.
- The listed software versions are the minimum supported requirements. Newer versions are supported unless otherwise noted.
- System Management Homepage (SMH), bundled in SysMgmtWeb, is optional, but it allows you to use SFM's EVWEB GUI component to view events handled by SFM on the host.
- The table lists items in the sequence in which they should be installed in case any of them are missing and require upgrade or installation.

Table 12.2 ***Required Software Components for HP-UX 11i v2***

Required Software	Version Required
HP-UX Operating Environment	• September, 2004 11i v2 OE (minimum OE release) • May, 2005 11i v2 OE (required for vPars enablement).
How to Download or Access Software: http://www.hp.com/go/hpux11ioe	
Operating System Patch Requirements **Note:** These six additional patches for HP-UX 11i v2 are *not* part of the Bundle11i and must be installed in addition to it.	BUNDLE 11i patch bundle B.11.23.0409.3 (September, 2004): • PHKL_36288 - 11.23 Cumulative diag2 driver and vPars enablement (use in place of PHKL_32653), requires a reboot • PHKL_34795 - 11.23 Cumulative IPMI driver patch, requires a reboot • PHSS_37552 - 1.0 Aries cumulative patch • PHSS_37947 - 1.0 linker + fdp cumulative patch • PHSS_35055 - aC++ Runtime (IA: A.06.10, PA: A.03.71) • PHSS_36345 - 11.23 Integrity Unwind Library
How to Download or Access Software: Go to http://www.hp.com/go/hpsc and sign in using your HP Passport account. In the left menu, click the **Patch management** link. Type the patch name into the Patch Management search box and click **GO**.	
OpenSSL Secure Network Communications Protocol	*A.00.09.07i.012 product bundle (December, 2006) or later*

Table 12.2 ***Required Software Components for HP-UX 11i v2, continued***

<table>
<tr><th>Required Software</th><th>Version Required</th></tr>
<tr><td colspan="2">How to Download or Access Software:
Available on Application Software Media from December, 2006 onwards
Alternatively, go to the following HP Software Depot location for the latest version: https://h20392.www2.hp.com/portal/swdepot/try.do?productNumber=OPENSSL11I</td></tr>
<tr><td colspan="2"></td></tr>
<tr><td>Online Diagnostics</td><td>B.11.23.10.05 HPUX 11.23 Support Tools Bundle (December, 2007) or later</td></tr>
<tr><td colspan="2">How to Download or Access Software:
Available on HWE0706 media
Alternatively, go to the following HP Software Depot location for the latest version: https://h20392.www2.hp.com/portal/swdepot/try.do?productNumber=B6191AAE</td></tr>
<tr><td colspan="2"></td></tr>
<tr><td>WBEMSvcs (WBEM Services Core Product)</td><td>A.02.05.08 WBEM Services CORE Product (December, 2007) or later</td></tr>
<tr><td colspan="2">How to Download or Access Software:
Available on Application Software Media from December, 2007 onwards
Alternatively, go to the following HP Software Depot location for the latest version: https://h20392.www2.hp.com/portal/swdepot/try.do?productNumber=WBEMSvcs</td></tr>
<tr><td colspan="2"></td></tr>
<tr><td>System Management Web (recommended for event monitoring, required for configuration collection)</td><td>A.2.2.7 HP-UX Web Based System Management User Interface (December 2007) or later</td></tr>
<tr><td colspan="2">System Management Web is recommended to take full advantage of SFM's EVWEB GUI component which allows viewing of events handled by SFM on the host.

If only SysMgmtHomepage version 2.2.6.2 is present on the server, then the following patch must also be applied:

For HP-UX 11i v2 (11.23) OE apply patch PHSS_36870

Patches can be obtained from the HP Support Center:

Go to http://www.hp.com/go/hpsc and sign in using your HP Passport account. In the left menu, click the Patch management link. Type the patch name into the Patch Management search box and click GO

This is a "sharfile" and must be unpacked by running it as a script ("sh PHSS_36870). Be sure to review the patch installation instructions in the enclosed text file (<code>PHSS_36870.text</code>)</td></tr>
<tr><td colspan="2">How to Download or Access Software:
Available on Application Software Media from September 2007 onwards
Alternatively, go to the following HP Software Depot location for the latest version: https://h20392.www2.hp.com/portal/swdepot/try.do?productNumber=SysMgmtWeb</td></tr>
<tr><td colspan="2"></td></tr>
<tr><td>System Fault Management (SFM)</td><td>B.07.01.01.yy System Fault Management (May, 2009) or later</td></tr>
</table>

Table 12.2 ***Required Software Components for HP-UX 11i v2, continued***

Required Software	Version Required
Important: It is *critical* that SFM is the final prerequisite software component installed or upgraded from this list.	
How to Download or Access Software: • Available on HWE0712 media • Alternatively, go to the following HP Software Depot location for the latest version: https://h20392.www2.hp.com/portal/swdepot/displayProductInfo.do?productNumber=SysFaultMgmt **Note:** See the *System Fault Management Release Notes* for more details and additional SFM prerequisites.	

Fulfilling Software and Patch Requirements for HP-UX 11i v3

Note: *Notes About the Following Table*

- WBEM Services, Online Diagnostics, and SysMgmtWeb are available on the Operating Environment (OE) media and can be selected for install during the SFM installation.
- The listed versions of the software are the minimum supported requirements. Newer versions are supported unless otherwise noted.
- System Management Homepage (SMH), bundled in SysMgmtWeb, is optional, but it allows you to use SFM's EVWEB GUI component to view events handled by SFM on the host.
- The table lists items in the sequence in which they should be installed in case any of them are missing and require upgrade or installation.

Table 12.3 ***Required Software Components for HP-UX 11i v3***

Required Software	Version Required
HP-UX Operating Environment	HP-UX 11i v3
How to Download or Access Software: http://www.hp.com/go/hpux11ioe	
Operating System Patch Requirements	• EVM-Event Mgr B.11.31 • Baseboard Management Controller (BMC) firmware version 05.21 or later • SysMgmtBase B.00.02.03 or later
How to Download or Access Software: Go to http://www.hp.com, and click the Support and Drivers tab. Then click the Download patches for HP-UX, Open VMS, Tru64 and MPE button on the right side of the web page. This brings you to the ITRC where you can search for the appropriate patch.	

Table 12.3 ***Required Software Components for HP-UX 11i v3, continued***

<table>
<tr><th>Required Software</th><th>Version Required</th></tr>
<tr><td>OpenSSL Secure Network Communications Protocol</td><td>A.00.09.07e.013 or later</td></tr>
<tr><td colspan="2">How to Download or Access Software:
Available on Application Software Media from September 2007 onwards
Alternatively, go to the following HP Software Depot location for the latest version:
https://h20392.www2.hp.com/portal/swdepot/try.do?productNumber=OPENSSL11I</td></tr>
<tr><td colspan="2"></td></tr>
<tr><td>Online Diagnostics: Diagnostic and Support Tools for HP-UX, including STM version A.49.10 or later and EMS version A.04.20 or later</td><td>B.11.31.01.yy or later (SysFaultMgmt version dependency.)</td></tr>
<tr><td colspan="2">How to Download or Access Software:
Available HP-UX 11i v3 - February 2007 media
Alternatively, go to the following HP Software Depot location for the latest version:
https://h20392.www2.hp.com/portal/swdepot/try.do?productNumber=B6191AAE</td></tr>
<tr><td colspan="2"></td></tr>
<tr><td>WBEMSvcs (WBEM Services Core Product)</td><td>A.02.05 or later</td></tr>
<tr><td colspan="2">How to Download or Access Software:
Available on Application Software Media from September 2007 onwards
Alternatively, go to the following HP Software Depot location for the latest version:
https://h20392.www2.hp.com/portal/swdepot/try.do?productNumber=WBEMSvcs</td></tr>
<tr><td colspan="2"></td></tr>
<tr><td>System Management Web (recommended for event monitoring, required for configuration collection)</td><td>A.2.2.4 HP-UX Web Based System Management User Interfaces (December, 2007) or later</td></tr>
<tr><td colspan="2">System Management Web is recommended to take full advantage of SFM's EVWEB GUI component which allows viewing of events handled by SFM on the host.

If only SysMgmtHomepage version 2.2.6.2 is present on the server, then the following patch must also be applied:

For HP-UX 11i v3 (11.31) OE apply patch PHSS_36871

Patches can be obtained from the HP Support Center:

Go to http://www.hp.com/go/hpsc and sign in using your HP Passport account. In the left menu, click the Patch management link. Type the patch name into the Patch Management search box and click GO

This is a "sharfile" and must be unpacked by running it as a script ("sh PHSS_36871). Be sure to review the patch installation instructions in the enclosed text file (<code>PHSS_36871.text</code>)</td></tr>
<tr><td colspan="2">How to Download or Access Software:
Available on Application Software Media from September 2007 onwards
Alternatively, go to the following HP Software Depot location for the latest version:
https://h20392.www2.hp.com/portal/swdepot/try.do?productNumber=SysMgmtWeb</td></tr>
<tr><td colspan="2"></td></tr>
</table>

Table 12.3 ***Required Software Components for HP-UX 11i v3, continued***

Required Software	Version Required
System Fault Management (SFM)	*C.07.10.08.yy HPUX System Fault Management or later*
It is *critical* that SFM is the final prerequisite software component installed or upgraded from this list.	
How to Download or Access Software: HP recommends installing SFM through the WBEMMgmtBundle, available at:https://h20392.www2.hp.com/portal/swdepot/displayProductInfo.do?productNumber=WBEMMgmtBundle **Note:** See the *System Fault Management Release Notes* for more details and additional SFM prerequisites.	

Verify System Fault Management is Operational

To verify the System Fault Management (SFM) component is operational on a HP-UX monitored device, complete the following steps:

1. Run the following command to verify the *HP WBEM Services for HP-UX* component is installed:

   # swlist | grep -i WBEM

   The output should look similar to the following:

```
# swlist | grep -i WBEM
  LVMProvider                    R11.23.008     CIM/WBEM Provider for LVM
  ProviderDefault                B.11.23.0706   Select WBEM Providers
  VMProvider                     A.03.00.76     WBEM Provider for Integrity VM
```

2. To list registered CIM providers and their current status to make sure that they are all enabled, run the following command:

   # cimprovider -l -s

3. Run the following command to check *OnlineDiag* is installed and make sure it is version B.11.11.16xx or later.

   # swlist | grep -i OnlineDiag

   The output should look similar to the following:

```
# swlist | grep -i OnlineDiag
  OnlineDiag                    B.11.31.03.06  HPUX 11.31 Support Tools Bundle, March 2008
```

4. Run the following command to check that OnlineDiag is reporting that *Event Monitoring is Currently Enabled* and that EMS version is A.04.20 or later and the STM is A.49.10 or later:

   # /etc/opt/resmon/lbin/monconfig

   The output should look similar to the following:

```
# /etc/opt/resmon/lbin/monconfig
============================================================================
===================        Event Monitoring Service       ===================
===================       Monitoring Request Manager      ===================
============================================================================

  EVENT MONITORING IS CURRENTLY ENABLED.
  EMS Version : A.04.20.31
  STM Version : D.03.00

============================================================================
==============      Monitoring Request Manager Main Menu      ==============
============================================================================

Note: Monitoring requests let you specify the events for monitors
      to report and the notification methods to use.

Select:
   (S)how monitoring requests configured via monconfig
   (C)heck detailed monitoring status
   (L)ist descriptions of available monitors
   (A)dd a monitoring request
   (D)elete a monitoring request
   (M)odify an existing monitoring request
   (E)nable Monitoring
   (K)ill (disable) monitoring
   (H)elp
   (Q)uit
   Enter selection: [s]
```

5. Select '**Q**' to exit the *EMS Monitoring Request Manager Main Menu*.
6. Disable EMS hardware monitors for HP-UX 11i v2 and 11i v3.
   a. Check if SFM is used for hardware monitoring:

      # /opt/sfm/bin/sfmconfig -w -q

   b. If EMS is enabled for monitoring, switch to SFM using the command:

      # /opt/sfm/bin/sfmconfig -w -s

7. Run the following command to determine the version of the optional, but recommended, SMH component is at least version A.2.2.6.2:

   # swlist SysMgmtWeb SysMgmtHomepage

   The output should look similar to the following:

```
# swlist SysMgmtWeb SysMgmtHomepage
# Initializing...
# Contacting target "hpux-server"...
#
# Target:  hpux-server:/
#
# SysMgmtWeb                                 A.2.2.8     HP-UX Web Based System Management User Interfaces
# SysMgmtWeb.SysMgmtHomepage                 A.2.2.8     HP-UX System Management Homepage - Web-Based User Interfaces
  SysMgmtWeb.SysMgmtHomepage.SMH-DOC         A.2.2.8     HP-UX System Management Homepage Help
  SysMgmtWeb.SysMgmtHomepage.SMH-DOC-COM     A.2.2.8     HP-UX System Management Homepage Help
  SysMgmtWeb.SysMgmtHomepage.SMH-PPAGES      A.2.2.8     HP-UX System Management Homepage Property Pages
  SysMgmtWeb.SysMgmtHomepage.SMH-PPAGES-COM  A.2.2.8     HP-UX System Management Homepage Property Pages (common files)
  SysMgmtWeb.SysMgmtHomepage.SMH-RUN         A.2.2.8     HP-UX System Management Homepage Runtime Core
  SysMgmtWeb.SysMgmtHomepage.SMH-SAMLOG      A.2.2.8     HP-UX System Management Homepage LogViewer Web Application
  SysMgmtWeb.SysMgmtHomepage.SMH-UILIB       A.2.2.8     HP-UX System Management Homepage User Interface Library
  SysMgmtWeb.SysMgmtHomepage.SMH-UILIB-COM   A.2.2.8     HP-UX System Management Homepage User Interface Library (common files)
  SysMgmtWeb.SysMgmtHomepage.SMH-XLAUNCH     A.2.2.8     HP-UX System Management Homepage Xlaunch Web Application
```

8. Change the startup mode for SMH so that the autostart URL mode is set to `OFF` and start on boot mode is set to `ON`:

   ```
   # /opt/hpsmh/lbin/hpsmh stop
   # /opt/hpsmh/bin/smhstartconfig -a off -b on
   # /opt/hpsmh/lbin/hpsmh start
   # /opt/hpsmh/bin/smhstartconfig
   ```

   ```
   HPSMH 'autostart url' mode.........: OFF
   HPSMH 'start on boot' mode.........: ON
   Start Tomcat when HPSMH starts.....: OFF
   ```

Create WBEM Users

Insight RS requires the WBEM Services user and password to communicate with the HP-UX monitored device. You can create a non-privileged user or, if using WBEM A.02.09.08 or later, create a privileged user in the Insight Remote configuration file.

If using HP-UX 11i v2, or v3, and a version of WBEM Services earlier than A.02.09.08, see "Creating Non-Privileged Users with cimauth" below.

If using HP-UX 11i v2 or v3 and WBEM Services A.02.09.08 or later, see "Creating WBEM Privileged Users with WBEM A.02.09.08 or Later" on page 119.

For details on SysFaultMgmt, including defining WBEM user/password accounts on HP-UX, refer to the *HP-UX System Fault Management Administration Guide* at: http://www.hp.com/go/hpux-diagnostics-sfm-docs.

Creating Non-Privileged Users with cimauth

It is possible to use a non-privileged account for WBEM communication. Create a user or use an existing non-privileged user. The specified username must represent a valid HP-UX user on the local host.

To create a non-privileged account on an HP-UX monitored device, complete the following steps:

1. Create a user, *hpirs* in this example, and assign the user to the *users* group:

   # useradd –g users hpirs

2. Set the password for user *hpirs*:

   # passwd hpirs (when prompted, provide and confirm the password)

3. Review the current CIM configuration, as follows:

   # cimconfig –l –p

   Example Output:

   ```
   sslClientVerificationMode=disabled
   enableSubscriptionsForNonprivilegedUsers=false
   shutdownTimeout=30
   authorizedUserGroups=
   enableRemotePrivilegedUserAccess=false
   ```

```
enableHttpsConnection=true
enableHttpConnection=false
```

4. Based on the above output, set the following variables in the CIM planned configuration:

   # cimconfig –s enableSubscriptionsForNonprivilegedUsers=true –p

   # cimconfig –s enableNamespaceAuthorization=true –p

5. Stop and start the CIM Server to set the configuration changes in the CIM current configuration:

   # cimserver –s

   # cimserver

6. Verify the settings in the CIM current configuration:

   # cimconfig –l -c

   Example Output:

   ```
   sslClientVerificationMode=disabled
   enableSubscriptionsForNonprivilegedUsers=true
   shutdownTimeout=30
   authorizedUserGroups=
   enableRemoteprivilegedUserAccess=true
   enableHttpsConnection=true
   enableNamespaceAuthroization=true
   enableHttpConnection=false
   ```

7. Add read and write authorizations for the user *hpirs* to each of the namespaces, root/cimv2, root/PG_InterOp, and root/PG_Internal:

   # cimauth –a –u *hpirs* –n root/cimv2 –R –W

   # cimauth –a –u *hpirs* –n root/PG_InterOp –R –W

   # cimauth –a –u *hpirs* –n root/PG_Internal –R –W

   # cimauth –a –u *hpirs* –n root/cimv2/npar –R –W

   # cimauth –a –u *hpirs* –n root/cimv2/vpar –R –W

8. Verify the user's authorizations:

   #cimauth -l

   Example Output:

   ```
   hpirs, root/PG_InterOp, "rw"
   hpirs, root/PG_Internal, "rw"
   hpirs, root/cimv2, "rw"
   ```

Creating WBEM Privileged Users with WBEM A.02.09.08 or Later

Specifying privileged users in WBEM release A.02.09.08 for HP-UX 11iv2 and 11iv3 servers has changed and no longer requires the use of `cimauth` commands. HP WBEM Services version A.02.09.08 and later supports Insight Remote Support configuration on HP-UX 11i v2 and HP-UX 11i v3 operating systems.

After installing WBEM version A.02.09.08 on previously functioning HP-UX 11iv2 and 11iV3 servers monitored by Insight Remote Support, monitoring functionality may be broken. This is primarily due to how privileged users are specified. In previous versions, root was the default user and additional users with root privileges had to be added and configured with the `cimauth` command on the HP-UX server. With WBEM release A.02.09.08 and above, this is now done with an Insight Remote Support configuration file, which contains a default user called `hp_irs`.

This update removes the need to provide root credentials for WBEM access or the creation of special users with root privileges and no system access to be used with WBEM.

To configure root privileges for Insight Remote Support users on an HP-UX system, complete the following steps:

1. Install HP WBEM Services version A.02.09.08 or later on HP-UX 11i v3 and 11i v2 operating systems.
2. On the HP-UX server, edit the Insight Remote Support configuration file located at: `/var/opt/wbem/hp_irs_users.conf` and type the required user name. By default, the `hp_irs` user name is added in the Insight Remote Support configuration file.

   Note: Only system administrators can modify or create user names in the Insight Remote Support configuration file. Ensure that the user name exists on the HP-UX system before configuring in the Insight Remote Support configuration file.

3. Change `enableSubscriptionsForNonprivilegedUsers` to true:

   `# cimconfig -s enableSubscriptionsForNonprivilegedUsers=true -p`

4. Stop the WBEM Services:

   `# cimserver -s`

5. Restart the WBEM Services:

   `# cimserver`

Note: Users configured in the Insight Remote Support configuration file can perform WBEM operations with root privileges. However, these users can still continue to have system privileges as defined in the HP-UX `/etc/passwd` file.

Important: For previous versions of WBEM and that have users configured with the `cimauth` command, WBEM Services can stop working after performing the `update-ux` command (updating to the September 2011 HP-UX version.) The WBEM bundle is in an *Installed* state when it should be in a *Configured* state. Run `swconfig` on all filesets to resolve this issue.

Once WBEM version A.02.09.08 is installed, update the WBEM credentials/protocols in Insight RS or HP SIM to reflect the new user `hp_irs`, or add the previously created users to the `hp_irs_users.conf` file.

For more information about WBEM, see the *HP WBEM Services for HP-UX System Administrator Guide*.

Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Add the WBEM Protocol to the Insight RS Console

Insight RS automatically associates the WBEM protocol configured in the Insight RS Console with the HP-UX server.

Option 1: Authenticate Using Username and Password

To configure WBEM in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the **Select and Configure Protocol** drop-down list, select **Web-Based Enterprise Management (WBEM)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Username and Password you have configured on your device. Use the user that was created previously in "Create WBEM Users" on page 117.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Option 2: Authenticate to HP-UX WBEM Using a Certificate

Important: Repeat these steps one per year. The Jetty certificate is valid for a year, and while it does automatically renew itself each year, the certificate needs to be moved to the HP-UX server after it renews. To see when the Jetty certificate expires, change the filename from `.pem` to `.cer` and then open the certificate to see property details.

To copy the certificate to the HP-UX server, complete the following steps:

1. On the Hosting Device, open a DOS window and export the public certificate to a file called `UCACMS.pem`, so it can be moved to the HP-UX server:

   `rsadmin cert –export -out c:\temp\UCACMS.pem`

2. FTP the `UCACMS.pem` file in ASCII format to the `cimserver_trust` folder on the HP-UX server:

   `# ftp <hp-ux_server_ip_address>`

   `ftp> cd /etc/opt/hp/sslshare/cimserver_trust/`

   `ftp> ASCII`

   `ftp> put UCACMS.pem`

   `ftp> quit`

3. From the Hosting Device, remotely log on to the HP-UX server.

4. Browse to the following directory:

   `cd /etc/opt/hp/sslshare/cimserver_trust/`

5. Type the following commands to set up the WBEM certificate authentication on the HP-UX server:

   a. Associate the Insight RS `UCACMS.pem` certificate with the root user:

      `# cimtrust -a -U root -f /etc/opt/hp/sslshare/cimserver_trust/UCACMS.pem -T s`

   b. Associate the HP-UX certificate with the root user:

      `# cimtrust -a -U root -f /etc/opt/hp/sslshare/cert.pem -T s`

   c. Check the current `cimom` values:

      `# cimconfig -l -c`

      ```
      enableAuditLog=false
      sslClientVerificationMode=optional
      idleConnectionTimeout=0
      enableSubscriptionsForNonprivilegedUsers=true
      socketWriteTimeout=20
      shutdownTimeout=30
      authorizedUserGroups=
      enableRemotePrivilegedUserAccess=true
      enableHttpsConnection=true
      enableNamespaceAuthorization=true
      enableHttpConnection=false
      ```

   d. Make sure the fields in red are defined as above. If they are not, change the values and validate them using the following commands:

      i. Set `sslClientVerificationMode` to `optional`:

         `# cimconfig -s sslClientVerificationMode=optional –p`

      ii. Set `enableHttpsConnection` to `true`:

`# cimconfig -s enableHttpsConnection=true -p`

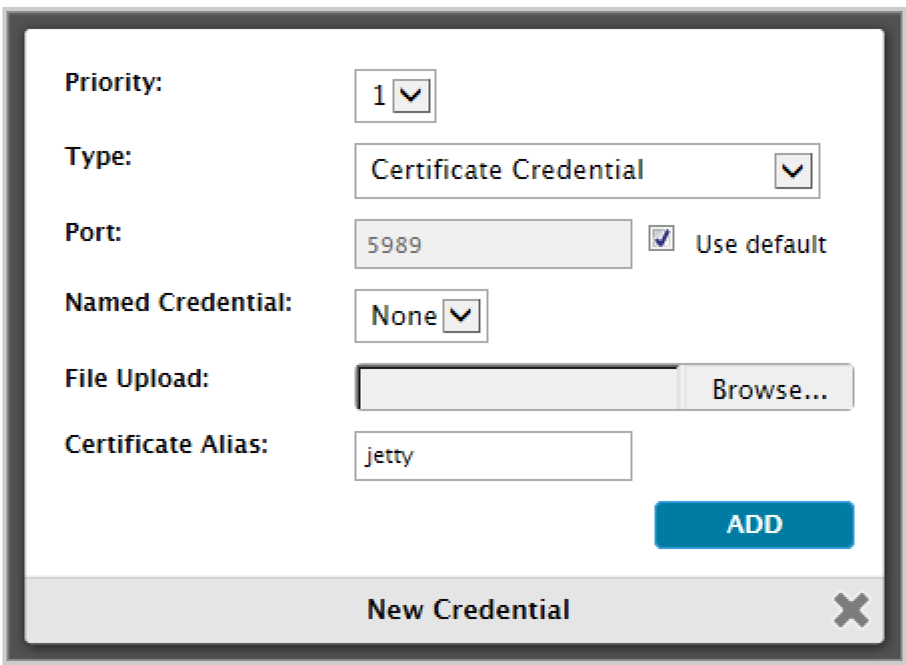
iii. Stop the `cimom` daemon:

`# cimserver -s`

iv. Start the `cimom` daemon:

`# cimserver`

6. In the Insight RS Console, add a WBEM certificate credential on the **Discovery** → **Credentials** tab.
   a. From the **Select and Configure Protocol** drop-down list, select **Web-Based Enterprise Management (WBEM)**.
   b. Click **New**. The New Credential dialog box appears.
   c. Select **Certificate Credential** from the **Type** drop-down list.
   d. Leave the **File Upload** field blank because the certificate is already in the certificate store and it is identified by using the alias name.
   e. Type the Certificate Alias of "jetty", which is the alias given to the certificate when it was exported above.



Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:

   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP address(es) of the devices to be discovered.
   d. Click **Add**.
4. Click **Start Discovery**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon ( ).

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following sections:

Verify Connectivity by Sending an SNMP Test Trap to the Hosting Device

Send a test event from the HP-UX server to verify communication between the HP-UX server and the Hosting Device. Execute one of the following commands on the HP-UX server:

```
# /etc/opt/resmon/lbin/send_test_event disk_em
```

or

```
# /opt/sfm/bin/sfmconfig -t -a
```

For details on SysFaultMgmt, including issuing test events, refer to the *HP-UX System Fault Management Administration Guide* at: http://www.hp.com/go/hpux-diagnostics-sfm-docs.

Verify Collections in the Insight RS Console

After discovery, a Server Basic Configuration Collection is automatically run for servers. Verify that this collection ran successfully on the Collection Services → Basic Collection Results tab in the Insight RS Console. Note that there is a delay between when discovery finishes and when the collection runs. For more information about scheduling collections, see the Insight RS Help.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services** and then click the **Basic Collection Results** tab.
3. Expand the Server Basic Configuration Collection section.
4. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears (). If it failed, an error icon appears ().

Chapter 13: Configuring Integrity Superdome 2 Servers

The Superdome 2 system complex has two independent interfaces that report event indications: the HP-UX partitions and the Superdome 2 Onboard Administrator (OA). HP-UX System Fault Manager (SFM) monitors the IO devices within the partition domain and generates WBEM indications when it determines an important event has occurred. The OA monitors the remainder of the system hardware and generates WS-Management indications when it determines that an important event occurs. To enable complete system monitoring, both the OA and the partitions must be monitored.

The HP-UX partitions and the OA can be discovered and monitored as individual components or by using OA functionality included in firmware bundle 3.6.44 and later. The OA can be configured to proxy HP-UX SFM WBEM indications and by doing so be the sole source for event reporting for the entire system complex. However, because Insight RS still needs to connect to the partitions to support configuration collections, the HP-UX partitions must still be discovered by Insight RS regardless of the method used to monitor the system for event indications.

The Superdome 2 OA can be configured to send alert messages using email, SNMP or WS-Man indications. While email can be used to directly notify your IT staff or SNMP can be used to send Traps to enterprise management applications (such as HP OpenView), Insight RS requires that WS-Man be used for event monitoring.

Important: If you upgraded from Insight RS 7.0.5, Superdome 2 servers were not supported in version 7.0.5. If you discovered your Superdome 2 servers in version 7.0.5 they appeared as unknown devices in the Insight RS Console. After you upgrade to version 7.1, they appear correctly as Superdome 2 servers, but Insight RS cannot monitor them until you complete the steps below and rediscover the Superdome 2 server.

Fulfill Configuration Requirements

To configure your Integrity Superdome 2 servers to be monitored by Insight RS, complete the following sections:

Table 13.1 *Integrity Superdome 2 Server Configuration Steps*

Task	Completed?
Make sure Insight RS supports your Integrity Superdome 2 server by checking the *HP Insight Remote Support Release Notes*.	
Verify Superdome 2 network settings and protocol status.	
Add the WBEM protocol to the Insight RS Console.	
Add the WS-Man protocol to the Insight RS Console.	
Discover the Superdome 2 server in the Insight RS Console.	
Verify the Superdome 2 and the partitions were discovered correctly.	
Send a test event to verify connectivity to the Hosting Device.	

Install and Configure Communication Software on Servers

To configure your monitored devices, complete the following sections:

Superdome 2 OA Configuration Verification

Connect to the Superdome 2 OA using Telnet, SSH or a web browser (HTTPS), and verify the network settings and protocol status of the OA.

Verify the Network Settings are correct, including the DNS server addresses for both the active and standby OAs, and verify the Protocol Status indicates that WS-Man is enabled. WS-Man is the supported event reporting protocol for the Superdome 2 OA. SNMP should be disabled, unless you require its use, because the SNMP Agents send redundant event notifications to Insight RS. HP recommends that Enclosure IP Mode is enabled.

Using Telnet or SSH

1. Telnet or SSH to the Onboard Administrator CLI.
2. Verify DNS is configured: type the **show network** command to show the Network Settings and Protocol Status of the OA.

Using a Web Browser

1. In a web browser, log on to the Onboard Administrator web interface.
2. In the left menu, select **Enclosure Information** → **Enclosure Settings** → **Enclosure TCP/IP Settings**.
3. Select **Network Access** and verify that WS-Management is enabled.
4. In the left menu, click the **Complex Information** link under Complex Overview. Click on the **Information** tab to show the Product Number and Serial Number.

Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Add Protocol Credentials and Start Discovery

You must add WBEM and WS-Man protocol credentials to the Insight RS Console so Insight RS can communicate with the Superdome 2 server.

To discover your monitored devices, complete the following sections:

Create a WBEM Protocol Credential in Insight RS Console

To configure WBEM in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. Selection the following filters:
   a. From the **Select Type** drop-down list, select **Server**.
   b. From the **Select Sub-Type** drop-down list, select **HP Integrity HPUX**.
   c. From the **Select and Configure Protocol** drop-down list, select **Web-Based Enterprise Management (WBEM)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Username and Password you have configured on your partitions. These are the credentials Insight RS will use to communicate with the HP-UX partitions.
6. Click **Add**.

Create a WS-Man Protocol Credential in the Insight RS Console

To configure WS-Man in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the **Select and Configure Protocol** drop-down list, select **Web Services Management Protocol (WSMAN)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Username and Password you have configured on your OA. These are the credentials Insight RS will use to communicate with the OA.
6. Click **Add**.

Discover the Superdome 2 Server in the Insight RS Console

To discover the Superdome 2 server from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.

   c. Type the IP addresses of the OAs and HP-UX partitions to be discovered.
   d. Click **Add**.
4. Click **Start Discovery**.

Note: If discovery is configured to discover standby OAs, it will populate the Devices screen with minimal information on these devices because although their network interfaces can be active and be reachable, they are not monitoring port 443 used for communication and therefore cannot provide Insight RS with information about themselves. Likewise because they are standby devices Insight RS will not be able to establish event monitoring subscriptions directly with them. If an OA failover should occur, the OA will pass the subscription to the new Monarch and Insight RS will continue to monitor the OA for event indications using that active subscription.

Verify Superdome 2 Discovery

After discovery completes, select **Devices** from the main menu. Verify that all the Superdome 2 OAs and HP-UX partitions previously configured for discovery were discovered and that the Status for those devices shows the success icon (✅).

Verify HP-UX Partitions

For HP-UX devices, verify the device information as detailed in the HP-UX server section. See "Configuring Integrity HP-UX Servers" on page 109.

Verify the Superdome 2 OA

To verify the device information for Superdome 2 OA devices, complete the following steps:

1. In the main menu, select **Devices**. Locate the Superdome 2 OA, and click its Device Name.
2. On the **Device** tab, expand the Status section and verify the Registration Status shows **Registered**, and the Enabled has the **Yes** option selected.
3. Expand the Hardware section and verify the Device Name, Category, Product, Vendor, IP Address, Acquired Serial Number and Acquired Product Number are populated correctly for the device.
4. Expand the Warranty & Contract section and verify the values listed are correct.

   Note: The OS Name and OS Version fields will be blank if viewed from the Operating System section.

5. Expand the Delivery section and verify the information is correct.
6. Click **Save Changes** to complete the discovery process for this device.

Note: Insight RS only monitors the Monarch (Active) OA. Insight RS subscribes to the OA that is active at the time of Insight RS discovery and uses the subscription to listen for events. If an

OA failover occurs, Insight RS continues to monitor the active subscription once it is passed by the Active OA to the Standby OA. The Insight RS Console will continue to show that it is actively monitoring the previously configured Active OA (even though that OA has failed).

Note: A Superdome 2 32s configuration consists of two enclosures joined into a single server complex. Each enclosure has a primary and a standby OA that manages and monitors that enclosure. Only one of the two active OAs in the Superdome 2 32s complex is the Monarch OA for the complex, and only it generates the WS-Man event indications produced by the Error Analysis Engine. As is the case with Superdome 2 16s complex configurations, Insight RS will only subscribe to the complex Monarch OA for WS-Man event indications and Insight RS will only display the Monarch OA on its Managed Entity page.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following sections:

Generating Test Events

Superdome 2 firmware version 2.2.27 or later supports the capability for an administrator to connect to the OA and generate a WS-Man test indication. Use the following procedure to generate a WS-Man test indication:

1. Telnet or SSH to the OA CLI interface and log on using a user account with administrator privileges.
2. Type the **test wsman** command.
3. In a web browser, log on to the Insight RS Console.
4. In the main menu, select **Service Events**. Click on the Date column header twice to sort the most recent event to the top, and verify that Insight RS received the test event.

"Configuring Integrity HP-UX Servers" on page 109 describes how to generate test events from each of the HP-UX partitions. Generate a test event and verify the HP-UX SFM test events have been received and processed by Insight RS.

Verify Collections in the Insight RS Console

After discovery, a Server Basic Configuration Collection is automatically run for servers. Verify that this collection ran successfully on the Collection Services → Basic Collection Results tab in the Insight RS Console. Note that there is a delay between when discovery finishes and when the collection runs. For more information about scheduling collections, see the Insight RS Help.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services** and then click the **Basic Collection Results** tab.
3. Expand the Server Basic Configuration Collection section.

4. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears (). If it failed, an error icon appears ().

Chapter 14: Configuring OpenVMS Servers

Important: Configuration collections for OpenVMS on AlphaServer are not available except when the OpenVMS server is part of a SAN collection.

Important: HP only provides limited remote support for customers using HP Alpha servers. As with standard support, HP Insight Remote Support version 7.x will attempt to discover these products as remotely monitored HP products and send information about failure events to HP.

However, if there are issues with the operation of the remote monitoring features of HP Insight Remote Support version 7.x for these specific products, including the remote discovery or the reporting of failure events to HP, then HP will resolve such issues in a commercially reasonable manner but without guarantee of an immediate response or a resolution. If a failure event is successfully received at HP, it is managed according to the support level agreement.

Fulfill Configuration Requirements

To configure your OpenVMS servers to be monitored by Insight RS, complete the following sections:

Table 14.1 ***OpenVMS Server Configuration Steps***

Task	Complete?
Make sure Insight RS supports your Integrity OpenVMS server by checking the *HP Insight Remote Support Release Notes*.	
Install and configure ELMC for OpenVMS on the OpenVMS server.	
Add ELMC to the Insight RS Console.	
Discover the OpenVMS server in the Insight RS Console.	
(Optional) Configure Dynamic Processor Resilience (for Integrity servers only).	

Install and Configure Communication Software on Servers

To configure your monitored devices, complete the following sections:

Fulfill ELMC System Requirements

OpenVMS monitored devices must meet the following requirements before ELMC is installed. In clusters, minimum requirements apply to each node in the cluster.

Fulfilling Hardware and Software Requirements

- Processor architecture:
  - Integrity (Itanium) Server
  - HP AlphaServer
- Operating system:
  - OpenVMS Itanium (Integrity) — 8.3-1H1 or higher.

    Note: Some supported OpenVMS platforms require a minimum version of 8.4. See http://h71000.www7.hp.com/openvms/hw_supportchart.html.

    Note: HP Sustaining Engineering maintains a schedule of support for the OpenVMS operating system. HP does not commit to supporting Insight Remote Support when installed on an operating system version that has exceeded its end-of-support date. For more information, go to: http://www.hp.com/hps/os/os_pvs_amap.html.

  - OpenVMS AlphaServer — 7.3-2 or higher
- Minimum 20,000 blocks free disk space.
- Connectivity: TCP/IP must be installed and running.

  Even if TCP/IP traffic to other machines has been disabled, the ability to resolve the local host name into an IP address must be enabled. Otherwise, the Director cannot handle ELMC message traffic correctly and fails to start.

  ELMC officially supports only one TCP/IP product for OpenVMS: HP TCP/IP Services for OpenVMS, version 5.4 or higher.

  Note: Other TCP/IP products may work as is, so the Insight Remote Support installation always completes regardless of what, if any, TCP/IP product is installed.

- LOCALHOST entry: For ELMC to operate correctly, the LOCALHOST entry must be defined in the OpenVMS TCP/IP HOSTS database. It is defined correctly by default, but it can be removed, which causes Insight Remote Support to fail.

  Type the following command:

  `$ TCPIP SHOW HOST /LOCAL`

  Look for LOCALHOST, which should have an IP address of 127.0.0.1. If LOCALHOST does not appear in the list, type the following command:

  `$ TCPIP SET HOST LOCALHOST /ADDRESS=127.0.0.1 /ALIAS=LOCALHOST`

  Type a ping command to verify that LOCALHOST was added:

  `$ TCPIP PING LOCALHOST`

  ```
  PING LOCALHOST (127.0.0.1): 56 data bytes

  64 bytes from 127.0.0.1: icmp_seq=0 ttl=64 time=0 ms
  ```

```
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0 ms
64 bytes from 127.0.0.1: icmp_seq=1 ttl=64 time=0 ms
```

After verifying that `LOCALHOST` responds to the ping command, you can use **Ctrl-C** or **Ctrl-Y** to stop pinging.

- System firmware: The prerequisite system firmware supports the logging of events according to the FRU Table Version 5 Specification, which is required for Insight Remote Support FRU configuration tree processing.

  All DSxx and ES40 systems must have firmware V5.7–4 or higher.

  All other products supported by Insight Remote Support shipped with a compatible firmware version.

  In general, users should take advantage of the latest improvements by obtaining the most recent firmware version available for their platforms.

Fulfilling Required Permissions and Access

Grant Permissions Required to Install ELMC

To install or uninstall ELMC, the user needs all of the following OpenVMS privileges:

- ALTPRI
- BUGCHK
- BYPASS
- CMKRNL
- DIAGNOSE
- IMPERSONATE
- NETMBX
- OPER
- SYSLCK
- SYSPRV
- TMPMBX

When uninstalling ELMC, the user who performs the uninstallation must use the same username originally used to install ELMC.

The **set process** command sets privileges for all cluster nodes only when the cluster is served by a single system disk. However, on a cluster with multiple system disks, you might choose to install ELMC on nodes served by system disks other than the one serving the node from which you are installing. In that case, **set process** does not set privileges on those other nodes (the nodes served by the other system disks), and ELMC does not install correctly on those other nodes.

To correctly install on clusters with multiple system disks, set up the required privileges as defaults (the privileges you get when logging on) on all nodes where you wish to install ELMC, instead of using the **set process** command.

See section "Installing the ELMC OpenVMS Software Package on the Monitored Device", for more precautions about one system disk versus multiple system disks.

Grant Permissions Required to Run ELMC

To execute any ELMC commands, the user needs all of the following OpenVMS privileges. Note that these are a subset of the privileges required to install, upgrade or uninstall ELMC:

- ALTPRI
- BUGCHK
- CMKRNL
- DIAGNOSE
- IMPERSONATE
- NETMBX
- SYSPRV
- TMPMBX

Grant Cluster Node Access to ELMC Install Directory

The installation package only can install itself on cluster nodes that have access to the target directory where you choose to install ELMC. Another way to explain this is that the nodes must mount the disk containing the target directory. This means that an installation might not always place ELMC on all cluster nodes, since all nodes might not be able to identify the place where ELMC is being installed.

The following illustrate this issue:

- **Cluster**: All nodes share the same system disk.

  Install Node: Any node.

  Install Target: The default location SYS$COMMON:[HP...].

  Result: ELMC installs itself for all nodes.

- **Cluster**: All but two nodes share system disk A. The other two nodes share system disk B.

  Install Node: A node that uses system disk A.

  Install Target: The default location SYS$COMMON:[HP...].

  Result: The other two nodes will not have ELMC.

  In the previous case, you can install ELMC one more time for the remaining two nodes by running the install from either node and again choosing the default location of SYS$COMMON:[HP...]. Consider this a completely separate ELMC installation from the first install on the majority of the nodes.

- **Cluster**: All but two nodes share system disk A. The other two nodes share system disk B. All nodes also mount a non-system disk C.

  Install Node: Any node.

Install Target: A directory on disk C, specified by you during the installation.

Result: ELMC installs itself for all nodes.

Note: In all cases the installation package also lets you choose only a subset of the nodes that can see the install location.

Archiving and Cleaning the Error Log

After Insight Remote Support is installed on the Hosting Device, it uses ELMC to analyze all events stored in the error log, which can result in high CPU usage over an extended period. To control this operation, you are encouraged to archive and clean the error log before installing. This reduces the size of the log and the time required for the initial scan.

Follow these guidelines for cleaning the error log. If ELMC is installed and running when you clean the log, you do not need to stop and restart the Director process. Also, do not stop and restart the `ERRFMT` system event logging process.

The default error log, typically `SYS$SYSROOT:[SYSERR]ERRLOG.SYS`, increases in size and remains on the system disk until the user explicitly renames or deleted it. When either occurs, the system creates a new, clean error log file after about 15 minutes.

Caution: After renaming or deleting the existing log, do not install ELMC until the new default log is present.

If you rename the log, the saved log can be analyzed at a later time.

Aside from starting with a clean log before installing Insight Remote Support, you may want to perform regular maintenance on the error log. One method is to rename `errlog.sys` on a daily basis. For example, you might rename `errlog.sys` to `errlog.old` every morning at 9:00. To free space on the system disk, you then can back up the renamed version to a different volume and delete the file from the system disk.

Verify Serial Number

On AlphaServer GS80, GS160, and GS320 systems, verify the serial number as directed before installing ELMC on the monitored device.

Certain GS80, GS160, and GS320 systems did not have their system serial number set correctly at the factory, and Insight RS only functions when the serial number is set correctly. Affected serial numbers will begin with the letter "G."

At the SRM console firmware prompt (the prompt when you first power the system on), check the serial number with the following command:

```
show sys_serial_num
```

The serial number shown should match the actual serial number on the model/serial number tag located in the power cabinet. If necessary, change the serial number with the following command:

```
set sys_serial_num
```

Type the six-character serial number provided on the tag in the power cabinet.

Note: This issue also can arise when multiple AlphaServers are ordered, because the factory may assign an identical serial number to each system. In this scenario, Insight RS does not work correctly because it requires that each AlphaServer have a unique number. If this is the case, uniquely identify each AlphaServer by appending –1, –2, –3, and so on, to the serial numbers when you use the set **sys_serial_num** command.

Note: Multiple partitions on the same AlphaServer always have the same serial number because they reside on the same machine. There are no Insight RS conflicts in this case, so do not attempt to assign unique serial numbers to different partitions on the same machine.

Install the ELMC OpenVMS Software Package on the Monitored Device

An OpenVMS cluster can contain nodes served by a single common system disk, or nodes served by multiple system disks. Any given node is served by only one system disk, but a system disk can serve one or more nodes. Each system disk contains its own PCSI database (product registry).

ELMC can be installed on a system disk or shared non-system disk. However, a shared non-system disk might be accessible by multiple nodes served by different system disks. This means that ELMC is not limited to being installed only on nodes served by one given system disk. A PCSI database, however, is limited to one system disk.

This scenario can generate discrepancies in the **product show product wccproxy** command. The command always shows ELMC as installed when run from a node served by the same system disk as the node from which ELMC was originally installed (the installing node). This is because the ELMC installer registers ELMC only into the PCSI database for the system disk serving the installing node, and not into any other PCSI databases. Two types of misleading information can result:

- If a node is served by the same system disk as the installing node, but the user did not add ELMC to that node, the command shows that ELMC is installed when it is not.
- Conversely, if a node is served by a different system disk from the installing node, and the user added ELMC to that node, the command does not show that ELMC is installed when it is.

If ELMC is already installed on the OpenVMS server, make sure it is version 6.2 or later. If it's older than version 6.2, it needs to be upgraded. To check the version of ELMC, run the following command: `wccproxy version`.

To install the ELMC software package, complete the following steps:

1. In the Insight RS Console, navigate to the **Administrator Settings** → **Software Updates** tab and select the *Event Log Monitoring Collector (ELMC)* software package.
2. On the **Available Version** tab, click **Download**.
3. When the download completes, click **Install**. The ELMC packages are placed in the `%HP_RS_DATA%\ELMC` folder. This folder defaults to `C:\ProgramData\HP\RS\DATA\ELMC`.

Note: The `ProgramData` folder is a hidden folder. To view this folder, set the folder options to show hidden folders.

4. Copy the OpenVMS Itanium ELMC software package (`ELMC[`*`version`*`]_OVMSI64.EXE`) to the OpenVMS monitored device. Place the `.exe` file in an empty directory. Make sure that:
   - There are no other kits in the directory, especially other versions of ELMC kits.
   - There are no old ELMC or WCCProxy files in the directory left over from previous operations.
5. Extract the ELMC installation files:

   `$ run ELMC[`*`version`*`]_OVMSI64.EXE`
6. Run the installation command and follow the prompts:

   `$ @wccproxy_install install`

   Note: The command executes the DCL script `wccproxy_install.com` in the current directory. Do not run the **product install wccproxy** command that would normally be used to install a PCSI-based product. This command aborts and prompts you to run the `wccproxy_install.com` script instead. Set your default directory to the one containing the file `wccproxy_install.com`, created by extracting the ELMC `.exe` file in the previous step.

   The kit will install and finish with no user prompting. When the DCL prompt ($) returns, the install has finished and the ELMC (WCCProxy) process will be running.

Configure Dynamic Processor Resilience

Dynamic Processor Resilience, also known as CPU Indictment, is the ability for HP Insight Remote Support to detect various error conditions occurring in the processor, or on the processor bus/links, and remove the processor from operation. On HP Integrity servers running HP OpenVMS, this capability is implemented by a combination of analysis within HP Insight Remote Support and Kernel CPU indictment services provided by the HP OpenVMS operating system. When a CPU is indicted, the kernel CPU indictment service will remove the processor from the active set. Additionally Insight Remote Support will log a service call. By removing the failing processor from the active set this prohibits the processor from failing fatally at some future time. Automatically logging the service call allows automatic proactive repair further aiding in prompt resolution of the problem.

To use Dynamic Processor Resilience, it needs to be configured on the OpenVMS server and also enabled in the Insight RS Console.

To enable Dynamic Processor Resilience on the OpenVMS server, complete the following steps:

1. Edit the `SYS$INDICTMENT_POLICY.COM` command procedure and change their values from 0 to 1:

   `$ DEFINE/SYSTEM/EXECUTIVE_MODE/NOLOG SYS$INDICT_START 1`

   `$ DEFINE/SYSTEM/EXECUTIVE_MODE/NOLOG SYS$INDICT_ALLOW_CPUS 1`

   This enables all CPUs to be eligible for Indictment except for CPU 0.

   Note: CPU 0 cannot be removed. If you wish to not have some CPU's eligible, refer to the

OpenVMS documentation for Indictment Service to learn how to modify this procedure to meet your needs.

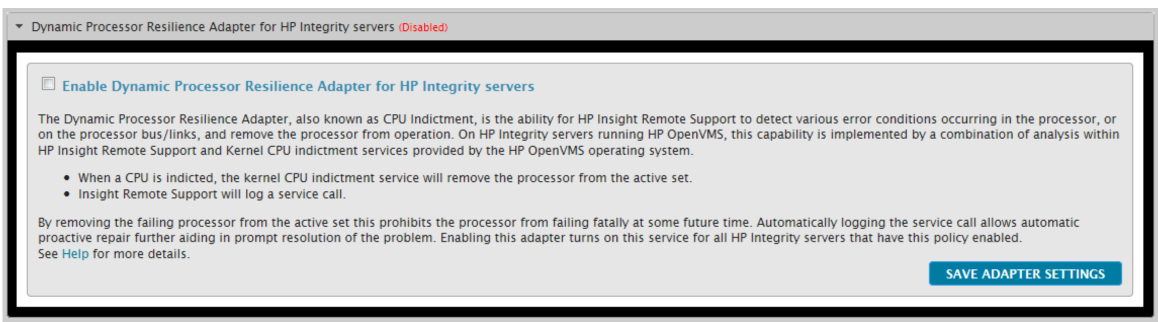
2. Reboot the server for the changes to take effect.

   When the Indictment Server is started, there will be an `INDICT_SERVER` process shown in the SHOW SYS output:

```
ES47S3_V8.3>>sho sys
OpenVMS V8.3 on node ES47S3 17-NOV-2008 15:29:15.48 Uptime 0 00:13:01
Pid Process Name State Pri I/O CPU Page flts Pages
0000041C INDICT_SERVER HIB 8 6 0 00:00:00.00 79 95
```

To enable the Dynamic Processor Resilience Adapter in the Insight RS Console, complete the following steps:

1. In the main menu, select **Administrator Settings**.
2. Click the **Integration Adapters** tab.
3. Click the **Dynamic Processor Resilience Adapter for HP Integrity servers** heading to expand the adapter pane.



4. Select the **Enable Dynamic Processor Resilience Adapter for HP Integrity servers** check box.
5. Click **Save Adapter Settings**.

Insight Remote Support enables the Dynamic Processor Resilience Adapter. (Enabled) now appears next to the Dynamic Processor Resilience Adapter to indicate it is enabled.

Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create an ELMC Protocol in the Insight RS Console

To configure ELMC in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the **Select and Configure Protocol** drop-down list, select **Event Log Monitoring Collector (ELMC)**.
4. Click **New**. The New Credential dialog box appears.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP address(es) of the devices to be discovered.
   d. Click **Add**.
4. Click **Start Discovery**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon ( ).

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following section:

Verify Collections in the Insight RS Console

After discovery, a Server Basic Configuration Collection is automatically run for servers. Verify that this collection ran successfully on the Collection Services → Basic Collection Results tab in the Insight RS Console. Note that there is a delay between when discovery finishes and when the collection runs. For more information about scheduling collections, see the Insight RS Help.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services** and then click the **Basic Collection Results** tab.
3. Expand the Server Basic Configuration Collection section.
4. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears (). If it failed, an error icon appears ().

Chapter 15: Configuring Tru64 UNIX Servers

Important: Configuration collections for Tru64 on AlphaServer are not available except when the Tru64 server is part of a SAN collection.

Important: HP only provides limited remote support for customers using HP Alpha servers. As with standard support, HP Insight Remote Support version 7.x will attempt to discover these products as remotely monitored HP products and send information about failure events to HP.

However, if there are issues with the operation of the remote monitoring features of HP Insight Remote Support version 7.x for these specific products, including the remote discovery or the reporting of failure events to HP, then HP will resolve such issues in a commercially reasonable manner but without guarantee of an immediate response or a resolution. If a failure event is successfully received at HP, it is managed according to the support level agreement.

Fulfill Configuration Requirements

To configure your Tru64 UNIX servers to be monitored by Insight RS, complete the following sections:

Table 15.1 ***Tru64 UNIX Server Configuration Steps***

Task	Complete?
Make sure Insight RS supports your Integrity Tru64 server by checking the *HP Insight Remote Support Release Notes*.	
Install and configure ELMC for Tru64 on the Tru64 UNIX server.	
Add ELMC to the Insight RS Console.	
Discover the Tru64 server in the Insight RS Console.	

Install and Configure Communication Software on Servers

To configure your monitored devices, complete the following sections:

Fulfill ELMC System Requirements

Tru64 UNIX monitored devices must meet the following requirements before ELMC is installed. In clusters, minimum requirements apply to each node in the cluster.

Hardware and Software Requirements

- Processor architecture: HP AlphaServer
- Operating system: Tru64 UNIX version 4.0F, 4.0G, 5.1A or higher

Note: HP Sustaining Engineering maintains a schedule of support for the Tru64 UNIX operating system. HP does not commit to supporting Insight Remote Support when installed on an operating system version that has exceeded its end-of-support date.

- Minimum 20 MB free disk space for installation of all components.
- TCP/IP services must be installed and running.
- Upgrade to V1.22 or higher of the Emulex (EMX) driver if you have an EMX LP6000, LP7000, or LP8000 adapter (KGPSA–xx) using the SLI2 programming interface. Use of an EMX driver prior to V1.22 may result in data not being processed correctly.
- System firmware: The prerequisite system firmware supports the logging of events according to the FRU Table Version 5 Specification, which is required for Insight Remote Support FRU configuration tree processing.
  - All DSxx and ES40 systems must have firmware V5.7–4 or higher.
  - All other systems (currently ES45, GSxx, and TS202c) ship with a compatible firmware version.

  In general, users should take advantage of the latest improvements by obtaining the most recent firmware version available for their platform.

Required Permissions and Access

To install, upgrade, or uninstall ELMC, you must be logged on as the root user. The `/usr/opt/hp/svctools` directory is owned by root, and has rwx (read, write, and execute) permissions for root (owner), and no permissions for any other user (group or world).

Archiving and Cleaning the Error Log

After Insight Remote Support is installed on the Hosting Device, it uses ELMC to analyze all events stored in the error log, which can result in high CPU usage over an extended period. To control this operation, you are encouraged to archive and clean the error log before installing. This reduces the size of the log and the time required for the initial scan.

Follow these guidelines for cleaning the error log. If ELMC is installed and running when you clean the log, you do not need to stop and restart the Director process. Also, do not stop and restart the `ERRFMT` system event logging process.

Tru64 UNIX Version 4.0F

1. Stop the binlogd process: **`# /sbin/init.d/binlog stop`**
2. If desired, move the original error log to any appropriate name, for example:

   `# mv /var/adm/binary.errlog /var/adm/binary.errlog.2002_06_11`

   Saved logs can be analyzed at a later time.
3. If you skipped step 2, remove the original error log: **`# rm /var/adm/binary.errlog`**
4. Restart the server. During restart, the server creates a new `binary.errlog` file containing a new configuration event. The server also restarts the `binlogd` process.

Tru64 UNIX Version 4.0G

1. Stop the binlogd process: **`# /sbin/init.d/binlog stop`**
2. If desired, move the original error log to any appropriate name, for example:

   `# mv /var/adm/binary.errlog /var/adm/binary.errlog.2002_06_11`

   Saved logs can be analyzed at a later time.
3. If you skipped step 2, remove the original error log: **`# rm /var/adm/binary.errlog`**
4. Restart the binlogd process: **`# /sbin/init.d/binlog start`**

Tru64 UNIX Version 5.A or Higher

A new feature can send a signal to binlogd to save the current log and create a new one without stopping the process. Follow the steps in section "Verify the binary.errlog CDSL" and section "Clear the Log with binlogd Running".

Verify the binary.errlog CDSL

In version 5.1A or higher, the binary error log `/var/adm/binary.errlog` should be a context-dependent symbolic link (CDSL) pointing to a file specific to each cluster node. This makes sure that the binlogd process on each node stores that node's events to its own node-specific error log `/var/cluster/members/`*<memb>*`/adm/binary.errlog`.

If the CDSL is ever deleted, binlogd recreates it as a regular, cluster-common file, which does not work correctly. To check your file, issue the command:

`# ls -l /var/adm/binary.errlog`

Correct output looks similar to the following:

```
>lrwxrwxrwx 1 root adm 43 Jun 11 12:54 /var/adm/binary.errlog -
>../cluster/members/<memb>/adm/binary.errlog
```

Incorrect output does not show the -> link indicator:

```
-rw-r----- 1 root adm 560 Jun 11 12:59 /var/adm/binary.errlog
```

If necessary, correct the file by performing the following steps:

1. Stop the binlogd process on all cluster nodes by issuing the following command on each node:

   `# /sbin/init.d/binlog stop`
2. Saved logs can be analyzed at a later time. If desired, move the original error log to any appropriate name, for example:

   `# cd /var/adm`

   `# mv binary.errlog binary.errlog.2002_06_11`
3. Issue similar move commands for any node-specific error logs you wish to save, for example:

 # mv /var/cluster/members/*<memb>*/adm/binary.errlog /var/cluster/members/*<memb>*/adm/binary.errlog.2002_06_11

 # mv /var/cluster/members/*<memb>*/adm/binlog.saved/binary.errlog.saved /var/cluster/members/*<memb>*/adm/binlog.saved/binary.errlog.saved.2002_06_11

4. Remove existing error logs, ignoring any `No such file or directory` errors:

   # rm /var/adm/binary.errlog

   # rm /var/cluster/members/*<memb>*/adm/binary.errlog

   # rm /var/cluster/members/*<memb>*/adm/binlog.saved/binary.errlog.saved

5. Create the CDSL:

   # mkcdsl /var/adm/binary.errlog

6. Restart the binlogd process on all cluster nodes by issuing the following command on each node:

   # /sbin/init.d/binlog start

Clear the Log with binlogd Running

For version 5.1A or higher, follow these steps on each cluster node that you want to clear:

1. Verify the `binary.errlog` CDSL section is complete as previously described.
2. If desired, keep any previously saved copy from being overwritten by moving it to any appropriate name, for example:

   # cd /var/cluster/members/member/adm/binlog.saved

   # mv binary.errlog.saved binary.errlog.2002_06_11

3. Cause binlogd to copy and clear the original error log:

   # kill -USR1 `cat /var/run/binlogd.pid`

   The previous command does not kill the `binlogd` process. Instead, it sends a signal to `binlogd` that causes it to copy `/var/adm/binary.errlog` to `/var/cluster/members/member/adm/binlog.saved`. Then, the original `/var/adm/binary.errlog` file gets recreated with only a configuration event. Note that `/var/adm/binary.errlog` is a CDSL that points to /var/cluster/members/*<memb>*/adm/binary.errlog.

For further details, including how to automate this kind of error log management, see the section on Managing the Binary Error Log File in the binlogd man page.

Verify Serial Number

On GS80, GS160, and GS320 systems, verify the serial number as directed before installing ELMC on the monitored device.

Certain GS80, GS160, and GS320 systems did not have their system serial number set correctly at the factory, and event rules on the Hosting Device only function when the serial number is set correctly. Affected serial numbers will begin with the letter "G."

At the SRM console firmware prompt (the prompt when you first power on the server), check the serial number with the following command:

```
show sys_serial_num
```

The serial number shown should match the actual serial number on the model/serial number tag located in the power cabinet. If necessary, change the serial number with the following command:

```
set sys_serial_num
```

Type the six-character serial number provided on the tag in the power cabinet.

Note: This issue also can arise when multiple AlphaServers are ordered, because the factory may assign an identical serial number to each system. In this scenario, event rules do not work correctly because they require that each AlphaServer have a unique number. If this is the case, uniquely identify each AlphaServer by appending –1, –2, –3, and so on, to the serial numbers when you use the set **sys_serial_num** command.

Multiple partitions on the same AlphaServer always have the same serial number because they reside on the same machine. There are no conflicts in this case, so do not attempt to assign unique serial numbers to different partitions on the same machine.

Installing the ELMC Tru64 UNIX Software Package

First, extract the ELMC Software Package, then install it.

To extract the ELMC installation kit, place the kit `.gz` file in a temporary directory and unzip it:

```
# gunzip ELMC_<version>.tar.gz
```

Then, extract the tar file. If there is already a "kit" subdirectory when you perform this command, be sure there are no previous ELMC kit files in this subdirectory before performing the command.

```
# tar -xvf ELMC_<version>.tar
```

This command creates a kit directory (if it does not already exist), and extracts the ELMC installation files.

Note: If installing in a TruCluster environment, make sure all nodes are up and running before proceeding.

When your current directory is the one in which you extracted the kit, type the following command to install the files for the ELMC WCCProxy.

```
# setld -l kit
```

Do not run `setld -D` to direct the ELMC installation to a non-default directory. The default directory is required for proper ELMC operation.

The kit will install and finish with no user prompting. When you are returned to the shell prompt (#), the install has finished and the wccproxy process will be running.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create an ELMC Protocol in the Insight RS Console

To configure ELMC in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the **Select and Configure Protocol** drop-down list, select **Event Log Monitoring Collector (ELMC)**.
4. Click **New**. The New Credential dialog box appears.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
    a. Click **New**.
    b. Select the **Single Address**, **Address Range**, or **Address List** option.
    c. Type the IP address(es) of the devices to be discovered.
    d. Click **Add**.
4. Click **Start Discovery**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon ().

    If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following section:

Verify Collections in the Insight RS Console

Configuration collections are not supported for this device type except when it is part of a SAN collection. If you have added this device to a SAN collection, you can manually run a SAN collection to verify the configuration.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services**.
3. Click the **Collection Schedules** tab.
4. In the List of Collection Schedules pane, select the **SAN Configuration Collection Schedule**. Information about the collection appears in the Collection Information pane. The On These Devices pane lists the devices the collection will run on.
5. In the Schedule Information pane, click **Run Now**.
6. When the collection completes, click the **SAN Storage Collection Results** tab.
7. Expand the SAN Configuration Collection section.
8. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears (). If it failed, an error icon appears ().

Chapter 16: Configuring IBM Servers

Insight Remote Support (RS) must be able to communicate with your IBM server before it can be monitored. Insight RS can communicate with IBM servers running Windows with SNMP. The following information describes how to install and configure the communication protocols and other recommended software components so that it can be monitored by Insight RS.

Insight RS supports the following IBM servers:

- IBM System x™ (xSeries®)
- IBM BladeCenter® chassis and BladeCenter® servers

Note: Insight RS supports Microsoft Windows Server 2008 and 2008 R2 operating systems for IBM servers.

Important: Configuration collections are not supported.

Fulfill Configuration Requirements

To configure your IBM servers to be monitored by Insight RS, complete the following sections:

Table 16.1 ***IBM Server Configuration Steps***

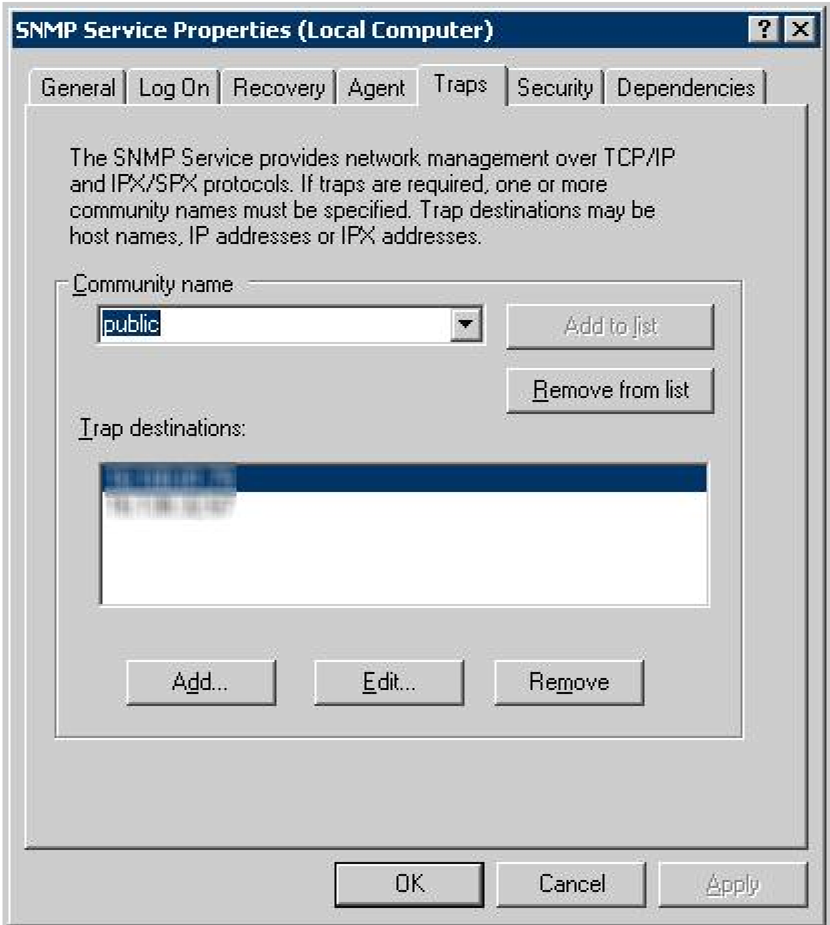
Task	Complete?
Make sure Insight RS supports your IBM server by checking the *HP Insight Remote Support Release Notes*.	
Configure the Windows SNMP service on the IBM server.	
Install IBM Director Agent on the IBM server.	
Configure the SNMP trap destination and SNMP read community string.	
Add the SNMP protocol to the Insight RS Console.	
Discover the IBM server in the Insight RS Console.	
Add the device's Support Type and Support Identifier to the Insight RS Console.	

Install and Configure Communication Software on Servers

To configure your monitored devices, complete the following sections:

Configure the Windows SNMP Service

1. On the monitored device, open the SNMP Service Properties window and select the **Traps** tab.



2. In the **Community name** field, type the SNMP community name and click **Add to list**.

Important: The community name must match the community name as configured in the Insight RS Console.

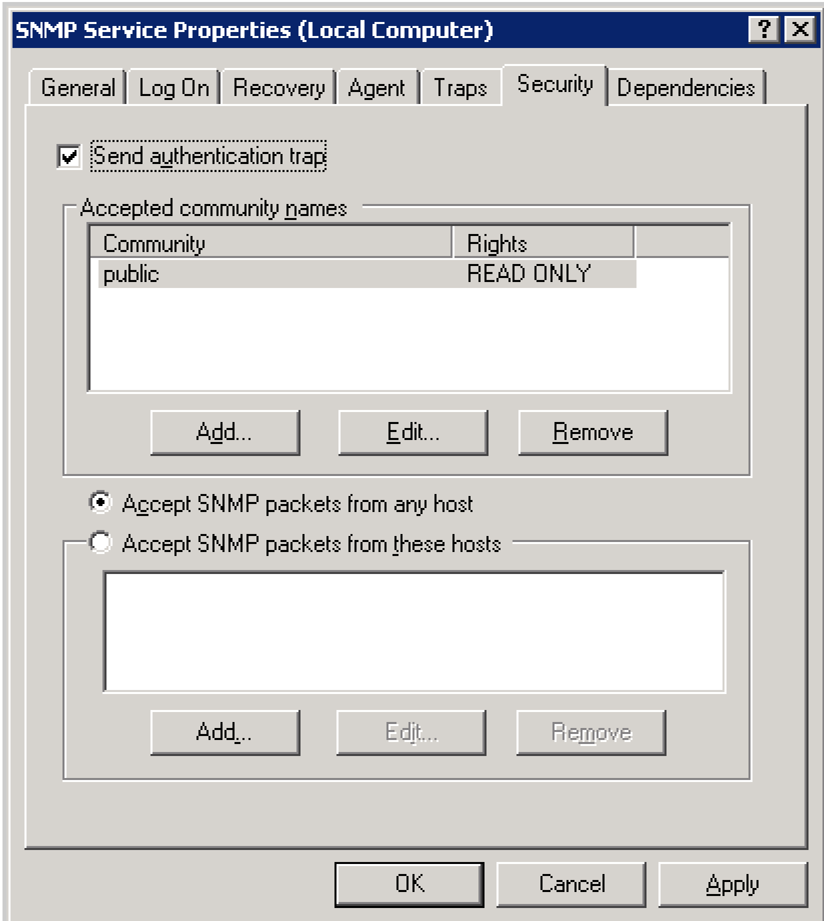
3. In the trap destinations field, add the Hosting Device to the list:

   a. To open the SNMP Service Configuration window, click **Add.**

   b. Type the host name or IP address of the Hosting Device.

   c. Click **Add**.

4. Click the **Security** tab and verify that the community name was added to the list of Accepted community names.



5. Select the **Send authentication trap** check box.
6. Click the **Accept SNMP packets from any host** option.
7. Click **OK** to save the settings.

Install and Configure SNMP

IBM Director Agent must be installed on the IBM server and the SNMP service must be configured. IBM Director Agent includes an SNMP agent.

On a Windows 2008 and 2008 R2 server the solution was tested with IBM Systems Director Common Agent version 6.1.1.

HP recommends installing the latest version of IBM Director Common Agent.

Installing IBM Director Agent

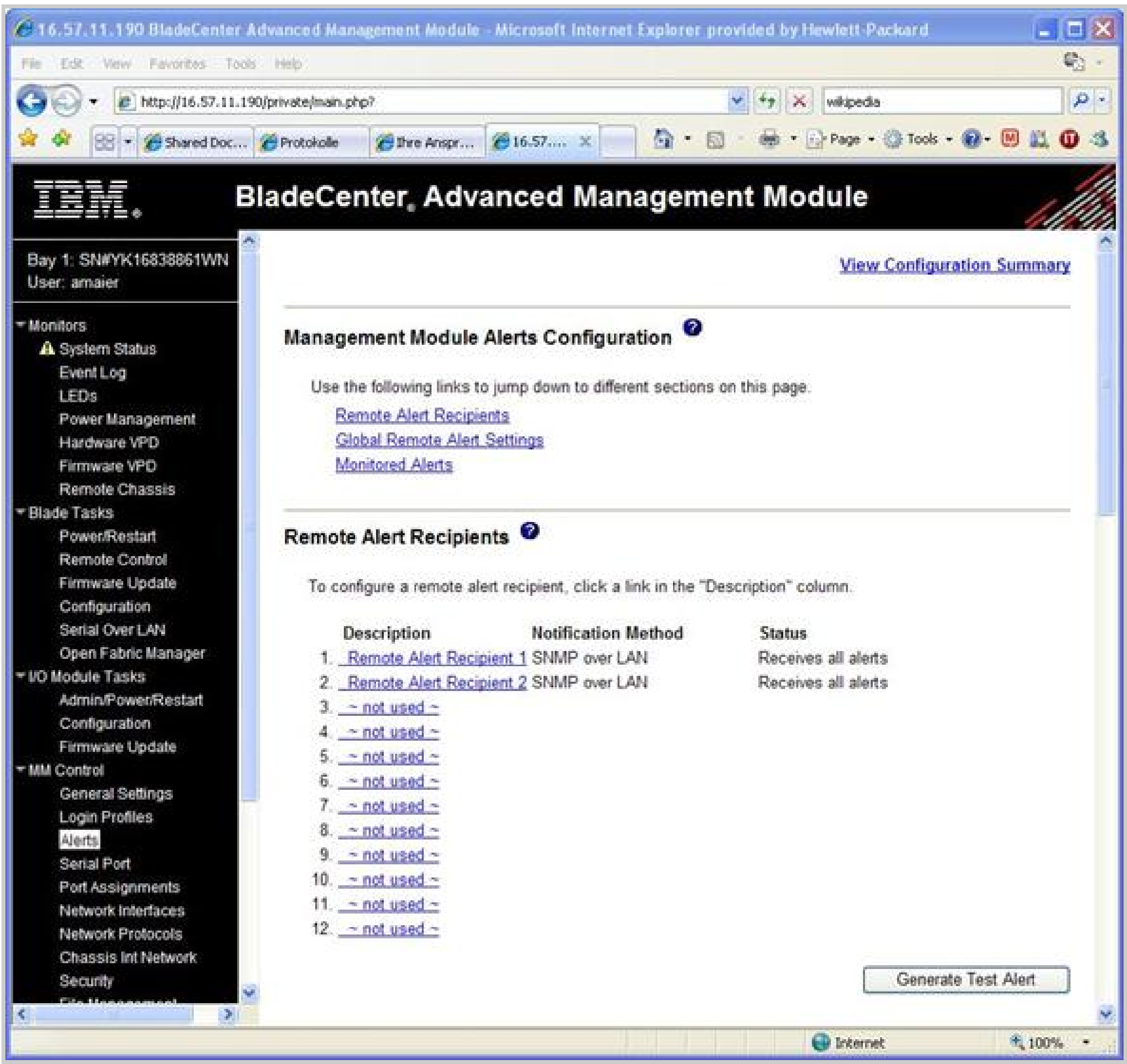
Download and install IBM Director Agent or IBM Systems Director Common Agent software for xSeries at: http://www-03.ibm.com/systems/software/director/downloads/agents.html.

Note: If a service processor is installed on the monitored server, complete the following steps:

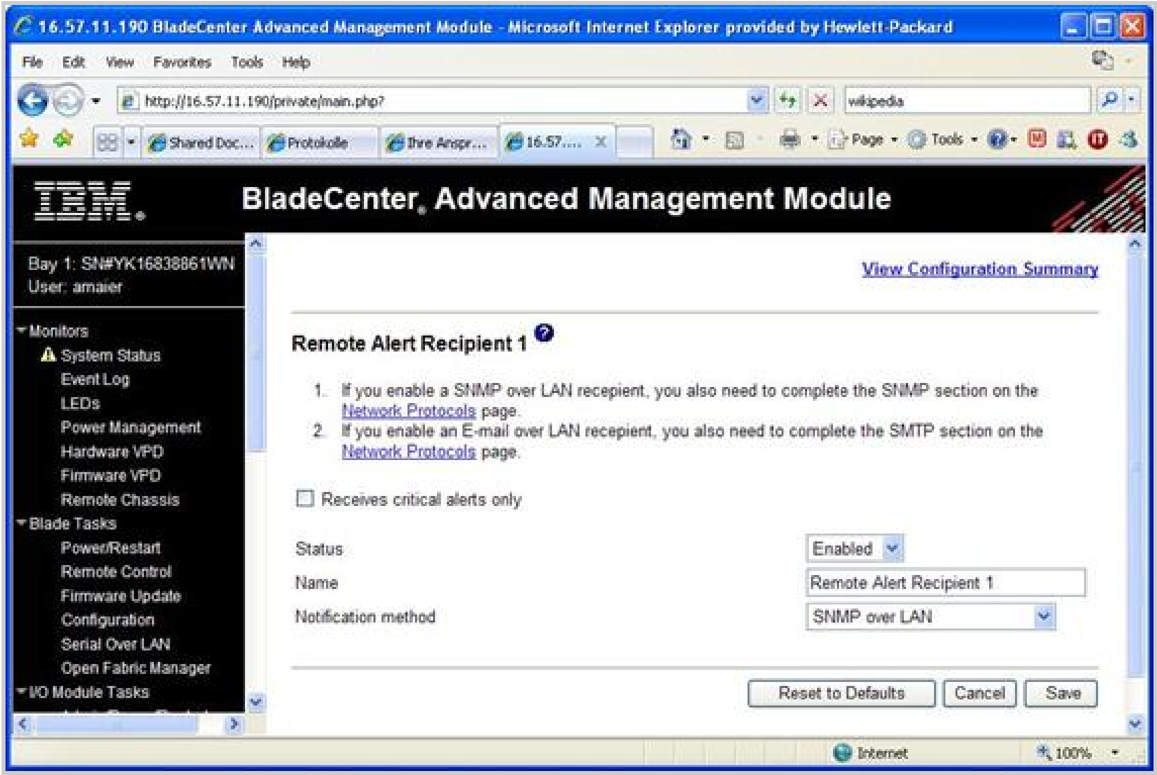
1. To configure the service processor's trap destination use the Management Processor Assistant task in the IBM Director console.
2. Add the IP address of the Hosting Device as a trap destination.
3. Restart the service processor.

Configuring the Management Module of the IBM BladeCenter® Chassis

1. Login into to the BladeCenter Management Module web page.
2. Select Alerts in the MM Control section of the left menu. The Alerts page appears.



3. Select one of the unused items from the Name column. The Remote Alert Recipient page appears.

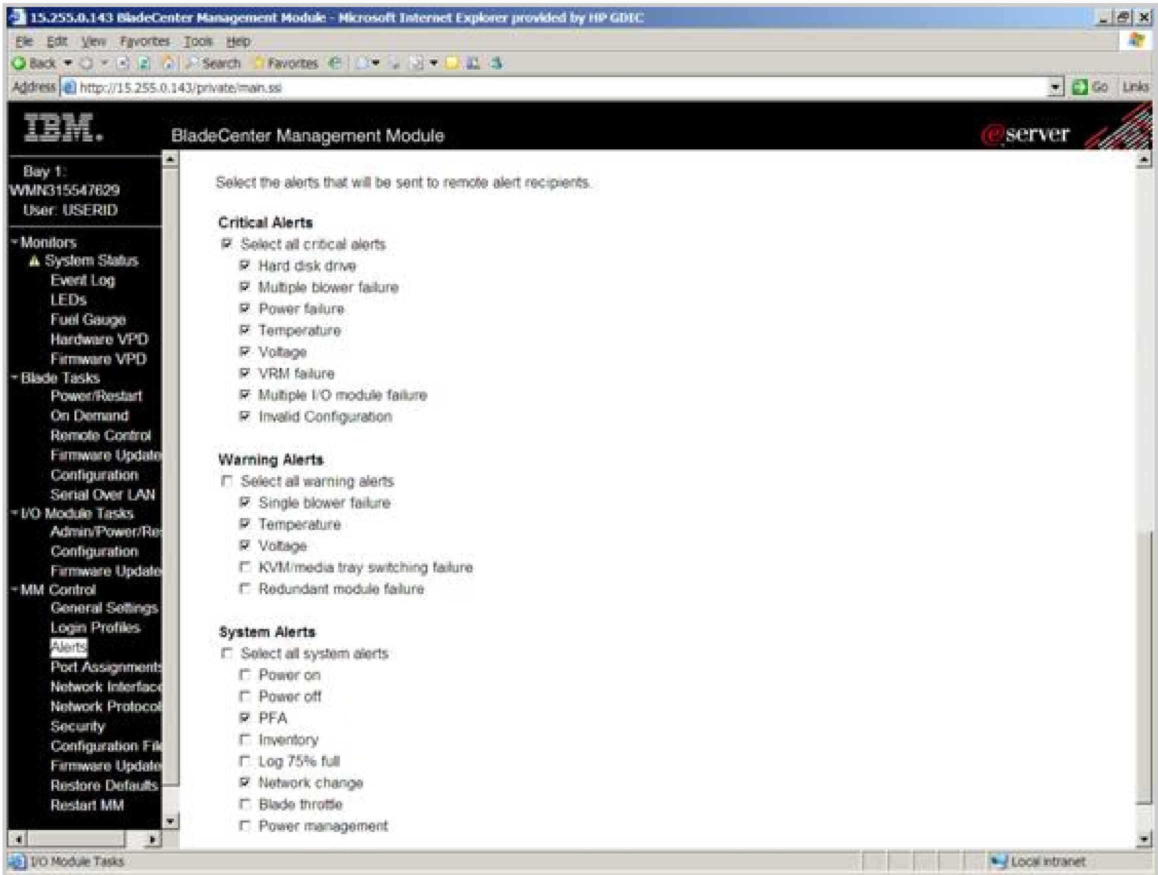


4. On the Remote Alert Recipient page:
   a. From the **Status** drop-down list, select **Enabled**.
   b. In the Name field, type a Remote Alert Recipient name. For example: Remote Alert Recipient 1
   c. From the **Notification method** drop-down list, select **SNMP**over LAN.
   d. In the IP address or host name field, type the Hosting Device IP address.
   e. Optional: In the E-mail address field, type an e-mail address, if you want to be notified about alerts by e-mail
5. Select **Save**.The Alerts page appears. The saved Remote Alert Recipient now appears on the Alerts page of the BladeCenter Management Module.
6. On the Alerts page scroll down to the Monitored Alerts section.
7. Select the following:
   a. All Critical Alerts
   b. From Warning Alerts:
      i. Single Chassis Cooling Device (Blower) failure
      ii. Temperature
      iii. Voltage
   c. From System Alerts:

i. PFA

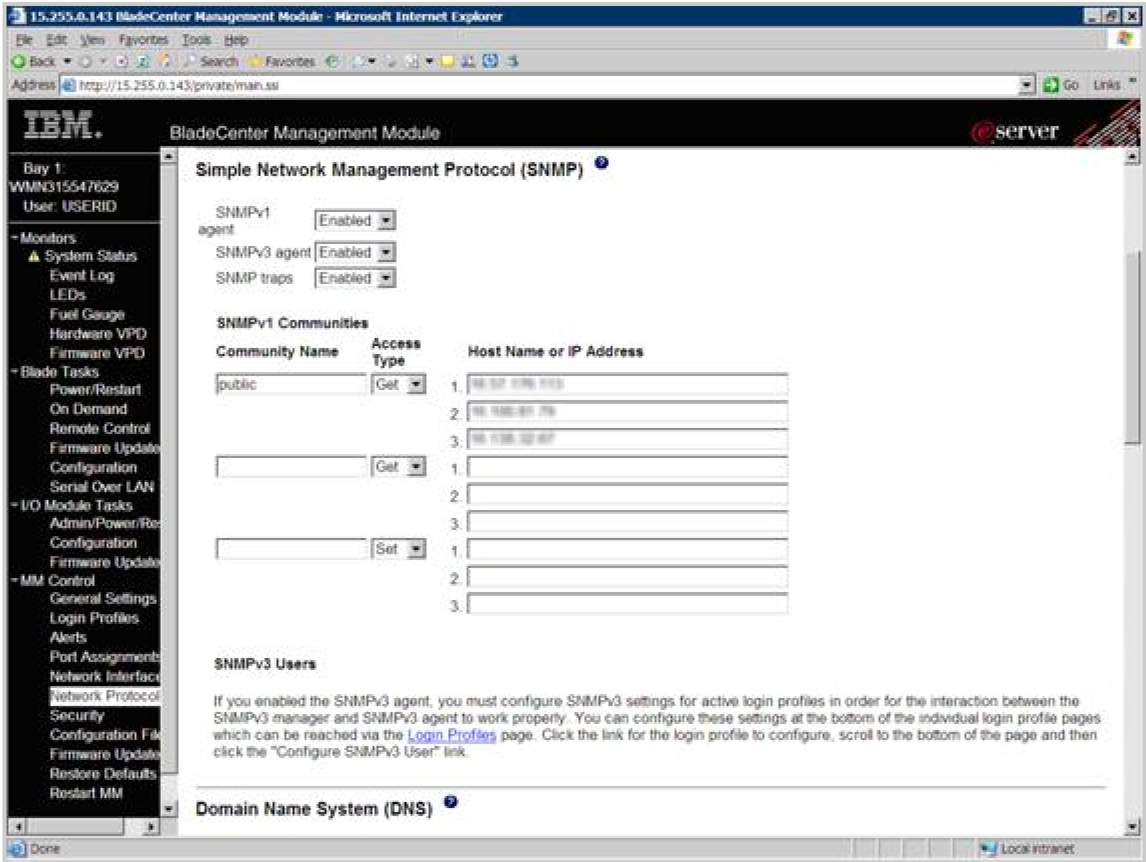
ii. Network change

8. Save your settings.



9. In the MM Control section of the left menu, select **Network Protocol**. The Network Protocol page appears.

10. Scroll down to SNMP.



11. From the **SNMPv1 agent** drop-down list, select **Enabled**.
12. Configure an SNMPv1 Community:
    a. In the Community Name field, type a SNMP community name.
    b. From the **Access Type** drop-down list, select **Get**.
    c. In the Host Name or IP Address field, type the Hosting Device host name or IP address.
13. Click **Save**.
14. Restart the BladeCenter Management Module:
    a. In the MM Control section of the left menu, select **Restart MM**.
    b. Click **Restart**.

Install IBM Device Drivers and Service Processor Firmware

To use an attached device, its device drivers must be installed. Make sure that the appropriate IBM server's service processors are installed.

Note:

- To determine which service processors are installed on an IBM server, refer to the *Implementing Systems Management Solutions using IBM Director* document at:

http://www.redbooks.ibm.com/redbooks/pdfs/sg246188.pdf

- A combined device driver for all service processors is available for each operating system at: http://www.ibm.com/systems/support/.
- No additional driver is needed for servers that have only an ISM processor installed.

Upgrade the service processor firmware to the latest version to ensure that all features of the service processors are available.

To install the IBM device drivers, complete the following steps:

1. Go to: www.ibm.com/systems/support.
2. From the top navigation bar, select **Support & downloads**.
3. From the **Choose support type** drop-down list, select **System x**.
4. From the Popular links list, select Software and device drivers.
5. Download and install the device driver for your server.
6. Reboot the server(if requested).

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create an SNMPv1 Protocol Credential in Insight RS Console

If your device's SNMP community string is set to `public` and your community access mode is `read only`, Insight RS automatically associates an SNMPv1 protocol with your device. If you use a different community string or use a nonstandard port, you must create an SNMPv1 protocol credential in the Insight RS Console.

To configure SNMPv1 in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the Select and Configure Protocol drop-down list, select **Simple Network Management Protocol Version 1 (SNMPv1)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Community String you have configured on your device.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP address(es) of the devices to be discovered.
   d. Click **Add**.
4. Click **Start Discovery**.

Configure Warranty and Contract Information

Monitoring support for non-HP servers requires special attention to server warranty and contract data. The serial number and product numbers that is discovered usually does not match the serial number and product number listed in the HP support contract.

To configure the serial number and product number, complete the following steps:

1. In the Insight RS Console, verify the server appears on the Devices page.

   Important: For IBM BladeCenter® chassis the Product column is empty by default.

2. Click the **Device Name** and on the **Device** tab, type the following information:

   Note: The HP Account Support team must add the entitlement data for IBM servers in the Insight RS Console.

   - Override Serial Number (as listed in the contract)
   - Override Product Number (as listed in the contract)
   - Support Type
   - Support Identifier

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon ( ).

If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Chapter 17: Configuring Dell PowerEdge Servers

Insight Remote Support (RS) must be able to communicate with your Dell PowerEdge server before it can be monitored. Insight RS can communicate with Dell PowerEdge servers with SNMP. The following information describes how to install and configure the communication protocols and other recommended software components so that it can be monitored by Insight RS.

Note: Insight RS supports Microsoft Windows Server 2008 and 2008 R2 operating systems for Dell PowerEdge servers.

Important: Configuration collections are not supported.

Note: If you have a RAID controller installed on your system and you plan to install the storage management function, ensure that the device drivers for each RAID controller are also installed.

Fulfill Configuration Requirements

To configure your Dell PowerEdge servers to be monitored by Insight RS, complete the following sections:

Table 17.1 ***Dell PowerEdge Server Configuration Steps***

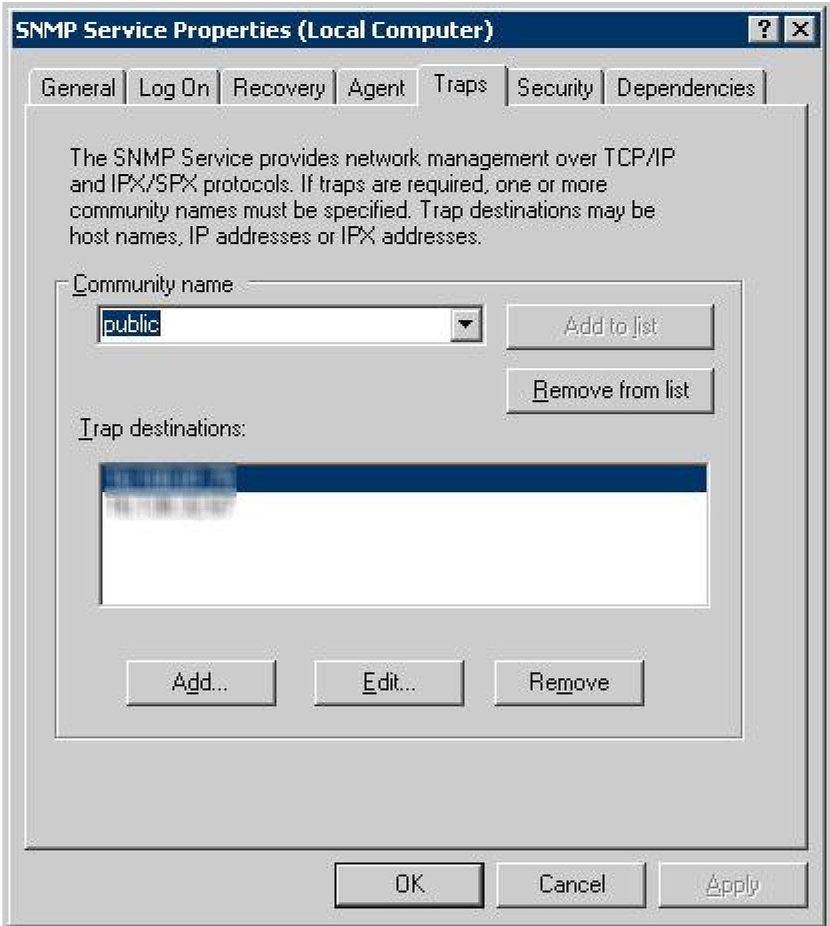
Task	Complete?
Make sure Insight RS supports your Dell PowerEdge server by checking the *HP Insight Remote Support Release Notes*.	
Configure the Windows SNMP service on the Dell PowerEdge server.	
Install Dell OpenManage Server Administrator on the Dell PowerEdge server.	
Configure the SNMP trap destination and SNMP read community string.	
Add the SNMP protocol to the Insight RS Console.	
Discover the Dell PowerEdge server in the Insight RS Console.	
Add the device's Support Type and Support Identifier to the Insight RS Console.	

Install and Configure Communication Software on Servers

To configure your monitored devices, complete the following sections:

Configure the Windows SNMP Service

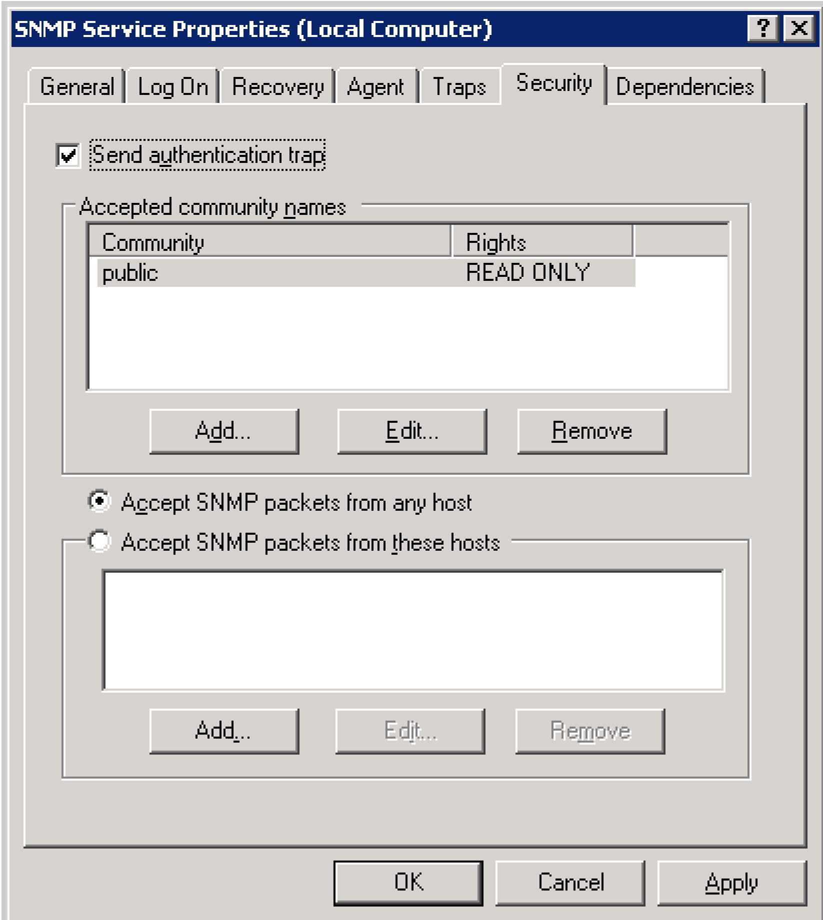
1. On the monitored device, open the SNMP Service Properties window and select the **Traps** tab.



2. In the **Community name** field, type the SNMP community name and click **Add to list**.

Important: The community name must match the community name as configured in the Insight RS Console.

3. In the trap destinations field, add the Hosting Device to the list:
   a. To open the SNMP Service Configuration window, click **Add.**
   b. Type the host name or IP address of the Hosting Device.
   c. Click **Add**.
4. Click the **Security** tab and verify that the community name was added to the list of Accepted community names.



5. Select the **Send authentication trap** check box.
6. Click the **Accept SNMP packets from any host** option.
7. Click **OK** to save the settings.

Install and Configure SNMP

Dell OpenManage Server Administrator must be installed on the Dell PowerEdge server and SNMP service must be configured.

On a Windows 2008 and 2008 R2 server the solution was tested with Dell OpenManage Server Administrator version 6.2.0.

HP recommends installing the latest version of Dell OpenManage Server Administrator.

Installing Dell OpenManage Server Administrator

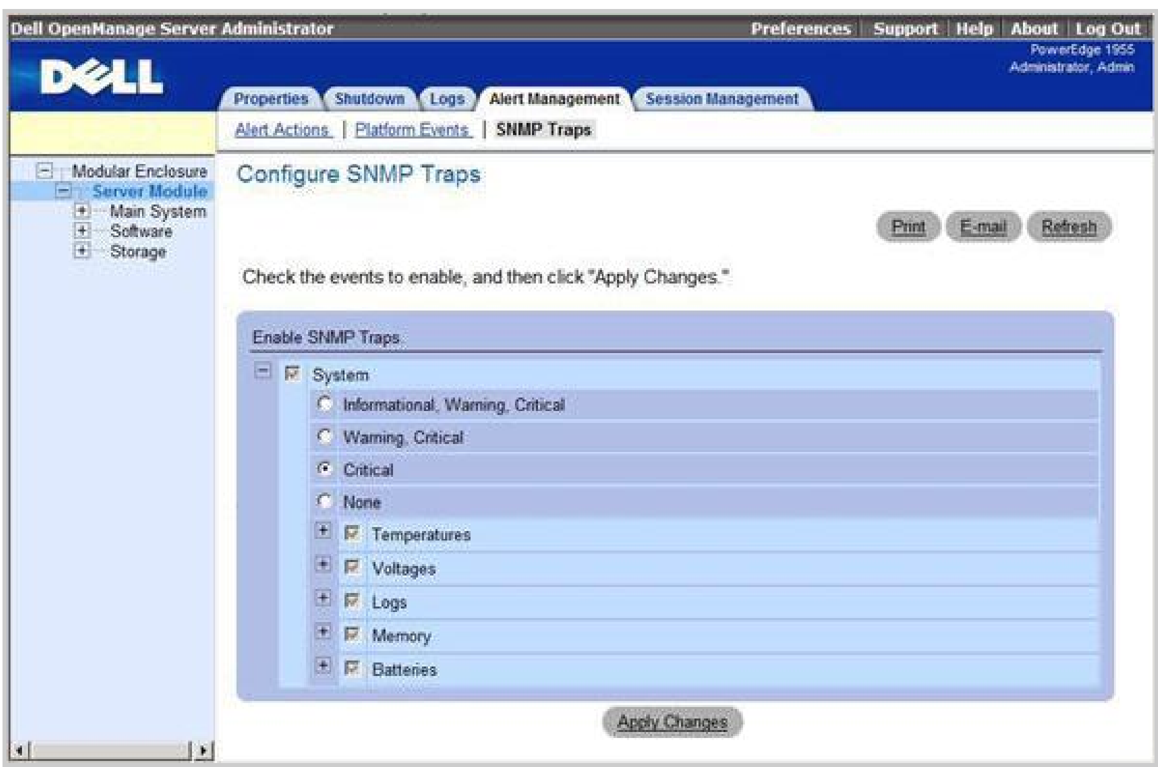
Download and install Dell OpenManage Server Administrator software. To download the software, go to: http://content.dell.com/us/en/enterprise/d/solutions/openmanage-server-administrator.

Note: You can verify the device driver under Windows by right-clicking **My Computer**, selecting **Manage**, and then clicking **Device Manager**.

See also *Dell OpenManage Installation and Security User's Guide*.

Configuring SNMP Traps in Dell OpenManage Server Administrator

1. Open Dell OpenManage Server Administrator.
2. Click the **Alert Management** tab.
3. Click the **SNMP Traps** link.
4. Select the following options:
   - System
   - Critical
   - Temperatures
   - Voltages
   - Logs
   - Memory
   - Batteries



5. Click **Apply Changes**.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create an SNMPv1 Protocol Credential in Insight RS Console

If your device's SNMP community string is set to `public` and your community access mode is `read only`, Insight RS automatically associates an SNMPv1 protocol with your device. If you use a different community string or use a nonstandard port, you must create an SNMPv1 protocol credential in the Insight RS Console.

To configure SNMPv1 in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the Select and Configure Protocol drop-down list, select **Simple Network Management Protocol Version 1 (SNMPv1)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Community String you have configured on your device.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP address(es) of the devices to be discovered.
   d. Click **Add**.
4. Click **Start Discovery**.

Configure Warranty and Contract Information

Monitoring support for non-HP servers requires special attention to server warranty and contract data. The serial number and product numbers that is discovered usually does not match the serial number and product number listed in the HP support contract.

To type the serial number and product number, complete the following steps:

1. In the Insight RS Console, verify the server appears on the Devices page.

   Important: For Dell PowerEdge chassis the Product column is empty by default.

2. Click the **Device Name** and on the **Device** tab, type the following information:

   Note: The HP Account Support team must configure the entitlement data for Dell Windows servers in the Insight RS Console.

   - Override Serial Number (as listed in the contract)
   - Override Product Number (as listed in the contract)
   - Support Type
   - Support Identifier

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon ( ).

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Chapter 18: Configuring BladeSystem c-Class Enclosures

HP BladeSystem c-Class Enclosures can be monitored by Insight Remote Support (RS). Configuring the BladeSystem c-Class Enclosure only enables Insight RS to monitor the enclosure itself. The blades installed in the BladeSystem c-Class Enclosure need to be configured separately to be monitored by Insight RS. See "Identify Required Communication Protocols and Software Components" on page 26 for information about configuring your individual blade types.

Configure BladeSystem c-Class Enclosures using one of two methods depending on the version of the Onboard Administrator (OA) firmware you are using:

Important: Do not attempt both configuration methods on the same Hosting Device.

- OA firmware version 3.60 or above, register Insight Remote Support through the OA.

  See "Register Remote Support Through the Onboard Administrator" on the next page.

  If using a primary and standby OA, ensure that the firmware version for both OAs is 3.60 or higher.
- OA firmware version prior to 3.55, configure Insight Remote Supportusing SNMP.

  See "Configure SNMP on the BladeSystem c-Class Enclosure" on page 169.

  If using a primary and standby OA, limit discovery to one of the OAs and not both.

If using OA firmware versions 3.55 or 3.56 HP recommends you upgrade to version 3.60.

Configuring BladeSystem c-Class Enclosures Through the OA

HP BladeSystem c-Class Enclosures can be monitored by Insight Remote Support (RS). Configuring the BladeSystem c-Class Enclosure only enables Insight RS to monitor the enclosure itself. The blades installed in the BladeSystem c-Class Enclosure need to be configured separately to be monitored by Insight RS. See "Identify Required Communication Protocols and Software Components" on page 26 for information about configuring your individual blade types.

Configuring through the OA requires firmware version 3.60 or above. If using a firmware version older than 3.55, see "Configuring BladeSystem c-Class Enclosures Using SNMP" on page 168. If using OA firmware versions 3.55 or 3.56 HP recommends you upgrade to version 3.60.

Important: When registering for Remote Support through the OA, make sure your trap destination is set to the same Insight RS Hosting Device as you register with for Remote Support. When the OA sends service events to the Hosting Device, it also generates SNMP traps that are sent to the SNMP trap destination. The SNMP trap destination needs to be the same Insight RS Hosting Device, otherwise duplicate events are reported to HP.

Fulfill Configuration Requirements

To configure your BladeSystem c-Class Enclosures through the OA to be monitored by Insight RS, complete the following sections:

Table 18.1 ***BladeSystem c-Class Enclosure Through the OA Configuration Steps***

Task	Complete?
Make sure Insight RS supports your BladeSystem c-Class enclosure by checking the *HP Insight Remote Support Release Notes*.	
Register for Remote Support through the Onboard Administrator.	
Verify the status of the BladeSystem c-Class enclosure in the Insight RS Console.	
Send a test event to verify connectivity between your BladeSystem c-Class enclosure and Insight RS.	

Configure Monitored Devices

To configure your monitored devices, complete the following section:

Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Register the OA and Verify Discovery

To discover your monitored devices, complete the following sections:

Register Remote Support Through the Onboard Administrator

If using OA firmware version 3.60 or above, you can enable Insight Remote Support for a BladeSystem c-Class Enclosure using the Onboard Administrator (OA). The OA is not responsible for reporting information about the blades installed in the BladeSystem c-Class Enclosure. The OA only reports enclosure issues, such as problems with fans, power supplies, and media bays.

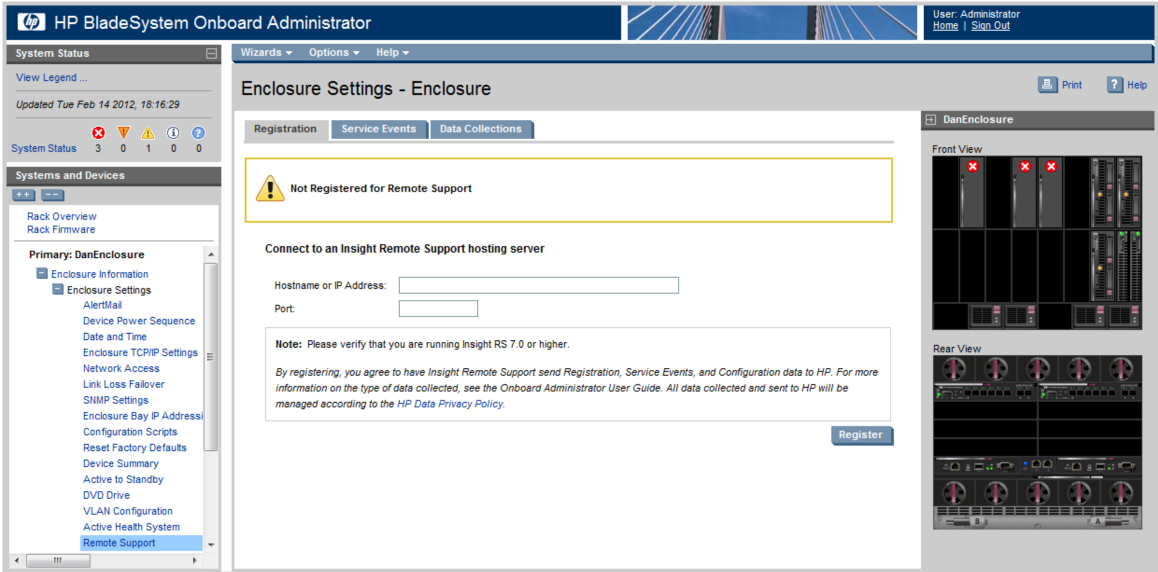
Important: If using a primary and standby OA, ensure that the firmware version for both OAs is 3.60 or higher.

Registering the enclosure does not register the individual blades installed in the enclosure. You need to configure each blade separately.

To register for Remote Support through he OA, complete the following steps:

1. In a web browser, log on to the HP BladeSystem Onboard Administrator (`https://`*`<OA hostname or IP address>`*).
2. Navigate to **Enclosure Information** → **Enclosure Settings** → **Network Access**, and click the **Protocols** tab.
3. Make sure the **Enable XML Reply** check box is selected. If this check box is not selected, the OA cannot be enabled in Insight RS.

4. Navigate to **Enclosure Information** → **Enclosure Settings** → **Remote Support**, and click the **Registration** tab.



5. Type a host name or IP address.
6. In the **Port** field, type *7906*.
7. Click **Register**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon ( ).

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following sections:

Sending a Test Event

After registering the c-Class enclosure with Insight Remote Support, send a test event to confirm the connection.

1. In the OA navigation menu, click **Enclosure Settings** → **Remote Support** → **Service Events**. The Service Events screen appears.
2. Click **Send Test Event**.

When the transmission is complete, the test event is listed in the Service Event Log and in the Insight RS Console.

The Time Generated column in the Service Event Log shows the date and time based on the configured enclosure time zone.

3. Check the Insight RS Console to verify that the test event arrived:
   a. Log on to the Insight RS Console.
   b. In the main menu, select **Devices**.
   c. Find the c-Class enclosure and click the device name.
   d. Click the **Service Events** tab. All service events submitted against the system are displayed here (even if you clear the Service Event Log).

      Insight RS converts the OA service event Time Generated value to the time zone of the browser used to access the Insight RS Console.

      Note: Test events are automatically closed by HP since no further actions are required.

Verify Collections in the Insight RS Console

After discovery, a Server Basic Configuration Collection is automatically run for servers. Verify that this collection ran successfully on the Collection Services → Basic Collection Results tab in the Insight RS Console. Note that there is a delay between when discovery finishes and when the collection runs. For more information about scheduling collections, see the Insight RS Help.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services** and then click the **Basic Collection Results** tab.
3. Expand the Server Basic Configuration Collection section.
4. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears (). If it failed, an error icon appears ().

Maintenance and Troubleshooting

Disable Monitoring of a BladeSystem c-Class Enclosure

There may be a reason that you need to temporarily disable a BladeSystem c-Class Enclosure so that is no longer recognized by Insight Remote Support. For example, for routine maintenance of the device or if the enclosure's warranty has expired.

Important: Disabling the c-Class Enclosure in the Insight RS Console does not unregister the device in the OA. For the OA to be aware that the device has been disabled, you must unregister the in the OA instead of disabling the device in the Insight RS Console.

Note the following:

- Disabling or deleting the c-Class enclosure in the Insight RS Console does not unregister the device in OA. For OA to be aware that an enclosure is disabled, you must use OA to unregister from Insight

Remote Support. Additionally, if the enclosure is deleted in the Insight RS Console and the OA sends an SNMP trap, Insight RS will trigger a re-discovery of the enclosure and possibly re-enable the device for remote support.

- Unregistering an enclosure through OA temporarily discontinues monitoring of the enclosure, but the enclosure is still listed in the Insight RS Console. To resume monitoring, use the OA software to re-register the enclosure.

To temporarily unregister the from Insight RS, complete the following steps:

1. In a web browser, log on to the HP BladeSystem Onboard Administrator (`https://<OA hostname or IP address>`).
2. Navigate to **Enclosure Information** → **Enclosure Settings** → **Remote Support**.
3. Click **Unregister**. The following message appears:

   `Are you sure you want to un-register and disable HP Insight Remote Support?`
4. Click **OK**. The following message appears:

   `Un-registration in progress. Please wait…`

   When the un-registration is finished, the Remote Support page shows the following message:

   `The enclosure is not registered.`

The system disables monitoring and collections on this device.

Configuring BladeSystem c-Class Enclosures Using SNMP

HP BladeSystem c-Class Enclosures can be monitored by Insight Remote Support (RS). Configuring the BladeSystem c-Class Enclosure only enables Insight RS to monitor the enclosure itself. The blades installed in the BladeSystem c-Class Enclosure need to be configured separately to be monitored by Insight RS. See "Identify Required Communication Protocols and Software Components" on page 26 for information about configuring your individual blade types.

If using firmware version 3.60 or above, HP recommends registering your c-Class Enclosure through the OA. For more information, see "Configuring BladeSystem c-Class Enclosures Through the OA" on page 164. If using OA firmware versions 3.55 or 3.56 HP recommends you upgrade to version 3.60.

Fulfill Configuration Requirements

To configure your BladeSystem c-Class Enclosures using SNMP to be monitored by Insight RS, complete the following sections:

Table 18.2 *BladeSystem c-Class Enclosure Using SNMP Configuration Steps*

Task	Complete?
Make sure Insight RS supports your BladeSystem c-Class enclosure by checking the *HP Insight Remote Support Release Notes*.	

Table 18.2 ***BladeSystem c-Class Enclosure Using SNMP Configuration Steps, continued***

Task	Complete?
Configure SNMP on the BladeSystem c-Class enclosure.	
Add the SNMP protocol to the Insight RS Console.	
Discover the BladeSystem c-Class enclosure in the Insight RS Console.	
Send a test event to verify connectivity between your BladeSystem c-Class enclosure and Insight RS.	

Install and Configure Communication Software on Enclosures

To configure your monitored devices, complete the following sections:

Configure SNMP on the BladeSystem c-Class Enclosure

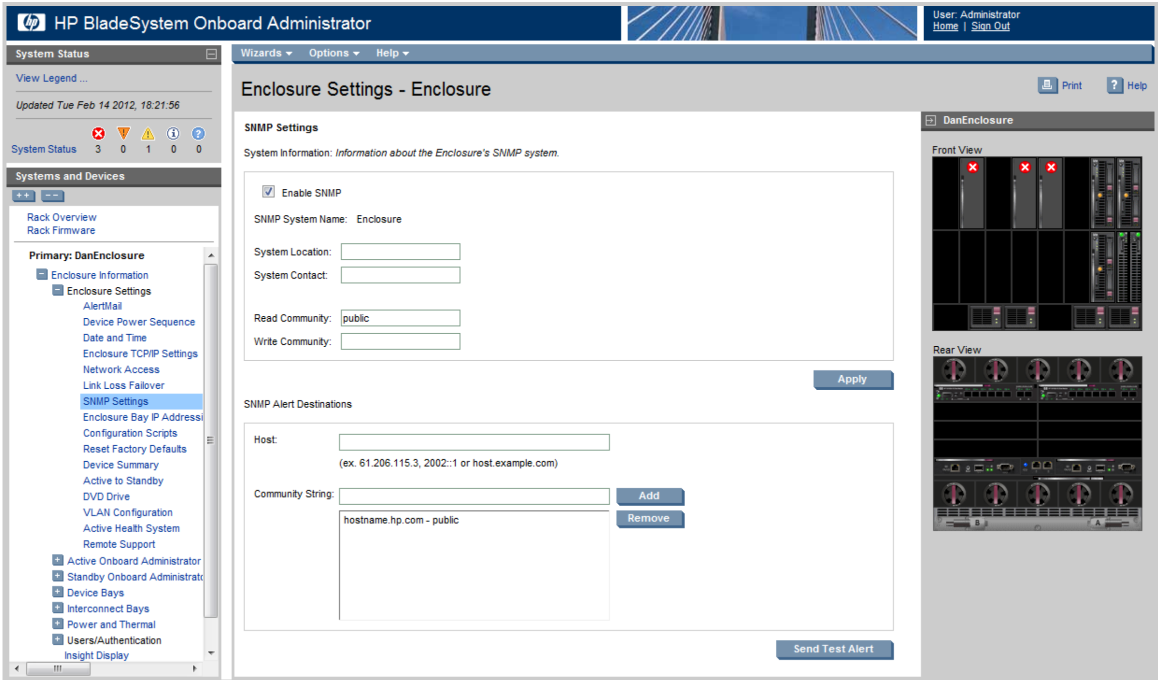
If using an OA firmware version below 3.55, you need to configure SNMP in the Onboard Administrator (OA) before the enclosure can be monitored by Insight RS.

Configuring the enclosure for Remote Support does not allow Insight RS to monitor the blades installed in the enclosure. You need to configure each blade separately.

Important: If using a primary and standby OA, limit discovery to one of the OAs and not both.

To configure SNMP, complete the following steps:

1. In a web browser, open the HP BladeSystem Onboard Administrator.
2. In the log on screen, type your user name and password and click **Sign In**.
3. Expand **Enclosure Information** → **Enclosure Settings**.
4. Click **SNMP Settings**.



5. In the System Information pane, select the **Enable SNMP** check box and complete the following steps:
   a. Add system location and contact information.
   b. Set Read and Write community strings.
   c. Click **Apply**.
6. In the SNMP Alert Destinations pane, complete the following steps:
   a. In the **Host** field, type the IP address of the Hosting Device.
   b. In the **Community String** field, type the SNMP community string for the Hosting Device.
   c. Click **Add**.

Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create an SNMPv1 Protocol Credential in Insight RS Console

If your device's SNMP community string is set to `public` and your community access mode is `read only`, Insight RS automatically associates an SNMPv1 protocol with your device. If you use a different community string or use a nonstandard port, you must create an SNMPv1 protocol credential in the Insight RS Console.

To configure SNMPv1 in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the Select and Configure Protocol drop-down list, select **Simple Network Management Protocol Version 1 (SNMPv1)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Community String you have configured on your device.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP address(es) of the devices to be discovered.
   d. Click **Add**.
4. Click **Start Discovery**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon ( ).

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following sections:

Verify Service Event Monitoring

To send a test event, complete the following steps:

1. In a web browser, open the HP BladeSystem Onboard Administrator.
2. In the log on screen, type your user name and password and click **Sign In**.
3. Expand **Enclosure Information** → **Enclosure Settings**.
4. Click **SNMP Settings**.
5. Click **Send Test Alert** to send a test SNMP trap to the Hosting Device.

Verify Collections in the Insight RS Console

After discovery, a Server Basic Configuration Collection is automatically run for servers. Verify that this collection ran successfully on the Collection Services → Basic Collection Results tab in the Insight RS Console. Note that there is a delay between when discovery finishes and when the collection runs. For more information about scheduling collections, see the Insight RS Help.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services** and then click the **Basic Collection Results** tab.
3. Expand the Server Basic Configuration Collection section.
4. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears ( ). If it failed, an error icon appears ( ).

Maintenance and Troubleshooting

Disable Monitoring of a BladeSystem c-Class Enclosure

There may be a reason that you need to temporarily disable a BladeSystem c-Class Enclosure so that is no longer recognized by Insight Remote Support. For example, for routine maintenance of the device or if the enclosure's warranty has expired.

To disable a device in Insight Remote Support, complete the following steps:

1. In the main menu, select **Devices**.
2. Click the **Device Summary** tab.
3. Select the check box in the far left column for the devices you want to disable.
4. Click **Actions** → **Disable Selected**, and click **OK** in the confirmation dialog box.

The system disables monitoring and collections on this device.

Chapter 19: Configuring P6000 and Enterprise Virtual Arrays

The P6000 Command View server can be either a standalone server with the Command View and SMI-S software installed (referred to as Server Based Management - SBM) or it can be built in to the EVA device itself, referred to as Array Based Management ABM.

- To configure a P6000 or EVA using Server Based Management, see "Configuring P6000 and Enterprise Virtual Arrays Using Server Based Management" below
- To configure a P6000 or EVA using Array Based Management, see "Configuring P6000 and Enterprise Virtual Arrays Using Array Based Management" on page 186

Configuring P6000 and Enterprise Virtual Arrays Using Server Based Management

The following information applies to both monitored devices and Hosting Devices that are hosting P6000 Command View. The ProLiant Windows system hosting P6000 Command View has the same system requirements and configuration for any ProLiant Windows server (see "Configuring ProLiant Windows Servers" on page 48). Be sure to set your WMI and/or SNMP credentials accordingly. Additionally, you must install the Event Log Monitoring Collector (ELMC) on the monitored device.

Before you install and configure Insight Remote Support:

1. Verify that P6000 Command View 9.4 or higher is installed on the Hosting Device.

   Go to the Windows Control Panel and make sure that HP P6000 Command View Software appears in the Add/Remove Programs window.
2. Make sure that the SMI-S EVA and SMI-S CIMOM components (the two components that comprise SMI-S) are installed on the Hosting Device.

   Verify the component is installed and running by checking the presence and status of the Service named `HP CIM Object Manager`.

Note: HP highly recommends that you enable the Email Adapter on the **Administrator Settings → Integration Adapters** tab and, at the minimum, select the following notifications: *Case Opened*, *Application Failure*, *Entitlement Expiration*, and *Device Change*.

Fulfill Configuration Requirements

To configure your P6000 and Enterprise Virtual Arrays using SBM to be monitored by Insight RS, complete the following sections:

Table 19.1 ***P6000 and Enterprise Virtual Array Using SBM Configuration Steps***

Task	Complete?
Make sure Insight RS supports your P6000 disk array by checking the *HP Insight Remote Support Release Notes*.	

Table 19.1 ***P6000 and Enterprise Virtual Array Using SBM Configuration Steps, continued***

Task	Complete?
Verify HP P6000 Command View is installed and configured on the Storage Management Server.	
Install ELMC on the Storage Management Server.	
Add ELMC to the Insight RS Console.	
Add P6000 Command View to the Insight RS Console.	
Discover the P6000 in the Insight RS Console.	
Verify the device's warranty and contract information in the Insight RS Console.	
Send a test event to verify connectivity between your P6000 and Insight RS.	

Install and Configure Communication Software on Arrays

To configure your monitored devices, complete the following sections:

Install P6000 Command View

Insight Remote Support and P6000 Command View software are supported to coexist on the Hosting Device providing the minimum hardware and software requirements are met. HP recommends co-hosting the P6000 Command View software on the Insight Remote Support Hosting Device. Environments with multiple instances of P6000 Command View can have one instance on the Hosting Device and other instances of P6000 Command View on other monitored devices.

Multiple instances of P6000 Command View can share the management role of common attached EVA arrays in the environment. This can be for the purpose of fault tolerance or load balancing, for example. Although there can be multiple instances of P6000 Command View capable of managing the EVA array, only one instance of P6000 Command View can be the active manager of that array. Other instances of P6000 Command View that are not active in managing the array are considered to be passive.

Note: Real Time Event Data and communication to the array is only capable when there is a connection to the P6000 Command View instance that is the active manager of the array. In order to configure a monitored device, you must install ELMC on any server that is running P6000 Command View that is capable of being the active manager of the array. Any P6000 Command View instance that is not the active manager of an EVA is considered to be a passive instance of P6000 Command View. Historic Event Data may be present on a passive instance if it had been the active manager in the past. This event data may be of use for manual analysis.

Important: A passive instance of P6000 Command View can see the array but cannot communicate with it. Only the active management instance is capable of communicating with the array. This is why you may see the array discovered but missing information about the array in some components.

To be supported by Insight Remote Support, the monitored device must have P6000 Command View installed (version 9.4 or later) and be managing at least one array. SMI-S is delivered with the P6000 Command View installation and is required because it serves as the communication path between Insight Remote Support and the array, using CIMOM for WBEM communications.

If the required P6000 Command View components are *not* installed, see "Documentation for P6000 Command View" on page 181 for more details.

Caution: Removing SMI-S from a host system also forces the removal of Command View for Tape Libraries (TL). The version of Command View TL must match the firmware used in the library.

If the version of SMI-S installed by Command View TL is older than the one being used by P6000 Command View, then install or upgrade P6000 Command View *after* Command View TL to avoid a conflict.

Additionally, the credentials used by each software application are overwritten by the other when it installs. Therefore, use the same account for both.

Fulfilling P6000 Command View System and Access Requirements

Access requirements:

- Supported versions of P6000 Command View use a secure logon (SSL) design.
- A Microsoft Windows account with the correct group privilege is required to access P6000 Command View.

Installing P6000 Command View adds two new group privileges to a system: HP Storage Admins (write access) and HP Storage Users (read access). P6000 Command View credentials are required when setting up Insight Remote Support, so verify that you have this information.

Installing HP SIM and P6000 Command View on the Same Server

Understanding HP SIM Port Setting Conflicts

The default port assignment of 5989 for WBEM/WMI and SMI-S has led to conflicts with HP SIM on the Hosting Device. The ability of Insight RS to interface with WBEM indications, WMI Mapper, and inclusion of HP-UX systems to the environment add more complexity to configuring the environment. HP-UX is hard coded to use port 5989. The following sections describe changes that address the port assignment issues and provide a new solution for configuring Insight Remote Support with P6000 Command View.

Note: If your Insight Remote Support environment is already configured and working, you may choose to leave settings as they are and implement the following steps only if you experience related problems.

The installation sequence for HP SIM and P6000 Command View on a Hosting Device affects the default ports settings. The best practice is to install HP SIM first and allow WMI Mapper to use the default of port 5989. The subsequent P6000 Command View installation can detect if port 5989 is in use, and can redirect to a different port.

To obtain a snapshot of the current port assignments in your environment, in a command window, execute the following command:

```
C:\ netstat –anb >netstat.txt
```

Installing HP SIM For the First Time

When installing HP SIM for the first time, the First Time Wizard includes a step to configure WBEM/WMI Mapper Proxy.

Keep the default setting of port 5989. After you complete the installation of HP SIM, add the port that P6000 Command View SMI-S will use to the `.xml` file shown in step 1 of the following procedure.

Follow these steps to identify the port that HP SIM will use to connect to SMI-S. In this example, port 60000 is used for the SMI-S HTTPS connection to the **interop** namespace.

1. Open the following file in a text editor:

   ```
   C:\Program Files\HP\Systems Insight Manager\config\identification\wbemportlist.xml
   ```

2. Add the following *red* lines of text to the end of the file.

   ```
   <port id="60000" protocol="https">
     <interopnamespacelist>
       <interopnamespace name="interop"/>
     </interopnamespacelist>
   </port>
   ```

   The following example shows the complete file after you add the port for SMI-S:

   ```
   <?xml version="1.0" encoding="UTF-8"?>
   <wbemportlist>
     <port id="5989" protocol="https">
       <cimnamespacelist>
         <cimnamespace name="root/cimv2"/>
         <cimnamespace name="vmware/esxv2"/>
         <cimnamespace name="root/hpq"/>
       </cimnamespacelist>
       <interopnamespacelist>
         <interopnamespace name="root/pg_interop"/>
         <interopnamespace name="root"/>
         <interopnamespace name="root/emulex"/>
         <interopnamespace name="root/qlogic"/>
         <interopnamespace name="root/ibm"/>
         <interopnamespace name="root/emc"/>
         <interopnamespace name="root/smis/current"/>
         <interopnamespace name="root/hitachi/dm51"/>
         <interopnamespace name="interop"/>
         <interopnamespace name="root/interop"/>
         <interopnamespace name="root/switch"/>
         <interopnamespace name="root/cimv2"/>
       </interopnamespacelist>
     </port>
     <port id="60000" protocol="https">
   ```

```
      <interopnamespacelist>
        <interopnamespace name="interop"/>
      </interopnamespacelist>
    </port>
  </wbemportlist>
```

3. Save the `wbemportlist.xml` file.
4. Restart the HP SIM service: right click on the HP Systems Insight Manager service and select **restart** from the drop-down list.

Correcting an Existing HP SIM Installation

Important: Do not use the following procedure if this is a first-time installation of HP SIM. Use this procedure only if HP SIM has already been installed and the WMI Mapper port has been changed to something (for example, 6989) other than the default (5989).

The following corrective steps are to be executed only if HP SIM has already been installed and the WMI Mapper port has been changed to something other than the default of 5989.

To restore the WMI Mapper port to the default port:

1. Stop the Pegasus WMI Mapper service.
2. In a text editor open the file: `C:\Program Files\The Open Group\WMI Mapper\cimserver_planned.conf`.
3. In the line that includes `httpsPort=6989`, change the port number to **`5989`**.

   Note: The port number may have been set to something other than 6989; it is used here as an example only.

4. Save the file.
5. Restart the Pegasus WMI Mapper service.
6. Run the `netstat` command and verify the change has been made.

   ```
   C:\ netstat -anb >netstat.txt
   ```

   Note: The changes made in the file `cimserver_planned.conf` will be applied to the `cimserver_current.conf` file after the service is restarted.

   The following is an example of the of the text in `cimserver_planned.conf` file.

   ```
   enableRemotePrivilegedUserAccess=true
   enableHttpsConnection=true
   enableHttpConnection=false
   sslCertificateFilePath=C:\hp\sslshare\cert.pem
   sslKeyFilePath=C:\hp\sslshare\file.pem
   httpsPort=5989
   ```

To change the WMI Mapper Proxy port in HP SIM:

1. Log in to HP SIM with Administrative privileges.
2. Verify the existing credentials for the Hosting Device are correct:
   a. Open the **All Systems** list and select the device serving as the HP SIM CMS from the list.
   b. Click the **Tools & Links** tab.
   c. Click the **System Credentials** link.
   d. Select the check box for the CMS instance of WMI Mapper that is running on the local host.
   e. Click **Edit system credentials**.
   f. Verify that the sign-in credentials exist and are correct for the CMS (preferred), or that there are advanced WBEM settings. Also, verify that the port number is 5989, and that the credentials are correct for the WMI Mapper.
3. From upper navigation bar select **Options** → **Protocol Settings** → **WMI Mapper Proxy**.
4. Select the CMS instance of WMI Mapper that is running on the local host.
5. Click **Edit**.
6. Enter **5989** in the **Port number** field.
7. Click **OK** to save the settings.

Restore Defaults to the wbemportlist.xml File

1. In a text editor open the file: `C:\Program Files\HP\Systems Insight Manager\config\identification\wbemportlist.xml`
2. Remove the following lines from the file:

```
<!-- WMI Mapper httpsPort=6989 -->
<port id="6989" protocol="https">
  <cimnamespacelist>
    <cimnamespace name="root/cimv2"/>
  </cimnamespacelist>
</port>
```

3. Add the following lines to the end of the file. In this example, the port number 60000 is used as the HTTPS port for SMI-S.

```
</port>
<port id="60000" protocol="https">
  <interopnamespacelist>
    <interopnamespace name="interop"/>
  </interopnamespacelist>
</port>
```

   The following example shows the complete file, with port number 60000 used as the HTTPS port for SMI-S:

```
<?xml version="1.0" encoding="UTF-8"?>
<wbemportlist>
```

```
    <port id="5989" protocol="https">
      <cimnamespacelist>
        <cimnamespace name="root/cimv2"/>
        <cimnamespace name="vmware/esxv2"/>
        <cimnamespace name="root/hpq"/>
      </cimnamespacelist>
      <interopnamespacelist>
        <interopnamespace name="root/pg_interop"/>
        <interopnamespace name="root"/>
        <interopnamespace name="root/emulex"/>
        <interopnamespace name="root/qlogic"/>
        <interopnamespace name="root/ibm"/>
        <interopnamespace name="root/emc"/>
        <interopnamespace name="root/smis/current"/>
        <interopnamespace name="root/hitachi/dm51"/>
        <interopnamespace name="interop"/>
        <interopnamespace name="root/interop"/>
        <interopnamespace name="root/switch"/>
        <interopnamespace name="root/cimv2"/>
      </interopnamespacelist>
    </port>
    <port id="60000" protocol="https">
      <interopnamespacelist>
        <interopnamespace name="interop"/>
      </interopnamespacelist>
    </port>
  </wbemportlist>
```

4. Save the file.
5. Restart the HP SIM service.
6. Select **Options** → **Identify Systems** to re-identify all WBEM/WMI systems in HP SIM to correct the port reference.

Installing and Configuring P6000 Command View After HP SIM

The P6000 Command View installation kit includes SMI-S as a component. At installation time, SMI-S assigns the following as default ports if they are available:

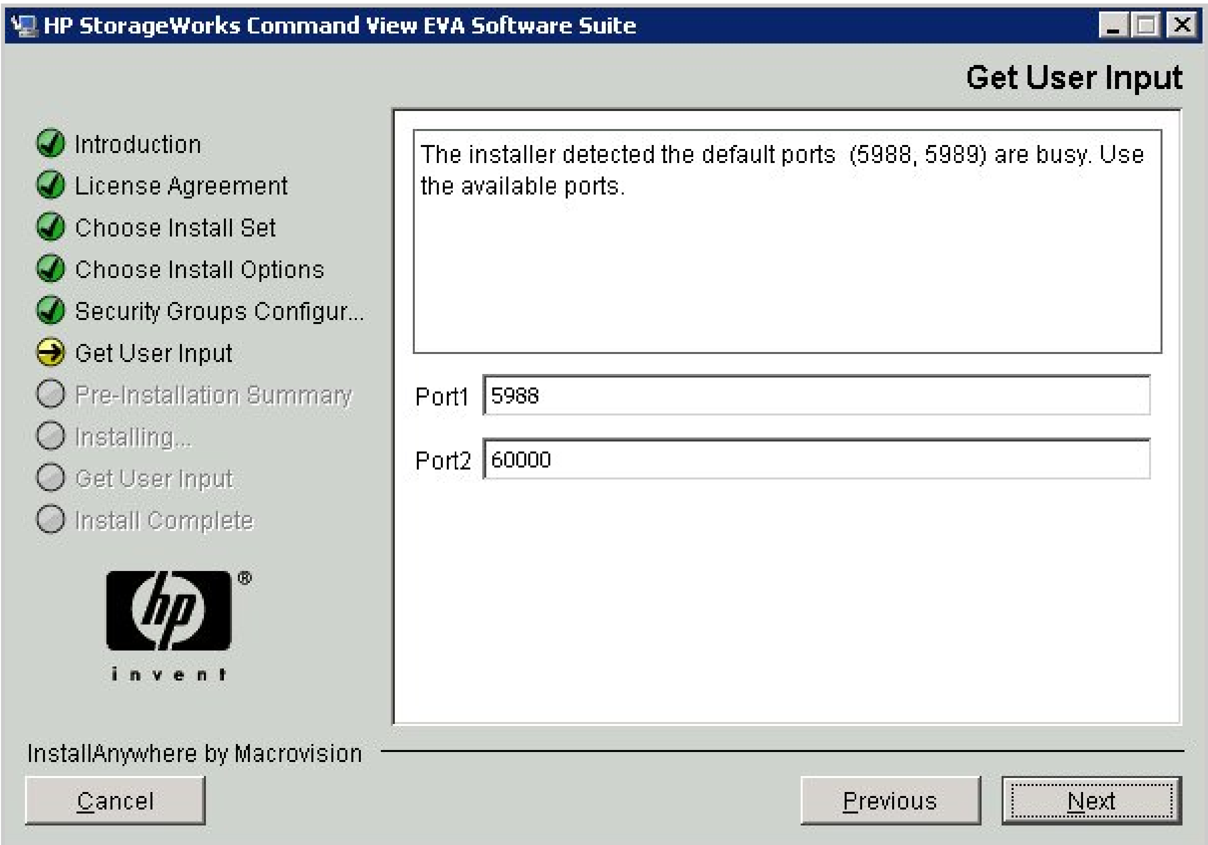
- HTTP port=5988
- HTTPS port=5989

If one or both of these ports are in use, the installation provides an additional configuration screen on which you can change the port numbers. The SMI-S installer suggests port 60000 if that port is not in use. SMI-S retains port 5988 if it does not conflict with another port number. SMI-S uses these ports when the HP CIM Object Manager (CIMOM) process starts.

Note: This screen appears only if a port conflict is detected, which is why HP recommends installing HP SIM before installing Command View.

The following figure shows an example of the SMI-S installation on an Insight RS Hosting Device that already has HP SIM installed with WMI Mapper:

Figure 19.1 ***Supplemental SMI-S Install Screen***



After the ports are set, you can verify them by executing the `netstat` command:

```
C:\ netstat -anb >netstat.txt
```

The following output appears:

```
TCP 0.0.0.0:5988 0.0.0.0:0 LISTENING 1712 [JavaWrapper.exe]
TCP 0.0.0.0:5989 0.0.0.0:0 LISTENING 460 [WMIServer.exe]
TCP 0.0.0.0:60000 0.0.0.0:0 LISTENING 1712 [JavaWrapper.exe]
```

In this example, SMI-S is set up to use ports 5988 and 60000.

To reset the port numbers if P6000 Command View is already installed:

1. Open the following file:

   `C:\Program Files\Hewlett-Packard\SMI-S\CXWSCimom\config\cxws.properties`

2. Locate the `cxws.http.port` and `cxws.https.port` entries and reset the to the following values:

```
cxws.http.port=5988
cxws.https.port=60000
```

3. Restart the HP CIMOM service.

Documentation for P6000 Command View

P6000 Command View documentation is available at: http://h18006.www1.hp.com/products/storage/software/cmdvieweva/index.html. In the Related links, click the **Technical Support / Manuals** link and then the **Manuals** link to view the list of P6000 Command View manuals.

The following documents are required for P6000 Command View installation and configuration:

- *HP P6000 Command View Suite Installation Guide*
- *HP P6000 Command View Suite Release Notes*
- *HP P6000 Enterprise Virtual Array Compatibility Reference*

For more information, visit the *HP P6000 Command View Software Suite QuickSpec Overview* at: http://h18004.www1.hp.com/products/quickspecs/12239_div/12239_div.html

Installing the ELMC Software Package on the P6000 Command View Server

ELMC must be installed on the P6000 Command View server to allow access to logged event data from the array. Previously this was referred to as a Storage Management Server (SMS).

ELMC is required on the P6000 Command View server so that events can be forwarded to Insight RS.

If ELMC is already installed on the P6000 Command View server, make sure it is version 6.2 or later. If it's older than version 6.2, it needs to be upgraded. To check the version of ELMC, run the following command: `wccproxy version`.

Note: When upgrading to ELMC version 6.4, the incorrect version number is displayed in the upgrade window. After performing the upgrade, the correct version number of 6.4 will show in the Programs and Features window and when you run the `wccproxy version` command.

To install the ELMC software package, complete the following steps:

1. In the Insight RS Console, navigate to the **Administrator Settings** → **Software Updates** tab and select the *Event Log Monitoring Collector (ELMC)* software package.
2. On the **Available Version** tab, click **Download**.
3. When the download completes, click **Install**. The ELMC packages are placed in the `%HP_RS_DATA%\ELMC` folder. This folder defaults to `C:\ProgramData\HP\RS\DATA\ELMC`.

   Note: The `ProgramData` folder is a hidden folder. To view this folder, set the folder options to show hidden folders.
4. Copy the appropriate Windows x86/x64 ELMC software package into a temporary directory on the P6000 Command View server.

5. Double-click the installer file to begin the install process. The kit will install and finish with no user prompting. No user configuration input is required to install the ELMC software package.

Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create a P6000 Command View Protocol in the Insight RS Console

To configure a P6000 protocol credential in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab
3. In the **Select and Configure Protocol** drop-down list, select **HP P6000 Command View**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Username and Password for the P6000 Command View instance.
6. Click **Add**.

HP recommends also creating credentials for the server that the P6000 Command View software is running on.

Create an ELMC Protocol in the Insight RS Console

To configure ELMC in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the **Select and Configure Protocol** drop-down list, select **Event Log Monitoring Collector (ELMC)**.
4. Click **New**. The New Credential dialog box appears.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the EVA in the Insight RS Console

An EVA is not discovered directly. The EVA is located when discovering the P6000 Command View server that is managing the EVA.

Important: When a new EVA array is added to an Insight RS environment that is already configured, a manual Discovery of the server running P6000 Command View that is managing the new array is required in order to detect the new EVA's warranty and contract information.

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP address(es) of the devices to be discovered.
   d. Click **Add**.
4. Click **Start Discovery**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon ( ).

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Warranty and Contract Information

Verify that the warranty and contract information was discovered correctly in the Insight RS Console:

1. In the Insight RS Console, navigate to **Devices** and click the EVA Device Name.
2. Expand the Hardware section, and make sure the **Acquired Serial Number** and **Acquired Product Number** are correct. If they are not correct, type the correct values in the **Override Serial Number** and **Override Product Number** fields and click **Save Changes**.

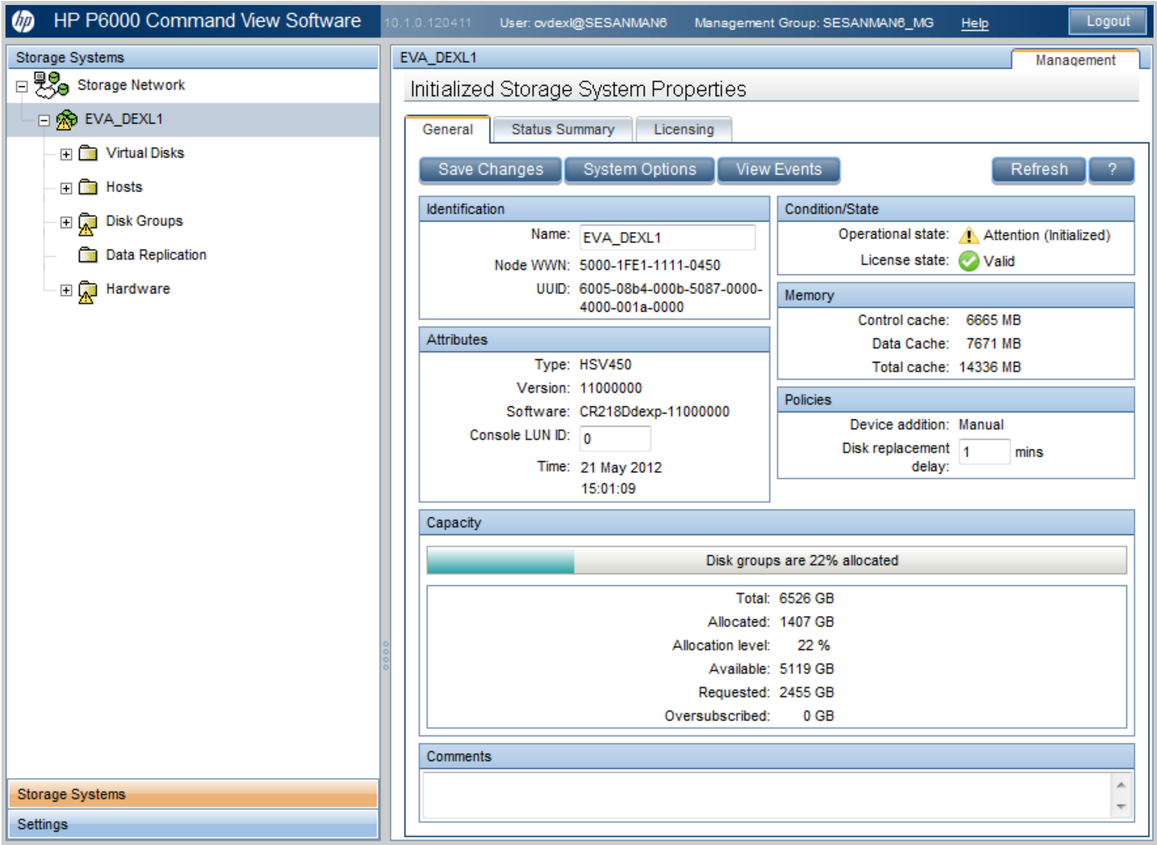
Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following sections:

Sending a Test Event

To send a test event from P6000 Command View, complete the following steps:

1. In a web browser, log on to P6000 Command View.
2. In the left menu, select the EVA.



3. On the **General** tab, click **System Options**.
4. On the System Options screen, under the **Service** heading, click **Perform remote service test**.
5. On the Perform Remote Service Test screen, click **Perform Remote Service Test**.
6. Click **OK**.
7. In the Insight RS Console, navigate to **Devices** → **Service Events** to verify that the event was received.

Verify Collections in the Insight RS Console

After discovery, a Storage Configuration Collection is automatically run for servers. Verify that this collection ran successfully on the Collection Services → SAN Storage Collection Results tab in the Insight RS Console. Note that there is a delay between when discovery finishes and when the collection runs. For more information about scheduling collections, see the Insight RS Help.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services** and then click the **SAN Storage Collection Results** tab.
3. Expand the Storage Configuration Collection section.
4. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears (). If it failed, an error icon appears ().

Maintenance and Troubleshooting

Enable User-Initiated Service Mode in P6000 Command View

User-Initiated Service Mode (UISM) provides a means of allowing Remote Support Diagnostic Service incidents to be suppressed or revealed only to the client, while allowing the Remote Support Diagnostic Technology to continue to monitor the device by writing a 'User Initiated Service Event' into the Device log. Using UISM will reduce the number of non-actionable events that are created when you are installing software or performing maintenance.

UISM is enabled through P6000 Command View. UISM will automatically transition back to normal mode after an adjustable time delay expires or by manually disabling the mode. Two new P6000 Command View management events will be created when UISM is enabled. The first will indicate the beginning of UISM and will contain a timeout value in minutes within the event description. The second will indicate the end of UISM.

Being in this mode has no effect on the functionality of P6000 Command View or event handling within P6000 Command View. This mode is solely for the benefit of external support tools, such as Insight Remote Support.

To enable User-Initiated Service Mode, complete the following steps:

1. Open P6000 Command View and select a device in the left menu tree.
2. On the System Options page, under the **Service** heading, select **Configure User-Initiated Service Mode (UISM)**.
3. On the Configure User Initiated Service Mode page, type a reason for the UISM in the **Reason** text area.
4. Modify the duration if necessary. The default is 30 minutes.
5. Click **Start service mode**.
6. In the pop-up window, click **Start Service Mode** to confirm that you want to enter UISM. The Configure User Initiated Service Mode page will be updated to show that UISM is active, the **Mode** will have changed, and the time left in the UISM will be displayed.
7. If you want to exit UISM before the duration has expired, you can click **Stop service mode**.

Perform a Remote Service Test in P6000 Command View

The Remote Service Test provides a method to confirm a successful remote support setup and to troubleshoot software communications problems. When a Remote Service Test is performed, Insight Remote Support will receive the event and pass it on to HP. HP will then send the event back to the user. This provides an end-to-end test that can be used test or troubleshoot the remote support software communication path. The remote service test has no other effect within P6000 Command View beyond logging the event and sending any configured notifications.

To perform a remote service test, complete the following steps:

1. Open P6000 Command View and select a device in the left menu tree.
2. On the System Options page, under the **Service** heading, select **Perform remote service test**.

3. On the Perform Remote Service Test page, click **Perform remote service test**.
4. Verify the test event was received in the Insight RS Console.

Configuring P6000 and Enterprise Virtual Arrays Using Array Based Management

EVA4400 and P6000 can be monitored using Array Based Management. The introduction of the EVA4400 included a new module to the HSV300 controller pair that installs Command View EVA on the array. This is called Array Based Management (ABM) which has a network interface for access across the LAN. The ABM is not capable of hosting SMI-S in the array, so SMI-S is required to be hosted elsewhere in the LAN to serve as a proxy connection to the array.

The P6000 Command View installation kit allows for SMI-S to be installed independently on a designated server. Install SMI-S on the Hosting Device or another supported server in the LAN.

Important: The ABM Fully Qualified Domain Name must not be the same as the EVA array name. Insight RS has no way to differentiate between two devices that have the same name.

Fulfill Configuration Requirements

To configure your P6000 and Enterprise Virtual Arrays using ABM to be monitored by Insight RS, complete the following sections:

Table 19.2 *P6000 and Enterprise Virtual Array Using ABM Configuration Steps*

Task	Complete?
Make sure Insight RS supports your P6000 disk array by checking the *HP Insight Remote Support Release Notes*.	
Add P6000 Command View to the Insight RS Console.	
Discover the P6000 in the Insight RS Console.	
Add the device's Support Type and Support Identifier to the Insight RS Console.	

Configure Monitored Devices

To configure your monitored devices, complete the following section:

Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create a P6000 Command View Protocol in the Insight RS Console

To configure a P6000 protocol credential in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab
3. In the **Select and Configure Protocol** drop-down list, select **HP P6000 Command View**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Username and Password for the P6000 Command View instance.

   Configure the credentials for the ABM that is running P6000 Command View and actively managing the array. If the array is not being managed by the instance of P6000 Command View you are pointing to, you will see the EVA array but Insight RS will not be able to obtain the Serial Number or Product Number information.
6. Click **Add**.

Discover the ABM in the Insight RS Console

If you are managing the array from the P6000 Command View instance running on the Array Based Management (ABM), then you need to discover the ABM.

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP address(es) of the devices to be discovered.
   d. Click **Add**.
4. Click **Start Discovery**.
5. After discovery completes successfully, browse to **Devices** in the main menu, and on the **Device Summary** tab, click the array's Device Name to open the Device Details screen.
6. On the **Device** tab, expand the Warranty & Contract section and type the device's Support Type and Support Identifier.
7. Click **Save Changes** to save the information.
8. Return to the **Devices** screen and verify that the Status of the device.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.

3. Make sure the Status column contains a success icon ().

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following section:

Verify Collections in the Insight RS Console

After discovery, a Storage Configuration Collection is automatically run for servers. Verify that this collection ran successfully on the Collection Services → SAN Storage Collection Results tab in the Insight RS Console. Note that there is a delay between when discovery finishes and when the collection runs. For more information about scheduling collections, see the Insight RS Help.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services** and then click the **SAN Storage Collection Results** tab.
3. Expand the Storage Configuration Collection section.
4. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears (). If it failed, an error icon appears ().

Chapter 20: Configuring StoreVirtual P4000 Storage Systems

HP Insight Remote Support software delivers secure remote support for your HP servers, storage, network, and SAN environments, which includes the P4000 SAN.

Note: HP StoreVirtual Storage is the new name for HP LeftHand Storage and HP P4000 SAN solutions. LeftHand Operating System (LeftHand OS) is the new name for SAN/iQ.

Important: Each P4000 Storage Node counts as 30 monitored devices within Insight RS.

Important: For every 100 P4000 devices being monitored by Insight RS, add 1 GB of free disk space to the Hosting Device requirements.

P4000 documentation is available at: http://www.hp.com/support/manuals. In the **Storage** section, click **Disk Storage Systems** and then select **HP LeftHand P4000 SAN Solutions**.

Fulfill Configuration Requirements

To configure your P4000 Storage Systems to be monitored by Insight RS, complete the following sections:

Table 20.1 ***P4000 Storage System Configuration Steps***

Task	Complete?
Make sure Insight RS supports your P4000 Storage System by checking the *HP Insight Remote Support Release Notes*.	
Install or upgrade CMC on the Hosting Device.	
Install or upgrade LeftHand OS on the P4000.	
Configure SNMP on the P4000.	
Add the P4000 SAN Solution (SAN/iQ) protocol to the Insight RS Console.	
Discover the P4000 Storage System in the Insight RS Console.	
Send a test event to verify connectivity between your P4000 Storage System and Insight RS.	

Install and Configure Communication Software on Storage Systems

To configure your monitored devices, complete the following sections:

Install and Configure CMC on the Hosting Device

You need to do the complete install of Centralized Management Console (CMC) to install the SNMP MIBs. If you do not modify the SNMP defaults that CMC uses, then SNMP should work with Insight Remote Support without modification.

Insight RS supports CMC 9.0 and higher (CMC 11.0 is recommended).

CMC 11.0 can be used to manage LeftHand OS 9.5, 10.x, and 11.0 storage nodes.

CMC, Windows version has the following requirements:

Table 20.2 ***Memory and storage requirements for CMC***

Memory Size	Free Disk Space
100 MB RAM during run-time	150 MB disk space for complete install

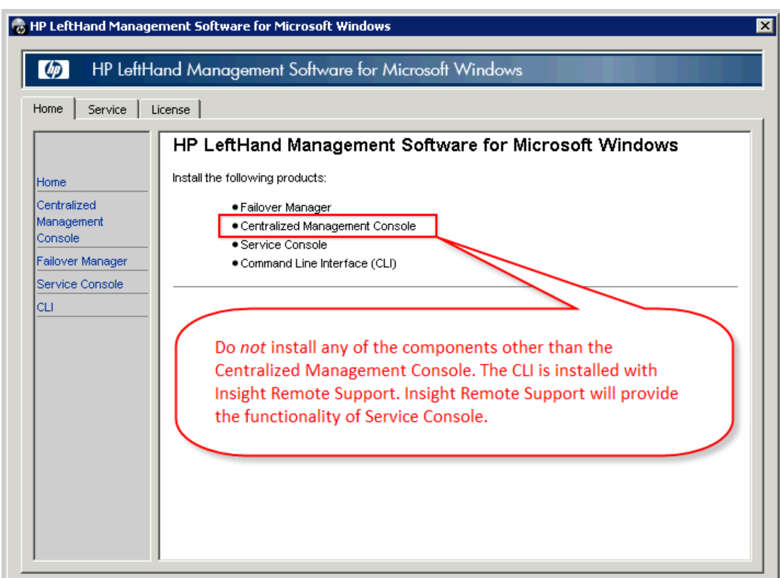
Note: CMC can be installed on the Hosting Device or it can be install on a separate system. If CMC is already installed on another system, it does not need to be installed on the Hosting Device.

Install the centralized management console (CMC) on the computer that you will use to administer the SAN. You need administrator privileges while installing the CMC.

1. Insert the P4000 Management SW DVD in the DVD drive. The installer should launch automatically. Or, navigate to the executable (`:\GUI\Windows\Disk1\InstData\VM\CMC_Installer.exe`)

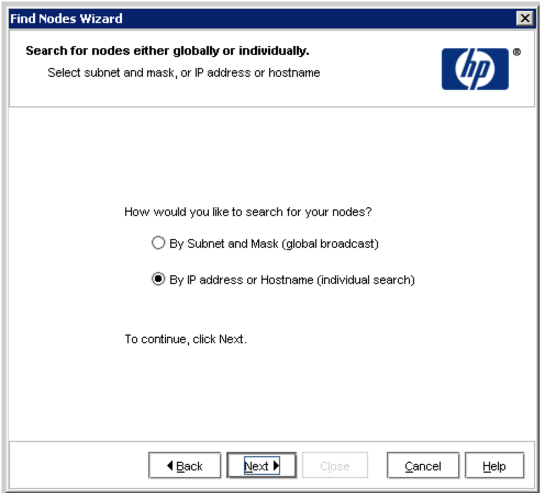
   Or download CMC at: http://www.hp.com/go/P4000downloads.

   Click the **Complete install** option, which is recommended for users that use SNMP.

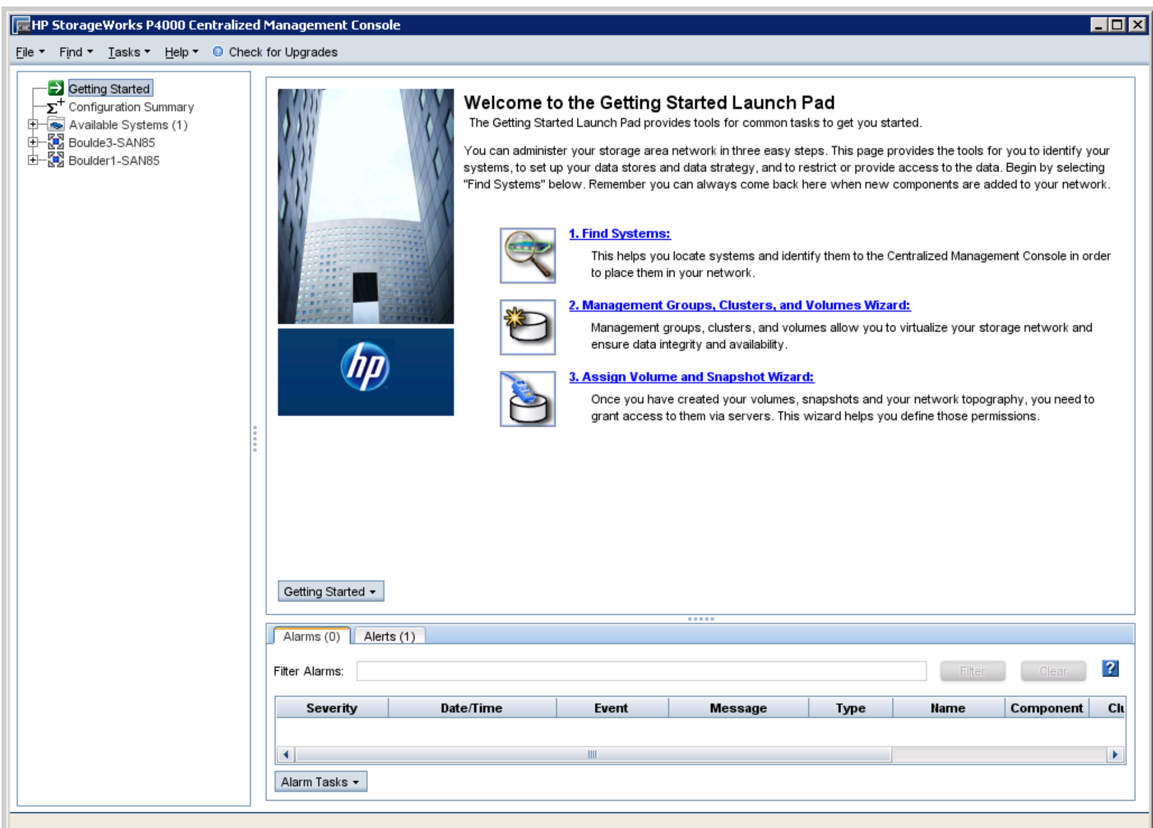
   

   Note: If you already have Service Console installed, you don't need to disable it. Insight Remote Support can co-exist with Service Console without conflict.

2. Use the Find Nodes wizard to discover the storage systems on the network, using either IP addresses or host names, or by using the subnet and gateway mask of the storage network.



The found storage systems appear in the available category in the CMC.



Upgrade LeftHand OS on the P4000 Storage Systems

Note: LeftHand Operating System (LeftHand OS) is the new name for SAN/iQ.

Use CMC to upgrade to LeftHand OS 9.5 or higher (LeftHand OS 11.0 is recommended) on the P4000 Storage Systems. If your P4000 Storage Systems already has LeftHand OS 9.5 installed, you do *not* need to perform this procedure.

Note: For LeftHand OS 9.5 devices, Patch Set 05 is required for all devices.

CMC version 9.0 and higher provide the option to download LeftHand OS upgrades and patches from HP. Once the current upgrades and patches are downloaded from HP to the CMC system, CMC can be used to update Management Groups or nodes with these changes.

When you upgrade the LeftHand OS software on a storage node, the version number changes. Check the current software version by selecting a storage node in the navigation window and viewing the Details tab window.

Note: Directly upgrading from LeftHand OS 9.5 to LeftHand OS 11.0 is not supported.

Important: When installing/upgrading LeftHand OS, do not modify the default SNMP settings. The default settings are used by Insight Remote Support, and communication between the P4000 Storage System and the Hosting Device will not function properly if the SNMP settings are modified. SNMP is enabled by default in LeftHand OS. In CMC 9.0 and higher, SNMP traps are modified at the Management Group level, not at the node level.

Understand Best Practices

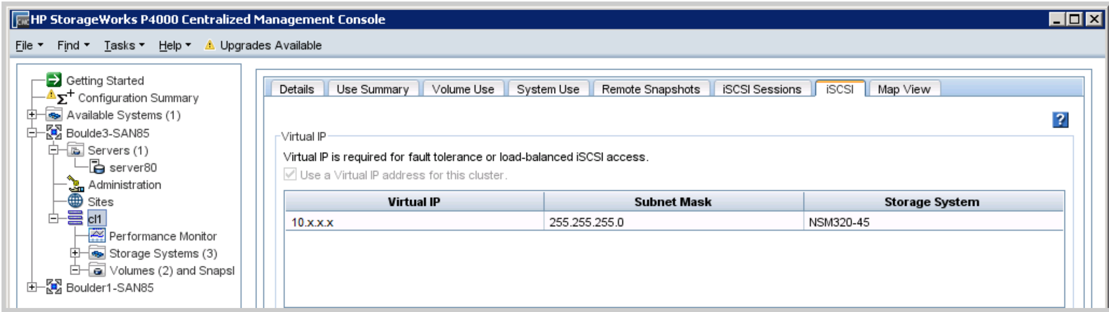
- **LSMD upgrade** — LSMD upgrade is required for upgrading from LeftHand OS 7.x.
- **Virtual IP addresses** — If a Virtual IP (VIP) address is assigned to a storage node in a cluster, the VIP storage node needs to be upgraded last. The VIP storage node appears on the cluster iSCSI tab, see figure "Find the Storage Node Running the VIP".

  a. Upgrade the non-VIP storage nodes that are running managers, one at a time.

  b. Upgrade the non-VIP non-manager storage nodes.

  c. Upgrade the VIP storage node.

Figure 20.1 ***Find the Storage Node Running the VIP***



- **Remote Copy** — If you are upgrading management groups with Remote Copy associations, you should upgrade the remote management groups first. If you upgrade the primary group first, Remote Copy may stop working temporarily, until both the primary management group and the remote group have finished upgrading. Upgrade the primary site immediately after upgrading the remote site. Refer to "How to Verify Management Group Version".

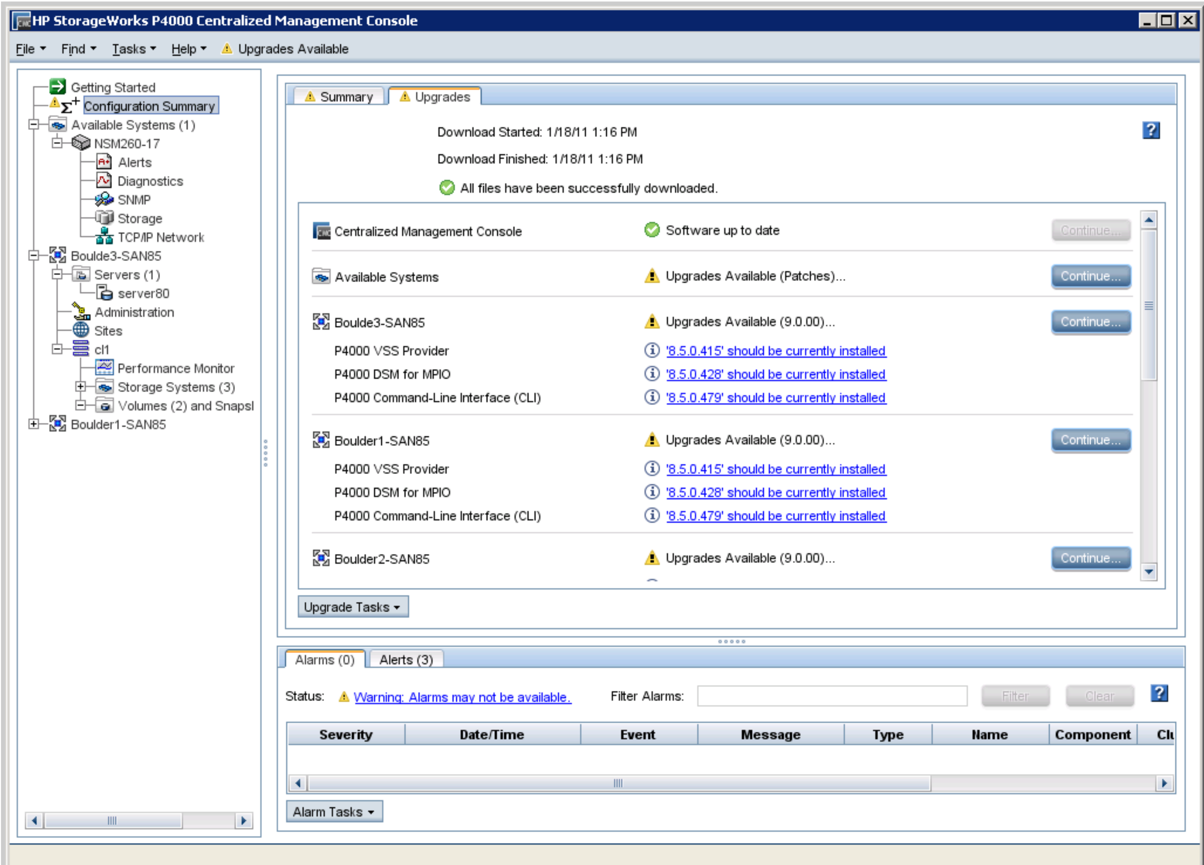
Select the Type of Upgrade

CMC supports two methods of upgrades, as shown in figure "Viewing the CMC Upgrade/Installation Window".

- One-at-a-time (recommended) - this is the **default and only method that should be used if the storage nodes exist in a management group**.
- Simultaneous (advanced) - this allows you to upgrade multiple storage nodes at the same time if they are not in a management group. Use this only for storage nodes in the Available pool.

Caution: Do not select "Simultaneous (advanced)" if your storage nodes are in a production cluster.

Figure 20.2 ***Viewing the CMC Upgrade/Installation Window***



Increase the Size of the OS Disk on the VSAs

Due to changes in the size of the VMware tools that get installed during software upgrades from pre-8.5 versions of LeftHand OS, you must increase the size of the OS disk before upgrading the VSA. Additional space requirements are necessary for future software releases, as well. Therefore, we recommend increasing the size of the OS disk to accommodate both requirements at this time.

Note: These instructions apply to VMware ESX Server. Other VMware products have similar instructions for extending a virtual disk. Please consult the appropriate VMware documentation for the product you are using.

To increase the OS disk size on the VSA, complete the following steps:

1. Using the CMC, power off the VSA.
2. Open the VI Client and select **VSA** → **Edit Settings** → **Hardware**.
3. Select Hard disk 1 (verify that the Virtual Device Node is SCSI (0:0)).
4. Under Disk provisioning, changed the provisioned size to 8 GB.
5. Click **OK**.

6. Repeat these steps for Hard disk 2 (verify the Virtual Device Node is SCSI (0:1))
7. Using the VI Client, power on the VSA.
8. Find the VSA in the CMC and apply the upgrade.

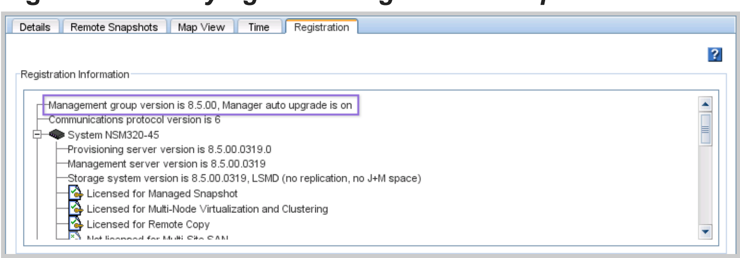
Verify Management Group Version

- When upgrading from version LeftHand OS 7.x to release 9.5 or higher, the management group version will not move to the new release version until all storage nodes in the management group (and in the remote management group if a Remote Copy relationship exists) are upgraded to release LeftHand OS 9.5 or higher.
- When upgrading from LeftHand OS version 7.x to release 9.5 or higher, the upgrade process validates the hardware identity of all of the storage nodes in the management group. If this validation fails for any reason, the management group version will not be upgraded to 9.5 or higher. For example, if a management group has a mix of platforms, some of which are unsupported by a software release; then only the supported platforms get upgraded successfully, the management group version will not be upgraded if the unsupported platforms remain in that management group.

To verify the management group version, complete the following steps:

1. In the CMC navigation window, select the management group, and then select the **Registration** tab. The management group version number is at the top of the Registration Information section, as shown in figure "Verifying the Management Group Version Number" below.

Figure 20.3 ***Verifying the Management Group Version Number***



Check for Patches

After you have upgraded LeftHand OS, use CMC to check for applicable patches required for your storage node. CMC 9.0 and higher shows available patches and can be used to download them from HP to the CMC system.

Configuring SNMP on the P4000 Storage System

Use the following procedure to verify your SNMP settings. If you did not modify the LeftHand OS SNMP settings when you installed/upgraded LeftHand OS then you should not need to make any updates during the following procedure.

With LeftHand OS, you can configure different severity Alert levels to send from the Management Group. Under Insight Remote Support, configure each Management Group to send v1 SNMP traps of `Critical` and `Warning` levels in `Standard` message text length.

LeftHand OS 9.5 and higher Management Groups configure the Hosting Device SNMP trap host destination once at the Management Group level. Even though you only configure SNMP once at the Management Group level under CMC 10.0 and higher, you need to configure SNMP on the Hosting Device to allow traps from all LeftHand OS nodes.

Note: For LeftHand OS Management Groups, users can configure the Hosting Device trap host destination P4000 CLI **createSNMPTrapTarget** command instead of using CMC. The **getGroupInfo** command will show the current SNMP settings and SNMP trap host destinations configured at the Management Group level.

Note: If using the P4000 CLI (CLIQ) **createSNMPTrapTarget** or **getGroupInfo** commands, the path environment variable is not updated when the P4000 CLI is installed with Insight RS. You will need to use the full path for the P4000 CLI, which is *`[InsightRS_Installation_Folder]`* `\P4000`.

To verify and/or update your SNMP settings, complete the following steps:

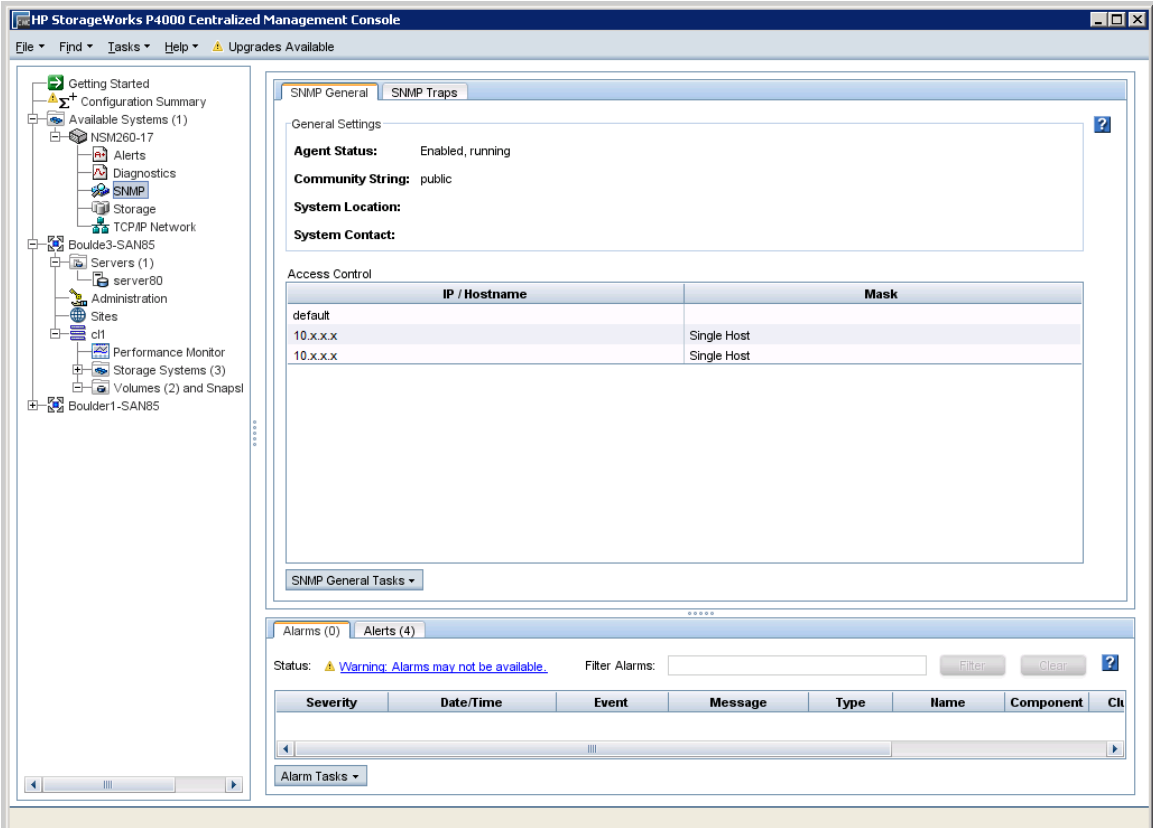
1. Open CMC.
2. Verify that SNMP is enabled for each storage system:

   Note: LeftHand OS ships by default with SNMP enabled for all storage systems and configured with the "Default" Access Control list.

   Note: LeftHand OS Management Groups configure the Hosting Device SNMP trap host destination once at the Management Group level. With LeftHand OS, you can configure different severity Alert levels to send from the Management Group. Under Insight Remote Support, configure each Management Group to send v1 SNMP traps of `Critical` and `Warning` levels in `Standard` message text length.

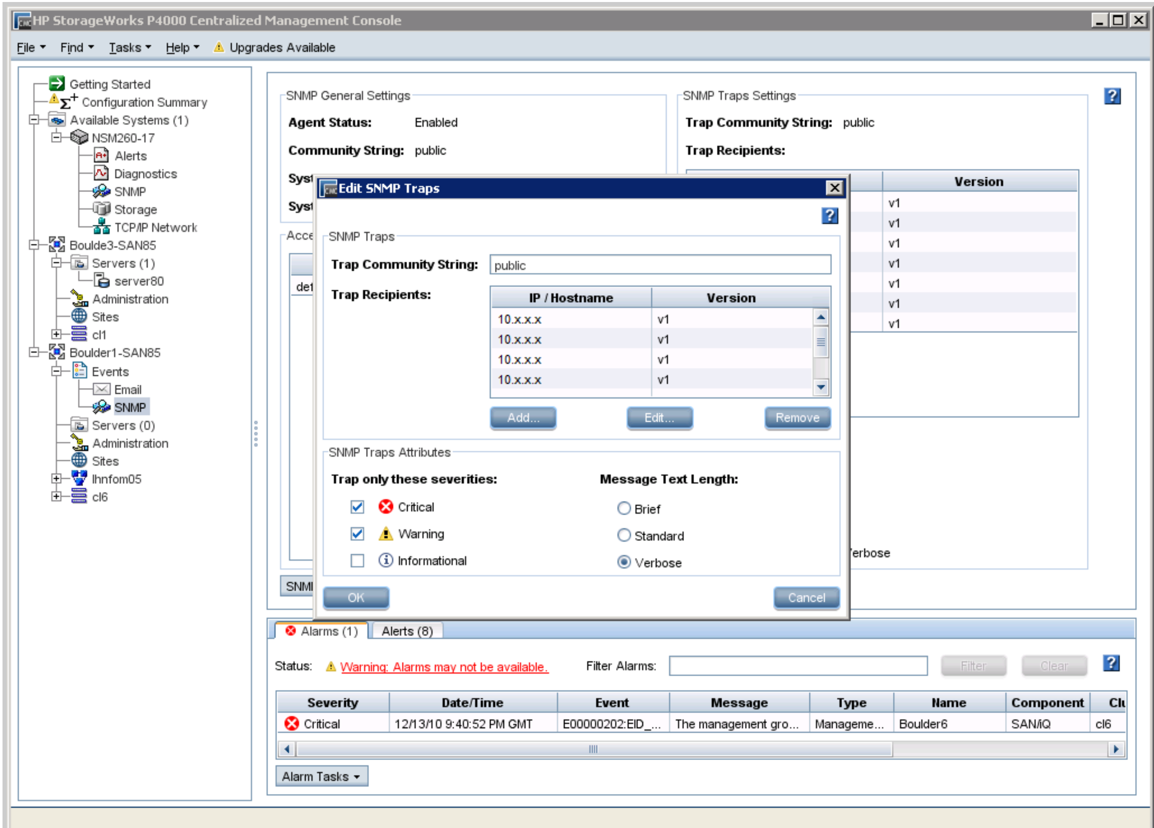
   a. Select **SNMP** in the left menu tree and open the **SNMP General** tab.
   b. Verify that the Agent Status is enabled.
   c. Verify that the P4000 Storage System SNMP Community String is set to `public` or the same value configured in the Insight RS Console for SNMP discovery.
   d. In the Access Control field, verify that either `Default` is listed or the Hosting Device host IP address is listed. The `Default` option configures SNMP to be accessed by the `public` community string for all IP addresses.

      Note: The SNMP settings on the P4000 Storage Systems need to match the SNMP settings on the Hosting Device.



3. Select **Alerts** in the left menu tree, and verify that alerts are configured with the "trap" option for each storage system.

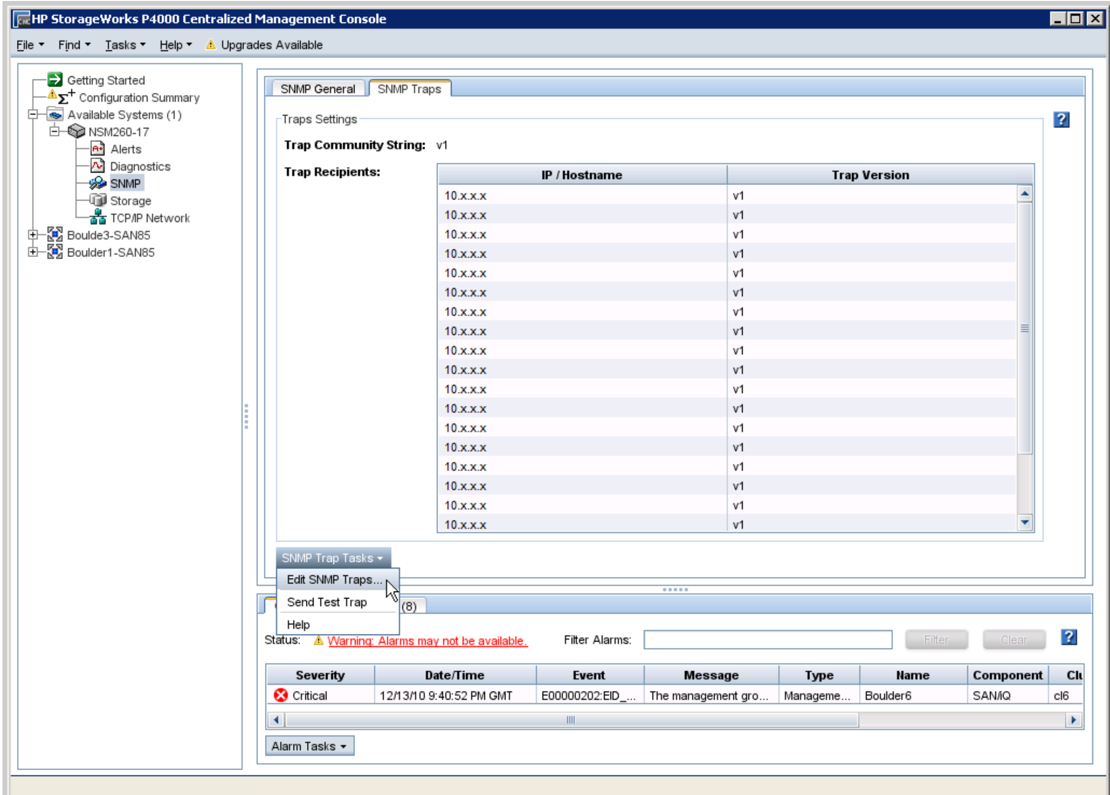
Note: LeftHand OS ships by default with traps set for all alert cases.



4. Add the Hosting Device IP address to the P4000 Storage System's SNMP trap send list. The Hosting Device IP address is needed to configure SNMP traps on each storage system. Note that in CMC 9.0 and higher, SNMP traps are configured at the Management Group level, not at the node level.

   a. Select **SNMP** in the left menu tree and open the **SNMP Traps** tab.

   b. Open the Edit SNMP Traps dialog by browsing to **SNMP Trap Tasks** → **Edit SNMP Traps**.



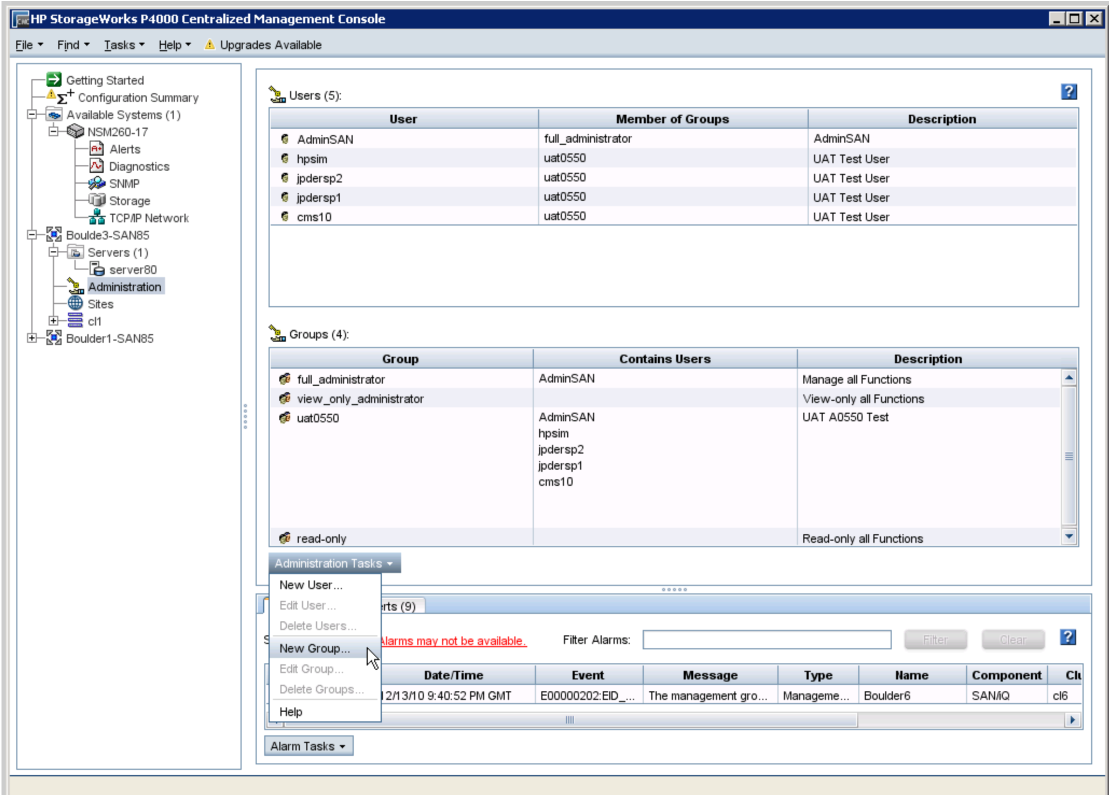
c. In the Edit SNMP Traps dialog, click **Add**.

d. In the Add IP or Hostname dialog, type the IP address or host name into the **IP or Hostname** field. Verify that the **Trap Version** is v1, and click **OK**.

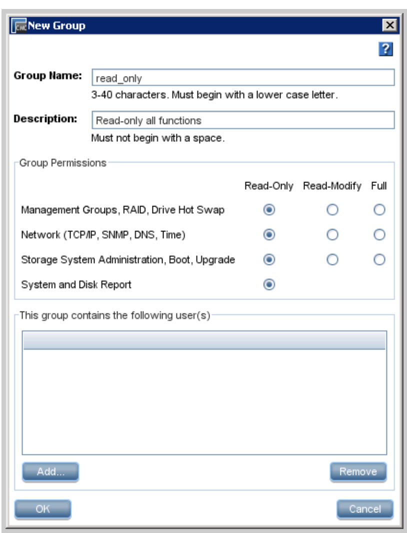
5. Repeat steps 3 to 4 for each P4000 Storage System. Alternatively, you could also configure one node using steps 3 to 4, then use the CMC copy node configuration option to copy the configuration to all other nodes.

6. Add an additional CMC user with read-only credentials. This is recommended if you don't want to have the Hosting Device system administrator to have create/delete control of the storage systems.

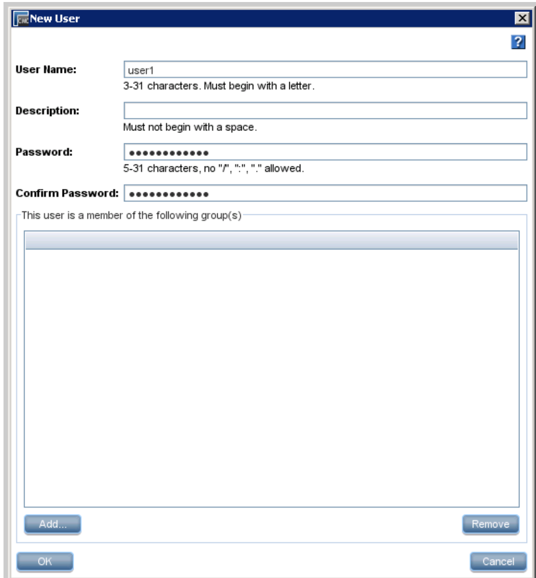
a. Select **Administration** in the left menu tree.



b. Create a user group with read-only access, if one does not already exist. Browse to **Administrative Tasks → New Group**.



c. Create a new user. First select the group you created in the previous step, then go to **Administrative Tasks → New User**. Type the User Name, Password, and click **Add** to add this user to the read-only user group you created.



Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Add P4000 SAN Solution (SAN/iQ) Protocol to the Insight RS Console

To add the P4000 SAN Solution (SAN/iQ) protocol, complete the following steps:

1. In a web browser, log on to the Insight RS Console (`https://`*<hosting_device_ip_or_fqdn>*`:7906`.)
2. Create a Named Credential for every set of Management Group credentials:
   a. In the Insight RS Console, navigate to **Company Information** → **Named Credentials**.
   b. Click **Add New Credential**.
   c. Type a Credential Name.
   d. In the **Protocol** drop-down list, select **P4000 SAN Solution (SAN/iQ)**.
   e. Type the Username and Password for the Management Group created in "Configuring SNMP on the P4000 Storage System" on page 195. These credentials can be either full LeftHand OS management group or read-only LeftHand OS management group credentials.
   f. Click **Save Info**.

3. Add Discovery credentials for the P4000 devices:
   a. In the Insight RS Console, navigate to **Discovery** → **Credentials**.
   b. Create discovery protocol credentials for each Named Credential created earlier.
      i. In the **Select and Configure Protocol** drop-down list, select **P4000 SAN Solution (SAN/iQ)**.
      ii. Click **New**.
      iii. In the **Named Credential** drop-down list, select the Named Credential created earlier.
      iv. Click **Add**.

Discover the P4000 in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP address(es) of the devices to be discovered.

   Note: When discovering new P4000 devices, do not include any P4000 Management Group Virtual IP (VIP) addresses. VIP addresses are created when you create and configure P4000 clusters. Instead, discover P4000 devices using individual IP addresses, or create IP discovery ranges excluding all P4000 VIP addresses.

   If you do discover a P4000 VIP address, you should delete the managed entity for the VIP address from Insight Remote Support Advanced before running any collections or capturing any test traps. After deleting the VIP address entity, re-discover the node using the actual IP address for the P4000 device.

   d. Click **Add**.
4. Click **Start Discovery**.

Note: Insight RS filters out iLO 4 but not iLO 2 or iLO 3 devices. If an iLO 2 or iLO 3 device is discovered, manually deleted them on the **Device Details** screen.

Verify Discovery

To verify the P4000 was discovered correctly, complete the following steps:

1. If synchronizing Insight RS with HP SIM, verify that the Device Name appears correctly on the **Devices** screen. If the devices appears as a Serial Number rather than as an IP address or host name, perform the following tasks:

   a. Click the Device Name, and on the **Device** tab, expand the Status section. Click **Delete Device**.
   b. On the **Credentials** tab, verify that the credentials assigned to the device are correct.
   c. On the **Administrator Settings → Integration Adapters** tab, expand the HP SIM Adapter section and select **Perform Manual Device Synchronization**.
   d. Click the **Single device from HP SIM** option and type the P4000 device's IP address or host name. Click **Save Adapter Settings**.
2. Verify that the appropriate protocols have been assigned to the P4000 device:
   a. In the Insight RS Console, navigate to **Devices** and click the P4000 Device Name.
   b. On the Credentials tab, verify that verify that the **P4000 SAN Solution (SAN/iQ)** and **SNMPv1** protocols have been assigned to the device.

Add Warranty and Contract Information

To add the P4000 device's warranty and contract information to Insight RS, complete the following steps:

1. In the Insight RS Console, navigate to **Devices** and click the P4000 Device Name.
2. Expand the Hardware section, and type the Serial Number in the **Override Serial Number** field and the Product Number in the **Override Product Number** field.
3. Click **Save Changes**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon (✔).

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

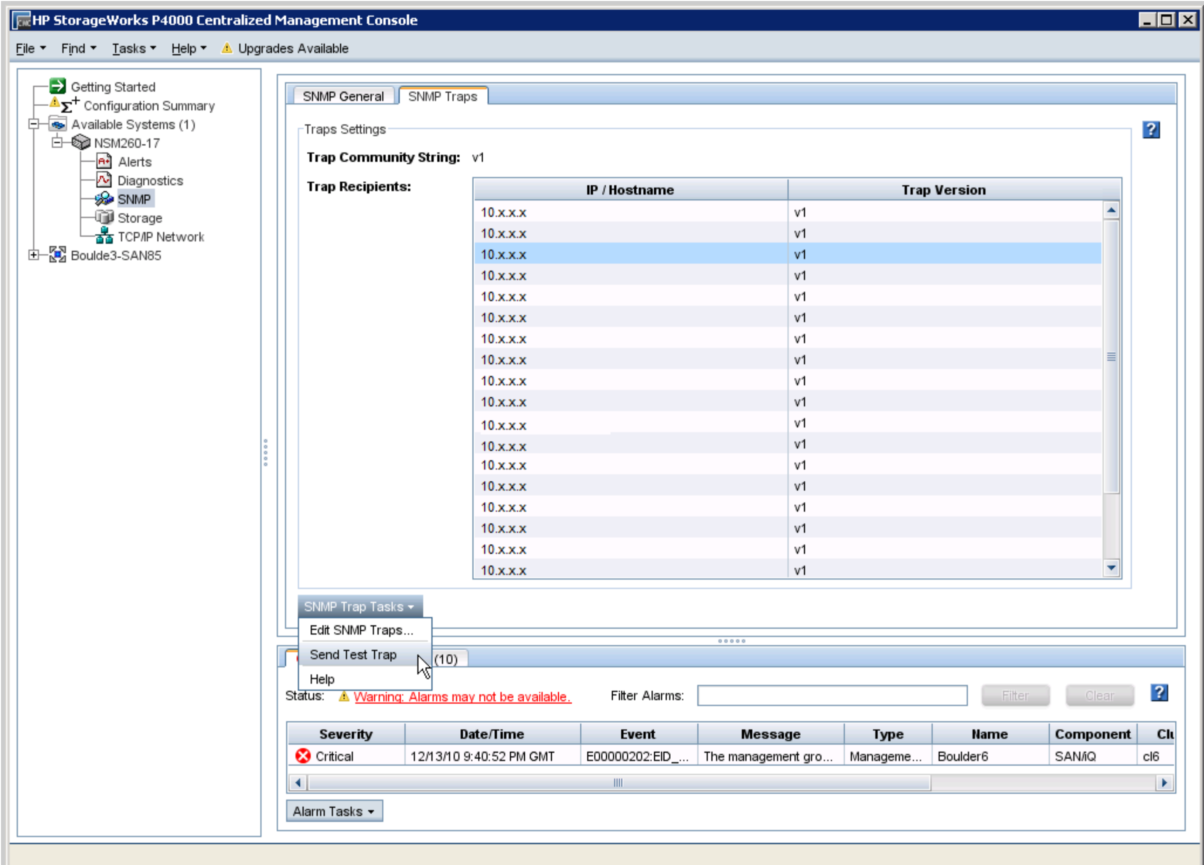
Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following sections:

Send a Test Event

To verify that the P4000 Storage System(s) are communicating with the Hosting Device, open CMC and select at least one storage node and send a test trap to the Hosting Device. Select **SNMP** in the left menu tree and open the **SNMP Traps** tab. In the **SNMP Trap Tasks** drop-down list, select **Send Test Trap**.

Note: In CMC, SNMP traps are configured at the Management Group level, not at the node level.



Go to Hosting Device configurations to verify that the test event was posted to the Hosting Device logs. The CMC read-only credentials were verified during device discovery when each LeftHand OS device was added to Hosting Device monitored listed.

Verify Collections in the Insight RS Console

After discovery, a StoreVirtual (P4000) Family Configuration Collection is automatically run for P4000 devices. Verify that this collection ran successfully on the Collection Services → Basic Collection Results tab in the Insight RS Console. Note that there is a delay between when discovery finishes and when the collection runs. For more information about scheduling collections, see the Insight RS Help.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services** and then click the **Basic Collection Results** tab.
3. Expand the P4000 Family Configuration Collection section.
4. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears (). If it failed, an error icon appears ().

Chapter 21: Configuring P2000 G3, MSA 1040, and MSA 2040 Modular Smart Arrays

Fulfill Configuration Requirements

To configure your P2000 G3, MSA 1040, and MSA 2040 Modular Smart Arrays to be monitored by Insight RS, complete the following sections:

Table 21.1 ***P2000, MSA 1040, and MSA 2040 and Modular Smart Array Configuration Steps***

Task	Complete?
Make sure Insight RS supports your Modular Smart Array by checking the *HP Insight Remote Support Release Notes*.	
Configure SNMP on the MSA.	
Configure WBEM on the MSA.	
Add the SNMP protocol to the Insight RS Console.	
Add the WBEM protocol to the Insight RS Console.	
Discover the Modular Smart Array server in the Insight RS Console.	

Install and Configure Communication Software on Arrays

To configure your monitored devices, complete the following sections:

Configure SNMP on Your Modular Smart Arrays

Each MSA monitored device has factory-installed SNMP on the device.

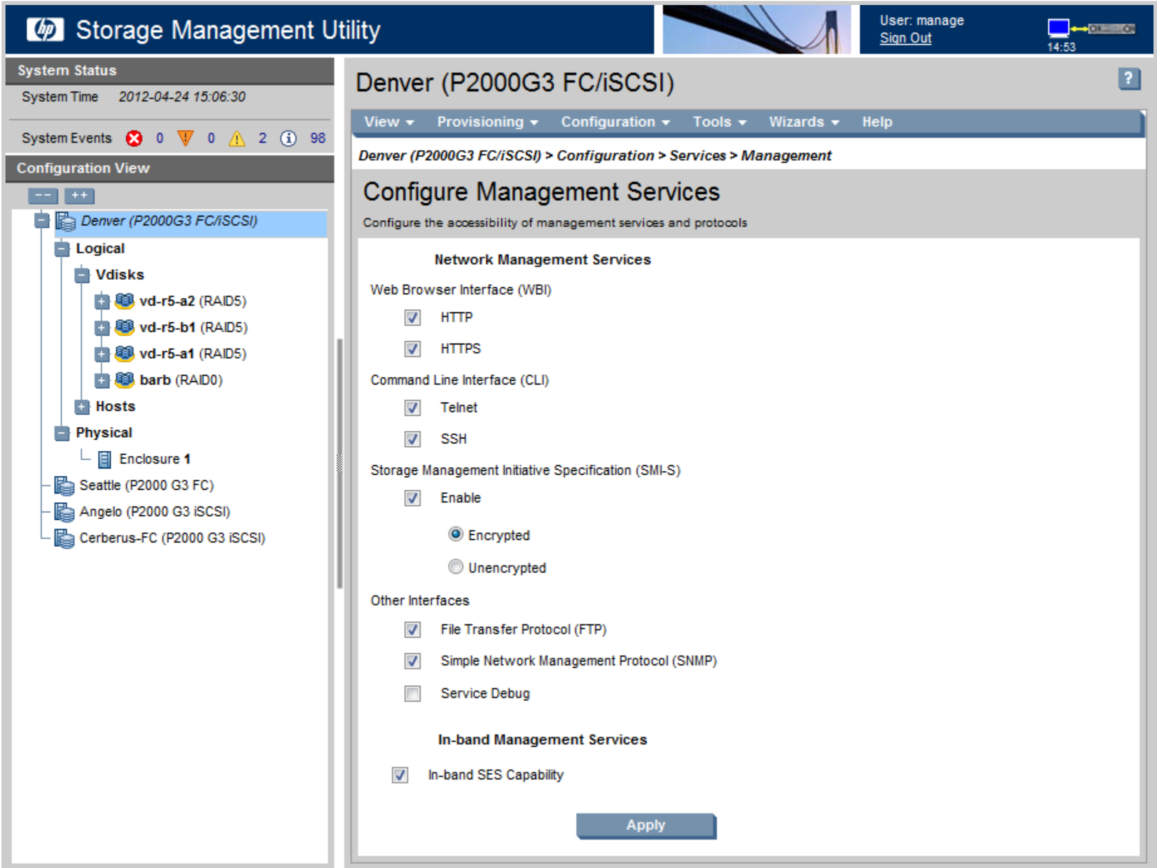
To access the Storage Management Utility for the MSA and configure SNMP for Insight Remote Support, complete the following steps:

1. Log on to the HP MSA Storage Management Utility (SMU).

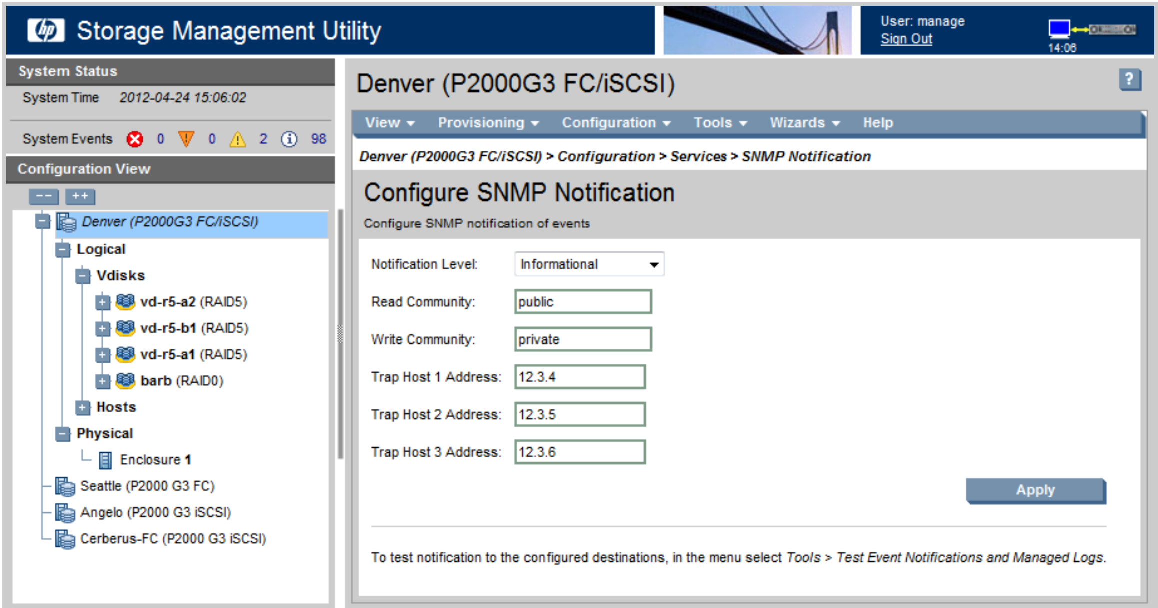
   Note: For basic information about setting up this tool and managing your credentials, consult the MSA Storage Management Utility documentation at: http://www.hp.com/support/manuals.

2. Once you have logged into the utility, select **Configuration** → **Services** → **Management**.
3. Make sure that SNMP is selected from the **Network Management Services** list and click **Apply**.

   Important: SNMP should be active by default. If for any reason it is not active, Insight Remote Support will not be supported. It is critical to verify SNMP is active.



4. From the menu select **Configuration** → **Services** → **SNMP Notification**.
5. On the **Configure SNMP Notification** page, modify the SNMP settings to include the Hosting Device as an SNMP Trap Host and select your **Notification Level** from the drop-down list (the Informational level is recommended).



6. Click **Apply** to commit your changes.
7. Exit the utility and discover your MSA in the Insight RS Console.

About WBEM on the P2000 G3, MSA 1040, and MSA 2040 Modular Smart Arrays

The SMI-S provider for the P2000 G3 MSA, MSA 1040, and MSA 2040 is integrated in the firmware and delivered with the P2000 G3, MSA 1040, and MSA 2040 series disk arrays. It provides a WBEM-based management framework. Insight RS can interact with this embedded provider directly and does not need an intermediate proxy provider.

No configuration of WBEM on the P2000 G3, MSA 1040, and MSA 2040 is necessary.

Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create an SNMPv1 Protocol Credential in Insight RS Console

If your device's SNMP community string is set to `public` and your community access mode is `read` `only`, Insight RS automatically associates an SNMPv1 protocol with your device. If you use a different community string or use a nonstandard port, you must create an SNMPv1 protocol credential in the Insight RS Console.

Important: MSAs with SNMPv3 enabled require SNMPv3 credentials be configured in the Insight RS Console. Failure to supply SNMPv3 credentials in the Insight RS Console results in Insight RS displaying only one of the controllers of those MSAs.

To configure SNMPv1 in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the Select and Configure Protocol drop-down list, select **Simple Network Management Protocol Version 1 (SNMPv1)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Community String you have configured on your device.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Create a WBEM Protocol Credential in Insight RS Console

To configure WBEM in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the **Select and Configure Protocol** drop-down list, select **Web-Based Enterprise Management (WBEM)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Username and Password used to access the Storage Management Utility interface on the MSA.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:

a. Click **New**.
b. Select the **Single Address**, **Address Range**, or **Address List** option.
c. Type the IP address(es) of the MSA devices to be discovered.

> **Important:** In a dual-controller system, you must discover both controller A and controller B if both exist. If you do not discover both controllers, events that originate from the controller that was not discovered could be lost.

d. Click **Add**.

4. Click **Start Discovery**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon ( ).

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following sections:

Verify Service Event Monitoring

To send a test event from the MSA, complete the following steps:

1. In a web browser, log on to the HP MSA Storage Management Utility (SMU).
2. Select **Tools** → **Send Test Notification**.
3. In the Send Test Notifications window, click **Send**.
4. Click **OK** to acknowledge that the message was sent.
5. Sign out of SMU.

Verify Collections in the Insight RS Console

After discovery, a Storage Configuration Collection is automatically run for servers. Verify that this collection ran successfully on the Collection Services → SAN Storage Collection Results tab in the

Insight RS Console. Note that there is a delay between when discovery finishes and when the collection runs. For more information about scheduling collections, see the Insight RS Help.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services** and then click the **SAN Storage Collection Results** tab.
3. Expand the Storage Configuration Collection section.
4. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears (). If it failed, an error icon appears ().

Chapter 22: Configuring MSA23xx G2 Modular Smart Arrays

Fulfill Configuration Requirements

To configure your MSA23xx G2 Modular Smart Arrays to be monitored by Insight RS, complete the following sections:

Table 22.1 ***P2000 and Modular Smart Array Configuration Steps***

Task	Complete?
Make sure Insight RS supports your Modular Smart Array by checking the *HP Insight Remote Support Release Notes*.	
Configure SNMP on the MSA.	
Configure WBEM on the MSA.	
Add the SNMP protocol to the Insight RS Console.	
Add the WBEM protocol to the Insight RS Console.	
Discover the Modular Smart Array server in the Insight RS Console.	

Install and Configure Communication Software on Arrays

To configure your monitored devices, complete the following sections:

Configure SNMP on Your MSA23xx G2 Modular Smart Arrays

Each MSA monitored device has factory-installed SNMP on the device.

To access the Storage Management Utility for the MSA23xx G2 and configure SNMP for Insight Remote Support, complete the following steps:

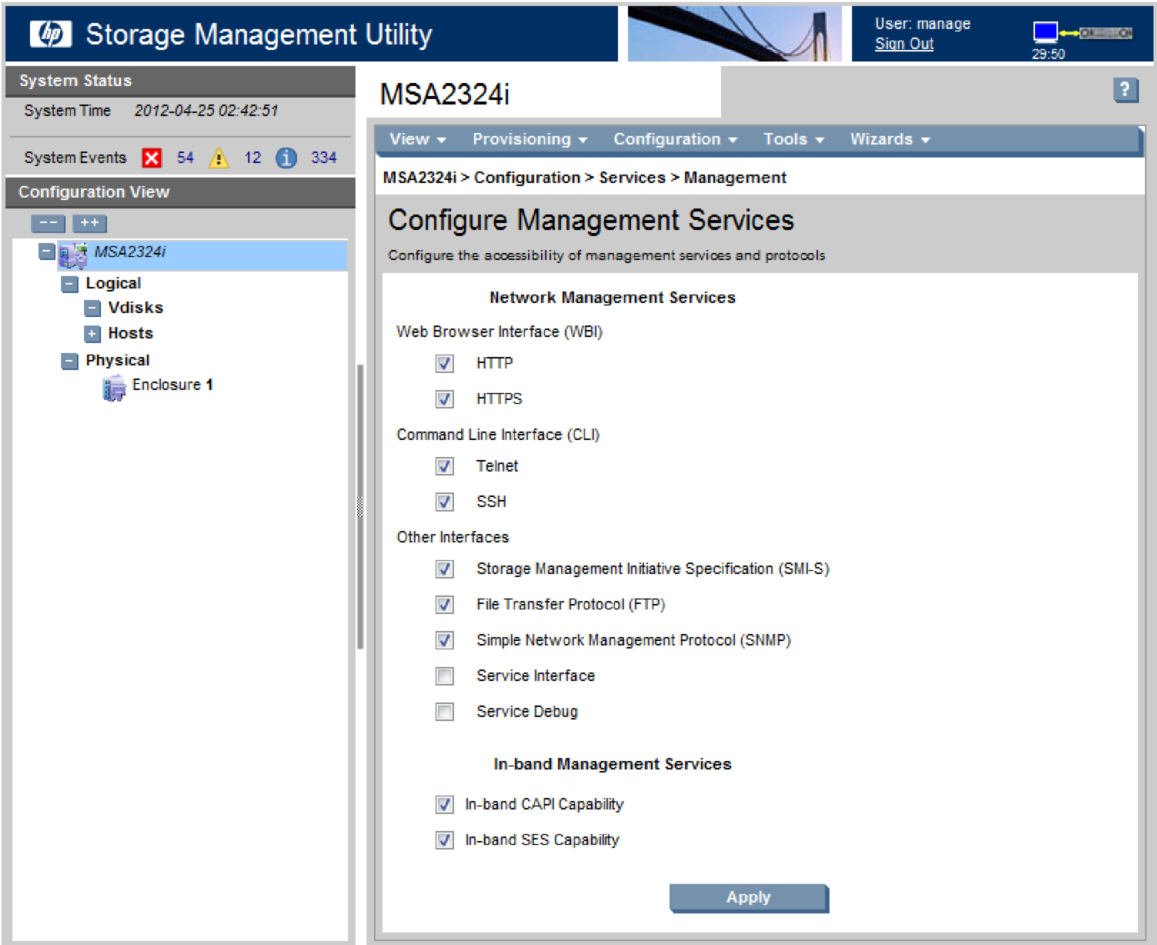
1. Log on to the HP MSA Storage Management Utility.

   Note: For basic information about setting up this tool and managing your credentials, consult the MSA Storage Management Utility documentation at: http://www.hp.com/support/manuals.

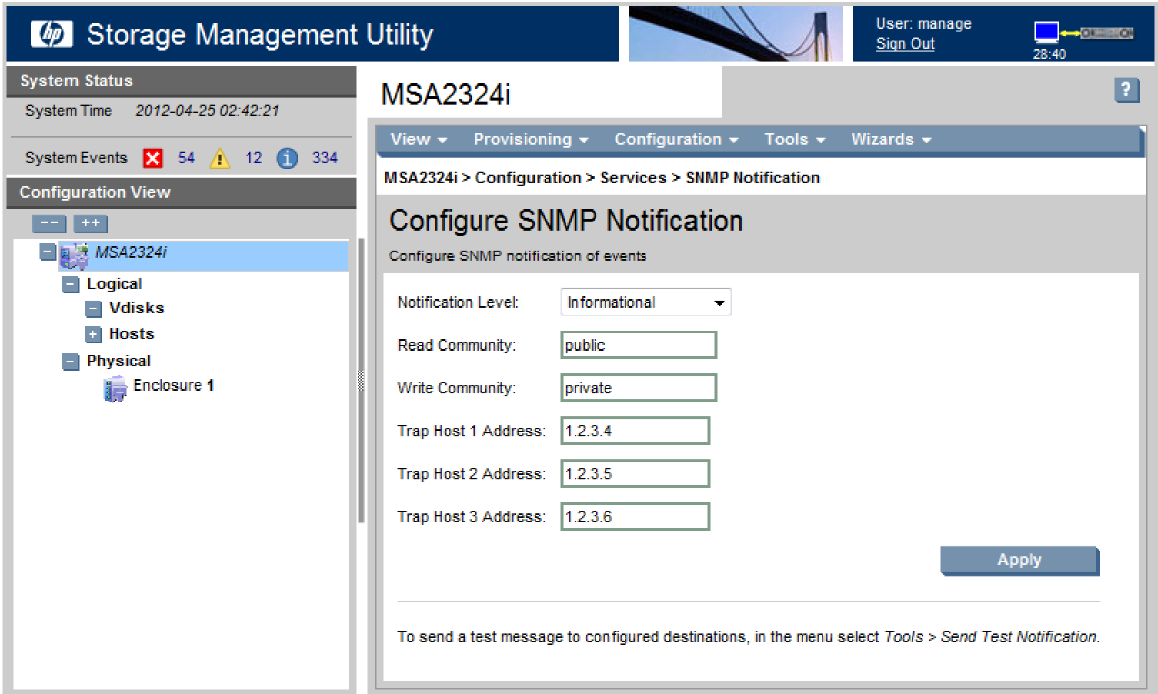
2. Once you have logged into the utility, select **Configuration** → **Services** → **Management**.
3. Make sure that SNMP is selected from the **Network Management Services** list, and click **Apply**.

   Important: SNMP should be active by default, but if it for any reason it is not, Insight

Remote Support will not be supported. It is critical to verify SNMP is active.



4. From the menu select **Configuration** → **Services** → **SNMP Notification**.
5. On the **Configure SNMP Notification** page, modify the SNMP settings to include the Hosting Device as an SNMP Trap Host and select your **Notification Level** from the drop-down list (the Informational level is recommended).



6. Click **Apply** to commit your changes.
7. Exit the utility and discover your 23xx in the Insight RS Console.

Install WBEM SMI-S Proxy Provider Software

The MSA2000 G2 SMI-S Proxy Provider software must be installed on a ProLiant host server that will manage your MSA23xx devices. It provides a WBEM-based management framework. Insight RS can interact with this proxy provider to gather collections on your MSA23xx devices.

To download and install the MSA2000 G2 SMI-S Proxy Provider software, complete the following steps:

1. In a web browser, navigate to the following website: http://www.hp.com/go/hpsc.
2. In the *Find support for your HP product* section, type your product family in the search box, for example, MSA 2000 G2. Click **Go**.
3. Select your product from the search results.
4. In the *Drivers, software & firmware for this product* section, click the **View and download all drivers, software and firmware for this product** link.
5. Select your product, then select the operating system.
6. In the *Downloads* table, click the software link for your operating system.

For more information about installing and configuring the SMI-S Proxy Provider, see the *HP MSA2000 G2 SMI-S Proxy Provider User Guide*.

Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create an SNMPv1 Protocol Credential in Insight RS Console

If your device's SNMP community string is set to `public` and your community access mode is `read only`, Insight RS automatically associates an SNMPv1 protocol with your device. If you use a different community string or use a nonstandard port, you must create an SNMPv1 protocol credential in the Insight RS Console.

To configure SNMPv1 in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the Select and Configure Protocol drop-down list, select **Simple Network Management Protocol Version 1 (SNMPv1)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Community String you have configured on your device.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Create a WBEM Protocol Credential in Insight RS

To configure WBEM in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the **Select and Configure Protocol** drop-down list, select **Web-Based Enterprise Management (WBEM)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Username and Password you have configured for your MSA23xx in the SMI-S Proxy Provider software.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address to be discovered.

   When discovering the MSA23xx G2, add the IP address of each MSA23xx G2 as well as the ProLiant server running the SMI-S Proxy Provider software. This ensures that the necessary event and collection services are configured.

   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP address(es) of the ProLiant server running the SMI-S Proxy Provider software and MSA23xx G2 devices to be discovered.
   d. Click **Add**.
4. Click **Start Discovery**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon ( ).

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following section:

Verify Collections in the Insight RS Console

After discovery, a Storage Configuration Collection is automatically run for servers. Verify that this collection ran successfully on the Collection Services → SAN Storage Collection Results tab in the Insight RS Console. Note that there is a delay between when discovery finishes and when the collection runs. For more information about scheduling collections, see the Insight RS Help.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services** and then click the **SAN Storage Collection Results** tab.
3. Expand the Storage Configuration Collection section.
4. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears (). If it failed, an error icon appears ().

Chapter 23: Configuring MSA2xxx G1 Modular Smart Arrays

Important: Configuration collections are not supported, except when the MSA2xxx G1 is part of a SAN collection.

Fulfill Configuration Requirements

To configure your MSA2xxx G1 Modular Smart Arrays to be monitored by Insight RS, complete the following sections:

Table 23.1 ***P2000 and Modular Smart Array Configuration Steps***

Task	Complete?
Make sure Insight RS supports your Modular Smart Array by checking the *HP Insight Remote Support Release Notes*.	
Configure SNMP on the MSA.	
Add the SNMP protocol to the Insight RS Console.	
Discover the Modular Smart Array server in the Insight RS Console.	

Install and Configure Communication Software on Arrays

To configure your monitored devices, complete the following sections:

Configure SNMP on Your MSA2xxx G1 Modular Smart Arrays

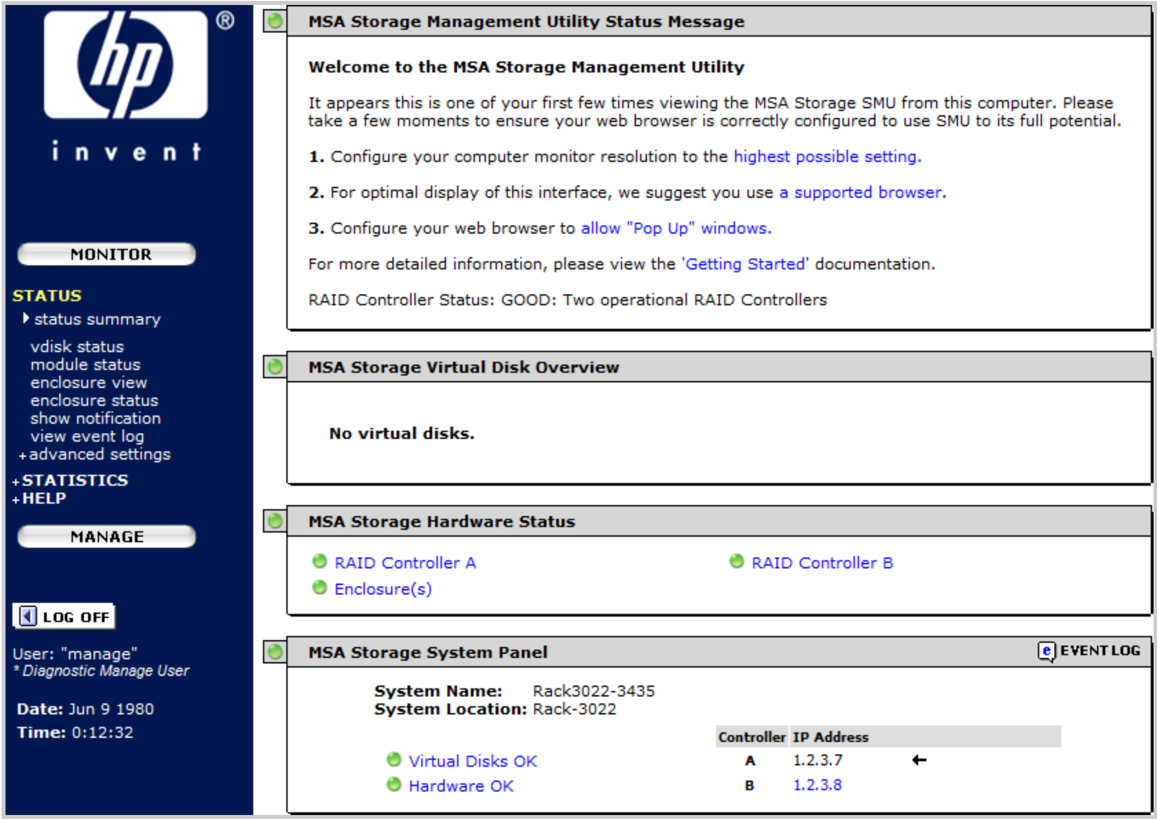
Each MSA monitored device has factory-installed SNMP on the device.

To access the Storage Management Utility for the MSA 2012 or 2112 and configure SNMP for Insight Remote Support, complete the following steps:

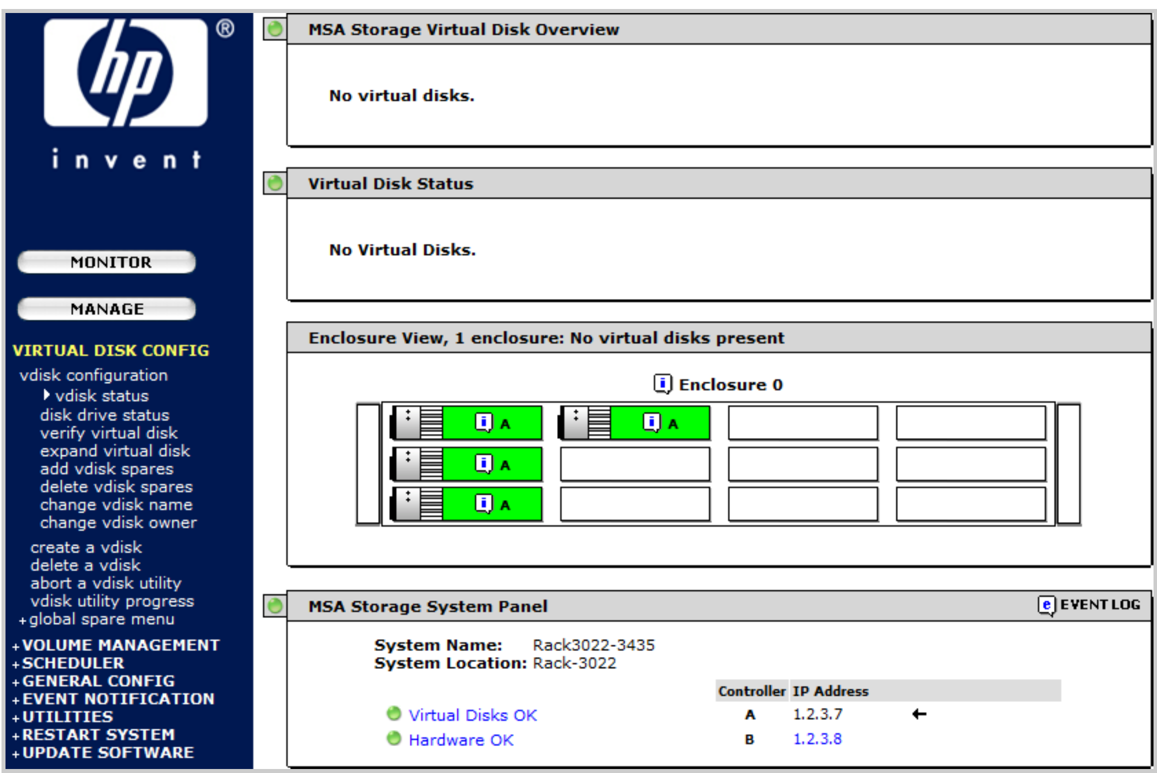
1. Log on to the HP MSA Storage Management Utility.

   Note: For basic information about setting up this tool and managing your credentials, consult the MSA Storage Management Utility documentation at: http://www.hp.com/support/manuals.

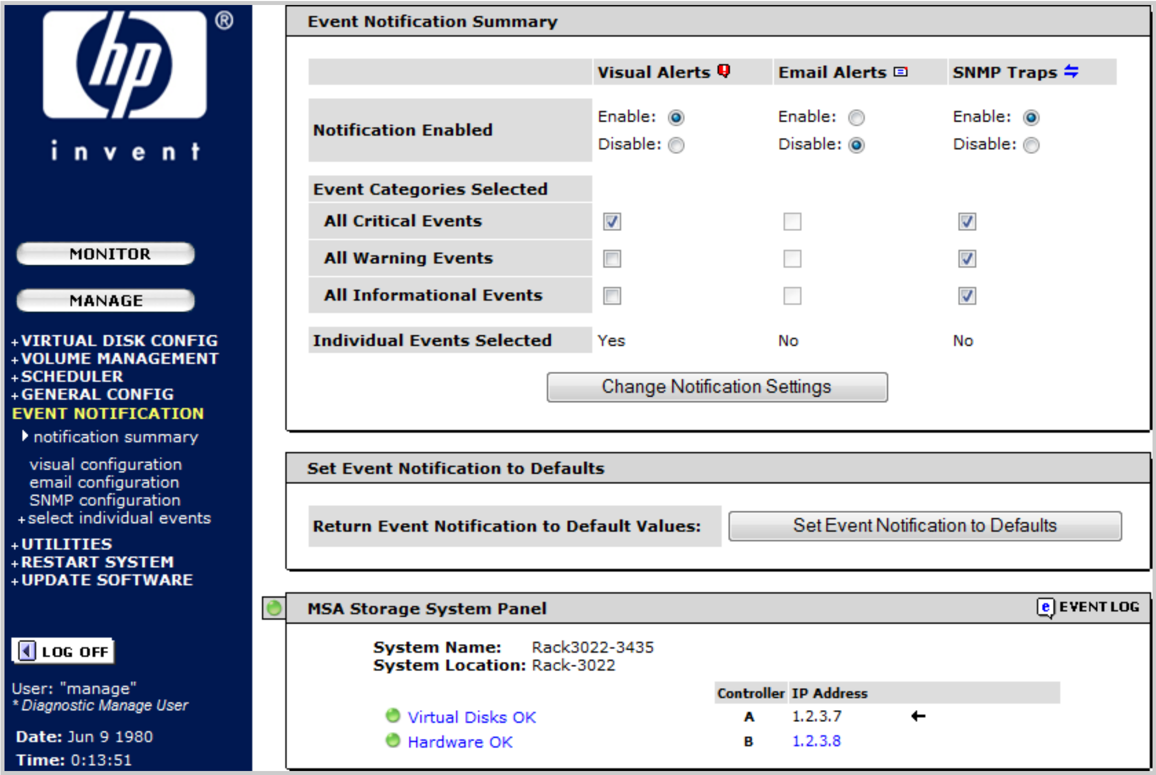
2. Select **Manage**.



3. In the expanded list select the **Event Notification** link.

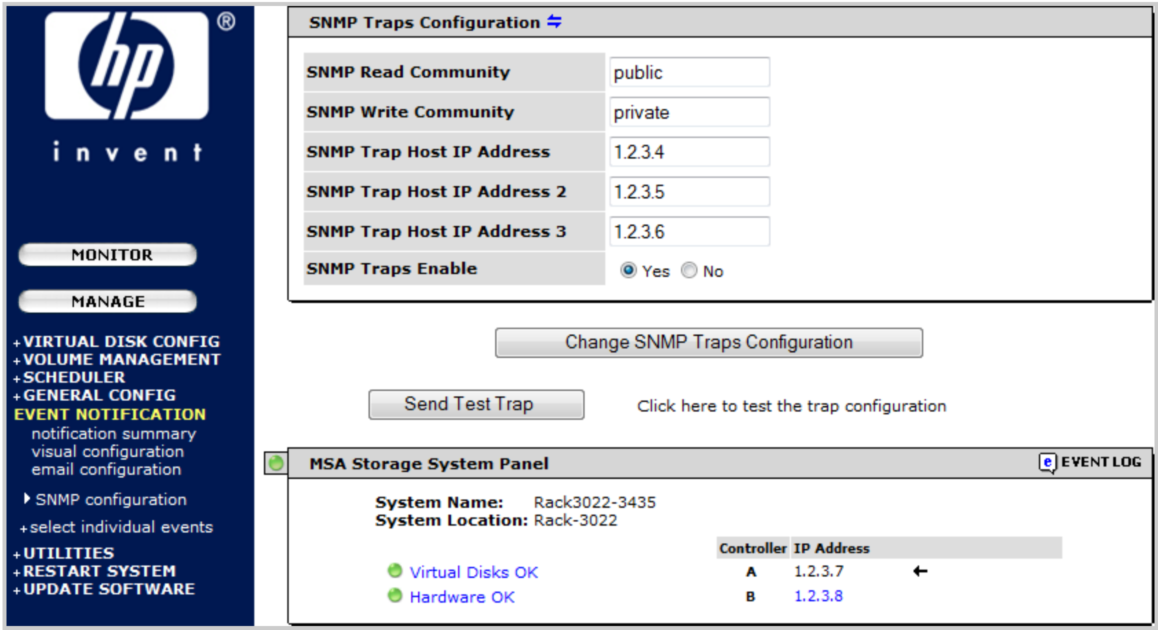


4. On the **Event Notification Summary** page, verify that the SNMP notification types (Visual Alerts, Email Alerts, and SNMP Traps) are enabled, and click **Change Notification Settings** to commit the changes.



5. Click the **SNMP configuration** link under the Event Notification heading in the left panel.

6. Modify the SNMP settings to include the Hosting Device as an SNMP Trap Host and make sure the **SNMP Traps Enabled** option is set to **Yes**.



7. Click **Change SNMP Traps Configuration** to commit your changes.

Note: If your Insight Remote Support Hosting Device is fully configured, you can send the test event from this page. If the Hosting Device is not yet configured, you can return to this page later to send the Test Event.

8. Exit the utility and discover your 2012/2112 in the Insight RS Console.

Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create an SNMPv1 Protocol Credential in Insight RS Console

If your device's SNMP community string is set to `public` and your community access mode is `read only`, Insight RS automatically associates an SNMPv1 protocol with your device. If you use a different community string or use a nonstandard port, you must create an SNMPv1 protocol credential in the Insight RS Console.

To configure SNMPv1 in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the Select and Configure Protocol drop-down list, select **Simple Network Management Protocol Version 1 (SNMPv1)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Community String you have configured on your device.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP address(es) of the devices to be discovered.
   d. Click **Add**.
4. Click **Start Discovery**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon (✔).

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following section:

Verify Collections in the Insight RS Console

Configuration collections are not supported for this device type except when it is part of a SAN collection. If you have added this device to a SAN collection, you can manually run a SAN collection to verify the configuration.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services**.
3. Click the **Collection Schedules** tab.
4. In the List of Collection Schedules pane, select the **SAN Configuration Collection Schedule**. Information about the collection appears in the Collection Information pane. The On These Devices pane lists the devices the collection will run on.
5. In the Schedule Information pane, click **Run Now**.
6. When the collection completes, click the **SAN Storage Collection Results** tab.
7. Expand the SAN Configuration Collection section.
8. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears ( ). If it failed, an error icon appears ( ).

Chapter 24: Configuring MSA15xx Modular Smart Arrays

Fulfill Configuration Requirements

To configure your MSA15xx Modular Smart Arrays to be monitored by Insight RS, complete the following sections:

Table 24.1 ***P2000 and Modular Smart Array Configuration Steps***

Task	Complete?
Make sure Insight RS supports your Modular Smart Array by checking the *HP Insight Remote Support Release Notes*.	
Configure SNMP on the MSA.	
Add the SNMP protocol to the Insight RS Console.	
Discover the Modular Smart Array server in the Insight RS Console.	

Install and Configure Communication Software on Arrays

To configure your monitored devices, complete the following sections:

Configure SNMP on Your MSA15xx Modular Smart Arrays

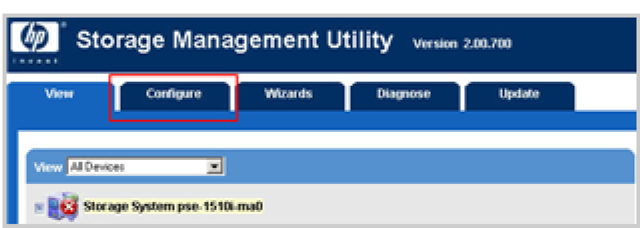
Each MSA monitored device has factory-installed SNMP on the device.

To access the Storage Management Utility for the MSA 1510i and configure SNMP for Insight Remote Support, complete the following steps:

1. Log on to the HP MSA Storage Management Utility.

   Note: For basic information about setting up this tool and managing your credentials, consult the MSA Storage Management Utility documentation at: http://www.hp.com/support/manuals.

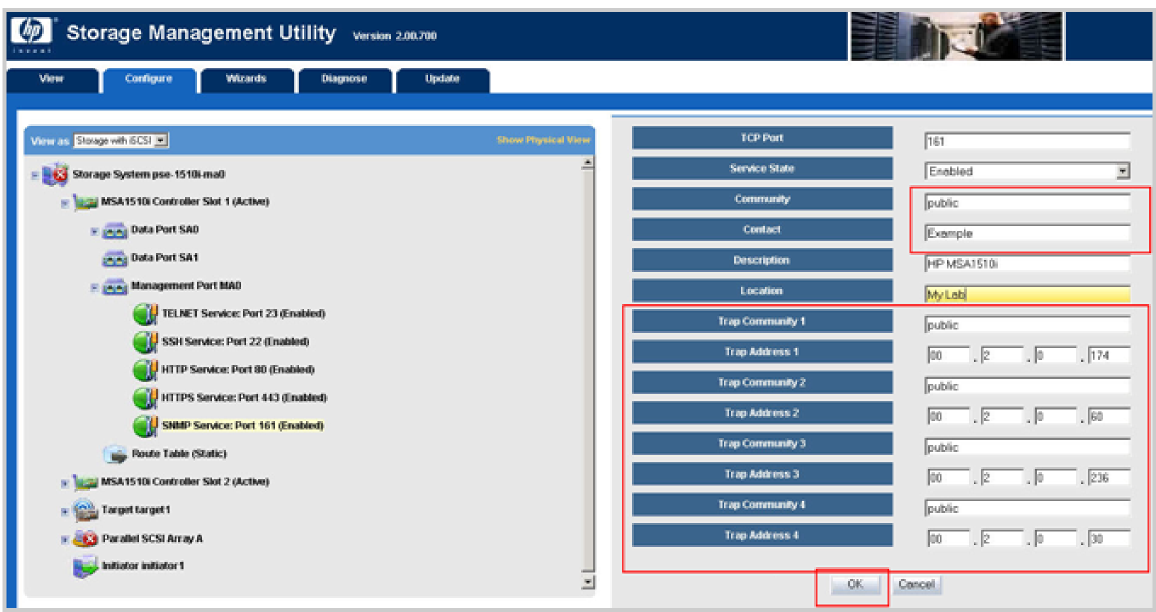
2. Once you have logged on to the utility, select the **Configure** tab.

3. On the **Configure** tab, expand the **Management Port** list for the MSA 1510i.

4. Select the **SNMP Service** link from the list.

5. Modify any necessary configuration fields and make sure that the IP address of the Hosting Device is listed as a **Trap Address**.
6. Click **OK** to commit your changes.



7. Exit the utility and discover your 1510i in the Insight RS Console.

Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create an SNMPv1 Protocol Credential in Insight RS Console

If your device's SNMP community string is set to `public` and your community access mode is `read only`, Insight RS automatically associates an SNMPv1 protocol with your device. If you use a different community string or use a nonstandard port, you must create an SNMPv1 protocol credential in the Insight RS Console.

To configure SNMPv1 in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the Select and Configure Protocol drop-down list, select **Simple Network Management Protocol Version 1 (SNMPv1)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Community String you have configured on your device.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP address(es) of the devices to be discovered.
   d. Click **Add**.
4. Click **Start Discovery**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon ().

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following section:

Verify Collections in the Insight RS Console

After discovery, a Storage Configuration Collection is automatically run for servers. Verify that this collection ran successfully on the Collection Services → SAN Storage Collection Results tab in the Insight RS Console. Note that there is a delay between when discovery finishes and when the collection runs. For more information about scheduling collections, see the Insight RS Help.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services** and then click the **SAN Storage Collection Results** tab.
3. Expand the Storage Configuration Collection section.
4. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears (). If it failed, an error icon appears ().

Chapter 25: Configuring StoreEasy Storage Systems

Insight Remote Support (RS) requires WMI to be installed and configured on your StoreEasy Storage system (formerly Commercial NAS) in order for Insight RS to communicate with your device for discovery, event monitoring, and collections.

WMI comes pre-installed on StoreEasy Storage systems. No additional configuration of WMI is necessary. However, you must add the WMI protocol credentials in the Insight RS Console for Insight RS to communicate with your device.

Fulfill Configuration Requirements

To configure your StoreEasy Storage system to be monitored by Insight RS, complete the following sections:

Table 25.1 ***StoreEasy Storage system Using WMI Configuration Steps***

Task	Complete?
Check the *HP Insight Remote Support Release Notes* to make sure your StoreEasy Storage system is supported.	
Add the WMI protocol to the Insight RS Console.	
Discover the StoreEasy Storage system in the Insight RS Console.	
Send a test event to verify connectivity between your StoreEasy Storage system and Insight RS.	

Configure StoreEasy Devices

To configure your monitored devices, complete the following section:

Configure Firewall

StoreEasy devices ship with Windows Firewall with Advanced Security enabled. To allow WMI through the Windows Firewall, you must enable additional inbound and outbound rules.

To configure Windows Firewall on the StoreEasy device, complete the following steps:

1. Open the Windows Firewall from the Server Manager.
2. Click on the **Tools** menu.
3. Select **Windows Firewall with Advanced Security**.
4. Select the **Inbound Rules** link from the left menu and make sure the following inbound rules are enabled:
   - File and Printer Sharing (Echo Request - ICMPv4-In)
   - Windows Management Instrumentation (DCOM-In)
   - Windows Management Instrumentation (WMI-In)

5. Select the **Outbound Rules** link from the left menu and make sure the following outbound rules are enabled:
   - File and Printer Sharing (Echo Request - ICMPv4-Out)
   - Windows Management Instrumentation (WMI-Out)

More information regarding security in Insight RS can be found in the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Note: You can confirm which protocols discovery associated with the StoreEasy device by clicking the StoreEasy device name on the **Devices** screen, and then selecting the **Credentials** tab. Confirm that WMI is one of the discovered credentials. SNMP should not be among the discovered credentials.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create a WMI Protocol Credential in Insight RS Console

WMI credentials must be set for each StoreEasy Storage system within the Insight RS Console before it can be monitored. If you change your WMI credentials at any time, you *must* modify the entry for those credentials in the Insight RS Console.

To configure WMI in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the **Select and Configure Protocol** drop-down list, select **Windows Management Instrumentation (WMI)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Username and Password you have configured on your device.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:

a. Click **New**.
b. Select the **Single Address**, **Address Range**, or **Address List** option.
c. Type the IP address(es) of the devices to be discovered.
d. Click **Add**.

4. Click **Start Discovery**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon ( ).

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following sections:

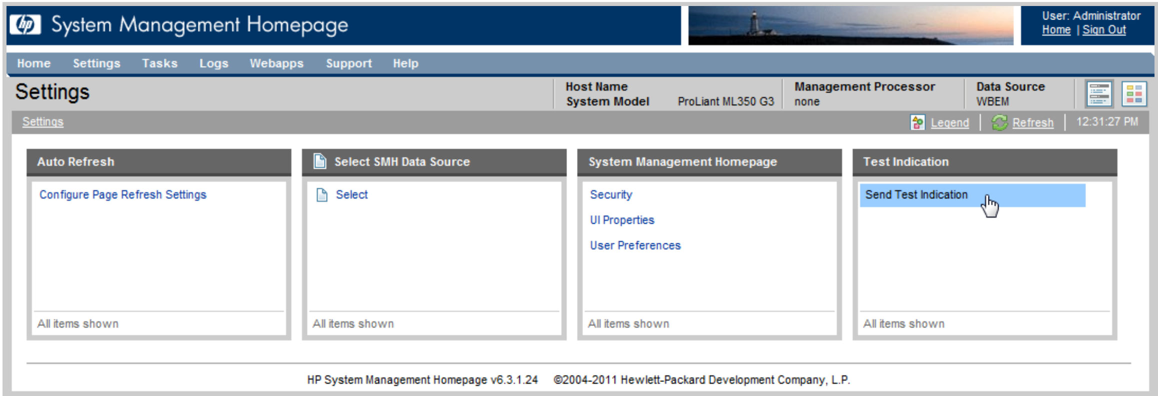
Send a WMI Test Indication to the Hosting Device

System Management Homepage (SMH) also comes pre-installed on your StoreEasy Storage system. It provides additional reporting capabilities on the device itself. While not mandatory for Insight Remote Support, SMH can be used to work with the WMI protocol such as sending test events.

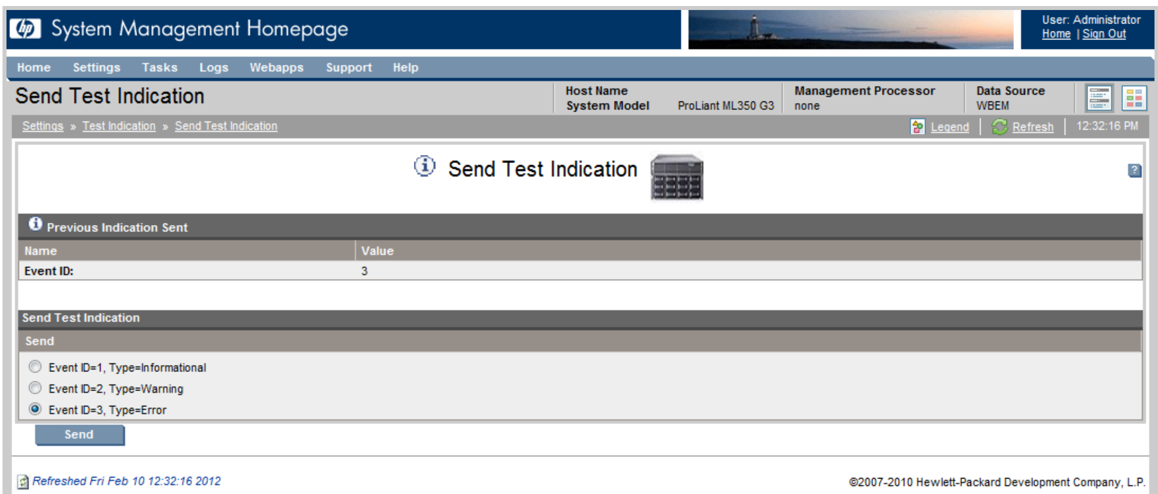
To verify connectivity from the device to Insight RS, send a WMI test indication to the Hosting Device and then verify the test indication was received in the Insight RS Console.

1. In a web browser, access System Management Homepage (SMH) on the monitored device: `https://[ipaddress]:2381`.
2. Log on using the administrator user name and password for the monitored device.

   If you are not prompted for a logon, check the upper right corner of the SMH interface and click the **Sign In** link. If you are not logged in as an administrator for the monitored device you will not have all of the relevant configuration options.
3. In the top menu bar, click **Settings**.
4. If you chose to install WMI with SPP, it will be set as your Data Source. In the Test Indication pane, click **Send Test Indication**.



5. In the **Send Test Indication** screen, select an Event ID type (any will work) and click **Send**.



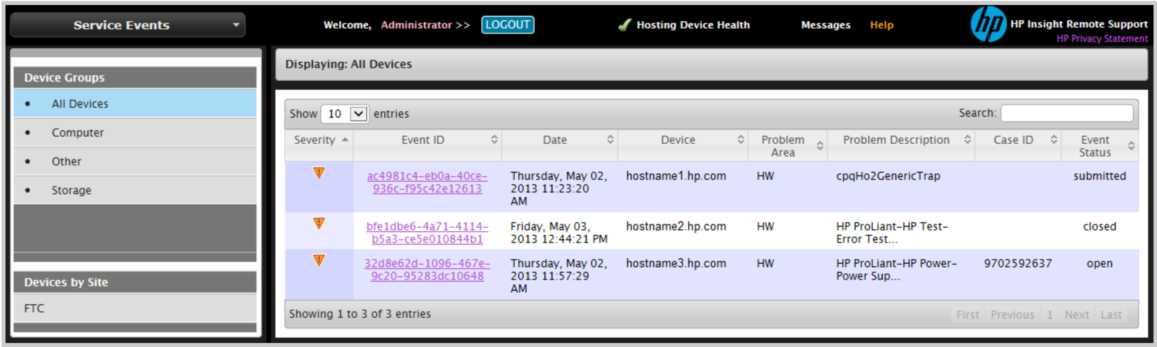
Viewing Test Events In the Insight RS Console

After sending the test indication, verify it arrived in the Insight RS Console.

To verify the test indication arrived, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Service Events**. If your monitored device is properly configured, the event

will appear in the Service Events Information pane.



Verify Collections

After discovery, a Server Basic Configuration Collection is automatically run for the StoreEasy Storage system. Verify that this collection ran successfully on the Collection Services → Basic Collection Results tab in the Insight RS Console. Note that there is a delay between when discovery finishes and when the collection runs. For more information about scheduling collections, see the Insight RS Help.

Also note that StoreEasy Storage systems use Server Basic Configuration Collections rather than Storage Configuration Collections.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services** and then click the **Basic Collection Results** tab.
3. Expand the Server Basic Configuration Collection section.
4. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears ( ). If it failed, an error icon appears ( ).

Chapter 26: Configuring StoreAll Network Storage Systems

Note: HP StoreAll Storage is the new name for HP IBRIX Storage.

Note the following limitations:

- For X9000 systems, the Insight Remote Support implementation is limited to hardware events.
- Some manual configurations require that the X9320 and X9300 nodes be recognized as a X9000 solution. See section "Configure Warranty and Contract Information" on page 235.

Fulfill Configuration Requirements

To configure your Network Storage Systems to be monitored by Insight RS, complete the following sections:

Table 26.1 ***Network Storage System Configuration Steps***

Task	Complete?
Make sure Insight RS supports your Network Storage Server by checking the *HP Insight Remote Support Release Notes*.	
Configure SNMP on the file server nodes.	
Restart Insight Management Agents.	
Add the SNMP protocol to the Insight RS Console.	
Discover the Network Storage Server in the Insight RS Console.	
Configure warranty and contract information in the Insight RS Console.	
Send a test event to verify connectivity between your Network Storage Server and Insight RS.	

Install and Configure Communication Software on Storage Systems

To configure your monitored devices, complete the following sections:

Configure SNMP on the File Serving Nodes

1. Run the `/sbin/hpsnmpconfig` script, which configures SNMP to integrate the Hosting Device running on the Windows server with the HP System Management Homepage on the file serving node. The script edits the `/etc/snmp/snmpd.conf` file, adding entries similar to the following:

```
# Following entries were added by HP Insight Management Agents at

# <date> <time> UTC <year>
```

```
dlmod cmaX /usr/lib64/libcmaX64.so
rwcommunity public 127.0.0.1
rocommunity public 127.0.0.1
rwcommunity public <Manager IP>
rocommunity public <Manager IP>
trapcommunity public
trapsink <Manager IP> public
syscontact
syslocation
# ---------------------- END --------------------
```

2. To add more than one SNMP Manager IP, copy the following lines:

```
rwcommunity public <Manager IP>
rocommunity public <Manager IP>
trapsink <Manager IP> public
```

3. After updating the snmpd.conf file, restart the snmpd service:

```
# service snmpd restart
```

For more information about the `/sbin/hpsnmpconfig` script, see "SNMP Configuration" in the hp-snmp-agents(4) man page. For information about HP System Management Homepage, go to: http://h18013.www1.hp.com/products/servers/management/agents/index.html.

Important: The `/opt/hp/hp-snmp-agents/cma.conf` file controls certain actions of the SNMP agents. You can add a trapIf entry to the file to configure the IP address used by the SNMP daemon when sending traps. For example, to send traps using the IP address of the eth1 interface, add the following:

trapIf eth1

Then restart the HP SNMP agents:

service hp-snmp-agents restart

For more information about the `cma.conf` file, see Section 3.1 of the *Managing ProLiant servers with Linux HOWTO* at: http://h20000.www2.hp.com/bc/docs/support/SupportManual/c00223285/c00223285.pdf.

Start or Restart Insight Management Agents

On the X9000, start or restart the following services:

- System Management Homepage:

```
# service hpsmhd restart
```

- SNMP Agent:

```
# service snmpd restart
```

- HP SNMP Agents:

  ```
  # service hp-snmp-agents start
  ```

  To ensure that `hp-snmp-agents` restarts when the system is rebooted, type the following command:

  ```
  # chkconfig hp-snmp-agents on
  ```

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create an SNMPv1 Protocol Credential in Insight RS Console

If your device's SNMP community string is set to `public` and your community access mode is `read only`, Insight RS automatically associates an SNMPv1 protocol with your device. If you use a different community string or use a nonstandard port, you must create an SNMPv1 protocol credential in the Insight RS Console.

To configure SNMPv1 in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the Select and Configure Protocol drop-down list, select **Simple Network Management Protocol Version 1 (SNMPv1)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Community String you have configured on your device.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP address(es) of the devices to be discovered.
   d. Click **Add**.
4. Click **Start Discovery**.

Configure Warranty and Contract Information

Configure the Insight RS Console to enable remote support for X9000 systems.

- **Custom field settings for X9300/X9320**

  Servers are discovered with their IP addresses. When a server is discovered, edit the system properties in the Insight RS Console. In the main menu, click **Devices**, then click the device name of your system.

  On the Device tab, in the Hardware section:

  - Type the X9000 enclosure product number as the Override Product Number
  - Type `MC DATACENTER` or `MC HYPERSCALE` as the Custom Delivery ID

  On the Device tab, in the Warranty & Contract section:

  - Type the appropriate Support Type and Support Identifier

- **Custom field settings for MSA Storage Management Utility**

  Configure SNMP settings on the MSA Storage Management Utility section. (For more information, see "Configuring SNMP event notification in SMU" in the *2300 Modular Smart Array Reference Guide* at: http://www.hp.com/support/manuals. On the Manuals page, select **Storage** → **Disk Storage Systems** → **MSA Disk Arrays** → **HP 2000sa G2 Modular Smart Array** or **HP P2000 G3 MSA Array Systems**.)

  A Modular Storage Array (MSA) unit should be discovered with its IP address. Once discovered, locate the Entitlement Information section of the Contract and Warranty Information page and update the following:

  - Type `MC DATACENTER` or `MC HYPERSCALE` as the Custom Delivery ID
  - Type the appropriate Support Type and Support Identifier

Note: For storage support on X9300 systems, do not set the Custom Delivery ID. (The MSA is an exception; the Custom Delivery ID is set as previously described.)

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon (✔).

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following sections:

Send a Test Trap

To determine whether the traps are working properly, send a generic test trap with the following command:

```
snmptrap -v1 -c public <Hosting Device IP> .1.3.6.1.4.1.232 <Managed System IP> 6
11003 1234 .1.3.6.1.2.1.1.5.0 s test .1.3.6.1.4.1.232.11.2.11.1.0 i 0
.1.3.6.1.4.1.232.11.2.8.1.0 s "X9000 remote support testing"
```

For example, if the Hosting Device IP address is 10.2.2.2 and the X9000 node is 10.2.2.10, type the following:

```
snmptrap -v1 -c public 10.2.2.2 .1.3.6.1.4.1.232 10.2.2.10 6 11003 1234
.1.3.6.1.2.1.1.5.0 s test .1.3.6.1.4.1.232.11.2.11.1.0 i 0 .1.3.6.1.4.1.232.11.2.8.1.0
s "X9000 remote support testing"
```

For Insight Remote Support, replace the Hosting Device IP address with the IP address of the Insight Remote Support server.

Verify Collections

After discovery, a Server Basic Configuration Collection is automatically run for the StoreAll Network Storage system. Verify that this collection ran successfully on the Collection Services → Basic Collection Results tab in the Insight RS Console. Note that there is a delay between when discovery finishes and when the collection runs. For more information about scheduling collections, see the Insight RS Help.

Also note that StoreAll Network Storage systems use Server Basic Configuration Collections rather than Storage Configuration Collections.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services** and then click the **Basic Collection Results** tab.
3. Expand the Server Basic Configuration Collection section.
4. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears ( ). If it failed, an error icon appears ( ).

Chapter 27: Configuring StoreOnce Backup (D2D) Systems

Fulfill Configuration Requirements

To configure your StoreOnce Backup (D2D) Systems to be monitored by Insight RS, complete the following sections:

Table 27.1 ***StoreOnce Backup (D2D) System Configuration Steps***

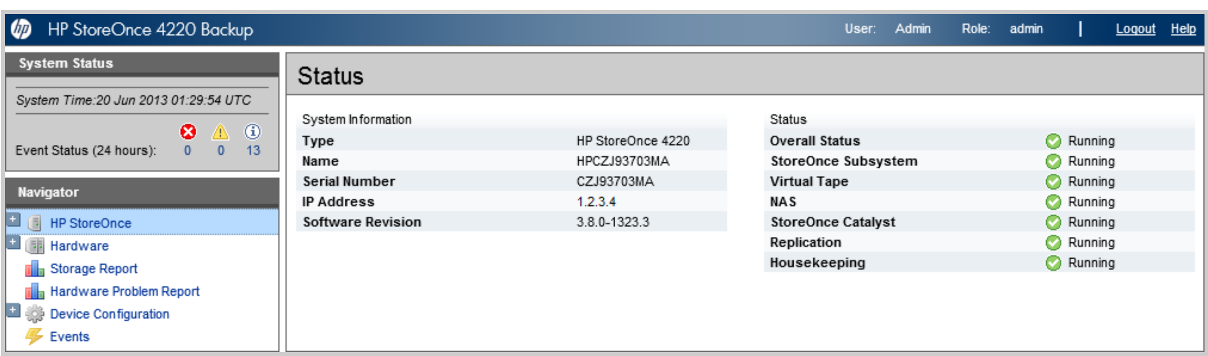
Task	Complete?
Make sure Insight RS supports your StoreOnce Backup (D2D) System by checking the *HP Insight Remote Support Release Notes*.	
Verify Firmware version on the StoreOnce Backup (D2D) System.	
Configure SNMP on the StoreOnce Backup (D2D) System.	
Add the SNMP protocol to the Insight RS Console.	
Discover the StoreOnce Backup (D2D) System in the Insight RS Console.	

Install and Configure Communication Software on Backup Systems

To configure your monitored devices, complete the following sections:

Verify the Firmware Version

Insight RS only supports StoreOnce Backup systems that use firmware version 2.1.03 or above. Check the firmware version in the StoreOnce Backup user interface as shown below. If your firmware version is older than version 2.1.03, update your firmware to a supported version.



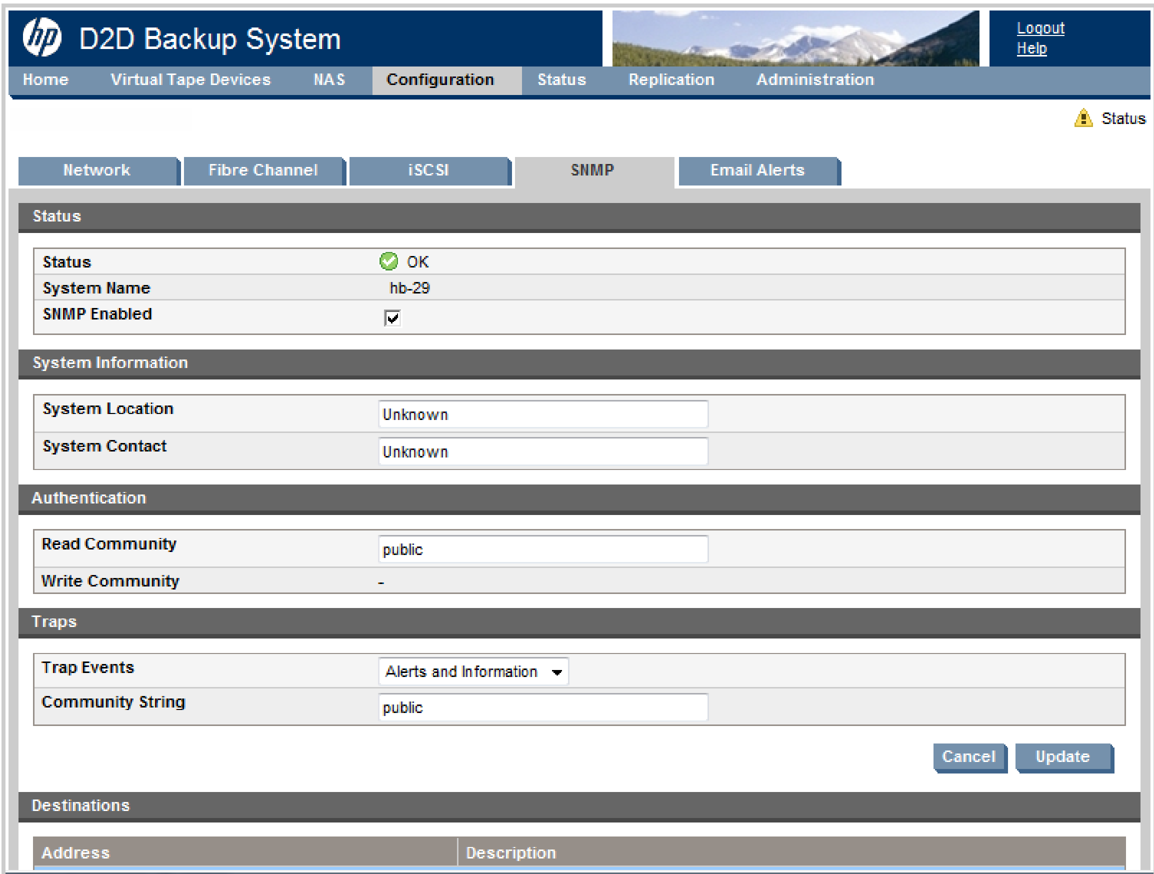
Configure SNMP on Your StoreOnce Backup (D2D) Systems

You must add the Hosting Device as a trap destination on your D2D Backup Systems for service events to reach Insight RS.

Configure SNMP for StoreOnce Gen2 Backup (D2D) Systems

To configure SNMP for StoreOnce Backup (D2D) Systems, complete the following steps:

1. Log on to the StoreOnce Backup System as "administrator" using `https://<StoreOnce_Backup_IP>`.
2. Click the **Configuration** tab.
3. On the **Device Configuration** pane, click the **SNMP** tab.
4. Click **Edit**, then enable the SNMP agent by selecting the **SNMP Enabled** check box.



5. Use the Hosting Device IP address to add a destination.

Configure SNMP for StoreOnce Gen3 B6200 and 2600/4200/4400 Backup Systems

To configure the StoreOnce Backup B6200 and 2600/4200/4400 Backup Systems to send traps to Insight Remote Support, complete the following steps:

1. Log on to the StoreOnce Backup System using the SSH protocol (Using "`putty.exe`" or a similar tool).

2. Check the current SNMP configuration by issuing the CLI command: `snmp show config`.
3. If the SNMP "State" displays as "off" run the CLI command: `snmp enable`.
4. To add your Hosting Device as a Trap Destination use: **snmp add trapsink** ***<IP address of Hosting Device>*** **events alert**.
5. Check that the current SNMP configuration shows the trap destination that you just configured by using: `snmp show config`.
6. Type `exit` to log off.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create an SNMPv1 Protocol Credential in Insight RS Console

If your device's SNMP community string is set to `public` and your community access mode is `read only`, Insight RS automatically associates an SNMPv1 protocol with your device. If you use a different community string or use a nonstandard port, you must create an SNMPv1 protocol credential in the Insight RS Console.

To configure SNMPv1 in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the Select and Configure Protocol drop-down list, select **Simple Network Management Protocol Version 1 (SNMPv1)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Community String you have configured on your device.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.

   c. Type the IP address(es) of the devices to be discovered.
   d. Click **Add**.
4. Click **Start Discovery**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon ( ).

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Chapter 28: Configuring Virtual Library Systems

Fulfill Configuration Requirements

To configure your Virtual Library Systems to be monitored by Insight RS, complete the following sections:

Table 28.1 ***Virtual Library System Configuration Steps***

Task	Complete?
Make sure Insight RS supports your Virtual Library System by checking the *HP Insight Remote Support Release Notes*.	
Configure SNMP on the Virtual Library System.	
Discover the Virtual Library System in the Insight RS Console.	
Send a test event to verify connectivity between your Virtual Library System and Insight RS.	

Install and Configure Communication Software on Virtual Library Systems

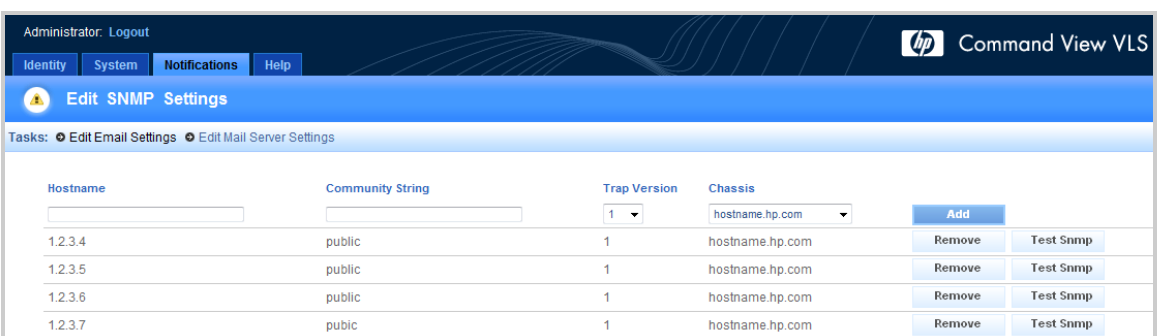
To configure your monitored devices, complete the following section:

Configure SNMP on the Virtual Library System

To support your Virtual Library System (VLS) Tape Libraries, you need to configure SNMP in Command View for Virtual Library Systems (VLS).

To configure SNMP on your VLS Tape Libraries, complete the following steps:

1. Log on to the Command View VLS web interface of each VLS node.
2. Click the **Notifications** tab.
3. Click the **Edit SNMP settings** task. Note that this task may be called **SNMP Setup**, depending on your version of Command View VLS.



4. In the **Hostname** field, type the Hosting Device IP address or FQDN.
5. In the **Community String** field, if you select a non-standard trap community name (for example, something other than `public`), verify the Insight RS Console recognizes this name.

6. The trap version is `1`.
7. Click **Add**.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create an SNMPv1 Protocol Credential in Insight RS Console

If your device's SNMP community string is set to `public` and your community access mode is `read only`, Insight RS automatically associates an SNMPv1 protocol with your device. If you use a different community string or use a nonstandard port, you must create an SNMPv1 protocol credential in the Insight RS Console.

To configure SNMPv1 in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the Select and Configure Protocol drop-down list, select **Simple Network Management Protocol Version 1 (SNMPv1)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Community String you have configured on your device.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP address(es) of the devices to be discovered.
   d. Click **Add**.
4. Click **Start Discovery**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon ( ).

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following sections:

Send a Text Event to Verify the Configuration

Note: Do not pull a redundant FRU to test the configuration.

If you have the latest VLS firmware and the latest version of Command View VLS, click the **Test SNMP** button on the Command View VLS SNMP configuration page to perform an end-to-end test. This test event will not result in a case being created at HP, but it will create a local email notification to confirm that the VLS SNMP and Insight Remote Support is configured correctly.

Verify Collections in the Insight RS Console

Configuration collections are not supported for this device type except when it is part of a SAN collection. If you have added this device to a SAN collection, you can manually run a SAN collection to verify the configuration.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services**.
3. Click the **Collection Schedules** tab.
4. In the List of Collection Schedules pane, select the **SAN Configuration Collection Schedule**. Information about the collection appears in the Collection Information pane. The On These Devices pane lists the devices the collection will run on.
5. In the Schedule Information pane, click **Run Now**.
6. When the collection completes, click the **SAN Storage Collection Results** tab.
7. Expand the SAN Configuration Collection section.

8. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears (✅). If it failed, an error icon appears (❎).

Chapter 29: Configuring StoreEver Tape Libraries

HP's physical tape libraries require a combination of their own web-based management interfaces and aggregated HP Command View for Tape Libraries (TL) library management software to work with Insight Remote Support (RS). Users configure each library to send SNMP traps for hardware events, such as fan failures, directly to the Insight RS server. Users configure Command View TL to send TapeAssure events over SMI-S such as poor drive margin. You need to discover Command View TL and the libraries separately.

Note: Insight RS does not support standalone (non-library) tape drives.

Fulfill Configuration Requirements

To configure Command View TL TapeAssure and your Tape Libraries to be monitored by Insight RS, complete the following sections:

Note: In future releases these configuration steps will be simplified.

Table 29.1 ***Tape Library Configuration Steps***

Task	Complete?
Make sure Insight RS supports your tape library by checking the *HP Insight Remote Support Release Notes*.	
Configure Command View TL TapeAssure. Note that Command View TL supplies the WBEM protocol for configuration collection support.	
Configure SNMPv1 traps on each tape library with the Hosting Device as the trap destination.	
Add the SNMP protocol to the Insight RS Console.	
Add the WBEM protocol to the Insight RS Console (for ESL and EML).	
Discover the tape library in the Insight RS Console.	
Verify the warranty and contract information in the Insight RS Console.	

Install and Configure Communication Software on Tape Libraries

To configure your monitored devices, complete the following sections:

Configure Command View TL TapeAssure

In addition to managing the tape libraries, Command View for Tape Libraries supplies the WBEM protocol that is necessary for Insight RS to gather collections on your tape libraries. During discovery of either the tape libraries or the Command View TL server, a WBEM protocol must be created in the Insight RS Console for the Command View TL instance on the server, and the managing server must be *actively* managing the tape libraries for collections to be gathered.

Note: HP recommends you use the latest version of Command View TL. Download the update at: http://www.hp.com/support/cvtl.

Important: Select Custom Installation to make sure the Command View TL SMI-S provider is installed.

Use the following procedure to configure Command View TL to allow Insight RS to register for TapeAssure events for Command View TL.

Configuring Command View TL 3.1 and Above

The Command View TL 3.1 CIMOM has user authentication enabled. With authentication enabled, the administrator of the Open Pegasus CIMOM has to create SMI-S users and provide access privileges for each user. Any current, or newly created Windows user can become a `cimuser`.

To add users and namespaces for Command View TL 3.1 CIMOM authentication complete the following steps:

1. Open a command window and change directory to: `C:\Program Files (x86)\Hewlett-Packard\Command View TL\op-cimom\bin`.
2. To create a `cimuser`, use the command:

   `cimuser –a –u` ***`<Windows username>`***

   The command asks for password and a confirmation. Provide the password of your choice.
3. Provide Read/Write access to the namespaces. In the TapeAssure Provider, all classes are implemented in two namespaces: `root/PG_Interop` and `root/hptl`. Both name spaces should be given read/write access. To provide access to the above name spaces to a selected user, use the command:

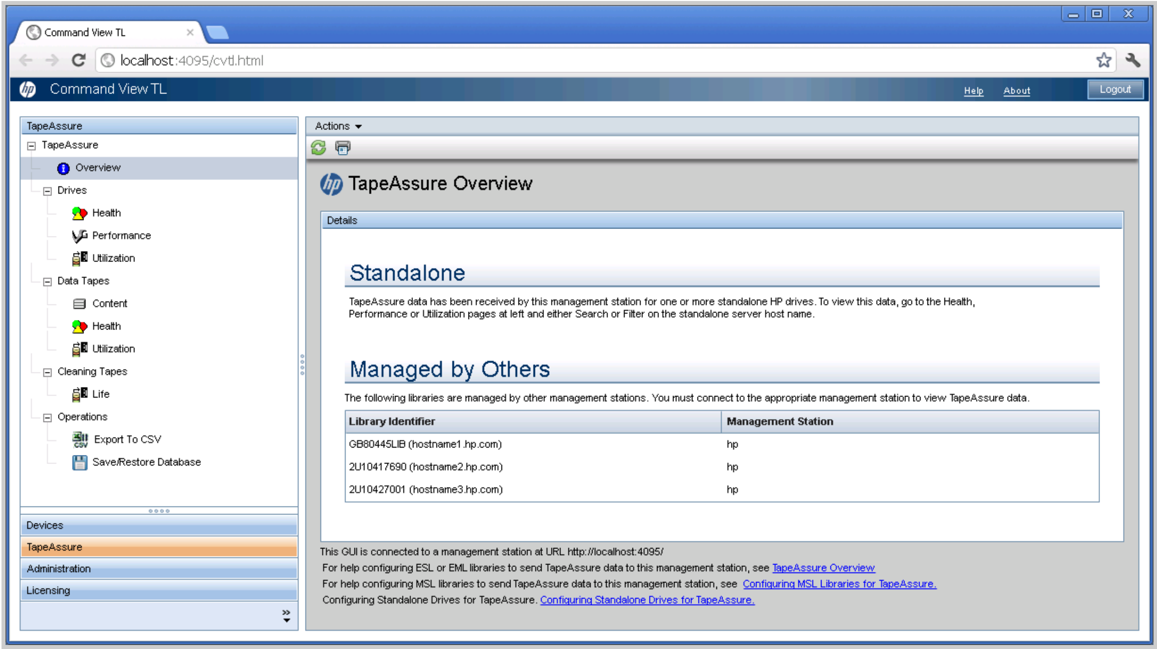
   `cimauth –a -u` ***`<cimuser>`*** **`-n`** ***`<namespace>`*** **`-R -W`**

Verifying Command View TL is Monitoring Your Tape Libraries

To configure Command View TL TapeAssure to monitor each library, complete the following steps:

1. In Command View TL, add the Tape Libraries to be monitored into the Command View TL Launcher window. **Library Selection** → **Actions** → **Add Library**.
2. Select the **TapeAssure** tab and check that the monitored libraries appear in the TapeAssure

Overview window.



Important: TapeAssure can also monitor standalone tape drives (those not in a tape library) and a host name appears instead of a library name; however, standalone drives cannot be configured for Insight RS.

Configure SNMP on Your Tape Libraries

You must add the Hosting Device as a trap destination on your tape libraries for service events to reach Insight RS.

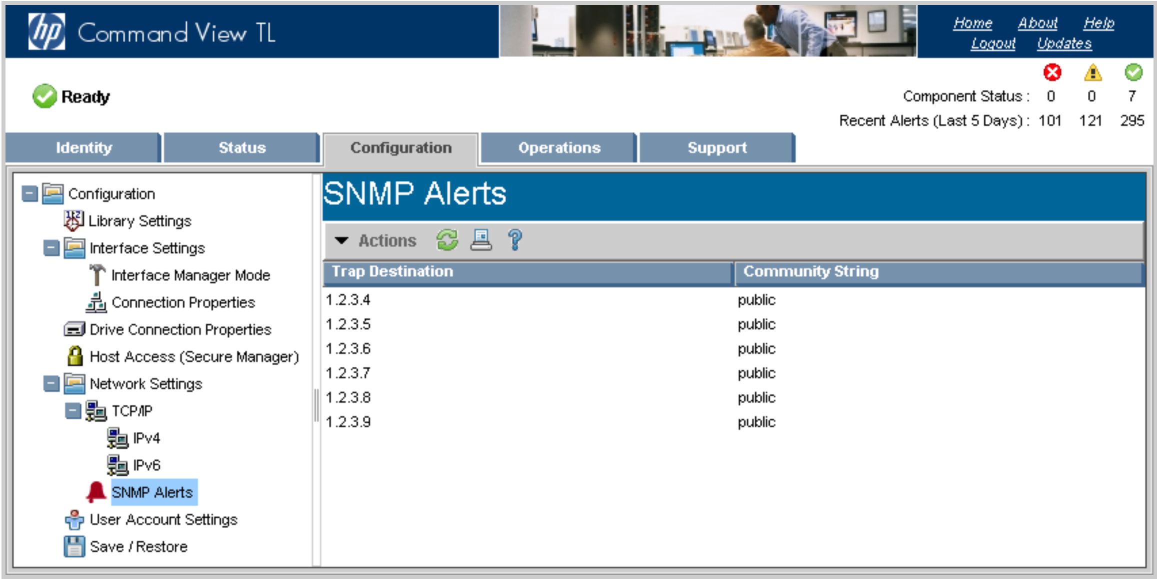
Configure Enterprise Systems Library E-series and Enterprise Modular Library

To allow Insight RS to register for your Enterprise Systems Library (ESL) E-series or Enterprise Modular Library (EML) library health events using SNMP, Command View TL must be configured.

To configure SNMP in Command View TL, complete the following steps:

1. Log on to the Command View TL web interface for each ESL-E or EML.
2. Click the **Configuration** tab.

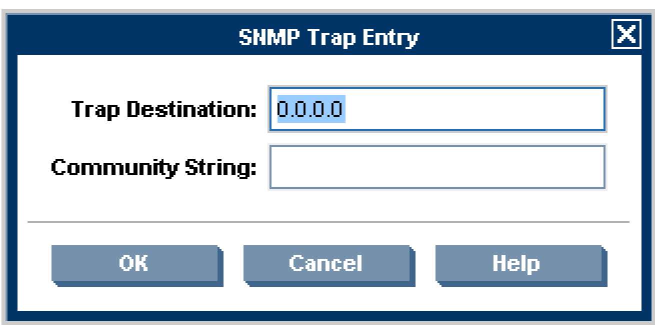
3. In the left panel, select **SNMP Alerts**. The current SNMP traps appear in the right panel.



4. Do one of the following:
   - In the right panel, right-click an SNMP trap, and then select **Add Trap Entry**.

     or
   - Select **Actions** → **Add Trap Entry**.

   The **SNMP Trap Entry** screen appears.



5. In the **Trap Destination** field, type the Hosting Device IP address or Fully Qualified Domain Name (FQDN).
6. Type `public` as the **Community String** unless you have defined another community string.
7. Click **OK**.

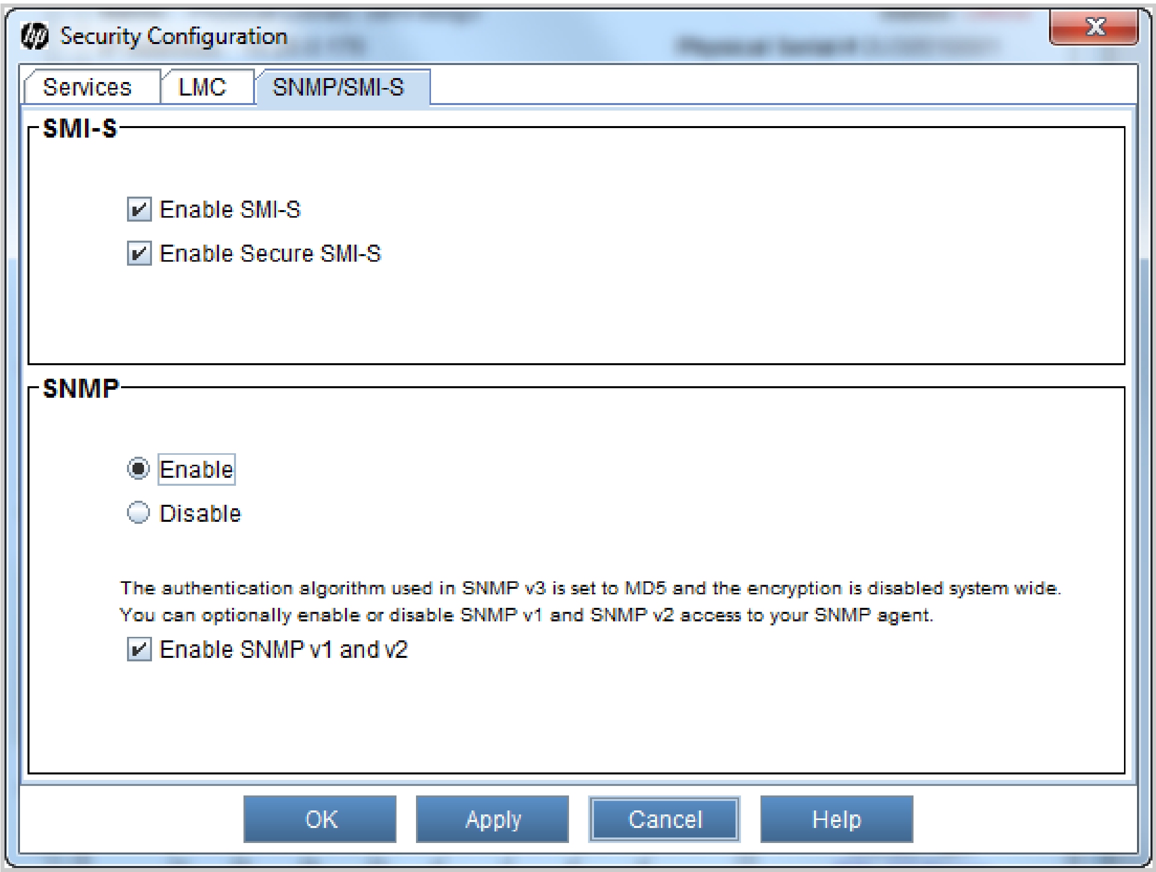
Configure Enterprise Systems Library G3 Series

Note: Basic configuration collections are *not* supported for HP StoreEver ESL G3 Tape Libraries.

To allow Insight RS to register for your Enterprise Systems Library (ESL) G3 library health events using SNMP, complete the following steps:

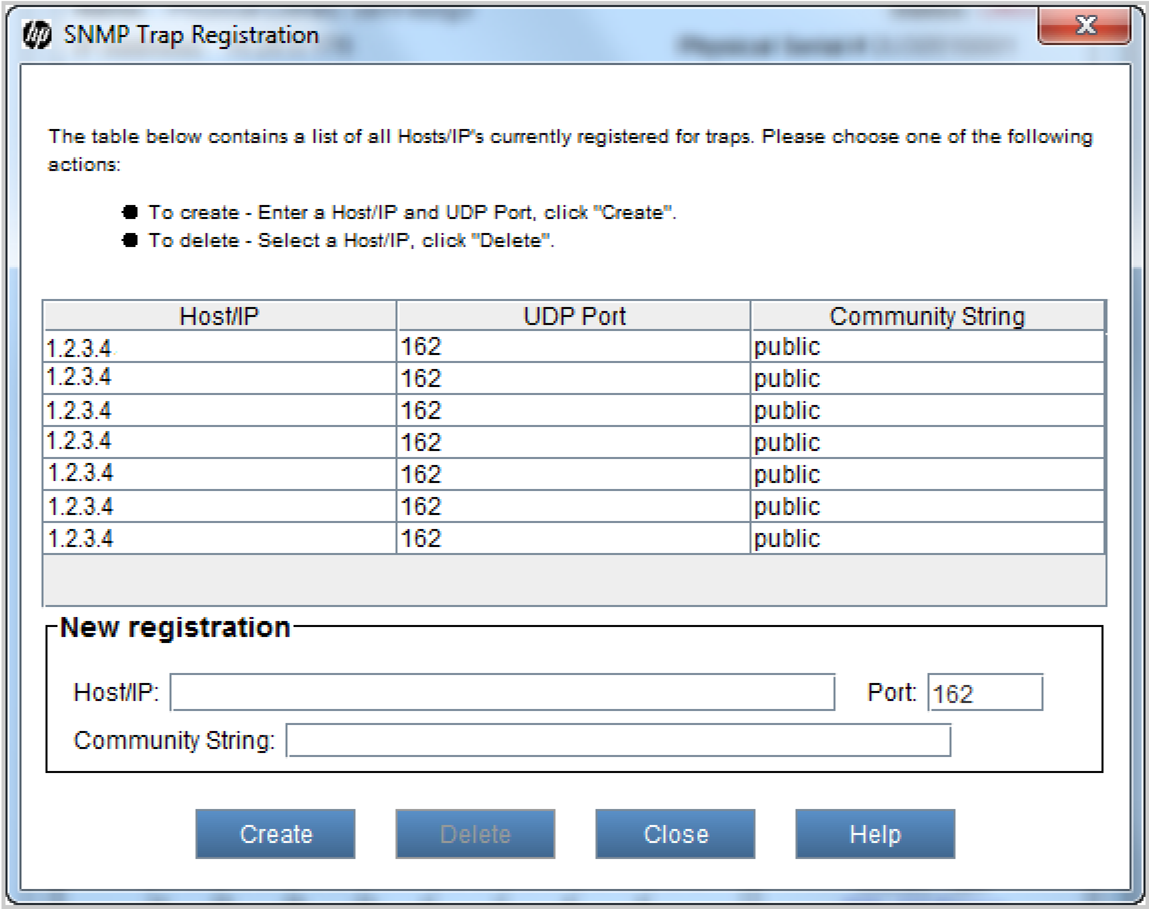
1. Log on to the ESL G3 user interface as `admin`.
2. In the console, go to **Setup** → **Network Configuration** → **Network Security Settings**.
3. In the Security Configuration screen, select the **SNMP/SMI-S** tab.

Note: Library health events are monitored with SNMP only. You do not need to enable SMI-S unless it is required for other Network Management software.



4. Click **OK**.
5. In the console, go to **Setup** → **Notifications** → **SNMP Trap Registration**.
6. In the SNMP Trap Registration screen, create an SNMPv1 trap, port 162 entry for the IP address of

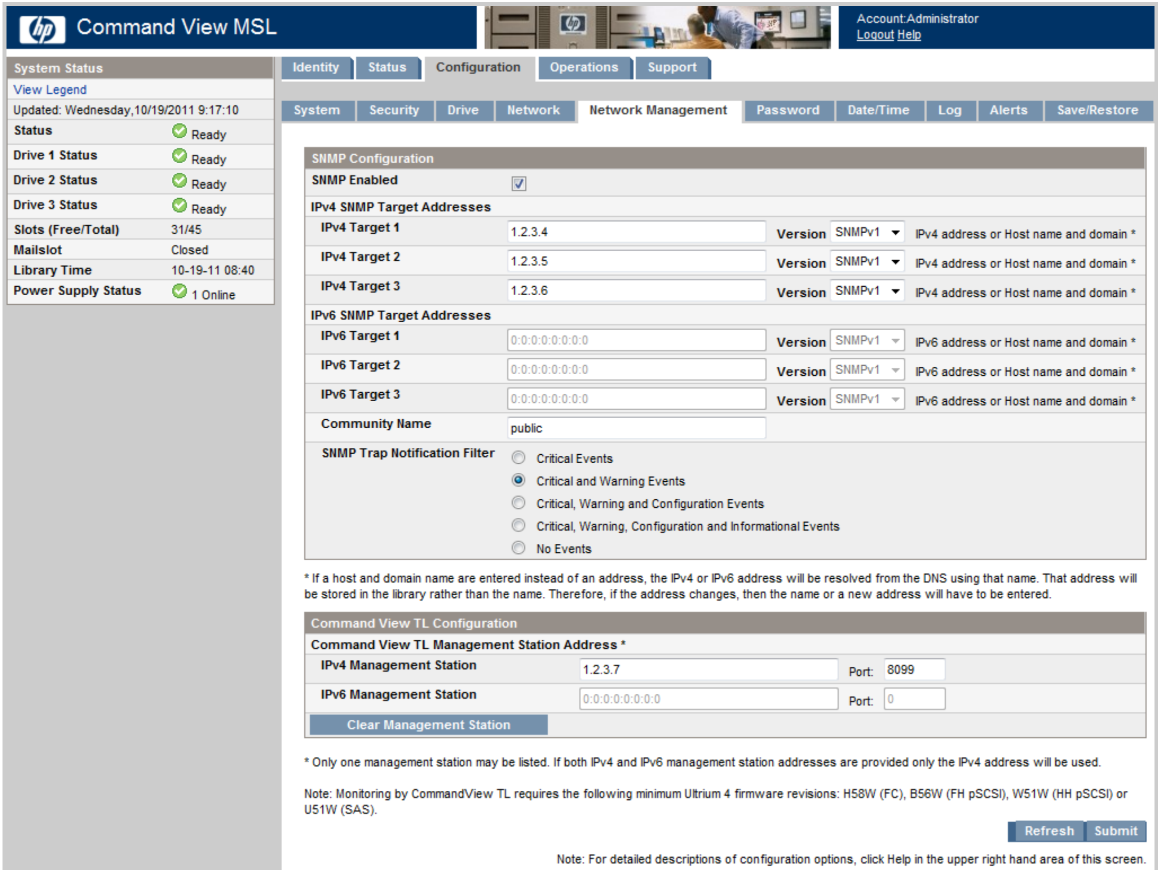
the Hosting Device. Use `public` for the community name. Click **Create**.



Configure Modular Systems Library G3 Series

To allow Insight RS to register for your Modular Systems Library (MSL) G3 library health events using SNMP, complete the following steps:

1. Log on to the Command View MSL web interface as `Administrator`.
2. Click the **Configuration** tab.
3. Click the **Network Management** tab.

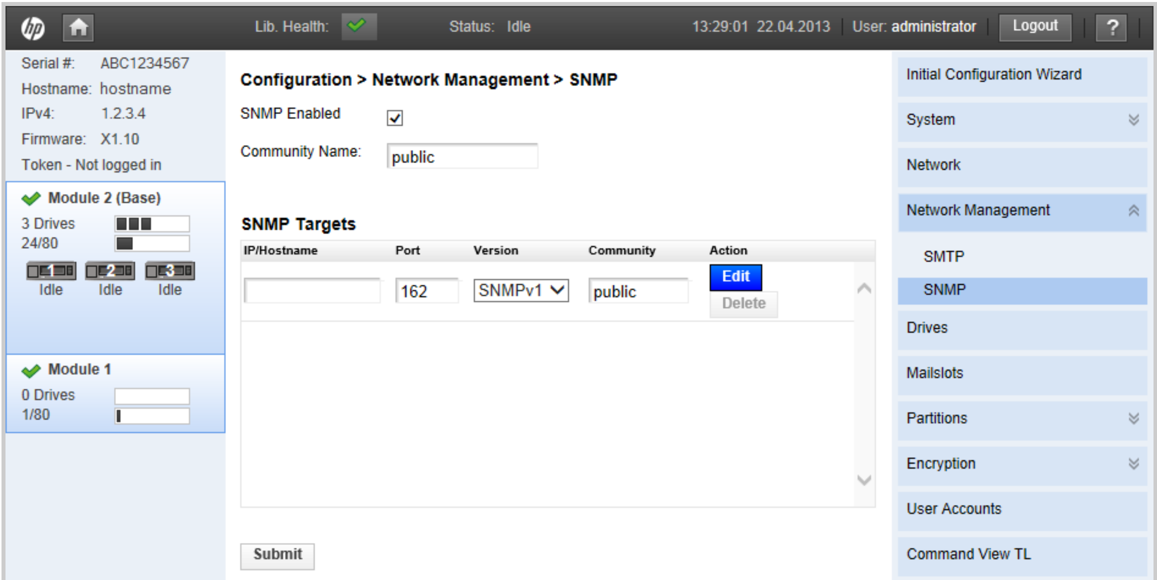


4. Verify the **SNMP Enabled** check box is selected.
5. In the **IPv4 Target** address field type the Hosting Device IP address or FQDN.
6. Verify **SNMPv1** is the version selected.
7. In the **SNMP Trap Notification Filter**, select **Critical and Warning Events** as the filter level.
8. Click **Submit**.

Configure Modular Systems Library MSL6480

To allow Insight RS to register for your Modular Systems Library (MSL) MSL6480 library health events using SNMP, complete the following steps:

1. Log on to the Log on to the MSL6480 web interface as `Administrator`.
2. Click **Configuration**.
3. Click the **Network Management** → **SNMP** tab.



4. Verify the **SNMP Enabled** check box is selected.
5. In the **SNMP Targets** table, click **Edit** next to a target without an IP/Hostname.
6. In the **IP/Hostname** field type the Hosting Device IP address or FQDN.
7. Verify **SNMPv1** is the version selected and enter the community string of the Hosting Device.
8. Click **Submit**.
9. Click **OK** in the confirmation dialogue box.

Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create an SNMPv1 Protocol Credential in Insight RS Console

If your device's SNMP community string is set to `public` and your community access mode is `read only`, Insight RS automatically associates an SNMPv1 protocol with your device. If you use a different community string or use a nonstandard port, you must create an SNMPv1 protocol credential in the Insight RS Console.

To configure SNMPv1 in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the Select and Configure Protocol drop-down list, select **Simple Network Management Protocol Version 1 (SNMPv1)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Community String you have configured on your device.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Create a WBEM Protocol Credential in Insight RS Console

For ESL and EML tape libraries, you must create a WBEM protocol in the Insight RS Console for the Command View TL server that monitors your tape libraries.

To configure WBEM in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the **Select and Configure Protocol** drop-down list, select **Web-Based Enterprise Management (WBEM)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Username and Password you have configured on your device.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Tape Libraries in Insight RS

Insight RS does not raise incidents for tape libraries or TapeAssure unless they are recognized by Insight RS. To make sure Insight RS raises incidents, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Type the IP addresses to be discovered:
   a. Type the IP address of the Command View TL host server. Discover the Command View TL host server individually by its IP address, making sure that when you create the Command View TL host discovery task, the CIMOM user credentials created in section "Configure Command View TL TapeAssure" on page 245 are used.
   b. Type the IP addresses of the tape libraries. Either discover each library individually or discover them within a range of IP addresses.
4. Click **Start Discovery**.

Verify Warranty and Contract Information

After device discovery, verify the warranty and contract information.

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Check that all of the monitored tape libraries and the Command View TL host(s) appear in the devices table.
4. Click the device name and expand the Hardware pane. Verify the correct product number and serial number for the tape library was discovered. Edit the individual devices if any appear to be missing any warranty and contract information.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon ( ).

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following section:

Verify Collections in the Insight RS Console

After discovery, a Storage Configuration Collection is automatically run for servers. Verify that this collection ran successfully on the Collection Services → SAN Storage Collection Results tab in the Insight RS Console. Note that there is a delay between when discovery finishes and when the collection runs. For more information about scheduling collections, see the Insight RS Help.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services** and then click the **SAN Storage Collection Results** tab.
3. Expand the Storage Configuration Collection section.
4. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears ( ). If it failed, an error icon appears ( ).

Chapter 30: Configuring StoreFabric B-Series Switches

Fulfill Configuration Requirements

To configure your StoreFabric B-Series Switches to be monitored by Insight RS, complete the following sections:

Note: Test trap support for B-series SAN switches requires the installation of Fabric Operating System v6.3.0b firmware or higher. For more details, refer to the *HP Fabric OS 6.3.0b Customer Advisory* at: http://h20000.www2.hp.com/bizsupport/TechSupport/Document.jsp?lang=en&cc=us&objectID=c01924282.

Table 30.1 ***StoreFabric B-Series Switch Configuration Steps***

Task	Complete?
Make sure Insight RS supports your B-Series switch by checking the *HP Insight Remote Support Release Notes*.	
Configure SNMPv1 on the B-Series switch.	
Add the SNMP protocol to the Insight RS Console.	
Discover the B-Series switch in the Insight RS Console.	
Verify the status of the B-Series switch in the Insight RS Console.	
Send a test event to verify connectivity between your B-Series switch and Insight RS.	

Install and Configure Communication Software on SAN Switches

To configure your monitored devices, complete the following sections:

Configure SNMP

Use the following procedure to configure SNMP on a B-Series switch so that it can be monitored with Insight Remote Support.

1. Display switch firmware version:

   ```
   > version
   Fabric OS: v6.1.0c
   ```

2. Display the current or default SNMPv1 settings:

   ```
   > snmpconfig --show snmpv1
   SNMPv1 community and trap recipient configuration:
   ```

```
Community 1: public (rw) No trap recipient configured yet
Community 2: public (rw) No trap recipient configured yet
Community 3: public (rw) No trap recipient configured yet
Community 4: public (ro) No trap recipient configured yet
Community 5: common (ro) No trap recipient configured yet
Community 6: FibreChannel (ro) No trap recipient configured yet
```

3. Set SNMPv1 community string and Trap recipient:

   > snmpconfig --set snmpv1

   ```
   SNMP community and trap recipient configuration:
   Community (rw): [public] gcc-public ! Create new community
   Trap Recipient's IP address : [0.0.0.0] 1.2.3.4
   Trap recipient Severity level : (0..5) [0] 2
   Trap recipient Port : (0..65535) [162] Community (rw): [public]
   Trap Recipient's IP address : [0.0.0.0] Community (rw): [public]
   Trap Recipient's IP address : [0.0.0.0] Community (ro): [public]
   Trap Recipient's IP address : [0.0.0.0] Community (ro): [common]
   Trap Recipient's IP address : [0.0.0.0] Community (ro): [FibreChannel]
   Trap Recipient's IP address : [0.0.0.0] Committing configuration...done.
   ```

   Note: The trap severity level is associated with each trap recipient IP address. The event trap level is in conjunction with the event's severity level. When an event occurs and if its severity level is at or below the set value, the SNMP event traps (`swEventTrap`, `swFabricWatchTrap`, and `connUnitEventTrap`) are sent to the trap recipients.

   By default, this value is set to 0, implying that no such traps are sent. The possible values are the following:

   - 0 none
   - 1 critical
   - 2 error
   - 3 warning
   - 4 informational
   - 5 debug

4. Display the new settings:

   > snmpconfig --show snmpv1

   ```
   SNMPv1 community and trap recipient configuration:
   Community 1: gcc-public (rw) Trap recipient: 1.2.3.4 Trap port: 162 Trap recipient
   Severity level: 2
   ```

```
Community 2: public (rw) No trap recipient configured yet
Community 3: public (rw) No trap recipient configured yet
Community 4: public (ro) No trap recipient configured yet
Community 5: common (ro) No trap recipient configured yet
Community 6: FibreChannel (ro) No trap recipient configured yet
```

5. Set SNMP access control:

   > snmpconfig --set accesscontrol

   ```
   SNMP access list configuration:
   Access host subnet area : [11.0.0.2] 1.2.3.4 ! set to the IP add of SIM
   Read/Write? (true, t, false, f): [true]
   Access host subnet area : [1.2.3.4] Read/Write? (true, t, false, f): [true] f
   Access host subnet area : [1.2.3.4] Read/Write? (true, t, false, f): [true] f
   Access host subnet area : [1.2.3.4] 0.0.0.0 ! remove this one Read/Write? (true,
   t, false, f): [true] f
   Access host subnet area : [1.2.3.4] Read/Write? (true, t, false, f): [true]
   Access host subnet area : [1.2.3.4] Read/Write? (true, t, false, f): [true]
   Committing configuration...done.
   ```

6. Remove the remaining IP address from the SNMP Access Control List if required:

   > snmpconfig --set accesscontrol

   ```
   SNMP access list configuration:
   Access host subnet area : [1.2.3.4] Read/Write? (true, t, false, f): [true]
   Access host subnet area : [1.2.3.4] 0.0.0.0 Read/Write? (true, t, false, f):
   [false] f
   Access host subnet area : [1.2.3.4] 0.0.0.0 Read/Write? (true, t, false, f):
   [false] f
   Access host subnet area : [0.0.0.0] Read/Write? (true, t, false, f): [false]
   Access host subnet area : [1.2.3.4] 0.0.0.0 Read/Write? (true, t, false, f):
   [true] f
   Access host subnet area : [1.2.3.4] 0.0.0.0 Read/Write? (true, t, false, f):
   [true] f
   Committing configuration...done.
   ```

7. Check the access to SNMP control list:

   > snmpconfig --show accesscontrol

   ```
   SNMP access list configuration:
   Entry 0: Access host subnet area 1.2.3.4 (rw)
   Entry 1: No access host configured yet
   ```

```
Entry 2: No access host configured yet
Entry 3: No access host configured yet
Entry 4: No access host configured yet
Entry 5: No access host configured yet
```

8. Set MIB Capability:

   > snmpconfig --set mibcapability

   ```
   The SNMP Mib/Trap Capability has been set to support FE-MIB SW-MIB FA-MIB HA-MIB
   SW-TRAP FA-TRAP HA-TRAP
   FA-MIB (yes, y, no, n): [yes]
   FICON-MIB (yes, y, no, n): [no]
   HA-MIB (yes, y, no, n): [yes] yes
   FCIP-MIB (yes, y, no, n): [no]
   ISCSI-MIB (yes, y, no, n): [no]
   SW-TRAP (yes, y, no, n): [yes]
   swFCPortScn (yes, y, no, n): [no] yes
   swEventTrap (yes, y, no, n): [no] yes
   swFabricWatchTrap (yes, y, no, n): [no] yes
   swTrackChangesTrap (yes, y, no, n): [no] yes
   FA-TRAP (yes, y, no, n): [yes]
   connUnitStatusChange (yes, y, no, n): [no] yes
   connUnitEventTrap (yes, y, no, n): [no] yes
   connUnitSensorStatusChange (yes, y, no, n): [no] yes
   connUnitPortStatusChange (yes, y, no, n): [no] yes
   SW-EXTTRAP (yes, y, no, n): [no] yes
   ```

9. Display the IP address of the switch, so it can be added to the Insight RS Console:

   > ipaddrshow

   ```
   SWITCH Ethernet IP Address: 1.2.3.4
   Ethernet Subnetmask: 1.2.3.4
   Fibre Channel IP Address: none
   Fibre Channel Subnetmask: none
   Gateway IP Address: 1.2.3.4
   DHCP: Off
   ```

10. Modify the switch `systemgroup` variables if required:

    > snmpconfig --set systemGroup

    ```
    Customizing MIB-II system variables ... At each prompt, do one of the following:
    ```

```
o to accept current value,
o enter the appropriate new value,
o to skip the rest of configuration, or
o to cancel any change.
To correct any input mistake: erases the previous character, erases the whole
line,
sysDescr: [Fibre Channel Switch.] Brocade 4/16 FC Switch
sysLocation: [End User Premise.] HP Building 11th floor
sysContact: [Field Support.] John Doe
authTrapsEnabled (true, t, false, f): [true]
Committing configuration...done.
```

Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create an SNMPv1 Protocol Credential in Insight RS Console

If your device's SNMP community string is set to `public` and your community access mode is `read only`, Insight RS automatically associates an SNMPv1 protocol with your device. If you use a different community string or use a nonstandard port, you must create an SNMPv1 protocol credential in the Insight RS Console.

To configure SNMPv1 in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the Select and Configure Protocol drop-down list, select **Simple Network Management Protocol Version 1 (SNMPv1)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Community String you have configured on your device.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP address(es) of the devices to be discovered.
   d. Click **Add**.
4. Click **Start Discovery**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon ( ).

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Warranty and Contract Information

Verify that the warranty and contract information was discovered correctly in the Insight RS Console:

1. In the Insight RS Console, navigate to **Devices** and click the switch Device Name.
2. Expand the Hardware section, and make sure the **Acquired Serial Number** and **Acquired Product Number** are correct. If they are not correct, type the correct values in the **Override Serial Number** and **Override Product Number** fields and click **Save Changes**.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following sections:

Send a Test Event

Since B-Series Fabric-OS v6.3.x was released, B-Series switches can emit test traps to ensure that the complete end-to-end event flow can be verified. The user must log on to the switch CLI admin interface and issue the following command:

Send SNMP traps to test the configuration:

1. Telnet or SSH to the B-Series switch.
2. Show the available traps you can send as tests:

   ```
   snmptraps --show
   ```

3. Send a blanket test trap for all tests:

   ```
   snmptraps --send
   ```

4. Or send a specific test trap:

   ```
   snmptraps --send --trap_name <trapname>
   ```

   For example:

   ```
   snmptraps --send -trap_name cp-status-change-trap -ip_address <hosting_device_ip>
   ```

   where *<hosting_device_ip>* is the IP address of the Hosting Device. If the switch reports that the test trap cannot be sent, make sure `snmpmibcaps` is configured to send HA traps. If it is configured correctly, this command should create a test event visible in the Insight RS Console and also submit an incident to HP.

Note: Not all traps result in an incident being delivered to the Insight RS Console. Not all traps are sent and not all are received. Some get received as ‘unregistered event’ if you have that option selected as a receive option in Events. The events that show up as ‘unregistered’ are not compiled in the SIM Mibs. FA-MIB’s generate a valid hardware event.

See the B-Series Switch documentation for more details.

Verify Collections in the Insight RS Console

After discovery, a Storage Configuration Collection is automatically run for servers. Verify that this collection ran successfully on the Collection Services → SAN Storage Collection Results tab in the Insight RS Console. Note that there is a delay between when discovery finishes and when the collection runs. For more information about scheduling collections, see the Insight RS Help.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services** and then click the **SAN Storage Collection Results** tab.
3. Expand the Storage Configuration Collection section.
4. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears ( ). If it failed, an error icon appears ( ).

Chapter 31: Configuring StoreFabric C-Series Switches

Fulfill Configuration Requirements

To configure your StoreFabric C-Series Switches to be monitored by Insight RS, complete the following sections:

Table 31.1 ***StoreFabric C-Series Switch Configuration Steps***

Task	Complete?
Make sure Insight RS supports your C-Series switch by checking the *HP Insight Remote Support Release Notes*.	
Configure SNMPv1 on the C-Series switch.	
Add the SNMP protocol to the Insight RS Console.	
Discover the C-Series switch in the Insight RS Console.	
Verify the status of the C-Series switch in the Insight RS Console.	
Send a test event to verify connectivity between your C-Series switch and Insight RS.	

Install and Configure Communication Software on SAN Switches

To configure your monitored devices, complete the following sections:

Configure SNMP

To configure C-Series (formerly Cisco) SNMP, complete the following steps:

1. Log on to the CLI interface with Telnet, or use Fabric Manager.
2. Use the CLI commands `config t` and `snmp-server host <IP> traps version 1 public udp-port 162` commands to add the Hosting Device IP address as a trap destination.
3. Choose to have all events returned as traps.
4. If you choose a non-standard trap community name (i.e. a name other than `public`), make sure this name is used in the Insight RS Console credentials settings.
5. Type the IP address of Hosting Device into one of the trap destination settings. You can also configure this with the C-Series Device Manager.
6. Discover the device in the Insight RS Console.
7. In the Insight RS Console, navigate to **Devices** and click the Device Name. Verify that the device information was discovered correctly.

Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create an SNMPv1 Protocol Credential in Insight RS Console

If your device's SNMP community string is set to `public` and your community access mode is `read only`, Insight RS automatically associates an SNMPv1 protocol with your device. If you use a different community string or use a nonstandard port, you must create an SNMPv1 protocol credential in the Insight RS Console.

To configure SNMPv1 in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the Select and Configure Protocol drop-down list, select **Simple Network Management Protocol Version 1 (SNMPv1)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Community String you have configured on your device.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP address(es) of the devices to be discovered.
   d. Click **Add**.
4. Click **Start Discovery**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon (✅).

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Warranty and Contract Information

Verify that the warranty and contract information was discovered correctly in the Insight RS Console:

1. In the Insight RS Console, navigate to **Devices** and click the switch Device Name.
2. Expand the Hardware section, and make sure the **Acquired Serial Number** and **Acquired Product Number** are correct. If they are not correct, type the correct values in the **Override Serial Number** and **Override Product Number** fields and click **Save Changes**.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following sections:

Send a Test Event

Note: Do not pull a redundant FRU to test the configuration.

Cisco has recently introduced a CLI test command to validate the device end-to-end event connectivity.

1. From the CLI prompt, issue the following command:

   `test pfm test-SNMP-trap power`
2. This causes the C-Series switch to send a `bad power supply` test event to the Hosting Device. If everything is configured correctly, this event results in an incident viewable at HP.

Note: You can also use **test pfm test-SNMP-trap fan**, but the `temp` type event, if used, will be ignored.

Note: The first Cisco test trap will be forwarded by Insight Remote Support to HP, but subsequent test traps will be suppressed for 24 hours before another test trap can be forwarded to HP.

Verify Collections in the Insight RS Console

After discovery, a Storage Configuration Collection is automatically run for servers. Verify that this collection ran successfully on the Collection Services → SAN Storage Collection Results tab in the Insight RS Console. Note that there is a delay between when discovery finishes and when the collection runs. For more information about scheduling collections, see the Insight RS Help.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services** and then click the **SAN Storage Collection Results** tab.
3. Expand the Storage Configuration Collection section.
4. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears (). If it failed, an error icon appears ().

Chapter 32: Configuring StoreFabric H-Series Switches

Fulfill Configuration Requirements

To configure your StoreFabric H-Series Switches to be monitored by Insight RS, complete the following sections:

Table 32.1 ***StoreFabric H-Series Switch Configuration Steps***

Task	Complete?
Make sure Insight RS supports your H-Series switch by checking the *HP Insight Remote Support Release Notes*.	
Configure SNMPv1 on the H-Series switch.	
Add the SNMP protocol to the Insight RS Console.	
Discover the H-Series switch in the Insight RS Console.	
Verify the status of the H-Series switch in the Insight RS Console.	

Install and Configure Communication Software on SAN Switches

To configure your monitored devices, complete the following sections:

Configure SNMP

Log on to the switch with Telnet and issue the following CLI commands to configure the HP SN6000, 8/20q and 2/8q FC switches to send SNMP traps to the Hosting Device.

1. Open the Admin session:

   `admin start`

2. Configure the SNMP trap destinations:

   `set setup snmp`

3. After configuration, type **y** to save and active the SNMP setup.
4. Close the Admin session:

   `admin end`

Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create an SNMPv1 Protocol Credential in Insight RS Console

If your device's SNMP community string is set to `public` and your community access mode is `read only`, Insight RS automatically associates an SNMPv1 protocol with your device. If you use a different community string or use a nonstandard port, you must create an SNMPv1 protocol credential in the Insight RS Console.

To configure SNMPv1 in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the Select and Configure Protocol drop-down list, select **Simple Network Management Protocol Version 1 (SNMPv1)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Community String you have configured on your device.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
    a. Click **New**.
    b. Select the **Single Address**, **Address Range**, or **Address List** option.
    c. Type the IP address(es) of the devices to be discovered.
    d. Click **Add**.
4. Click **Start Discovery**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon (✅).

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Warranty and Contract Information

Verify that the warranty and contract information was discovered correctly in the Insight RS Console:

1. In the Insight RS Console, navigate to **Devices** and click the switch Device Name.
2. Expand the Hardware section, and make sure the **Acquired Serial Number** and **Acquired Product Number** are correct. If they are not correct, type the correct values in the **Override Serial Number** and **Override Product Number** fields and click **Save Changes**.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following section:

Verify Collections in the Insight RS Console

After discovery, a Storage Configuration Collection is automatically run for servers. Verify that this collection ran successfully on the Collection Services → SAN Storage Collection Results tab in the Insight RS Console. Note that there is a delay between when discovery finishes and when the collection runs. For more information about scheduling collections, see the Insight RS Help.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services** and then click the **SAN Storage Collection Results** tab.
3. Expand the Storage Configuration Collection section.
4. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears (✅). If it failed, an error icon appears (❎).

Chapter 33: Configuring ProVision-based Networking Switches

ProVision-based networking switches (formerly E-series/ProCurve) require SNMP for discovery and event monitoring. ProVision-based switches ship with SNMP installed and enabled.

You can use either Telnet or SSH for configuration collections. Telnet is enabled on ProVision-based switches by default. If you want to use SSH for configuration collections, you must enable it (see section "Enabling SSH").

Fulfill Configuration Requirements

To configure your ProVision-based networking switches to be monitored by Insight RS, complete the following sections:

Table 33.1 ***ProVision-based Networking Switch Configuration Steps***

Task	Complete?
Make sure Insight RS supports your ProVision-based switch by checking the *HP Insight Remote Support Release Notes*.	
Create Operator/Manager passwords on the switch.	
Configure SSH on the switch.	
Configure SNMP on the switch.	
Add the Telnet or SSH and SNMP protocols to the Insight RS Console.	
Discover the ProVision-based switch in the Insight RS Console.	

Install and Configure Communication Software on Switches

To configure your monitored devices, complete the following sections:

Create Operator/Manager Passwords

It is required that you set a manager password.

To set a manager password, complete the following steps:

1. Type the **password all** command. You are prompted to type manager and operator passwords:

   ```
   # password all
   ```

2. Type the **password manager** command to set the manager password:

   ```
   # password manager
   ```

Configure SSH

Generate the Public/Private key for SSH

1. Type the **config** command to enable the global configuration mode:

   # config

2. Type the **crypto** command to generate an RSA key pair, which includes one public RSA key and one private RSA key.

   # crypto key generate ssh rsa

Enable SSH

You only need to enable SSH if you want to use it instead of Telnet. If you choose to use Telnet, you can skip this section.

To enable SSH, execute the following commands from the switch command line:

1. Type the **config** command to enable the global configuration mode:

   # config

2. Enable the SSH server:

   # ip ssh

3. Type the write memory command to save the configuration changes:

   # write mem

For more information about ProVision-based switches, please refer to the ProVision-based switch documentation at: http://h17007.www1.hp.com/us/en/products/switches/index.aspx.

Verify Telnet/SSH Communication

To verify that Telnet or SSH communication is working properly, complete the following steps:

1. From the Hosting Device, connect to the ProVision-based switch using the protocol you want to verify. Wait until the connection is established.
2. Type the `username` and `password` when prompted. If any of the following messages do occur then, it is treated to be a login failure:
   - `Login incorrect`
   - `User authorization failure`
   - `Invalid login name`
   - `Too many users logged on already`
3. When the banner or login data is received, send a carriage return (\n) to make sure that the host is

ready.

4. Type the `exit` command to disconnect.

Configure SNMP Communication

To verify that SNMP communication is working properly, complete the following steps:

1. Type the **show version** command to determine which software version your ProVision-based switch is running:

   `# sho ver`

2. Type the **config** command to enable the global configuration mode:

   `# config`

3. The output should show a version number similar to *x*.14.60, where *x* will be a letter such as K.

4. Use the **snmp-server** command to type the Hosting Device's IP address. Depending on the version of the firmware on the ProVision-based switch, use one of the following versions of the command:

Firmware version	Command
14.60 and lower	**# snmp-server host <*IP address*> public not-info**
14.61 and higher	**# snmp-server host <*IP address*> community public trap-level not-info**

5. Verify that the settings you just typed were saved:

   `# sho run`

6. Type the write memory command to save the configuration changes:

   `# write mem`

Configure SSH Public Key Authentication

Option 1: Using the Hosting Device Certificate

Prerequisites

- *TFTP server* — used to move the public key to the ProVision-based switch.

Copy the Certificate to the Switch

To copy the certificate to the ProVision-based switch, complete the following steps:

1. On the Hosting Device, export the Hosting Device public key. The alias name is “jetty”:

   rsadmin cert -export -keycomment manager@[*IP_of_switch*] -sshkey -out [*file_path_name*]

where *file_path_name* is a path that the TFTP server can see.

2. Telnet or SSH to the switch using username/password authentication.
3. Enter configuration mode:

   configure
4. Turn off filetransfer using SSH:

   no ip ssh filetransfer
5. Enable the TFTP client:

   tftp client
6. Move the public key to the switch:

   copy tftp pub-key-file <*tftp_server_ip*> <*public_key_file*> manager
7. Enable public key authentication:

   aaa authentication ssh login public-key
8. Enable user/password authentication:

   aaa authentication ssh enable local
9. Write the configuration and public key to memory:

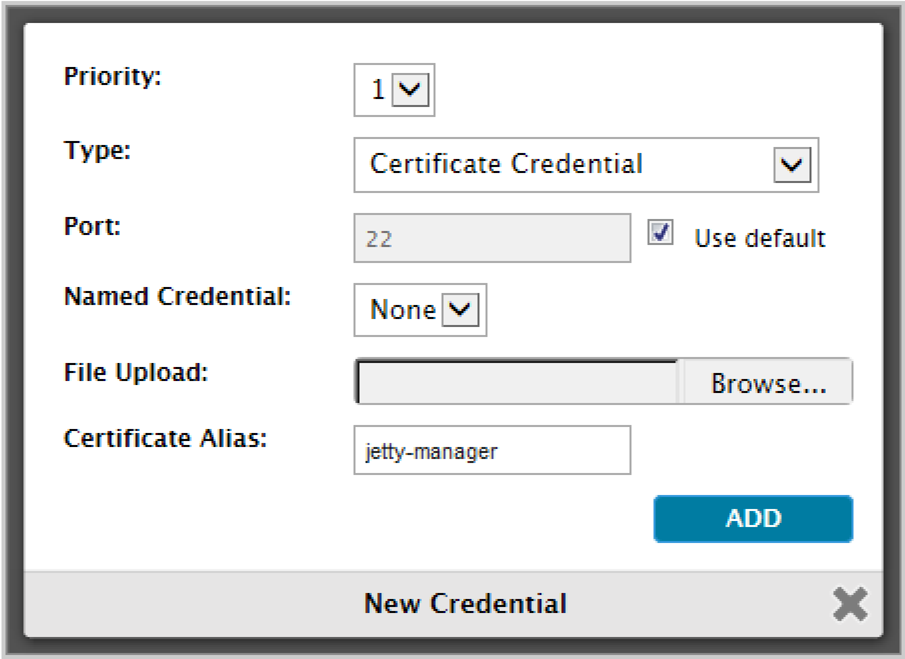
   wr mem

Create an SSH Protocol Credential in the Insight RS Console

To configure SSH in the Insight RS Console, complete the following steps:

1. In the Insight RS Console, add a SSH certificate credential on the **Discovery** → **Credentials** tab.
   a. From the **Select and Configure Protocol** drop-down list, select **Secure Shell (SSH)** and click **New**.
   b. From the **Type** drop-down list, select **Certificate Credential**.
   c. Leave the **File Upload** field blank because the certificate is already in the certificate store and it is identified by using the alias name.
   d. Type the Certificate Alias of “jetty-manager”, which is the alias given to the certificate when it was

exported above.



2. Click **Add**.

Option 2: Using Other Certificates

Prerequisites

- *PuTTYgen* — used to generate a key pair if required.
- *TFTP server* — used to move the public key to a ProVision-based switch.

Copy the Certificate to the Switch

To copy the certificate to the ProVision-based switch, complete the following steps:

1. Use PuTTYgen to create a key pair.
2. Change the public key comment field in PuTTYgen to `manager@IP` for operator account access where IP is the IP of the switch or `manager@ip` for manager account access.
3. Copy the public key to a file path that your TFTP server can see. (Refer to the TFTP server documentation and server configuration for more information.)
4. In PuTTYgen, select **Conversions** → **Export OpenSSH Key** to export the private key. Don't set a pass phrase. Name the private key `PCPrivate.pem`.
5. Telnet or SSH to the switch using username/password authentication.
6. Enter configuration mode:

   ```
   configure
   ```

7. Turn off filetransfer using SSH:

   `no ip ssh filetransfer`

8. Enable the TFTP client:

   `tftp client`

9. Move the public key to the switch:

   copy tftp pub-key-file <*tftp_server_ip*> <*public_key_file*> manager

10. Enable public key authentication:

    `aaa authentication ssh login public-key`

11. Enable user/password authentication:

    `aaa authentication ssh enable local`

12. Write the configuration and public key to memory:

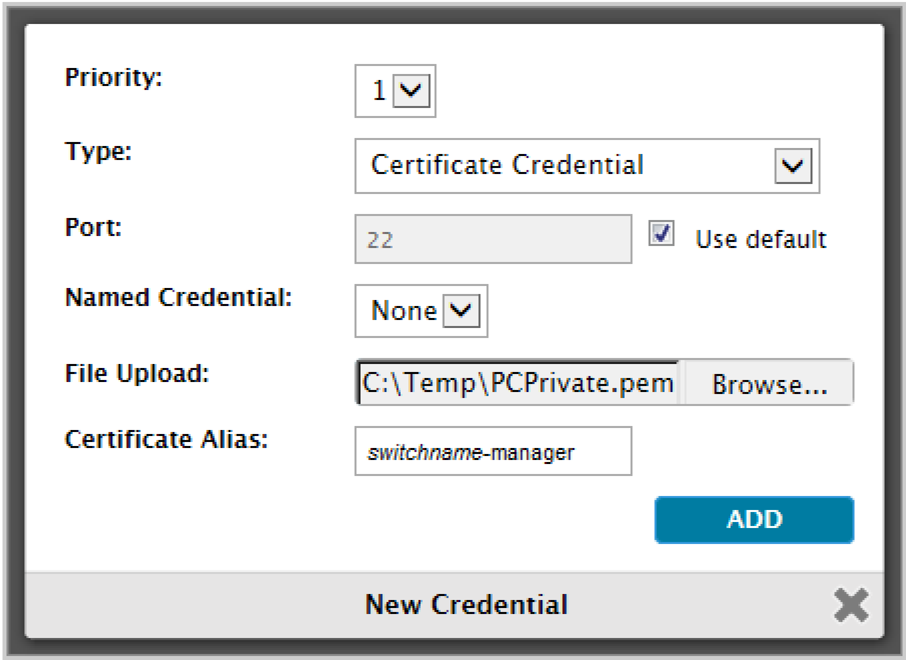
    `wr mem`

Create an SSH Protocol Credential in the Insight RS Console

To configure SSH in the Insight RS Console, complete the following steps:

1. In the Insight RS Console, add a SSH certificate credential on the **Discovery** → **Credentials** tab.
   a. From the **Select and Configure Protocol** drop-down list, select **Secure Shell (SSH)** and click **New**.
   b. From the **Type** drop-down list, select **Certificate Credential**.
   c. Browse to the private key file, `PCPrivate.pem`.
   d. Type the Certificate Alias. The alias will take the form of *switchname*-manager. The SSH protocol

uses the *-manager* portion of the alias as the user name during log on.



2. Click **Add**.

Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create an SNMPv1 Protocol Credential in Insight RS Console

If your device's SNMP community string is set to `public` and your community access mode is `read only`, Insight RS automatically associates an SNMPv1 protocol with your device. If you use a different community string or use a nonstandard port, you must create an SNMPv1 protocol credential in the Insight RS Console.

To configure SNMPv1 in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.

3. From the Select and Configure Protocol drop-down list, select **Simple Network Management Protocol Version 1 (SNMPv1)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Community String you have configured on your device.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP address(es) of the devices to be discovered.
   d. Click **Add**.
4. Click **Start Discovery**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon ().

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following section:

Verify Collections in the Insight RS Console

After discovery, a Network Configuration Collection is automatically run for servers. Verify that this collection ran successfully on the Collection Services → Basic Collection Results tab in the Insight RS

Console. Note that there is a delay between when discovery finishes and when the collection runs. For more information about scheduling collections, see the Insight RS Help.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services** and then click the **Basic Collection Results** tab.
3. Expand the Network Configuration Collection section.
4. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears (). If it failed, an error icon appears ().

Chapter 34: Configuring ComWare-based Networking Switches

ComWare-based networking switches (formerly A-Series or H3C/3COM) require SNMP for discovery and event monitoring.

Insight RS supports Intelligent Resilient Framework (IRF) technology, and can detect the following IRF alerts:

- Power supply failures
- Fans and fan tray failures
- Module and board failures
- Over-temperature conditions

Important: Make sure all IRF stack member switches have a valid warranty or contract. If the master switch fails, the member switch assigned to be the new master switch must also have a valid warranty or contract for the stack to continue to be entitled for remote support. If the stack has not been built yet, you can verify entitlement by discovering each switch individually. If the stack has already been built you will need to verify each has a valid warranty or contract manually.

Fulfill Configuration Requirements

To configure your ComWare-based networking switches to be monitored by Insight RS, complete the following sections:

Table 34.1 ***ComWare-based Networking Switch Configuration Steps***

Task	Complete?
Make sure Insight RS supports your ComWare-based switch by checking the *HP Insight Remote Support Release Notes*.	
Configure Telnet or SSH on the switch.	
Configure SNMP traps on the switch.	
Add SNMP protocol credentials to the Insight RS Console for monitoring.	
Add Telnet or SSH protocol credentials to the Insight RS Console for collections.	
Discover the ComWare-based switch in the Insight RS Console.	
Verify SNMP communication.	

Install and Configure Communication Software on Switches

To configure your monitored devices, complete the following sections:

Configure Telnet or SSH

To configure trap destinations to point to Insight Remote Support, you need to use Telnet or SSH (or use a serial console) to access the CLI. You also need to configure Telnet or SSH to enable collection services. Use the below procedures to configure Telnet or SSH.

Configuring Telnet

From a serial console, complete the following steps:

1. Enter system view, and enable the Telnet service.

   ```
   <Sysname> system-view
   [Sysname] telnet server enable
   ```

2. Configure an IP address for VLAN interface 1. This address will serve as the destination of the Telnet connection.

   ```
   [Switch] interface vlan-interface 1
   [Switch-Vlan-interface1] ip address <ip_address> 255.255.255.0
   [Switch-Vlan-interface1] quit
   ```

3. Set the authentication mode for the user interfaces to AAA.

   ```
   [Switch] user-interface vty 0 4
   [Switch-ui-vty0-4] authentication-mode scheme
   ```

4. Enable the user interfaces to support Telnet.

   ```
   [Switch-ui-vty0-4] protocol inbound telnet
   [Switch-ui-vty0-4] quit
   ```

5. Create local user `manager`, and set the user command privilege level to 3

   ```
   [Switch] local-user manager
   [Switch-luser-client001] password simple <password>
   [Switch-luser-client001] service-type telnet
   [Switch-luser-client001] authorization-attribute level 3
   [Switch-luser-client001] quit
   ```

Configuring SSH Version 2

From a serial console, complete the following steps:

1. ```
   <Switch> system-view
   [Switch] public-key local create rsa
   [Switch] public-key local create dsa
   [Switch] ssh server enable
   ```

2. Configure an IP address for VLAN interface 1. This address will serve as the destination of the SSH connection.
   ```

```
[Switch] interface vlan-interface 1
[Switch-Vlan-interface1] ip address <ip_address> 255.255.255.0
[Switch-Vlan-interface1] quit
```

3. Set the authentication mode for the user interfaces to AAA.

```
[Switch] user-interface vty 0 4
[Switch-ui-vty0-4] authentication-mode scheme
```

4. Enable the user interfaces to support SSH.

```
[Switch-ui-vty0-4] protocol inbound ssh
[Switch-ui-vty0-4] quit
```

5. Create local user `manager`, and set the user command privilege level to 3.

```
[Switch] local-user manager
[Switch-luser-client001] password simple <password>
[Switch-luser-client001] service-type ssh
[Switch-luser-client001] authorization-attribute level 3
[Switch-luser-client001] quit
```

Verifying Telnet/SSH Communication

To verify that Telnet or SSH communication is working properly, complete the following steps:

1. From the Hosting Device, connect to the ComWare-based switch using the protocol you want to verify. Wait until the connection is established.
2. Type the `username` and `password` when prompted.

   A successful log on confirms that Telnet/SSH is configured properly.

Setup SNMP Traps

From a serial console, or through SSH or Telnet, complete the following steps:

1.
```
<Sysname> system-view
[Sysname] snmp-agent sys-info version v1
[Sysname] snmp-agent community read public
```

2. Enable SNMP traps, and set the Hosting Device as the trap destination. Use `public` as the community name.

```
[Sysname] snmp-agent trap enable
[Sysname] snmp-agent target-host trap address udp-domain <hosting_device_ip>
params securityname public v1
```

Saving the Current Configuration

From a serial console, complete the following steps:

1. Save your configuration changes by issuing the **save** command.
2. When prompted that the "current configuration will be written to the device", type **Y**.
3. Press **Enter** to leave the existing filename unchanged.
4. When prompted to overwrite the existing `*.cfg` file, type **Y**.

A confirmation message appears when the current configuration saves successfully.

Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create an SNMPv1 Protocol Credential in Insight RS Console

If your device's SNMP community string is set to `public` and your community access mode is `read only`, Insight RS automatically associates an SNMPv1 protocol with your device. If you use a different community string or use a nonstandard port, you must create an SNMPv1 protocol credential in the Insight RS Console.

To configure SNMPv1 in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the Select and Configure Protocol drop-down list, select **Simple Network Management Protocol Version 1 (SNMPv1)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Community String you have configured on your device.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Create a Telnet or SSH Protocol in the Insight RS Console

Create a Telnet or SSH protocol credential in the Insight RS Console. You do not need to configure both.

Create a Telnet Protocol in the Insight RS Console

To configure Telnet in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the **Select and Configure Protocol** drop-down list, select **Telnet**.
4. Click **New**. The New Credential dialog box appears.

Insight RS creates the protocol credential and it appears in the credentials table.

Create an SSH Protocol Credential in Insight RS Console

To configure SSH in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the **Select and Configure Protocol** drop-down list, select **Secure Shell (SSH)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Username and Password you have configured on your device.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

Important: ICMP is used by Insight RS for discovery. Make sure you have ICMP enabled on your device.

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP address(es) of the devices to be discovered.
   d. Click **Add**.
4. Click **Start Discovery**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon ( ).

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Warranty and Contract Information

Verify that the warranty and contract information was discovered correctly in the Insight RS Console:

1. In the Insight RS Console, navigate to **Devices** and click the switch Device Name.
2. Expand the Hardware section, and make sure the **Acquired Serial Number** and **Acquired Product Number** are correct. If they are not correct, type the correct values in the **Override Serial Number** and **Override Product Number** fields and click **Save Changes**.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following sections:

Verifying SNMP Communication

There is not a way to manually verify the SNMP configuration without generating an event, which can only be accomplished by performing an action like pulling a fan tray or disconnecting a power supply.

Note: Do not pull a redundant FRU to test the configuration.

You can verify SNMP communication by reviewing the running configuration from the ComWare-based switch command line. Look for the `snmp-agent` section in the output and verify that the community string is correct and the trap destination is the Hosting Device.

To display the running configuration, run the following command:

```
> display current-configuration
```

```
#
 snmp-agent
 snmp-agent community read public
 snmp-agent sys-info version v1
 snmp-agent target-host trap address udp-domain <hosting_device_ip> params
securityname public v1
#
```

If any of the settings are incorrect, use the **`undo`** command to remove the setting, for example, **`undo snmp-agent community read public`**. Then return to "Setup SNMP Traps" on page 280 and reconfigure the setting that was incorrect.

Verify Collections in the Insight RS Console

After discovery, a Network Configuration Collection is automatically run for servers. Verify that this collection ran successfully on the Collection Services → Basic Collection Results tab in the Insight RS Console. Note that there is a delay between when discovery finishes and when the collection runs. For more information about scheduling collections, see the Insight RS Help.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services** and then click the **Basic Collection Results** tab.
3. Expand the Network Configuration Collection section.
4. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears ( ). If it failed, an error icon appears ( ).

Chapter 35: Configuring Uninterruptable Power Supplies

Uninterruptable Power Systems (UPS) come with either a Management Module or a Network Module installed, depending on the model. Use the appropriate configuration procedure below for your model:

- To configure a UPS that uses a Management Module, see "Configuring UPS Management Modules" below.
- To configure a UPS that uses a Network Module, see "Configuring UPS Network Modules" on page 288.

Configuring UPS Management Modules

The Uninterruptable Power Systems (UPS) Management Module is an option in several HP Rack-Mountable and Tower UPS units, and these units can be monitored by Insight Remote Support through the module.

Important: Configuration collections are not supported.

Fulfill Configuration Requirements

To configure your UPS Management Modules to be monitored by Insight RS, complete the following sections:

Table 35.1 ***UPS Management Module Configuration Steps***

Task	Complete?
Make sure Insight RS supports your UPS Management Module by checking the *HP Insight Remote Support Release Notes*.	
Configure SNMP on the UPS Management Module and set the Hosting Device as a trap receiver.	
Add the SNMP protocol to the Insight RS Console.	
Discover the UPS Management Module in the Insight RS Console.	
Send a test event to verify connectivity between your UPS Management Module and Insight RS.	

Install and Configure Communication Software on Management Modules

To configure your monitored devices, complete the following sections:

Configure SNMP

SNMP can be setup in the SNMP Traps screen in the UPS Management Module web interface.

For more details about configuring SNMP, see the *HP UPS Management Module User Guide*.

To configure SNMP trap notifications, complete the following steps:

1. Log on to the UPS Management Module web interface.
2. Click the **Setup** tab, and in the left menu, click **Event Notifications**.
3. Add the Insight RS Hosting Device as an SNMP trap recipient on the SNMP Traps tab.
   a. Click the **SNMP Traps** tab. This screen enables administrators to configure SNMP trap event notifications.
   b. To enable SNMP traps for a server, select the **Enable** option.
   c. Enter the IP address of the Hosting Device in the **IP Address** field.
   d. Enter the community string of the Hosting Device in the **Community** field.
   e. Click **Save Settings**.
4. Configure the management module to send event notifications to the Insight RS Hosting Device on the Events tab.
   a. Click the **Events** tab. This screen enables administrators to define the event notifications, emails, or SNMP traps the management module sends for each event.
   b. For each event description listed, select the **Enabled** check box to indicate that email notifications or SNMP traps are sent for that event. To enable all events, click the **Email** check box and the **SNMP Trap** check box at the top of each column.
   c. For each email and SNMP trap enabled, enter the number of minutes that should pass between the occurrence of an alert condition and the sending of the notification.

   Note: If the event clears before the delay time has expired, then the event notification is not sent.

   d. Click **Save Settings**.

Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create an SNMPv1 Protocol Credential in Insight RS Console

If your device's SNMP community string is set to `public` and your community access mode is `read only`, Insight RS automatically associates an SNMPv1 protocol with your device. If you use a different community string or use a nonstandard port, you must create an SNMPv1 protocol credential in the Insight RS Console.

To configure SNMPv1 in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.

3. From the Select and Configure Protocol drop-down list, select **Simple Network Management Protocol Version 1 (SNMPv1)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Community String you have configured on your device.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP address(es) of the devices to be discovered.
   d. Click **Add**.
4. Click **Start Discovery**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon ( ).

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following section:

Send a Test Trap

To send a test trap, complete the following steps:

1. Log on to the UPS Management Module web interface.
2. Click the **Setup** tab.
3. In the left menu, click **Event Notifications**, then the **SNMP Traps** tab.

4. Click **Send Test Trap** to send a test SNMP trap.
5. Verify that Insight Remote Support received the test trap.

Configuring UPS Network Modules

The Uninterruptable Power Systems (UPS) Network Module is an option in several HP Rack-Mountable and Tower UPS units, and these units can be monitored by Insight Remote Support through the module.

Important: Configuration collections are not supported.

Fulfill Configuration Requirements

To configure your UPS Network Modules to be monitored by Insight RS, complete the following sections:

Table 35.2 ***UPS Network Module Configuration Steps***

Task	Complete?
Make sure Insight RS supports your UPS Network Module by checking the *HP Insight Remote Support Release Notes*.	
Configure SNMP on the UPS Network Module and set the Hosting Device as a trap receiver.	
Add the SNMP protocol to the Insight RS Console.	
Discover the UPS Network Module in the Insight RS Console.	
Send a test event to verify connectivity between your UPS Network Module and Insight RS.	

Install and Configure Communication Software on Network Modules

To configure your monitored devices, complete the following sections:

Configure SNMP

SNMP can be setup in the SNMP Settings screen in the UPS Network Module web interface. The SNMP Settings screen allows an administrator to configure SNMP settings for computers that use the HP Power MIB to request information from the UPS Network Module.

For more details about configuring SNMP, see the *HP UPS Network Module User Guide*.

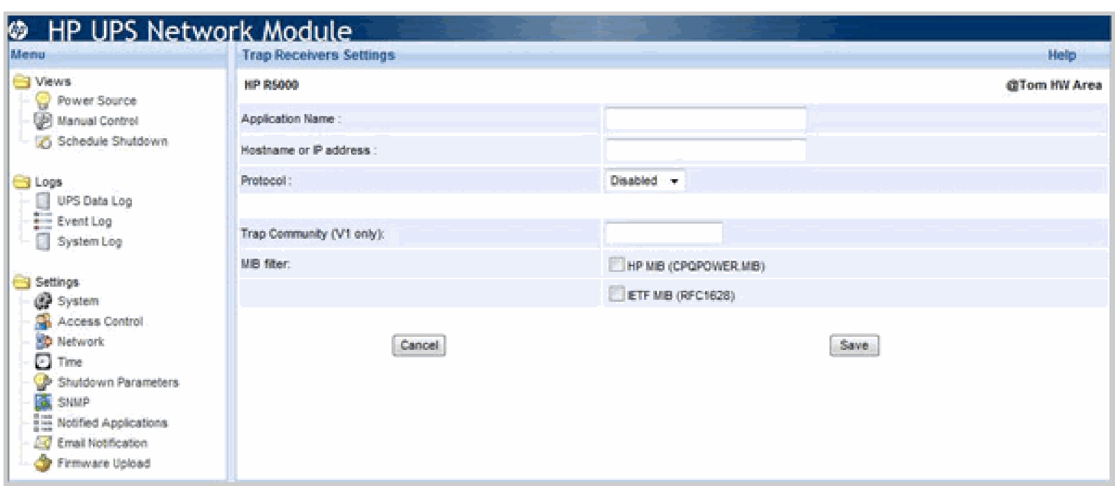
To configure SNMP, complete the following steps:

1. Log on to the UPS Network Module web interface.
2. In the left menu, click **SNMP**.

3. Select the SNMP version from the SNMP Version drop-down list.
4. Fill in the appropriate fields.
5. Click **Save**.
6. Set the Hosting Device as a trap receiver UPS Network Module web interface. The Trap Receivers Settings screen allows an administrator to configure management applications to receive SNMP traps from the UPS Network Module. SNMP management applications, such as Insight Remote Support, can receive notifications from the UPS Network Module.

   To configure the Hosting Device as an SNMP trap recipient, complete the following steps:

   a. In the left menu, click **Notified Applications**, then click **Add Trap Receiver**. Configure up to three applications to receive SNMP traps from the UPS Network Module.



   b. Type the name of the application, such as Insight RS, in the **Application Name** field. HP recommends adding “SNMP” or “Trap” to the name to for easy monitoring.

c. Type the host name or the IP address of the management server on which the application is running in the **Hostname or IP address** field.

d. Select the SNMP version from the **Protocol** drop-down menu.

e. If you selected SNMPv1, type the community string in the **Trap Community** field.

f. Select the check box for the appropriate MIB:

   - **HP MIB (cpqpower.mib)**—The HP Power MIB
   - **IETF MIB (RFC1628)**—A standard UPS MIB

g. Click **Save**. The application information appears on the Notified Applications screen.

Configure Firewall and Port Settings

For a complete list of firewall and port requirements for monitored devices, refer to the *HP Insight Remote Support Security White Paper* at: http://www.hp.com/go/insightremotesupport/docs.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create an SNMPv1 Protocol Credential in Insight RS Console

If your device's SNMP community string is set to `public` and your community access mode is `read only`, Insight RS automatically associates an SNMPv1 protocol with your device. If you use a different community string or use a nonstandard port, you must create an SNMPv1 protocol credential in the Insight RS Console.

To configure SNMPv1 in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the Select and Configure Protocol drop-down list, select **Simple Network Management Protocol Version 1 (SNMPv1)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Community String you have configured on your device.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:

a. Click **New**.

b. Select the **Single Address**, **Address Range**, or **Address List** option.

c. Type the IP address(es) of the devices to be discovered.

d. Click **Add**.

4. Click **Start Discovery**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon ( ).

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

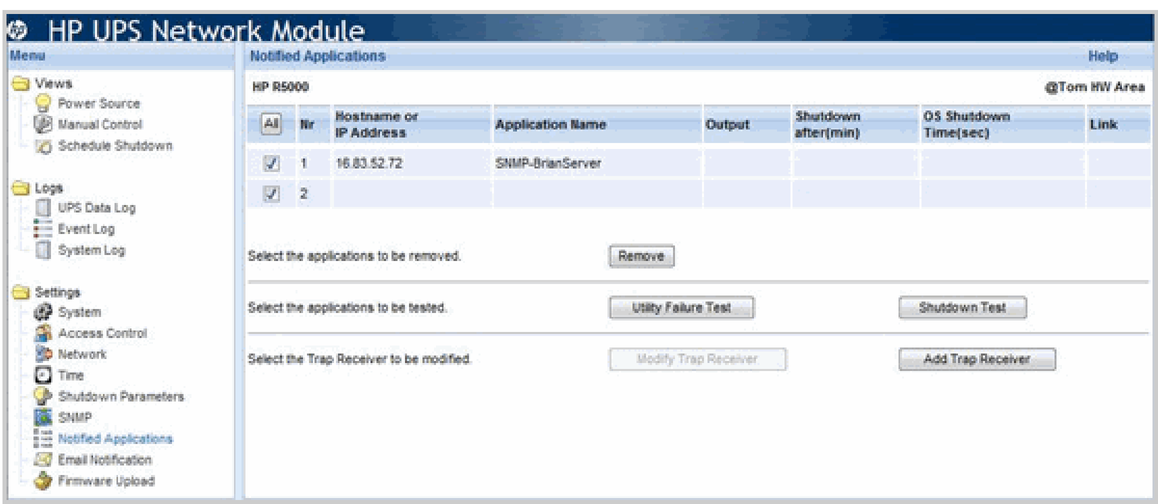
Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following section:

Send a Test Trap

To send a test trap, complete the following steps:

1. Log on to the UPS Network Module web interface.
2. In the left menu, click **Notified Applications**, then click **Add Trap Receiver**. Up to three applications can be configured to receive SNMP traps from the UPS Network Module.



3. Click **Utility Failure Test**. The UPS Network Module sends a Utility failure trap, and then sends a

Utility restored trap 30 seconds later.

4. Verify that Insight Remote Support received the test trap.

Chapter 36: Configuring Modular Cooling Systems

Modular Cooling Systems are monitored through the management module that is installed on the Modular Cooling System. Insight Remote Support automatically detects management modules as part of the device discovery process. The management module should be installed and running before attempting discovery.

Important: Configuration collections are not supported.

Fulfill Configuration Requirements

To configure your Modular Cooling Systems to be monitored by Insight RS, complete the following sections:

Table 36.1 ***Modular Cooling System Configuration Steps***

Task	Complete?
Make sure Insight RS supports your Modular Cooling System by checking the *HP Insight Remote Support Release Notes*.	
Configure SNMP on the Modular Cooling System and set the Hosting Device as a trap receiver	
Add the SNMP protocol to the Insight RS Console.	
Discover the Modular Cooling System in the Insight RS Console.	
Send a test event to verify connectivity between your Modular Cooling System and Insight RS.	

Install and Configure Communication Software on Modular Cooling Systems

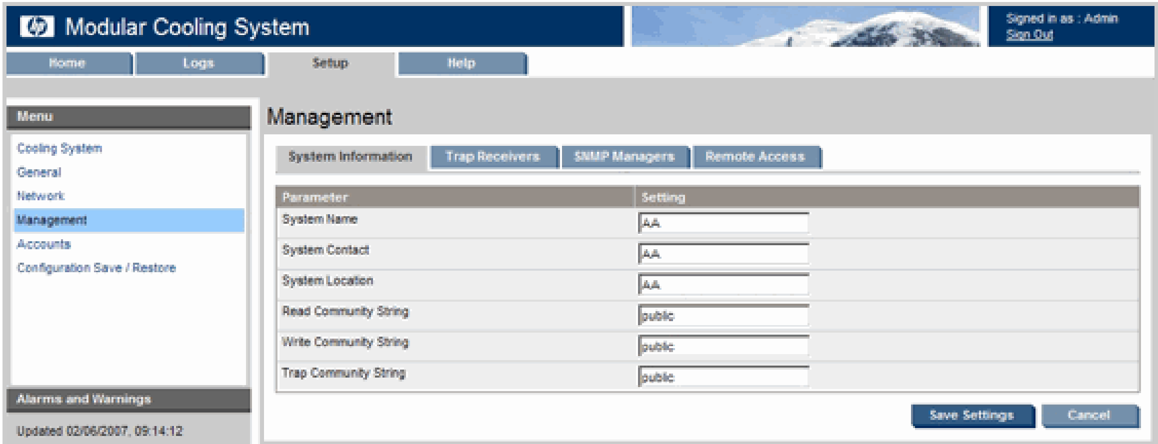
To configure your monitored devices, complete the following section:

Configure SNMP on the Modular Cooling System

SNMP must be configured on the Modular Cooling System so that Insight Remote Support can communicate with it.

To configure SNMP, complete the following steps:

1. Log on to the Modular Cooling System web interface.
2. Navigate to **Setup** → **Management** → **System Information**. The information configured on this screen included with SNMP traps sent by the management module.

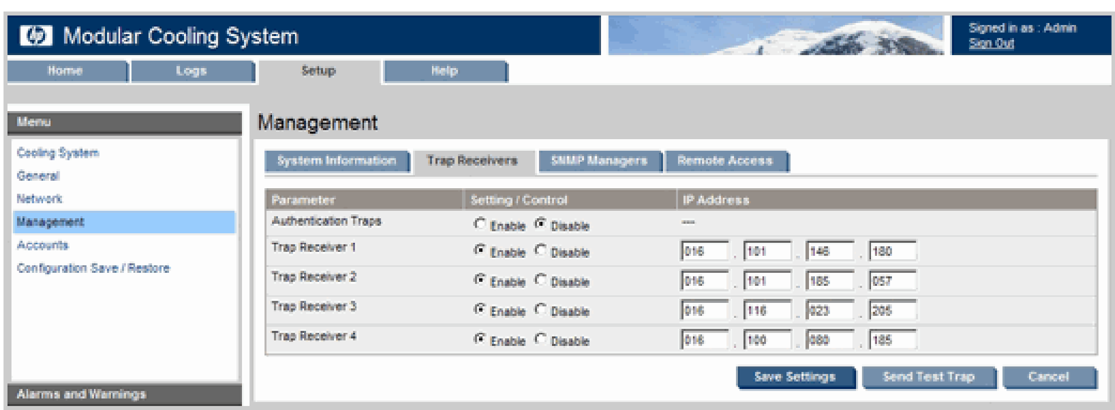


3. Type the name of the management module in the System Name field.
4. Type the name of the contact person in the System Contact field.
5. Type the name of the location in the System Location field.
6. Type the Read community string.
7. Type the Write community string.
8. Type the Trap community string.
9. Click **Save Settings** to save the information.
10. In the Modular Cooling System web interface, set the Hosting Device as a trap receiver. The Trap Receivers screen enables the Admin to add information for servers that should receive SNMP traps from the management module.

    To configure the Hosting Device to receive SNMP traps, complete the following steps:

    a. Log on to the Modular Cooling System web interface.

    b. Navigate to **Setup** → **Management** → **Trap Receivers**.



    c. Click to enable or disable SNMP authentication traps from the Authentication Traps option.

    d. Enable an SNMP trap receiver from the Trap Receivers 1 through 4 options.

e. Type the IP address for up to four trap recipients in the IP Address fields.
f. Click **Save Settings** to save the information.

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create an SNMPv1 Protocol Credential in Insight RS Console

If your device's SNMP community string is set to `public` and your community access mode is `read only`, Insight RS automatically associates an SNMPv1 protocol with your device. If you use a different community string or use a nonstandard port, you must create an SNMPv1 protocol credential in the Insight RS Console.

To configure SNMPv1 in the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. From the Select and Configure Protocol drop-down list, select **Simple Network Management Protocol Version 1 (SNMPv1)**.
4. Click **New**. The New Credential dialog box appears.
5. Type the Community String you have configured on your device.
6. Click **Add**.

Insight RS creates the protocol credential and it appears in the credentials table.

Discover the Device in the Insight RS Console

To discover the device from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP address(es) of the devices to be discovered.
   d. Click **Add**.
4. Click **Start Discovery**.

Verify Discovery and Device Status

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon (✅).

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following section:

Send a Test Trap

To send a test trap to the Hosting Device, complete the following steps:

1. Log on to the Modular Cooling System web interface.
2. Navigate to **Setup** → **Management** → **Trap Receivers**.
3. Click **Send Test Trap** to send a test SNMP trap to all enabled trap receivers.

Chapter 37: Configuring VMware vCenter Servers

Insight RS can collect virtual machine configuration information for your IT environment from your vCenter installation. Insight RS automatically discovers any VMware vCenter installations when the ProLiant Windows server it is installed on is discovered as long as the proper security credentials are configured.

Insight RS supports collections for vCenter virtual machine guests, but does not support monitoring of the virtual machine guests.

Note: vCenter installations on a virtual machine are not supported.

Communication with a vCenter installation requires the configuration of security credentials within the Insight RS Console. Once discovered, information about the virtual machines on any server in your environment can be collected and sent automatically to HP on a scheduled basis. The vCenter cluster uses the same warranty and contract information as the ProLiant server it's installed on. When the ESX and ESXi servers are discovered, Insight RS does not have the ability to tell which vCenter cluster manages them.

The vCenter collection gathers information about the VM Guests and how they are configured on the VMware vCenter cluster. There must be a Proactive Care contract on at least one of the ESX or ESXi devices that make up the cluster. This is an environment wide contract based on a single server entitlement, not an entitlement against each of the servers on which virtual machines are installed.

The vCenter cluster appears in the Insight RS Console with a name of the form: `VC@`*`<hostname>`*.

Note: Although vCenter installations are visible in the Insight RS Console, they are not monitored for events.

Insight RS does not differentiate between multiple DataCenter objects or between multiple cluster objects being managed by a given vCenter instance. Insight RS 7.1 does not have hierarchical capability and therefore cannot represent these constructs. All hosts from all clusters are added to Insight RS without differentiating the various clusters in which they may be configured.

Insight RS can use CIM ticket authentication on ESXi clusters managed by vSphere, which offers the following advantages:

- It allows Insight RS to communicate with the ESXi hosts when they are in lockdown mode.

  Important: Insight RS supports ESXi hosts in lockdown mode when they are managed by vCenter in a cluster configuration. Standalone ESXi hosts managed by vCenter are not supported.

- It allows Insight RS to discover ESXi hosts when it discovers the vCenter server without having to enter WBEM credentials for each ESXi host; the CIM ticket is used for the credentials instead.

To enable CIM ticketing for ESXi devices, whether in lockdown mode or not, you must provide the credentials for the VMware vCenter cluster to the discovery tasks (see "Create a VMWare VirtualCenter Web Service Interface Protocol Credential in the Insight RS Console" on the next page.) If any of the ESXi hosts are Gen8 ProLiant devices, then appropriate RIBCL credentials must also be provided prior to discovery.

Important: The CIM ticket capability only applies to ESXi, not to ESX clusters. When Insight RS discovers the vCenter server, it will also discover ESX hosts, but you must enter the SNMP credentials in Insight RS before discovery for them to be monitored correctly.

Fulfill Configuration Requirements

To configure your VMware vCenter servers to be monitored by Insight RS, complete the following sections:

Table 37.1 ***ProLiant VMware vCenter Server Configuration Steps***

Task	Complete?
Make sure Insight RS supports your ProLiant VMware ESX server by checking the *HP Insight Remote Support Release Notes*.	
Add VMware vCenter Protocol Credentials in the Insight RS Console.	
Discover the ProLiant vCenter server in the Insight RS Console.	
Verify the status of the ProLiant VMware ESX server in the Insight RS Console.	

Add Protocol Credentials and Start Discovery

To discover your monitored devices, complete the following sections:

Create a VMWare VirtualCenter Web Service Interface Protocol Credential in the Insight RS Console

For the vCenter cluster to be discovered, you need to add the credentials you use to log into vCenter. If these credentials are not added, the vCenter cluster will not be discovered. Adding this credential enables CIM ticketing for ESXi devices, whether in lockdown mode or not.

Important: If any of the ESXi hosts are Gen8 ProLiant devices, then appropriate RIBCL credentials must also be provided prior to discovery.

To configure VMWare VirtualCenter Web Service Interface in the Insight RS Console, complete the following steps:

1. Log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Credentials** tab.
3. In the **Select and Configure Protocol** drop-down list, select **VMWare VirtualCenter Web Service Interface**.
4. Click **New**. The New Credential dialog box appears.

5. Type the username and password you use to log on to vCenter.
6. Click **Add**.

Discover the ProLiant vCenter Server in the Insight RS Console

To discover the ProLiant vCenter server from the Insight RS Console, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Discovery** and click the **Sources** tab.
3. Expand the IP Addresses section and add the IP address for your device:
   a. Click **New**.
   b. Select the **Single Address**, **Address Range**, or **Address List** option.
   c. Type the IP address(es) of the devices to be discovered. Use the IP address of the ProLiant server where vCenter is installed, and as long as you have the vCenter credentials configured in the Insight RS Console, vCenter will be discovered as well.
   d. Click **Add**.
4. Click **Start Discovery**.

Note: The vCenter discovery causes the ESX and ESXi devices it manages to be discovered as long as proper credentials are provided. For ESXi, proper credentials could be WBEM or could be the vCenter CIM ticket. Make sure the ESX and ESXi devices have proper credentials configured.

Verify the Status of ProLiant vCenter Server in the Insight RS Console

To verify the device was discovered correctly, complete the following steps:

1. In a web browser, log on to the Insight RS Console.
2. In the main menu, select **Devices** and click the **Device Summary** tab.
3. Make sure the Status column contains a success icon ( ).

   If there is no success icon, determine which column has the problem and consult the Online Help for troubleshooting information.

Verify Communication Between Monitored Device and Insight RS

To verify communication between your monitored device and Insight RS, complete the following section:

Verify Collections in the Insight RS Console

After discovery, a VMware vCenter Server Collection is automatically run for vCenter servers. Verify that this collection ran successfully on the Collection Services → Basic Collection Results tab in the Insight RS Console. Note that there is a delay between when discovery finishes and when the collection runs. For more information about scheduling collections, see the Insight RS Help.

1. Log on to the Insight RS Console.
2. In the main menu, select **Collection Services** and then click the **Basic Collection Results** tab.
3. Expand the VMware vCenter Server Collection section.
4. Locate the entry for your device and check the Result column. If the collection was successful, a success icon appears ( ). If it failed, an error icon appears ( ).

Glossary

A

Agentless Management Service (AMS)

iLO 4 Agentless Management uses out-of-band communication for increased security and stability. With Agentless Management, health monitoring and alerting is built into the system and begins working the moment a power cord is connected to the server. This feature runs on the iLO hardware, independent of the operating system and processor. The separately installed AMS collects additional operating system data.

C

CDID

See Custom Delivery ID.

Central Management Server (CMS)

The CMS is a system in the management domain that executes the HP Systems Insight Manager software. All central operations within HP Systems Insight Manager are initiated from this system.

Centralized Management Console (CMC)

The CMC is used to configure and manage the P4000 Storage Systems.

CLIQ

Legacy term for the LeftHand OS command line interface. See P4000 CLI.

cluster

A cluster is a grouping of storage nodes that create the storage pool from which you create volumes.

collection schedules

The frequency in which collections are run. The schedule is configurable in the Insight RS Console.

collections

The term within HP SIM for grouping system or event searches. While HP SIM uses the term collection in reference to a group of monitored devices, Insight Remote Support uses the term configuration collection in reference to data collected from a monitored device. This data is communicated to HP for proactive analysis.

configuration collection

HP Insight Remote Support uses the term configuration collection in reference to data collected from a monitored device. This data is sent to HP for proactive analysis.

Custom Delivery ID

The Custom Delivery ID is a free text field that can be individually populated for each monitored device. Unless a specific value is specified within the documentation relating to configuring a particular device, the field should be left blank. In some very specific circumstances, this field may also be populated by an HP Representative to allow customized handling/routing of reported incidents. In those instances, the field must be populated with a unique value associated with the customized handling that is required. Determination of the unique value must be done by the HP Representative working with the HP Automation Team TS that sets up the customizations. Failure to follow this guidance could result in incorrectly handled incident reports.

D

device groups

Configurable groups of devices within the Insight RS Console that helps users organize the devices in their environment.

discovery

A feature within a management application that finds and identifies network objects. In HP management applications, discovery finds and identifies all the devices within a specified network range.

E

Enterprise Virtual Array (EVA)

An EVA is a high performance, high capacity and high availability virtual RAID storage solution for high-end enterprise environments.

entitlement

The process of authorizing a request for support based on the contents of warranty or support contracts held by the customer, normally with respect to a specific Object of Service (OOS) such as a hardware or software component. Your level of entitlement is determined by your HP Support contract. Contact your HP Account Team for more details.

event

A general term for all types of notifications from one process to another.

Event Log Monitoring Collector (ELMC)

ELMC provides error condition detection of the event log and communicates these events to Insight RS.

H

hardware event

A specific type of event that suggests that a specific hardware component may be in trouble. Hardware events may result in a service event.

Health Check

The LeftHand Networks Health Check Utility is used to send monitoring log file information from customer sites to LeftHand Networks for troubleshooting and proactive health monitoring. See Service Console.

Hosting Device

The Hosting Device is a supported Windows ProLiant server that hosts the Insight Remote Support software. When used with HP Systems Insight Manager, the Hosting Device can act as a Central Management Server (CMS).

Hosting Device Setup Wizard

Step-by-step screens that helps users with the initial configuration of their Hosting Device.

HP Authorized Reseller/Distributor

Channel partners who sell hardware and services.

HP Authorized Service Partner

Channel partners who deliver services and/or installation service on HP’s behalf.

HP Care Pack Services

HP Care Packs can be purchased with HP products and services to upgrade or extend standard warranties with enhanced support packages. They reduce downtime risks with support levels from basic to mission-critical. For many products, post-warranty HP Care Pack Services are also available when an original warranty expires.

HP Passport (HPP)

HP Passport single sign-in service lets you use one user ID and password of your choice to sign-in to all HP Passport-enabled Web sites.

I

identification

An aspect of the discovery process that identifies the management protocol and type of system.

iLO

Integrated Lights-Out. Embedded server management technology that delivers web-based remote management that is always available.

iLO Remote Insight Board Command Language (RIBCL)

Communication protocol required for Insight Remote Support to communicate with ProLiant Gen8 servers.

Insight Online

HP Insight Online provides one-stop personalized, secure access to support the devices in your IT environment. It is integrated into HP Support Center for your IT staff who deploy, manage and support systems, plus HP Authorized Channel Partners who support your IT infrastructure. Insight Online can automatically discover devices remotely monitored by HP (requires Insight Remote Support 7.0 or later). Depending on your support model you or your HP Authorized Channel Partner can easily organize your devices into groups and have the flexibility to efficiently monitor, track and service your HP devices.

Insight Remote Support

HP Insight Remote Support provides proactive remote monitoring, diagnostics, and troubleshooting to help improve the availability of supported HP servers and storage systems in your data center. HP Insight Remote Support reduces cost and complexity through support of systems. HP Insight Remote Support securely communicates hardware incident information through your firewall and/or web proxy to the HP Data Center for reactive support. Additionally, based on your support agreement, system information can be collected for proactive analysis and services

Insight RS Console

The Insight Remote Support user interface that is installed on the Hosting Device.

Insight RSA

Insight Remote Support Advanced. Previous version of Insight Remote Support that integrates with HP SIM to provide proactive remote monitoring, diagnostics, and troubleshooting of devices.

M

management group

A collection of one or more storage nodes which serves as the container within which you cluster storage nodes and create volumes for storage.

Management Information Base (MIB)

The data specification for passing information using the SNMP protocol. An HP MIB is also a database of managed objects accessed by network management protocols.

management protocol

A set of protocols, such as WBEM, HTTP, or SNMP, used to establish communication with discovered systems.

monitored device

Any device monitored by HP Insight Remote Support, such as servers, storage systems, and switches. To monitor a device, some type of management protocol (for example, SNMP or WBEM) must be present on the device.

Monitored Device Setup Wizard

Step-by-step screens that helps users discover devices in their environment to be monitored. Users specify the range of devices to be discovered and the corresponding credentials.

P

P4000 CLI

The P4000 CLI is the command line interface that is used to interface with the P4000 Storage Systems from the Hosting Device. The P4000 CLI is installed with Insight Remote Support. Note that the P4000 CLI is sometimes referred to as cliq, which is the name of the command used within the P4000 CLI.

R

Remote Support Eligible Systems

Systems that are eligible for Remote Support, and when enabled will submit events to the HP Data Center for incident resolution. Systems must also be entitled to Remote Support, otherwise submitted events will be closed. You can verify that an eligible system is actually supported by using the Remote Support Entitlement Check.

Remote Support Entitlement Check (RSEC)

The RSEC is a check against the HP entitlement datastore for the current obligation status of a particular system. The Entitlement window displays the results of the Remote Support Entitlement Check.

RIBCL

See iLO Remote Insight Board Command Language.

S

Service Console

The Service Console is the legacy software that enabled remote hardware and software support for P4000 Storage Systems. This functionality is now provided by Insight Remote Support. See Health Check.

service event

Insight RS monitors a customer’s hardware environment for service events that require action from the Service Provider (HP Channel Partner and/or HP services team) or customer. For the most part, actionable service events require replacing a failed Field Replaceable Unit (FRU) or Customer Replaceable Unit (CRU) in the monitored hardware. Non-actionable events are filtered out, and actionable service events are forwarded to HP. Insight RS monitors hardware for hard failures from a major component on the hardware such as CPU, disk, memory, and power supply, and these will trigger a service event communication to HP. In addition, Insight RS monitors hardware for soft errors, and in some instances, when the number of soft errors exceeds a specific threshold, an event is sent to HP. Further details regarding what classifies as an actionable service event is HP confidential information.

Simple Network Management Protocol (SNMP)

One of the management protocols supported by Insight Remote Support. Traditional management protocol used extensively by networking systems and most servers. MIB-2 is the standard information available consistently across all vendors.

SNMP trap

Asynchronous event generated by an SNMP agent that the system uses to communicate a fault.

Storage Area Network (SAN)

A SAN is a network of storage devices and the initiators that store and retrieve information on those devices, including the communication infrastructure. In large enterprises, a SAN connects multiple servers to a centralized pool of disk storage. Compared to managing hundreds of servers, each with their own disks, SANs improve system administration.

Storage Management Server (SMS)

A system on which HP Enterprise Virtual Array (EVA) software is installed, including HP P6000 Command View and HP Replication Solutions Manager, if used. It is a dedicated management server that runs EVA management software exclusively.

System Fault Management (SFM or SysFaultMgmt)

SFM is the HP-UX fault management solution that implements WBEM standards. SFM integrates with other manageability applications like HP SIM, HP SMH, and other WBEM-based clients.

system health

Health status is an aggregate status all of the status sources (which can be SNMP, WBEM, and HTTP) with the most critical status being displayed.

System Management Homepage (SMH)

System Management Homepage (SMH) is a web-based interface that consolidates and simplifies single system management for HP servers on HP-UX, Linux, and Windows operating systems.

Systems Insight Manager (SIM)

SIM is a unified server and storage management platform. From a single management console, administrators can manage their complete HP server and storage environment with a secure management tool set.

V

virtual node

The P4000 Virtual SAN Appliance uses captive server disk drives to build a virtual iSCSI SAN consisting of Virtual Nodes that create the storage pool from which virtualized volumes are created. Virtual Nodes can be discovered and managed in the same manner as physical Storage Nodes.

volume

A logical entity that is made up of storage on one or more storage nodes. It can be used as raw data storage or it can be formatted with a file system and used by a host or file server.

W

Web-Based Enterprise Management (WBEM)

This industry initiative provides management of systems, networks, users, and applications across multiple vendor environments. WBEM simplifies system management, providing better access to software and hardware data that is readable by WBEM client applications.

Windows Management Instrumentation (WMI)

Microsoft's implementation of Web-Based Enterprise Management (WBEM).

workflow case

The specific HP obligation to send a replacement part for a failed customer part.

Index